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**Department of Defense
Fiscal Year (FY) 2022 Budget Estimates**

May 2021



Navy

Justification Book Volume 1 of 1

Shipbuilding and Conversion, Navy

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The estimated cost of this report for the Department of the Navy (DON) is \$90,349. The estimated total cost for supporting the DON budget justification material is approximately \$3,919,738 during the 2021 fiscal year. This includes \$84,638 in supplies and \$3,835,100 in labor.

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Navy • Budget Estimates FY 2022 • Procurement

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Department of Defense Appropriations Act, 2022

Shipbuilding and Conversion, Navy

For expenses necessary for the construction, acquisition, or conversion of vessels as authorized by law, including armor and armament thereof, plant equipment, appliances, and machine tools and installation thereof in public and private plants; reserve plant and Government and contractor-owned equipment layaway; procurement of critical, long lead time components and designs for vessels to be constructed or converted in the future; and expansion of public and private plants, including land necessary therefore, and such lands and interests therein, may be acquired, and construction prosecuted thereon prior to approval of title.

In all: \$22,571,059 to remain available for obligation until September 30, 2033: *Provided*, That additional obligations may be incurred after September 30, 2033, for engineering services, tests, evaluations, and other such budgeted work that must be performed in the final stage of ship construction: *Provided further*, That none of the funds provided under this heading for the construction or conversion of any naval vessel to be constructed in shipyards in the United States shall be expended in foreign facilities for the construction of major components of such vessel: *Provided further*, That none of the funds provided under this heading shall be used for the construction of any naval vessel in foreign shipyards.

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Department of the Navy
 FY 2022 President's Budget
 Exhibit P-1 FY 2022 President's Budget
 Total Obligational Authority
 (Dollars in Thousands)

10 May 2021

Appropriation -----	FY 2020 Actual* -----	FY 2021 Enacted** -----	FY 2022 Request -----
Shipbuilding and Conversion, Navy	21,243,451	23,268,880	22,571,059
Total Department of the Navy	21,243,451	23,268,880	22,571,059

P-122BAS: FY 2022 President's Budget (Total Base Published Version), as of May 10, 2021 at 09:45:11

*Includes Division A, Title IX and X of the Consolidated Appropriations Act, 2020 (P.L. 116-93), Division F, Title IV and V from the Further Consolidated Appropriations Act, 2020 (P.L. 116-94) and the Coronavirus Aid, Relief, and Economic Security Act (P.L. 116-136).

** Includes Division C, Title IX and Division J, Title IV of the Consolidated Appropriations Act, 2021 (P.L. 116-260).

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Department of the Navy
 FY 2022 President's Budget
 Exhibit P-1 FY 2022 President's Budget
 Total Obligational Authority
 (Dollars in Thousands)

10 May 2021

Appropriation: Shipbuilding and Conversion, Navy

Budget Activity -----	FY 2020 Actual*	FY 2021 Enacted**	FY 2022 Request -----
01. Fleet Ballistic Missile Ships		4,122,199	4,646,980
02. Other Warships	18,523,203	15,400,770	14,490,780
03. Amphibious Ships	611,100	1,960,801	129,273
05. Auxiliaries, Craft, and Prior-Year Program Costs	2,109,148	1,785,110	3,304,026
Total Shipbuilding and Conversion, Navy	21,243,451	23,268,880	22,571,059

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*Includes Division A, Title IX and X of the Consolidated Appropriations Act, 2020 (P.L. 116-93), Division F, Title IV and V from the Further Consolidated Appropriations Act, 2020 (P.L. 116-94) and the Coronavirus Aid, Relief, and Economic Security Act (P.L. 116-136).

** Includes Division C, Title IX and Division J, Title IV of the Consolidated Appropriations Act, 2021 (P.L. 116-260).

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Department of the Navy
 FY 2022 President's Budget
 Exhibit P-1 FY 2022 President's Budget
 Total Obligational Authority
 (Dollars in Thousands)

10 May 2021

Appropriation: 1611N Shipbuilding and Conversion, Navy

Line No	Item Nomenclature	Ident Code	FY 2020 Actual* Quantity Cost	FY 2021 Enacted** Quantity Cost	FY 2022 Request Quantity Cost	S e c
----	-----	-----	-----	-----	-----	-
Budget Activity 01: Fleet Ballistic Missile Ships						

Fleet Ballistic Missile Ships						
1	OHIO Replacement Submarine	A		1 (15,030,533)		U
	Less: Advance Procurement (PY)			(-6,227,811)		U
	Less: Subsequent Full Funding (FY)			(-5,933,698)		U
			-----	-----	-----	
				2,869,024		
	Subsequent Full Funding for FY 2021				3,003,000	
2	OHIO Replacement Submarine					
	Advance Procurement (CY)			1,253,175	1,643,980	U
	C (FY 2021 for FY 2024) (M)			(1,110,652)		
	C (FY 2021 for FY 2026) (M)			(93,034)		
	C (FY 2021 for FY 2027) (M)			(140)		
	C (FY 2021 for FY 2028) (M)			(1,070)		
	C (FY 2021 for FY 2029) (M)			(140)		
	C (FY 2021 for FY 2030) (M)			(140)		
	C (FY 2021 for FY 2031) (M)			(140)		
	C (FY 2021 for FY 2032) (M)			(140)		
	C (FY 2021 for FY 2033) (M)			(140)		
	C (FY 2021 for FY 2034) (M)			(140)		
	C (FY 2021 for FY 2035) (M)			(47,439)		
	C (FY 2022 for FY 2024) (M)				(1,251,428)	
	C (FY 2022 for FY 2026) (M)				(149,899)	
	C (FY 2022 for FY 2027) (M)				(77,973)	
	C (FY 2022 for FY 2028) (M)				(10,667)	
	C (FY 2022 for FY 2029) (M)				(2,241)	
	C (FY 2022 for FY 2030) (M)				(892)	
	C (FY 2022 for FY 2031) (M)				(893)	
	C (FY 2022 for FY 2032) (M)				(892)	
	C (FY 2022 for FY 2033) (M)				(892)	
	C (FY 2022 for FY 2034) (M)				(892)	
	C (FY 2022 for FY 2035) (M)				(147,311)	
			-----	-----	-----	
	Total Fleet Ballistic Missile Ships			4,122,199	4,646,980	

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** Includes Division C, Title IX and Division J, Title IV of the Consolidated Appropriations Act, 2021 (P.L. 116-260).

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Department of the Navy
FY 2022 President's Budget
Exhibit P-1 FY 2022 President's Budget
Total Obligational Authority
(Dollars in Thousands)

10 May 2021

Appropriation: 1611N Shipbuilding and Conversion, Navy

Line No	Item Nomenclature	Ident Code	FY 2020 Actual* Quantity Cost	FY 2021 Enacted** Quantity Cost	FY 2022 Request Quantity Cost	S e c
----	-----	----	-----	-----	-----	-
Budget Activity 02: Other Warships						

Other Warships						
3	Carrier Replacement Program	A	786,682			U
	Subsequent Full Funding for FY 2018		1,062,000	958,933	1,068,705	
4	CVN-81	A	1 (12,324,167)			U
	Less: Subsequent Full Funding (FY)		(-11,896,349)			U
			-----	-----	-----	
			427,818			
	Subsequent Full Funding for FY 2020			1,606,432	1,299,764	
5	Virginia Class Submarine	B	2 (7,483,677)	2 (7,215,253)	2 (6,915,757)	U
	Less: Advance Procurement (PY)		(-2,118,496)	(-2,612,040)	(-2,666,517)	U
			-----	-----	-----	
			5,365,181	4,603,213	4,249,240	
6	Virginia Class Submarine					
	Advance Procurement (CY)		2,969,552	2,173,187	2,120,407	U
	C (FY 2020 for FY 2021) (M)		(1,112,629)			
	C (FY 2020 for FY 2022) (M)		(1,562,935)			
	C (FY 2020 for FY 2023) (M)		(293,988)			
	C (FY 2021 for FY 2022) (M)			(833,091)		
	C (FY 2021 for FY 2023) (M)			(1,340,096)		
	C (FY 2022 for FY 2023) (M)				(611,920)	
	C (FY 2022 for FY 2024) (M)				(1,508,487)	

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Department of the Navy
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(Dollars in Thousands)

10 May 2021

Appropriation: 1611N Shipbuilding and Conversion, Navy

Line No	Item Nomenclature	Ident Code	FY 2020 Actual* Quantity Cost	FY 2021 Enacted** Quantity Cost	FY 2022 Request Quantity Cost	S e c
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7	CVN Refueling Overhauls	A	1 (5,853,456)			U
	Less: Advance Procurement (PY)		(-749,870)			U
	Less: Subsequent Full Funding (FY)		(-3,987,171)			U
	Less: Other Procurement, Navy (PY)		(-481,789)			U
			-----	-----	-----	
			634,626			
	Subsequent Full Funding for FY 2020			1,531,153	2,456,018	
8	CVN Refueling Overhauls					
	Advance Procurement (CY)		16,900	17,384	66,262	U
	C (FY 2020 for FY 2025) (M)		(16,900)			
	C (FY 2021 for FY 2025) (M)			(17,384)		
	C (FY 2022 for FY 2025) (M)				(66,262)	
9	DDG 1000	A	155,944	78,205	56,597	U
10	DDG-51	A	3 (5,428,955)	2 (4,016,478)	1 (2,401,719)	U
	Less: Advance Procurement (PY)		(-363,660)	(-796,635)	(-384,932)	U
			-----	-----	-----	
			5,065,295	3,219,843	2,016,787	
11	DDG-51					
	Advance Procurement (CY)		744,028	159,297		U
	C (FY 2020 for FY 2021) (M)		(632,014)			
	C (FY 2020 for FY 2022) (M)		(112,014)			
	C (FY 2021 for FY 2022) (M)			(159,297)		
12	Littoral Combat Ship	A				
	Completion PY Shipbuild for FY 2016		14,000			
13	FFG-Frigate	A	1 1,281,177	1 1,053,123	1 1,087,900	U
14	FFG-Frigate					
	Advance Procurement (CY)				69,100	U
	C (FY 2022 for FY 2023) (M)				(69,100)	
			-----	-----	-----	
Total	Other Warships		18,523,203	15,400,770	14,490,780	

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Department of the Navy
FY 2022 President's Budget
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Total Obligational Authority
(Dollars in Thousands)

10 May 2021

Appropriation: 1611N Shipbuilding and Conversion, Navy

Line No	Item Nomenclature	Ident Code	FY 2020 Actual* Quantity Cost	FY 2021 Enacted** Quantity Cost	FY 2022 Request Quantity Cost	S e c
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Budget Activity 03: Amphibious Ships						

Amphibious Ships						
15	LPD Flight II	A	(524,100)	1 (1,957,637)		U
	Less: Advance Procurement (PY)			(-247,100)		U
	Less: Subsequent Full Funding (FY)			(-584,736)		U
			-----	-----	-----	
			524,100	1,125,801		
	Subsequent Full Funding for FY 2021				60,636	
16	LPD Flight II					
	Advance Procurement (CY)			2,000		U
	C (FY 2021 for FY 2023) (M)			(1,000)		
	C (FY 2021 for FY 2025) (M)			(1,000)		
17	Expeditionary Sea Base (ESB)	A				
	Completion PY Shipbuild for FY 2016		38,000			
18	Expeditionary Sea Base (ESB)					
	Advance Procurement (CY)			73,000		U
	C (FY 2021 for FY 2036) (M)			(73,000)		
19	LHA Replacement	A		(500,000)	(918,637)	U
	Less: Advance Procurement (PY)				(-350,000)	U
	Less: Subsequent Full Funding (FY)				(-500,000)	U
			-----	-----	-----	
				500,000	68,637	
20	Expeditionary Fast Transport (EPF)	A		1 260,000		U
	Completion PY Shipbuild for FY 2019		49,000			
			-----	-----	-----	
	Total Amphibious Ships		611,100	1,960,801	129,273	

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Department of the Navy
FY 2022 President's Budget
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(Dollars in Thousands)

10 May 2021

Appropriation: 1611N Shipbuilding and Conversion, Navy

Line No	Item Nomenclature	Ident Code	FY 2020 Actual* Quantity	Cost	FY 2021 Enacted** Quantity	Cost	FY 2022 Request Quantity	Cost	S e c
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Budget Activity 05: Auxiliaries, Craft, and Prior-Year Program Costs									

Auxiliaries, Craft and Prior Yr Program Cost									
21	TAO Fleet Oiler	A	2	(1,056,261)		(93,000)	1	(668,184)	U
	Less: Advance Procurement (PY)			(-75,046)		(-73,000)			U
				-----		-----		-----	
				981,215		20,000		668,184	
	Completion PY Shipbuild for FY 2018			3,700					
22	TAO Fleet Oiler								
	Advance Procurement (CY)			73,000				76,012	U
	C (FY 2020 for FY 2021) (M)			(73,000)					
	C (FY 2022 for FY 2023) (M)							(76,012)	
23	Tagos Surtass Ships	A					1	434,384	U
24	Towing, Salvage, and Rescue Ship (ATS)	A	2	150,282	2	157,790	2	183,800	U
25	LCU 1700	A	4	83,670	5	87,395	4	67,928	U
26	Outfitting	A		695,992		752,005		655,707	U
27	Ship to Shore Connector	A	1	65,000			2	156,738	U
28	Service Craft	A		56,289		244,147		67,866	U
29	LCAC SLEP	A			3	56,461	2	32,712	U
30	Auxiliary Vessels (Used Sealift)	A			2	60,000	5	299,900	U

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Department of the Navy
FY 2022 President's Budget
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(Dollars in Thousands)

10 May 2021

Appropriation: 1611N Shipbuilding and Conversion, Navy

Line No	Item Nomenclature	Ident Code	FY 2020 Actual* Quantity Cost	FY 2021 Enacted** Quantity Cost	FY 2022 Request Quantity Cost	S e c
----	-----	-----	-----	-----	-----	-
31	Completion of PY Shipbuilding Programs	A		407,312	660,795	U
	AUX (MEMO NON ADD)			(59,900)	(66,700)	U
	CVN (MEMO NON ADD)			(71,000)	(291,000)	U
	CVN RCOH (MEMO NON ADD)			(186,200)	(158,800)	U
	LCS (MEMO NON ADD)				(44,860)	U
	JHSV (MEMO NON ADD)			(50,000)		U
	DDG (MEMO NON ADD)			(9,634)	(45,753)	U
	LPD 17 (MEMO NON ADD)			(30,578)	(53,682)	U
			-----	-----	-----	
	Total Auxiliaries, Craft, and Prior-Year Program Costs		2,109,148	1,785,110	3,304,026	
			-----	-----	-----	
	Total Shipbuilding and Conversion, Navy		21,243,451	23,268,880	22,571,059	

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19	03	01	3041	LHA Replacement.....	Volume 1 - 271
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Oceanographic Ships	5087	992	05	01.....	Volume 1 - 359
Outfitting	5110	26	05	01.....	Volume 1 - 317
Service Craft	5113	28	05	01.....	Volume 1 - 337
Ship to Shore Connector	5112	27	05	01.....	Volume 1 - 331
TAGOS Surtass Ships	5030	23	05	01.....	Volume 1 - 297
TAO Fleet Oiler	5025	21	05	01.....	Volume 1 - 287
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Towing, Salvage, and Rescue Ship (ATS)	5035	24	05	01.....	Volume 1 - 305
Virginia Class Submarine	2013	5	02	01.....	Volume 1 - 97
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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy										Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 01: Fleet Ballistic Missile Ships / BSA 1: Fleet Ballistic Missile Ships							P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine					
ID Code (A=Service Ready, B=Not Service Ready): A				Program Elements for Code B Items: N/A				Other Related Program Elements: 0603595N, 0603570N				
Line Item MDAP/MAIS Code: 444												
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (<i>Units in Each</i>)	-	-	1	-	-	-	-	-	-	-	-	-
Gross/Weapon System Cost (<i>\$ in Millions</i>)	0.000	0.000	15,030.533	0.000	0.000	0.000	-	-	-	-	-	-
Less PY Advance Procurement (<i>\$ in Millions</i>)	-	0.000	-	-	-	-	-	-	-	-	-	-
Less Subsequent Year Full Funding (<i>\$ in Millions</i>)	-	-	5,933.698	-	-	-	-	-	-	-	-	-
Less AP Transfer to NSBDF (<i>\$ in Millions</i>)	-	-	6,227.811	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (<i>\$ in Millions</i>)	0.000	0.000	2,869.024	0.000	0.000	0.000	-	-	-	-	-	-
Plus Subsequent Year Full Funding (<i>\$ in Millions</i>)	-	-	-	3,003.000	-	3,003.000	-	-	-	-	-	-
Full Funding TOA (<i>\$ in Millions</i>)	-	-	2,869.024	3,003.000	-	3,003.000	-	-	-	-	-	-
Plus CY Advance Procurement (<i>\$ in Millions</i>)	-	-	1,253.175	1,643.980	-	1,643.980	-	-	-	-	-	-
Plus AP Transfer to NSBDF (<i>\$ in Millions</i>)	4,808.363	1,820.927	-	-	-	-	-	-	-	-	-	-
Total Obligation Authority (<i>\$ in Millions</i>)	4,808.363	1,820.927	4,122.199	4,646.980	0.000	4,646.980	-	-	-	-	-	-
<i>(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)</i>												
Plus Outfitting and Post Delivery (<i>\$ in Millions</i>)	-	0.000	-	-	-	-	-	-	-	-	-	-
Total (<i>\$ in Millions</i>)	4,808.363	1,820.927	4,122.199	4,646.980	-	4,646.980	-	-	-	-	-	-
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	-	-	15,030.533	-	-	-	-	-	-	-	-	-
Description: MISSION: Strategic Deterrence. The COLUMBIA Class Program is an Acquisition Category (ACAT) ID Major Defense Acquisition Program (MDAP) to design, construct, and deliver a replacement for the OHIO Class Fleet Ballistic Missile Submarines (SSBNs), which begins retiring at a rate of one per year beginning in 2027. The mission of the COLUMBIA SSBN is to maintain an appropriate state of readiness to assist in deterring nuclear attack on the United States and its allies. In the event deterrence should fail, the force must be capable of launching missiles against pre-planned or adaptively planned targets. To fulfill this mission COLUMBIA SSBNs must be capable of performing extended strategic deterrent patrols without requiring assistance or replenishment. It does not have a requirement for additional capabilities or other missions unrelated to survivable strategic nuclear deterrence. Armament: Torpedo Tubes Ballistic Missile Tubes Major Electronics: Trident D5 Strategic Weapons System Command, Control, Communications and Intelligence System - Open System Architecture - Twenty-three Subsystems												

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy		Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 01: Fleet Ballistic Missile Ships / BSA 1: Fleet Ballistic Missile Ships		P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine	
ID Code (A=Service Ready, B=Not Service Ready): A	Program Elements for Code B Items: N/A	Other Related Program Elements: 0603595N, 0603570N	
Line Item MDAP/MAIS Code: 444			
Note: FY2022 increases from FY2021 are due to updated program cost estimate. In August 2020, the COLUMBIA Class program completed its Construction Authorization In Progress Review (IPR). The Milestone Decision Authority (MDA) approved the program for entry into Build I construction on 28 August 2020, pending FY21 congressional authorization and appropriation. As a result of the IPR, the budget submission updates SSBN826-837 requirements to align to the NAVSEA 05C's 2020 cost estimate plus additional funding for SSBN826 based on the CAPE Lead Ship Authorization (LSA) cost estimate. Estimate updates were primarily driven by updates to shipyard construction performance and material escalation. Additional changes include FY21 Congressional Add (+\$130M) for supplier development, Integrated Enterprise Plan (IEP) Advance Procurement rephase (net zero across entire profile), and changes to the program due to rate adjustments. The IEP re-phase is needed to accelerate ship delivery schedules to reduce the strategic deterrence coverage gaps during transition from OHIO SSBNs to CLB SSBNs and the transition from D5LE to D5LE2 missiles. It is crucial to maintain force structure necessary to meet USSTRATCOM requirements for Sea-Based Strategic Deterrence (SBSD). FY2021 was the year of authorization for the lead ship, USS COLUMBIA, and was a new start program due to FY2021 being the first year of incremental full funding. FY2022 is the second year of full funding for lead ship, USS COLUMBIA. The budget request supports detailed design and construction of Contractor Furnished Equipment (CFE) and Government Furnished Equipment (GFE) systems to build, test, outfit, and deliver SSBN826. Funding also supports Continuous Production of Missile Tubes, Advance Construction (AC), Economic Order Quantity (EOQ) for Multi-Program Procurement, Continuous Production of Shipyard Manufactured Items, and Supplier Development to reduce COLUMBIA Class construction schedule risk executed in accordance with enhanced acquisition authorities contained within the U.S.C. 2218a. Funds also supports Advance Procurement (AP) and AC efforts for SSBN827. Funds received in October of 2020 will help to ensure the COLUMBIA Program meets program schedules and the components will meet in yard need dates to support on time construction start and delivery. The Navy executed a Build I contract for the first two COLUMBIA Class submarines (SSBN826 and SSBN827), detailed below. This strategy provides industrial base stability, production efficiencies, and cost savings when compared to an annual procurement with options cost estimate. A Contract Modification for ongoing design/advance construction efforts was awarded on 22 Jun 2020, which also included the Build 1 Option for the First Two Ships. This is a Pre-Priced Option for the two ships, SSBN 826 and SSBN 827, and associated design/support efforts. This was a modification of the current IPPD contract (N00024-17-C-2117), and is in line with the program's approved Acquisition Strategy. Since the FY21 submission, the COLUMBIA Class Program continues investment in initiatives to reduce construction schedule risk and enable cost savings (Multi-Program Material Procurement (MPMP), Continuous Production (CP) of Missile Tubes/Outfitting, Ordnance Systems and Shipyard Manufactured Components, Economic Order Quantity efforts, and Advance Construction). Increased investment in Strategic Weapons System (SWS) Shipboard Systems EOQ and Continuous Production leverages additional opportunities to mitigate obsolescence and construction schedule risks, and realize additional cost savings outside of the FYDP. The Navy has also funded additional IEP initiatives (MPMP, CP, and Production Backup Units) which resulted in a class cost savings of \$420M. Details are contained in the P-10. The first Missile Tube Module (MTM) was funded through RDTEN PE 0603595N for \$584M.			

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy			Date: May 2021																																																																																																																									
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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy		Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1		P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine	
Cost Categories (†) indicates the presence of a P-8a	FY 2021		Total Cost (\$ M)
	Qty (Each)		
Plan Costs	1		6,557.555
Basic Construction/Conversion			5,297.871
Change Orders			299.365
Electronics (†)			358.141
Propulsion Equipment			1,700.896
Hull, Mechanical, and Electrical (HM&E) (†)			174.348
Ordnance (†)			564.115
Other Cost			78.242
Total Ship Estimate			15,030.533
Less Subsequent Full Funding FY 2022			3,003.000
Less Subsequent Full Funding FY 2023			2,930.698
Less AP Transfer to NSBDF FY 2017			773.138
Less AP Transfer to NSBDF FY 2018			802.288
Less AP Transfer to NSBDF FY 2019			3,016.029
Less AP Transfer to NSBDF FY 2020			1,636.356
Net P-1 Funding			2,869.024
Remarks: (1) Ordnance estimates include full funding and advance procurement only. (2) Lockheed Martin Space Systems will be awarded a FY21 cost-plus contract for Strategic Weapon System (SWS) subsystem integration efforts and procurement of service support equipment, special test vehicles and test assemblies from the COLUMBIA Plans line. (3) Major deltas in SSBN 826 caused by shipbuilder performance (Plans and Basic), material escalation (Basic) accounting for class construction startup efforts split between SSBN826 and SSBN827 (Basic), Ordnance and Other; these reflect the updated program cost estimate. (4) SSBN826 includes \$155M fund to the CAPE Lead Ship Authorization (LSA) cost estimate as directed by the Milestone Decision Authority (MDA).			

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Exhibit P-27, Ship Production Schedule: PB 2022 Navy				Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1			P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine		
Ship	Shipbuilder	Fiscal Year	Contract Award	Start of Construction	Delivery Date
SSBN 826 ⁽¹⁾	General Dynamics Electric Boat	2021	Oct 2020	Oct 2020	Oct 2027
Footnotes: ⁽¹⁾ A Contract Modification for ongoing design/advance construction efforts was awarded on 22 Jun 2020, which included the Build I Option for the First Two Ships. This is a Pre-Priced Option for the two ships, SSBN 826 and SSBN 827 (as noted in the P40). The start of construction dates reflect when Electric Boat starts full construction on each ship. The Lead Ship was authorized to start construction in October 2020 and Navy exercised the option for Build I (SSBN 826 and SSBN 827) in November 2020.					

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1	P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine	
Electronics	FY 2021	
	Qty (Each)	Total Cost (\$ M)
P-35 Items		
Sonar	1	95.117
Combat Control	1	15.512
CANES	1	16.306
Electronic Warfare	1	21.804
Photonics	1	22.104
Universal Modular Masts (UMM)	1	9.761
Exterior Communications	1	39.219
P-35 Items Subtotal		219.823
Major Items		
System Level Activities	1	46.642
RADAR	1	3.452
Navigation	1	2.075
CWITT	1	25.151
Non-Propulsion Electronics System, Systems Engineering and Integration (NPES SE&I)	1	27.119
BRR-6 System	1	17.916
BST-1 System	1	8.100
External Countermeasures, IFF	1	7.863
Major Items Subtotal		138.318
Total Electronics		358.141

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1	P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine	
Hull, Mechanical, and Electrical (HM&E)	FY 2021	
	Qty (Each)	Total Cost (\$ M)
P-35 Items		
Propulsor	1	98.235
Advanced Carbon Dioxide Removal Unit (ACRU)	1	9.877
P-35 Items Subtotal		108.112
Other Cost Elements		
HM&E Installation and testing		11.411
T&E		7.547
SUPSHIP responsible material		1.304
Naval Foundry Propeller Center (NFPC)		45.974
Other Cost Elements Subtotal		66.236
Total Hull, Mechanical, and Electrical (HM&E)		174.348

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy		Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1		P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine	
Ordnance	FY 2021		
	Qty (Each)	Total Cost (\$ M)	
P-35 Items			
Strategic Weapons System (SWS) Launcher	1	275.294	
SWS Fire Control	1	101.572	
SWS Navigation	1	79.836	
P-35 Items Subtotal		456.702	
Major Items			
SWS System Level Activities		40.313	
SWS Reentry Subsystem		4.114	
SWS Guidance Subsystem		3.891	
SWS Missile Integration Activities		19.135	
SWS Test Instrumentation Subsystem		39.960	
Major Items Subtotal		107.413	
Total Ordnance		564.115	
Remarks:			
(1) Since the FY2021 submission, all costs associated with initial load out of the ship (outfitting) and Demonstration and Shakedown Availability related efforts (Post Delivery) have been removed and the corresponding budget information/requests are documented in the SCN Outfitting and Post Delivery Budget Line Item 5110.			
(2) Budget estimates for SWS Missile Integration Activities (was \$40.808M in FY2021 - now \$39.960M) and SWS Test Instrumentation Subsystem (remained \$19.35M from FY 21) were swapped in error in the FY2021 submission and corrected in this submission.			
(3) Overall, the P-8a submission is \$79M less in this submission than in FY2021 because FY2021 included a line for Demonstration and Shakedown Availability (System Certification) for \$64.1M which has been removed / transferred to the appropriate Post Delivery budget line items. Other adjustments are attributed to the removal / transfer of outfitting costs to appropriate budget line items and miscellaneous alignments to the updated program cost estimate.			
4) This line does not fund any efforts related to the development and procurement of the TRIDENT II D5 Life Extension (D5LE) Missile or TRIDENT II D5 Life Extension 2 (D5LE2) Missile (however, CLB SWS is dependent on these efforts to ensure the system maintains demonstrated performance and remains survivable while facing a dynamic threat environment until COLUMBIA end of life.)			

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1				P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine			
Equipment Item: Sonar						PARM Code: N/A	
P-35 Category				FY 2021			
				Qty (Each)		Total Cost (\$ M)	
Major Hardware				1		60.875	
Technical Engineering Services						17.597	
Other Costs						16.645	
Total				1		95.117	
Description: The COLUMBIA ELECTRONICS budget funds procurement of the Non-Propulsion Electronics System (NPES), which is the collection of command and control subsystems required to provide all of the non-strategic warfighting, navigation, communications, and crew information technology functions. The Sonar P-35 item contains acoustic sensors and processing electronics required to detect, classify, track and localize surface and submerged contacts. This includes Large Aperture Bow (LAB) array hydrophones, outboard electronics bottles, the Low Cost Conformal Array (LCCA), the AN/WSQ-9 signal analysis system, acoustic intercept and miscellaneous single purpose function sensors, total ship monitoring system (TSMS) , two (2) tactical towed arrays, high frequency sail array, and inboard signal conditioning and display electronics capacity to support both GFE sensors and the CFE Large Vertical Array (LVA). The SONAR P-35 includes engineering services required to produce the sonar logistics products, perform equipment installation, provide construction window field support/troubleshooting/repair, and support platform compatibility engineering, integration, and testing. Other costs include software, initial spares, ship systems data, and SONAR program management labor.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2021	SSBN 826	Lockheed Martin	C/CPIF	Apr 2019	Option	1	60.875
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2021	SSBN 826	Oct 2027	42	32	Aug 2021		
Competition/Second Source Initiatives: N/A							
Remarks: The SONAR P-35 exhibit includes funding for multiple subsystem contracts, field activities, and support efforts, not just system hardware. The single award date shown represents when the placement for preponderance of equipment is needed to meet contractual obligations to the shipbuilder in support of off-hull outfitting and testing of lead ship Command and Control Service Module ("COATS period"). Above does not contain award and delivery dates for every component, including those required earlier than the COATS period. The unit cost in the contract data section only reflects the major prime contractor HW/SW procurement and system integration cost, specifically Lockheed Martin: A-RCI, TSMS, AI&R, LWLCCA (CPIF), NSMA: WSQ-9 (FPI), and L-3: TB-34,TB-29 (CPIF).							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1				P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine			
Equipment Item: Combat Control						PARM Code: N/A	
P-35 Category				FY 2021			
				Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>		
Major Hardware				1	4.886		
Technical Engineering Services					3.490		
Other Costs					7.136		
Total				1	15.512		
Description: The COLUMBIA ELECTRONICS budget funds procurement of the Non-Propulsion Electronics System (NPES), which is the collection of command and control subsystems required to provide all of the non-strategic warfighting, navigation, communications, and crew information technology functions. The Combat Control P-35 item contains processing electronics and attack center controls and displays required to support launch of non-strategic defensive weapons and fuse all contact data with own-ship navigational positioning. The hardware suite contains combat control processing and display electronics, and electronics specific to cyber security protection and monitoring of the entire NPES. The Combat Control P-35 includes engineering services required to produce the logistics products, perform equipment installation, provide construction window field support/troubleshooting/repair, and support platform compatibility engineering, integration, and system testing. Other costs include the extensive suite of individually procured software applications (Tactical Control, Weapons Control, Common Infrastructure Services, Voyage Management System, Information Assurance, On-Board Team Trainer Master Controller, Integrated System Maintenance Tool), initial spares, ship systems data, and Combat Control program management labor.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity <i>(Each)</i>	Unit Cost <i>(\$ M)</i>
FY 2021	SSBN 826	Competitive	C/CPIF	Apr 2021	New	1	4.866
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2021	SSBN 826	Oct 2027	42	32	Aug 2021		
Competition/Second Source Initiatives: N/A							
Remarks: The Combat Control P-35 exhibit includes funding for the H/W suite, multiple software contracts, field activities, and support efforts, not just system hardware. The single award date shown represents when the placement for preponderance of equipment is needed to meet contractual obligations to the shipbuilder in support of off-hull outfitting and testing of lead ship Command and Control Service Module ("COATS period"). Above does not contain award and delivery dates for every component, including those required earlier than the COATS period. The unit cost in the contract data section only reflects the H/W prime contractor cost.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1				P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine			
Equipment Item: CANES						PARM Code: N/A	
P-35 Category				FY 2021			
				Qty (Each)		Total Cost (\$ M)	
Major Hardware				1		4.484	
Technical Data and Documentation						4.810	
Other Costs						7.012	
Total				1		16.306	
Description: The COLUMBIA ELECTRONICS budget funds procurement of the Non-Propulsion Electronics System (NPES), which is the collection of command and control subsystems required to provide all of the non-strategic warfighting, navigation, communications, and crew information technology functions. This P-35 covers the procurement requirements for the Consolidated Afloat Networks and Enterprise Services (CANES) crew information technology hardware suite and the non-tactical data processing system (NTDPS) software. The hardware suite contains CANES processing and display electronics and peripherals (laptops, printers, data storage), and the software suite is inclusive of the NTDPS applications. The CANES P-35 includes engineering services required to produce the logistics products, perform equipment installation, provide construction window field support/troubleshooting/repair, and support platform compatibility engineering, integration, and system testing. Other costs include the suite of individually procured NTDPS software applications, initial spares, ship systems data, and CANES program management labor.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2021	SSBN 826	NEIS	Various	Mar 2021	Option	1	4.484
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2021	SSBN 826	Oct 2027	67	12	Mar 2021		
Competition/Second Source Initiatives: N/A							
Remarks: The CANES P-35 exhibit includes funding for the H/W suite, software contracts, field activities, and support efforts in addition to system hardware. The required award date shown represents initial contract award to support Command and Control Service Module ("COATS period"). There are subsequent contract awards for the rest of the equipment to support the construction schedule.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1				P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine			
Equipment Item: Electronic Warfare						PARM Code: N/A	
P-35 Category				FY 2021			
				Qty (Each)		Total Cost (\$ M)	
Major Hardware				1		11.992	
Technical Engineering Services						5.342	
Other Costs						4.470	
Total				1		21.804	
Description: The COLUMBIA ELECTRONICS budget funds procurement of the Non-Propulsion Electronics System (NPES), which is the collection of command and control subsystems required to provide all of the non-strategic warfighting, navigation, communications, and crew information technology functions. The EW P-35 item contains the processing electronics and software required to detect surface borne electromagnetic energy and classify threat emissions spanning a variety of signal types and frequency range. This includes the AN/BLQ-10 signal processing, display and control system and TYPE 20/24 antenna below deck processing electronics. The EW P-35 includes engineering services required to produce the EW logistics products, perform equipment installation, provide construction window field support/troubleshooting/repair, and support platform compatibility engineering, integration, and system testing. Other costs include software, initial spares, ship systems data, and EW program management labor.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2021	SSBN 826	LM RMS	Various	Dec 2020	Option	1	11.992
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2021	SSBN 826	Oct 2027	70	12	Dec 2020		
Competition/Second Source Initiatives: N/A							
Remarks: The EW P-35 exhibit includes total funding for the contracts, field activities, and support efforts in addition to system hardware. The required award date shown represents initial contract award to support Command and Control Service Module ("COATS period"). There are subsequent contract awards for the rest of the equipment to support the construction schedule.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1				P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine			
Equipment Item: Photonics						PARM Code: N/A	
P-35 Category				FY 2021			
				Qty (Each)		Total Cost (\$ M)	
Major Hardware				1		12.820	
Technical Engineering Services						5.195	
Other Costs						4.089	
Total				1		22.104	
Description: The COLUMBIA ELECTRONICS budget funds procurement of the Non-Propulsion Electronics System (NPES), which is the collection of command and control subsystems required to provide all of the non-strategic warfighting, navigation, communications, and crew information technology functions. The Photonics P-35 item contains the two (2) imaging masts, inboard processing electronics and software required to support surface and submerged periscope operations, and includes visual and infrared (IR) imaging, RF signal communications, and radar early warning. The Photonics P-35 includes engineering services required to produce the logistics products, perform equipment installation, provide construction window field support/troubleshooting/repair, and support platform compatibility engineering and integration. Other costs include software, initial spares, ship systems data, and Photonics program management labor.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2021	SSBN 826	LM RMS	C/CPIF	Dec 2020	Option	1	12.820
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2021	SSBN 826	Oct 2027	70	12	Dec 2020		
Competition/Second Source Initiatives: N/A							
Remarks: The Photonics P-35 exhibit includes funding for multiple contracts, field activities, and support efforts, not just system hardware. The unit cost in the contract data section only reflects the mast (1 shipset, \$4,220 unit cost) prime contractor HW/SW procurement (1 shipset, \$8,600 unit cost) and system integration cost. The contracts are with LM (CPIF Option) and L3-KEO (CPIF Option). The required award date shown represents initial contract award to support Command and Control Service Module ("COATS period"). There are subsequent contract awards for the rest of the equipment to support the construction schedule.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1				P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine			
Equipment Item: Universal Modular Masts (UMM)						PARM Code: N/A	
P-35 Category				FY 2021			
				Qty (Each)		Total Cost (\$ M)	
Major Hardware				1		6.979	
Technical Engineering Services						2.050	
Other Costs						0.732	
Total				1		9.761	
Description: The COLUMBIA ELECTRONICS budget funds procurement of the Non-Propulsion Electronics System (NPES), which is the collection of command and control subsystems required to provide all of the non-strategic warfighting, navigation, communications, and crew information technology functions. The UMM P-35 item contains the hardware costs for six (6) telescopic mechanical mast assemblies custom sized for Columbia Class Sail installation, and required to raise and lower the Photonics and Exterior Communications mast heads. The UMM P-35 includes engineering services required to produce the logistics products, perform equipment installation, provide construction window field support/troubleshooting/repair, and support platform compatibility engineering and integration. Other costs include initial spares, ship systems data, and UMM program management labor.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2021	SSBN 826	L3-KEO	SS/FFP	Feb 2021	Option	1	6.979
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2021	SSBN 826	Oct 2027	53	21	Aug 2021		
Competition/Second Source Initiatives: N/A							
Remarks: The UMM P-35 exhibit includes funding for the prime contract, services contract, and field activities.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1				P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine			
Equipment Item: Exterior Communications						PARM Code: N/A	
P-35 Category				FY 2021			
				Qty (Each)	Total Cost (\$ M)		
Major Hardware				1	24.512		
Technical Engineering Services					7.648		
Other Costs					7.059		
Total				1	39.219		
Description: The COLUMBIA ELECTRONICS budget funds procurement of the Non-Propulsion Electronics System (NPES), which is the collection of command and control subsystems required to provide all of the non-strategic warfighting, navigation, communications, and crew information technology functions. The Exterior Communications Systems (ECS) provides the secure off-hull connectivity vital to ship operation and mission success. The ECS P-35 item contains two (2) multi-function antennas, one (1) high data rate antenna, two (2) floating wire antennas, below deck digital modular radios, cryptology, routers, and operator display and control electronics. The ECS P-35 includes engineering services required to produce the ECS logistics products, perform equipment installation, provide construction window field support/troubleshooting/repair, and support platform compatibility engineering and integration. Other costs include software, initial spares, ship systems data, and ECS program management labor.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2021	SSBN 826	Various	Various	Various	Option	1	24.512
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2021	SSBN 826	Oct 2027	79	9	Jun 2020		
Competition/Second Source Initiatives: N/A							
Remarks: The ECS P-35 exhibit includes funding for multiple subsystems, contracts, field activities, and support efforts, not just system hardware. The preponderance of components are submarine fleet common, and procured as Government-Off-The-Shelf (GOTS) components using existing PEO C4I contracts. The required award date shown represents initial contract award to support Command and Control Service Module ("COATS period"). There are subsequent contact awards for rest of equipment to support construction schedule.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1				P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine			
Equipment Item: Propulsor						PARM Code: N/A	
P-35 Category				FY 2021			
				Qty (Each)	Total Cost (\$ M)		
Major Hardware				1	84.643		
Technical Engineering Services					13.592		
Total				1	98.235		
Description: The propulsor consists of Ni-Al-bronze components and assemblies, a monel bearing support structure, other engineered components and supporting items for corrosion protection, noise monitoring, etc. The purpose of the propulsor is to generate proper thrust to propel the ship at a rated speed within the approved limits of torque and shaft RPM, while at the same time meeting acoustic and structural requirements. This design is unique to the COLUMBIA Class. The propulsor consists of a large quantity of government supplied material and contracts for specific engineered components.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2021	SSBN 826	Various	Various	Sep 2019	New	1	84.643
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2021	SSBN 826	Oct 2027	52	44	Oct 2019		
Competition/Second Source Initiatives: Each contract planned to support manufacture of SSBN 826 propulsor components is intended to be a competitive award.							
Remarks: Naval Surface Warfare Center Carderock Division (NSWCCD) awarded the lead ship Bearing Support Structure contract in Sep 2019 to General Atomics. NSWCCD will award separate contracts for Ropeguard, Plugs, Hydrodynamic Shaping Material (HYSHMA), and Tailcone in FY20-22 to support deliveries to SSBN 826. The Navy will deliver the Government Furnished Equipment (GFE) propulsor components to Electric Boat in FY23 and FY24. Small adjustments were made based on latest cost estimates/performance.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1				P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine			
Equipment Item: Advanced Carbon Dioxide Removal Unit (ACRU)						PARM Code: N/A	
P-35 Category				FY 2021			
				Qty (Each)	Total Cost (\$ M)		
Major Hardware				1	5.237		
Technical Engineering Services					2.340		
Other Costs					2.300		
Total				1	9.877		
Description: The Advanced Carbon Dioxide (CO2) Removal Unit (ACRU) takes the place of legacy CO2 removal equipment used on previous classes and will be the system for COLUMBIA Class Submarines. The ACRU uses a solid sorbent material vice the hazardous liquid amine used in existing CO2 scrubbers. The ACRU is Government Furnished Equipment (GFE) for the first two hulls and will be procured to support contractor integration schedules and required in year need dates. The solid sorbent material will be procured separately from the ACRUs. On-board spares, drawing updates, program management and Government technical and contractual oversight are not included in this cost.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2021	SSBN 826	Various	Various	Sep 2020	New	1	5.237
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2021	SSBN 826	Oct 2027	55	34	Sep 2020		
Competition/Second Source Initiatives: Each contract planned to support manufacture of SSBN 826 propulsor components is intended to be a competitive award.							
Remarks: NSWC Philadelphia is executing contract efforts for lead ship and second ship ACRU procurement. Initial values from FY21 submission did not include support through delivery; updated values include these costs in Technical Engineering and Other Costs. Hardware costs increased due to updated expected contract costs.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1				P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine			
Equipment Item: Strategic Weapons System (SWS) Launcher						PARM Code: N/A	
P-35 Category				FY 2021			
				Qty (Each)		Total Cost (\$ M)	
Major Hardware				1		215.327	
Technical Engineering Services						59.967	
Total				1		275.294	
Description: The COLUMBIA Strategic Weapon System (SWS) will consist of multiple subsystems comprised of tactical and non-tactical subsystems which work as a single unit to prepare, launch, and deliver the warheads to their predetermined targets. The SWS Launcher Subsystem includes the equipment necessary to store and launch the TRIDENT II (D5) Missile when the submarine is submerged or surfaced. This SWS Launcher Subsystem consists of a Launch Tube Group, Missile Handling Equipment, a Vertical Support Group, an Umbilical Retractor Group, a Closure Group, a Missile Ejector Group, and security locks. The SWS Launcher P-35 includes program management and engineering services required to produce the logistics products, support equipment installation, provide construction window field support/troubleshooting/repair, and support platform compatibility engineering, integration, and system testing. Since the FY 2021 submission, all costs associated with initial load out of the ship (outfitting) and Demonstration and Shakedown Availability related efforts (Post Delivery) have been removed and the corresponding budget information/requests are documented in the SCN Outfitting and Post Delivery Budget Line Item 5110.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2021	SSBN 826	Various	Various	Mar 2019	New	1	215.327
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2021	SSBN 826	Oct 2027	74	29	Mar 2019		
Competition/Second Source Initiatives: N/A							
Remarks: Northrop Grumman System Corporation - Marine Systems was awarded a contract in March 2019 to procure the preponderance of Launcher Subsystem components for SSBN 826. Naval Surface Warfare Center Crane Division will award separate contracts for associated support equipment in FY19-21 to support deliveries to SSBN 826. The unit cost in the contract data section reflects the aggregate hardware procurement and system integration cost.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1				P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine			
Equipment Item: SWS Fire Control						PARM Code: N/A	
P-35 Category				FY 2021			
				Qty <i>(Each)</i>		Total Cost <i>(\$ M)</i>	
Major Hardware				1	80.742		
Technical Engineering Services					20.830		
Total				1	101.572		
Description: The COLUMBIA Strategic Weapon System (SWS) will consist of multiple subsystems comprised of tactical and non-tactical subsystems which work as a single system to prepare, launch, and deliver the warheads to their predetermined targets. The SWS Fire Control Subsystem includes equipment necessary to launch the TRIDENT II (D5) Missile; monitor and control the missile environment; provide checkout and fault isolation capabilities; and provide equipment and network infrastructure capable of collecting, recording, processing, on-loading, and off-loading SWS data to support weapon system performance and accuracy evaluation. The SWS Fire Control P-35 includes engineering services required to produce the logistics products, support equipment installation, provide construction window field support/troubleshooting/repair, and support platform compatibility engineering, integration, and system testing. Since the FY 2021 submission, all costs associated with initial load out of the ship (outfitting) and Demonstration and Shakedown Availability related efforts (Post Delivery) have been removed and the corresponding budget information/requests are documented in the SCN Outfitting and Post Delivery Budget Line Item 5110.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity <i>(Each)</i>	Unit Cost <i>(\$ M)</i>
FY 2021	SSBN 826	General Dynamics Mission Systems	SS/CPIF	Jun 2020	New	1	80.742
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2021	SSBN 826	Oct 2027	65	27	Jun 2020		
Competition/Second Source Initiatives: N/A							
Remarks: General Dynamics Missions Systems was awarded a contract in June 2020 to procure the preponderance of Fire Control Subsystem components for SSBN 826 and provide subsystem integration efforts, The unit cost in the contract data section reflects the aggregate hardware procurement and system integration cost.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1				P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine			
Equipment Item: SWS Navigation						PARM Code: N/A	
P-35 Category				FY 2021			
				Qty (Each)		Total Cost (\$ M)	
Major Hardware				1		59.809	
Technical Engineering Services						20.027	
Total				1		79.836	
Description: The COLUMBIA Strategic Weapon System (SWS) will consist of multiple subsystems comprised of tactical and non-tactical subsystems which work as a single unit to prepare, launch, and deliver the warheads to their predetermined targets. The SWS Navigation Subsystem includes inertial and non-inertial equipment necessary to determine submarine position which is critical in support of SWS accuracy as well as platform tactical navigation requirements. The SWS Navigation P-35 includes engineering services required to produce the logistics products, support equipment installation, provide construction window field support/troubleshooting/repair, and support platform compatibility engineering, integration, and system testing. Since the FY 2021 submission, all costs associated with initial load out of the ship (outfitting) and Demonstration and Shakedown Availability related efforts (Post Delivery) have been removed and the corresponding budget information/requests are documented in the SCN Outfitting and Post Delivery Budget Line Item 5110.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2021	SSBN 826	Lockheed-Martin RMS / General Dynamics Mission Systems	SS/CPIF	Apr 2020	Option	1	59.809
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2021	SSBN 826	Oct 2027	55	26	Jan 2021		
Competition/Second Source Initiatives: N/A							
Remarks: Non-inertial navigation components for SSBN 826, as well as Navigation Subsystem integration efforts, will be procured as options to an existing Lockheed Martin Rotary and Mission Systems awarded in April 2020. Additional priced contract will be awarded to Lockheed Martin Rotary and Mission Systems in FY21 to continually produce inertial navigation components for all SSBN platforms.							

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Exhibit P-10, Advance Procurement Requirements Analysis (page 1 - Budget Funding Justification): PB 2022 Navy						Date: May 2021				
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1				P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine						
First System (2022) Award Date: October 2020		First System (2022) Completion Date: October 2027			Interval Between Systems: 0 Months					
Cost Elements		Production Leadtime (Months)	When Required* (Months)	FY 2020 (\$ M)	FY 2021 (\$ M)	FY 2022 (\$ M)	FY 2023 (\$ M)	FY 2024 (\$ M)	FY 2025 (\$ M)	FY 2026 (\$ M)
PLANS										
Lead Ship Design (1)		12-60	Various	727.262	-	0.000	-	-	-	-
Supplier Development - SSBN 826 (2)		12-60	Various	145.400	-	0.000	-	-	-	-
Supplier Development - SSBN 827 (2)		-	-	-	85.000	0.000	-	-	-	-
Total: PLANS				872.662	85.000	-	-	-	-	-
BASIC CONSTRUCTION (3) - SHIPBUILDER PROCURED LLTM										
SSBN 826		24-42	Various	460.469	-	0.000	-	-	-	-
SSBN 827		24-54	Various	-	136.000	225.550	-	-	-	-
SSBN 828		24-54	Various	-	-	0.000	-	-	-	-
SSBN 829		24-54	Various	-	-	0.000	-	-	-	-
SSBN830		24-54	Various	-	-	0.000	-	-	-	-
SSBN831		24-54	Various	-	-	0.000	-	-	-	-
Total: BASIC CONSTRUCTION (3) - SHIPBUILDER PROCURED LLTM				460.469	136.000	225.550	-	-	-	-
BASIC CONSTRUCTION (4) - MISSILE TUBE CONTINUOUS PRODUCTION & OUTFITTING										
SSBN 827		36-42	Various	64.270	-	29.470	-	-	-	-
SSBN 828		36-42	Various	19.477	86.288	90.522	-	-	-	-
SSBN 829		36-42	Various	-	-	57.399	-	-	-	-
SSBN 830		36-42	Various	-	-	0.000	-	-	-	-
SSBN 831		36-42	Various	-	-	0.000	-	-	-	-
SSBN 832		36-42	Various	-	-	0.000	-	-	-	-
SSBN 833		36-42	Various	-	-	0.000	-	-	-	-
Total: BASIC CONSTRUCTION (4) - MISSILE TUBE CONTINUOUS PRODUCTION & OUTFITTING				83.747	86.288	177.391	-	-	-	-
BASIC CONSTRUCTION (5) - ADVANCE CONSTRUCTION										
SSBN 826		24-42	Various	148.380	-	0.000	-	-	-	-
SSBN 827		24-42	Various	-	27.468	87.060	-	-	-	-
SSBN 828		24-42	Various	-	-	0.000	-	-	-	-
SSBN 829		24-42	Various	-	-	0.000	-	-	-	-
SSBN 830		24-42	Various	-	-	0.000	-	-	-	-

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Exhibit P-10, Advance Procurement Requirements Analysis (page 1 - Budget Funding Justification): PB 2022 Navy						Date: May 2021				
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1				P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine						
First System (2022) Award Date: October 2020		First System (2022) Completion Date: October 2027			Interval Between Systems: 0 Months					
Cost Elements		Production Leadtime (Months)	When Required* (Months)	FY 2020 (\$ M)	FY 2021 (\$ M)	FY 2022 (\$ M)	FY 2023 (\$ M)	FY 2024 (\$ M)	FY 2025 (\$ M)	FY 2026 (\$ M)
SSBN 831		24-42	Various	-	-	0.000	-	-	-	-
Total: BASIC CONSTRUCTION (5) - ADVANCE CONSTRUCTION				148.380	27.468	87.060	-	-	-	-
BASIC CONSTRUCTION (6) - EOQ IN SUPPORT OF MULTI-PROGRAM PROCUREMENT										
SSBN 827		24-42	Various	60.447	30.866	0.000	-	-	-	-
SSBN 829		24-42	Various	-	-	0.000	-	-	-	-
SSBN 830		24-42	Various	-	-	0.000	-	-	-	-
SSBN 837 - Production Backup Units		24-42	Various	13.896	47.299	146.419	-	-	-	-
Total: BASIC CONSTRUCTION (6) - EOQ IN SUPPORT OF MULTI-PROGRAM PROCUREMENT				74.343	78.165	146.419	-	-	-	-
BASIC CONSTRUCTION (7) - SHIPYARD MANUFACTURED ITEMS CONTINUOUS PRODUCTION										
SSBN 827		12-36	Various	5.808	7.093	12.229	-	-	-	-
SSBN 828		12-36	Various	-	4.530	8.210	-	-	-	-
SSBN 829		12-36	Various	-	-	4.567	-	-	-	-
SSBN 830		12-36	Various	-	0.930	0.000	-	-	-	-
SSBN 831		12-36	Various	-	-	0.930	-	-	-	-
SSBN 832		12-36	Various	-	-	0.000	-	-	-	-
SSBN 833		12-36	Various	-	-	0.000	-	-	-	-
SSBN 834		12-36	Various	-	-	0.000	-	-	-	-
SSBN 835		12-36	Various	-	-	0.000	-	-	-	-
Total: BASIC CONSTRUCTION (7) - SHIPYARD MANUFACTURED ITEMS CONTINUOUS PRODUCTION				5.808	12.553	25.936	-	-	-	-
NUCLEAR PROPULSION PLANT EQUIPMENT (8)										
SSBN 826		30-72	Various	-	-	0.000	-	-	-	-
SSBN 827 (In support of AC)		30-72	Various	-	816.737	797.262	-	-	-	-
SSBN 828 (In Support of AC)		30-72	Various	-	-	0.000	-	-	-	-
SSBN 829 (In Support of AC)		30-72	Various	-	-	0.000	-	-	-	-
SSBN 830 (In support of AC)		30-72	Various	-	-	0.000	-	-	-	-
SSBN 831 (In Support of AC)		30-72	Various	-	-	0.000	-	-	-	-
Total: NUCLEAR PROPULSION PLANT EQUIPMENT (8)				-	816.737	797.262	-	-	-	-

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Exhibit P-10, Advance Procurement Requirements Analysis (page 1 - Budget Funding Justification): PB 2022 Navy						Date: May 2021				
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1				P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine						
First System (2022) Award Date: October 2020		First System (2022) Completion Date: October 2027			Interval Between Systems: 0 Months					
Cost Elements		Production Leadtime (Months)	When Required* (Months)	FY 2020 (\$ M)	FY 2021 (\$ M)	FY 2022 (\$ M)	FY 2023 (\$ M)	FY 2024 (\$ M)	FY 2025 (\$ M)	FY 2026 (\$ M)
HM&E (9)										
SSBN 826 (In support of AC)		24-42	Various	41.948	-	0.000	-	-	-	-
SSBN 827 (In Support of AC)		24-42	Various	-	3.100	42.857	-	-	-	-
SSBN 828 (In support of AC)		24-42	Various	-	-	0.000	-	-	-	-
SSBN 829 (In Support of AC)		24-42	Various	-	-	0.000	-	-	-	-
SSBN 830 (In Support of AC)		24-42	Various	-	-	0.000	-	-	-	-
SSBN 831 (In Support of AC)		24-42	Various	-	-	0.000	-	-	-	-
Total: HM&E (9)				41.948	3.100	42.857	-	-	-	-
ORDNANCE SWS SHIPBOARD SYSTEMS (10) - LLTM										
SSBN 826 (In Support of AC)		12-48	Various	83.783	-	0.000	-	-	-	-
SSBN 827 (In Support of AC)		12-48	Various	9.947	1.912	46.882	-	-	-	-
SSBN 828 (In Support of AC)		12-48	Various	-	-	12.857	-	-	-	-
SSBN 829 (In Support of AC)		12-48	Various	-	-	11.355	-	-	-	-
SSBN 830 (In Support of AC)		12-48	Various	-	-	0.000	-	-	-	-
SSBN 831 (In Support of AC)		12-48	Various	-	-	0.000	-	-	-	-
SSBN 832 (In Support of AC)		12-48	Various	-	-	0.000	-	-	-	-
SSBN 833 (In Support of AC)		12-48	Various	-	-	0.000	-	-	-	-
Total: ORDNANCE SWS SHIPBOARD SYSTEMS (10) - LLTM				93.730	1.912	71.094	-	-	-	-
ORDNANCE SWS SHIPBOARD SYSTEMS (11) - ECONOMIC ORDER QUANTITY										
SSBN 826		12-24	Various	0.224	-	0.000	-	-	-	-
SSBN 827		12-24	Various	0.478	0.140	0.892	-	-	-	-
SSBN 828		12-24	Various	0.478	0.140	0.892	-	-	-	-
SSBN 829		12-24	Various	0.478	0.140	0.892	-	-	-	-
SSBN 830		12-24	Various	0.224	0.140	0.891	-	-	-	-
SSBN 831		12-24	Various	0.224	0.140	0.891	-	-	-	-
SSBN 832		12-24	Various	0.224	0.140	0.892	-	-	-	-
SSBN 833		12-24	Various	0.224	0.140	0.893	-	-	-	-
SSBN 834		12-24	Various	0.224	0.140	0.892	-	-	-	-
SSBN 835		12-24	Various	0.224	0.140	0.892	-	-	-	-

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Exhibit P-10, Advance Procurement Requirements Analysis (page 1 - Budget Funding Justification): PB 2022 Navy						Date: May 2021			
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1				P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine					
First System (2022) Award Date: October 2020		First System (2022) Completion Date: October 2027			Interval Between Systems: 0 Months				
Cost Elements	Production Leadtime (Months)	When Required* (Months)	FY 2020 (\$ M)	FY 2021 (\$ M)	FY 2022 (\$ M)	FY 2023 (\$ M)	FY 2024 (\$ M)	FY 2025 (\$ M)	FY 2026 (\$ M)
SSBN 836	12-24	Various	0.224	0.140	0.892	-	-	-	-
SSBN 837	12-24	Various	0.224	0.140	0.892	-	-	-	-
Total: ORDNANCE SWS SHIPBOARD SYSTEMS (11) - ECONOMIC ORDER QUANTITY			3.450	1.540	9.811	-	-	-	-
ORDNANCE SWS SHIPBOARD SYSTEMS (12) - CONTINUOUS PRODUCTION									
SSBN 827	12-24	Various	-	2.336	1.129	-	-	-	-
SSBN 828	12-24	Various	-	2.076	37.418	-	-	-	-
SSBN 829	12-24	Various	-	-	3.760	-	-	-	-
SSBN 830	12-24	Various	-	-	9.776	-	-	-	-
SSBN 831	12-24	Various	-	-	0.420	-	-	-	-
SSBN 832	12-24	Various	-	-	0.000	-	-	-	-
SSBN 833	12-24	Various	-	-	0.000	-	-	-	-
SSBN 834	12-24	Various	-	-	0.000	-	-	-	-
SSBN 835	12-24	Various	-	-	0.000	-	-	-	-
SSBN 836	12-24	Various	-	-	0.000	-	-	-	-
SSBN 837	12-24	Various	-	-	0.000	-	-	-	-
-	12	Various	-	-	0.000	-	-	-	-
Total: ORDNANCE SWS SHIPBOARD SYSTEMS (12) - CONTINUOUS PRODUCTION			-	4.412	52.503	-	-	-	-
ELECTRONICS (13)									
SSBN 826	12-24	Various	28.890	-	0.000	-	-	-	-
SSBN 827 (EOQ in support of Multi-Program Procurement)	12-24	Various	7.500	-	0.000	-	-	-	-
SSBN 827	12-24	Various	-	-	8.097	-	-	-	-
SSBN 828	12-24	Various	-	-	0.000	-	-	-	-
SSBN 829	12-24	Various	-	-	0.000	-	-	-	-
SSBN 830	12-24	Various	-	-	0.000	-	-	-	-
Total: ELECTRONICS (13)			36.390	-	8.097	-	-	-	-
Total Advance Procurement/Obligation Authority			1,820.927	1,253.175	1,643.980	-	-	-	-
*Note: "When Required" is the number of months required before ship delivery.									

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Exhibit P-10, Advance Procurement Requirements Analysis (page 2 - Budget Funding Justification): PB 2022 Navy					Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1				P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine			
Cost Elements	FY 2022						
	Production Leadtime <i>(Months)</i>	When Required* <i>(Months)</i>	Unit Cost <i>(\$ M)</i>	Contract Forecast Date	2022 Qty <i>(Each)</i>	For FY	Total Cost Request <i>(\$ M)</i>
PLANS							
Supplier Development - SSBN 827 (2)	-	-	-		-		0.000
Total: PLANS							-
BASIC CONSTRUCTION (3) - SHIPBUILDER PROCURED LLTM							
SSBN 827	24-54	Various	-	Oct 2021	-	2024	225.550
SSBN 828	24-54	Various	-		-	2026	0.000
SSBN 829	24-54	Various	-		-	2028	0.000
SSBN830	24-54	Various	-		-	2028	0.000
SSBN831	24-54	Various	-		-	2028	0.000
Total: BASIC CONSTRUCTION (3) - SHIPBUILDER PROCURED LLTM							225.550
BASIC CONSTRUCTION (4) - MISSILE TUBE CONTINUOUS PRODUCTION & OUTFITTING							
SSBN 827	36-42	Various	-	Oct 2021	-	2024	29.470
SSBN 828	36-42	Various	-	Oct 2021	-	2026	90.522
SSBN 829	36-42	Various	-	Oct 2021	-	2027	57.399
SSBN 830	36-42	Various	-		-	2028	0.000
SSBN 831	36-42	Various	-		-	2029	0.000
SSBN 832	36-42	Various	-		-	2030	0.000
SSBN 833	36-42	Various	-		-		0.000
Total: BASIC CONSTRUCTION (4) - MISSILE TUBE CONTINUOUS PRODUCTION & OUTFITTING							177.391
BASIC CONSTRUCTION (5) - ADVANCE CONSTRUCTION							
SSBN 827	24-42	Various	-	Oct 2021	-	2024	87.060
SSBN 828	24-42	Various	-		-	2026	0.000
SSBN 829	24-42	Various	-		-	2027	0.000
SSBN 830	24-42	Various	-		-	2028	0.000
SSBN 831	24-42	Various	-		-	2029	0.000
Total: BASIC CONSTRUCTION (5) - ADVANCE CONSTRUCTION							87.060
BASIC CONSTRUCTION (6) - EOQ IN SUPPORT OF MULTI-PROGRAM PROCUREMENT							
SSBN 827	24-42	Various	-		-	2024	0.000
SSBN 829	24-42	Various	-		-	2027	0.000

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Exhibit P-10, Advance Procurement Requirements Analysis (page 2 - Budget Funding Justification): PB 2022 Navy				Date: May 2021			
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1				P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine			
Cost Elements	FY 2022						
	Production Leadtime (Months)	When Required* (Months)	Unit Cost (\$ M)	Contract Forecast Date	2022 Qty (Each)	For FY	Total Cost Request (\$ M)
SSBN 830	24-42	Various	-		-	2027	0.000
SSBN 837 - Production Backup Units	24-42	Various	-	Oct 2021	-	2035	146.419
Total: BASIC CONSTRUCTION (6) - EOQ IN SUPPORT OF MULTI-PROGRAM PROCUREMENT							146.419
BASIC CONSTRUCTION (7) - SHIPYARD MANUFACTURED ITEMS CONTINUOUS PRODUCTION							
SSBN 827	12-36	Various	-	Oct 2021	-	2024	12.229
SSBN 828	12-36	Various	-	Oct 2021	-	2026	8.210
SSBN 829	12-36	Various	-	Oct 2021	-	2027	4.567
SSBN 830	12-36	Various	-		-	2028	0.000
SSBN 831	12-36	Various	-	Oct 2021	-	2029	0.930
SSBN 832	12-36	Various	-		-	2030	0.000
SSBN 833	12-36	Various	-		-	2031	0.000
SSBN 834	12-36	Various	-		-	2032	0.000
SSBN 835	12-36	Various	-		-	2033	0.000
Total: BASIC CONSTRUCTION (7) - SHIPYARD MANUFACTURED ITEMS CONTINUOUS PRODUCTION							25.936
NUCLEAR PROPULSION PLANT EQUIPMENT (8)							
SSBN 827 (In support of AC)	30-72	Various	-	Oct 2021	-	2024	797.262
SSBN 828 (In Support of AC)	30-72	Various	-		-	2026	0.000
SSBN 829 (In Support of AC)	30-72	Various	-		-	2027	0.000
SSBN 830 (In support of AC)	30-72	Various	-		-	2028	0.000
SSBN 831 (In Support of AC)	30-72	Various	-		-	2029	0.000
Total: NUCLEAR PROPULSION PLANT EQUIPMENT (8)							797.262
HM&E (9)							
SSBN 827 (In Support of AC)	24-42	Various	-	Oct 2021	-	2024	42.857
SSBN 828 (In support of AC)	24-42	Various	-		-	2026	0.000
SSBN 829 (In Support of AC)	24-42	Various	-		-	2028	0.000
SSBN 830 (In Support of AC)	24-42	Various	-		-	2029	0.000
SSBN 831 (In Support of AC)	24-42	Various	-		-	2029	0.000
Total: HM&E (9)							42.857

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Exhibit P-10, Advance Procurement Requirements Analysis (page 2 - Budget Funding Justification): PB 2022 Navy					Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1				P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine			
Cost Elements	FY 2022						
	Production Leadtime <i>(Months)</i>	When Required* <i>(Months)</i>	Unit Cost <i>(\$ M)</i>	Contract Forecast Date	2022 Qty <i>(Each)</i>	For FY	Total Cost Request <i>(\$ M)</i>
ORDNANCE SWS SHIPBOARD SYSTEMS (10) - LLTM							
SSBN 827 (In Support of AC)	12-48	Various	-	Oct 2021	-	2024	46.882
SSBN 828 (In Support of AC)	12-48	Various	-	Oct 2021	-	2026	12.857
SSBN 829 (In Support of AC)	12-48	Various	-	Oct 2021	-	2027	11.355
SSBN 830 (In Support of AC)	12-48	Various	-		-	2029	0.000
SSBN 831 (In Support of AC)	12-48	Various	-		-	2029	0.000
SSBN 832 (In Support of AC)	12-48	Various	-		-	2030	0.000
SSBN 833 (In Support of AC)	12-48	Various	-		-	2032	0.000
Total: ORDNANCE SWS SHIPBOARD SYSTEMS (10) - LLTM							71.094
ORDNANCE SWS SHIPBOARD SYSTEMS (11) - ECONOMIC ORDER QUANTITY							
SSBN 827	12-24	Various	-	Oct 2021	-	2024	0.892
SSBN 828	12-24	Various	-	Oct 2021	-	2026	0.892
SSBN 829	12-24	Various	-	Oct 2021	-	2027	0.892
SSBN 830	12-24	Various	-	Oct 2021	-	2028	0.891
SSBN 831	12-24	Various	-	Oct 2021	-	2029	0.891
SSBN 832	12-24	Various	-	Oct 2021	-	2030	0.892
SSBN 833	12-24	Various	-	Oct 2021	-	2031	0.893
SSBN 834	12-24	Various	-	Oct 2021	-	2032	0.892
SSBN 835	12-24	Various	-	Oct 2021	-	2033	0.892
SSBN 836	12-24	Various	-	Oct 2021	-	2034	0.892
SSBN 837	12-24	Various	-	Oct 2021	-	2035	0.892
Total: ORDNANCE SWS SHIPBOARD SYSTEMS (11) - ECONOMIC ORDER QUANTITY							9.811
ORDNANCE SWS SHIPBOARD SYSTEMS (12) - CONTINUOUS PRODUCTION							
SSBN 827	12-24	Various	-	Oct 2021	-	2024	1.129
SSBN 828	12-24	Various	-	Oct 2021	-	2026	37.418
SSBN 829	12-24	Various	-	Oct 2021	-	2027	3.760
SSBN 830	12-24	Various	-	Oct 2021	-	2028	9.776
SSBN 831	12-24	Various	-	Oct 2021	-	2029	0.420
SSBN 832	12-24	Various	-		-	2030	0.000

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Exhibit P-10, Advance Procurement Requirements Analysis (page 2 - Budget Funding Justification): PB 2022 Navy	Date: May 2021
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Cost Elements	FY 2022						
	Production Leadtime <i>(Months)</i>	When Required* <i>(Months)</i>	Unit Cost <i>(\$ M)</i>	Contract Forecast Date	2022 Qty <i>(Each)</i>	For FY	Total Cost Request <i>(\$ M)</i>
SSBN 833	12-24	Various	-		-	2031	0.000
SSBN 834	12-24	Various	-		-	2032	0.000
SSBN 835	12-24	Various	-		-	2033	0.000
SSBN 836	12-24	Various	-		-	2034	0.000
SSBN 837	12-24	Various	-		-	2035	0.000
-	12	Various	-		-		0.000
Total: ORDNANCE SWS SHIPBOARD SYSTEMS (12) - CONTINUOUS PRODUCTION							52.503
ELECTRONICS (13)							
SSBN 827 (EOQ in support of Multi-Program Procurement)	12-24	Various	-		-	2024	0.000
SSBN 827	12-24	Various	-	Oct 2021	-	2024	8.097
SSBN 828	12-24	Various	-		-	2026	0.000
SSBN 829	12-24	Various	-		-	2027	0.000
SSBN 830	12-24	Various	-		-	2028	0.000
Total: ELECTRONICS (13)							8.097
Total Advance Procurement/Obligation Authority							1,643.980

Description:
 GENERAL NOTE: FY2022 increases are driven by updated program cost estimate. In support of the August 2020 Lead Ship Authorization In Progress Review, the program completed an updated cost estimate update in 2020 and cost deltas are reflected in the FY2022 budget submission. As a result, SSBN 826-837 requirements are updated for NAVSEA 05C's 2020 cost estimate and the CAPE Lead Ship Authorization cost estimate, per direction from the Milestone Decision Authority. Estimate updates were primarily driven by updates to shipyard construction performance and material escalation. Program Advance Procurement profiles were updated and aligned with the 2020 cost estimate update and the Advance Construction profile reflects the Integrated Enterprise Plan (IEP) Advance Procurement rephase of the 2020 cost estimate. The IEP re-phase is needed to accelerate ship delivery schedules to reduce the strategic deterrence coverage gaps during transition from OHIO SSBNs to CLB SSBNs and the transition from D5LE to D5LE2 missiles. It is crucial to maintain force structure necessary to meet USSTRATCOM requirements for Sea-Based Strategic Deterrence (SBSD).

(1) PLANS: COLUMBIA Class Lead Design Yard and program office support for the detail design for the Common Missile Compartment, Strategic Weapons System, Propulsion Plant, and Rest of Ship. The Program achieved OSD MDA assigned 83% design maturity target in May 2020 (the requirement to start construction) and was 81.2% design disclosure completion for the entire ship at construction start (October 2020). This design maturity target was necessary to achieve the aggressive 7 year lead ship construction time for the lead ship and subsequent construction spans for the Class, which is required to support Strategic Deterrent mission requirements - there is no margin to patrol dates. Detail design activities also support critical engineering analysis and risk reduction efforts. This line reflects adjustments made based on the award of the Integrated Product and Process Development (IPPD) contract.

(2) SUPPLIER DEVELOPMENT: COLUMBIA Class, in coordination with Congress, continues to execute Supplier Development to de-risk construction schedules, strengthen submarine industrial base health, ensure on-time delivery of quality components and increase sub-vendors' ability to accept shipbuilder outsourced work. These funds execute supplier improvement and facilitization actions to increase capability of existing suppliers, increase capacity where needed, and qualify new suppliers in market spaces with capacity challenges. Initiatives include developing alternate sources for high-risk suppliers, developing sources for complex parts traditionally built at the shipbuilders that have been identified for strategic outsourcing, performing process and facility improvements with existing sources, and qualifying existing

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Exhibit P-10, Advance Procurement Requirements Analysis (page 2 - Budget Funding Justification): PB 2022 Navy		Date: May 2021
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sources for more stringent requirements. Investment in the sub-vendor supplier base is required to promote facilities investment and vendor qualification to ensure adequate industrial capacity, and reduce COLUMBIA Class construction schedule risk. FY2021 includes \$85M funded via congressional add for supplier development.		
(3) BASIC CONSTRUCTION - SHIPBUILDER PROCURED LLTM: Funding is required to support long lead time shipbuilder procured material (for example the Weapons Handling, Air Conditioning Unit, Diesel Generator Set, non-nuclear propulsion equipment, and Reverse Osmosis Unit). These and other components are required early in the construction phase to meet the delivery schedule. 3 Year AP is required for long lead shipbuilder procured equipment for ships 2-12 as determined through actual production spans on recently-completed prototype equipment and updated shipbuilder required-in-yard need dates to support advance construction and serial production of follow-on COLUMBIA hulls. Without these efforts the construction schedules are not supported. SSBN 827, 829, and 830 LLTM scope is also reflected in EOQ to support multiprogram procurement to mitigate construction risk and strengthen submarine industrial base (see footnote #6 - funds are reflected in this category). SSBN 828 and 831 do not currently have funds for EOQ in support of multiprogram procurement and therefore the Shipbuilder LLTM appears to be much larger for 828 and 831 but it is not. (4) BASIC CONSTRUCTION - MISSILE TUBE CONTINUOUS PRODUCTION & OUTFITTING: COLUMBIA Class continues to execute Continuous Production of Missile Tubes to reduce schedule risk, improve manufacturing efficiencies, improve vendor learning, maintain critical production skills, and reduce costs by leveraging high-volume procurements. These benefits increase schedule margin and reduce risk to follow ship deliveries, while also achieving cost reduction savings. Missile Tubes produced for SSBN 826 are funded through RDT&E,N Program Element 0603595N, Project Number 3220. Updates for every hull were required for Missile Tube Outfitting to support construction schedules. Missile Tube Outfitting funding provides labor in support of outfitting Missile Tubes procured under Continuous Production for the COLUMBIA Class. In addition to reducing risk to COLUMBIA on-time ship delivery, this effort is estimated to generate savings throughout the class build through workforce level loading, minimizing the effects of gaps or large variation in demand and procurement efficiencies. (5) BASIC CONSTRUCTION - ADVANCE CONSTRUCTION: Advance Construction (AC) efforts are to de-risk construction schedule and improve probability of on-time delivery. AC begins construction activities in key areas to gain schedule margin and reduce controlling path risks. AC key areas include, but are not limited to, the Bow (Sections 1A and 1B in Supermodule 1 that includes the forward Ballast Tanks and Hemi-head), Stern (Sections 9B and 9C in Supermodule 6 that includes the X-Stern and aft Ballast Tanks and Hemi-Head) and Common Missile Compartment (CMC) adjacent areas contained in Supermodule 2 that include the Missile Compartment Control Module (MCCM). AC is normally, but not limited to, structural fabrication areas that have sufficient design maturity and material availability to begin construction and some outfitting. These areas include MCCM Deck Module Fabrication, Mid-Span Tank complex and Foundation Fabrication, and Missile Compartment Forward Bulkhead and S2C Hull Cylinder Fabrication. AC efforts improve efficiency by smoothing workload at Quonset Point and capture efficiencies. Since the FY2021 submission budgets were updated in accordance with the updated 2020 program estimate to support construction schedules. These changes are required to support the Integrated Enterprise Plan (IEP) construction schedules which enable ability to deliver ships earlier to reduce strategic coverage gaps. (6) BASIC CONSTRUCTION - EOQ IN SUPPORT OF MULTI-PROGRAM PROCUREMENT (MPMP): COLUMBIA continues to execute MPMP to align shipbuilder procured material procurements with COLUMBIA Class funding with funds budgeted for VIRGINIA Class (LI 2013) and CVN (LI 2001) for common components and vendors, where applicable. Advance Procurement (AP) required to execute coordinated material procurements to align COLUMBIA Build II (SSBN828-830) Multi-Program Material Procurement and VIRGINIA Class (VCS) material placements for common components and vendors. Appreciable risk mitigation is provided for COLUMBIA Class construction schedules through acceleration of material procurement, ensuring increased readiness to support construction need dates and through supplier base management and workload leveling. MPMP strengthens the industrial base to improve material availability and avoid construction delays due to late material. COLUMBIA Class Production Backup Units procure critical long lead time material to serve as rolling construction assets throughout COLUMBIA Class construction to be consumed in the event of unforeseen issues from late material or obsolescence to reduce schedule risk and avoid production disruptions from late material, and ensure Required In Yard (RIY) dates are met for compressed shipbuilding schedules. Since FY2021, funding was added/phasing adjusted to support executing additional Production Backup Unit (PBU) efforts to focus on Yellow and Green vendors to strengthen the sub-tier industrial base, and continue to reduce COLUMBIA Class schedule risk to the IEP schedules through supplier base management. Material not consumed during construction will be consumed on the 12th hull if not used. PBUs also strengthen critical at-risk vendors in the sub-tier industrial base, reducing the risk of future obsolescence issues and production disruptions. FY2021 includes \$45M funded via congressional add for supplier development. (7) BASIC CONSTRUCTION - SHIPYARD MANUFACTURED ITEMS CONTINUOUS PRODUCTION: COLUMBIA Class continues to execute Continuous Production of selected Shipyard Manufactured Items (US Code 10 2218a subsection (i)) to reduce schedule risk, improve manufacturing efficiencies, improve learning, maintain critical production skills, and reduce costs by leveraging high-volume procurements. Continuous production of items with long lead time material requirements which are difficult to manufacture helps ensure critical material is available to support construction schedules. These benefits will increase schedule margin and reduce risk to follow ship deliveries, while also achieving cost reduction savings through workforce level loading and minimizing the effects of large variations in demand and procurement efficiencies. Examples of Shipyard Manufactured Items include Spherical Air Flasks, Hull Valves, Logistics Escape Trunks, and Torpedo Tubes. (8) NUCLEAR PROPULSION PLANT EQUIPMENT: Nuclear Propulsion Plant Equipment AP is required to fund long-lead time propulsion plant equipment to support the COLUMBIA Class' implementation of advanced modular construction methods, and meet shipbuilder required RIY dates in support of advance construction and serial production of follow-on COLUMBIA hulls. The component fabrication and delivery timeline is different than that of the VIRGINIA Class submarines due to COLUMBIA component production spans and shipbuilder production needs. Without three-year AP the construction schedules are not supported		

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Exhibit P-10, Advance Procurement Requirements Analysis (page 2 - Budget Funding Justification): PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 01 / 1		P-1 Line Item Number / Title: 1045 / COLUMBIA Class Submarine
(9) HM&E: Hull Mechanical & Engineering AP is required to align the Propulsor procurement and production schedule with COLUMBIA Class Advance Construction schedule acceleration and support the overall ship schedules. Additionally FY2021 AP properly aligns funds with GFE procurement of the ACRU and supports program cost and schedule risk reduction by procuring shipsets one and two concurrently (potential savings of 5% to 10% is anticipated). Changes since the FY2021 submission include funding for Continuous Production of Propulsors for SSBN 827-831 material and labor. This enables level loading of Naval Foundry and Propeller Center resources, and will provide needed schedule execution to mitigate the risk of supporting the accelerated shipbuilder construction schedules per the IEP. Funding supports COLUMBIA Class construction, reducing risk of construction schedule delays and cost growth, thereby ensuring nation's strategic deterrence requirements are met. (10) ORDNANCE SWS SHIPBOARD SYSTEMS - LLTM: Ordnance Strategic Weapons Systems (SWS) Shipboard Systems AP is required to fund Long Lead Time Material (LLTM) for the Strategic Weapon System (SWS) shipboard components to meet COLUMBIA (CLB) Class Government Furnished Equipment (GFE) Required-In-Yard (RIY) dates to support construction activities for CLB. AP is required to procure selected SWS Launcher, Fire Control and Navigation Subsystem components including launch tube assemblies and construction support equipment, launch tube umbilical housings, launcher ejector group and vertical support group equipment; Fire Control power and network equipment and sub-assemblies; and Fire Control and Navigation common infrastructure. The SWS shipboard components have a lead time of 12-48 months and are required by the shipbuilder early in the construction phase to meet the ship delivery schedule. (Note: SWS components requiring 48 month lead time for SSBN 826 are procured through RDT&E,N Program Element 0603595N, Project Number 3220 in support of lead ship Missile Tube Module prototyping.) A delay to these LLTM purchases will impact the U.S. Navy's ability to meet the shipbuilder's RIY dates for tactical hardware resulting in increased schedule risk to construction and ship delivery. Since the FY2021 submission, required-in-yard dates for Launcher, Fire Control and Navigation subsystem components were reevaluated and subsequent rephasing was required to minimize risk to meet the COLUMBIA construction schedules. This line does not fund any efforts related to the development and procurement of the TRIDENT II D5 Life Extension (D5LE) Missile or TRIDENT II D5 Life Extension 2 (D5LE2) Missile (however, CLB SWS is dependent on these efforts to ensure the system maintains demonstrated performance and remains survivable while facing a dynamic threat environment until COLUMBIA end of life.) (11) ORDNANCE SWS SHIPBOARD SYSTEMS - EOQ: Ordnance SWS Shipboard Systems EOQ funding is required for large lot procurements of SWS shipboard components to mitigate significant obsolescence, requalification and vendor risk while ensuring SWS homogeneity and consistent SWS system performance across all COLUMBIA Class hulls. Examples include SWS Launcher Subsystem Detonator Power Assembly electronics components and SWS Fire Control Subsystem components such as network switch memory and flash devices, Navigation Timing Interface Module oscillators and display units. Since the FY2021 submission, funding for SWS Shipboard Systems EOQ increased to leverage additional opportunities to mitigate obsolescence and construction schedule risks. FY2020-2022 represent amounts required for contracts awarded in the third quarter of FY20. This line does not fund any efforts related to the development and procurement of the TRIDENT II D5 Life Extension (D5LE) Missile or TRIDENT II D5 Life Extension 2 (D5LE2) Missile. (12) ORDNANCE SWS SHIPBOARD SYSTEMS- CONTINUOUS PRODUCTION: Ordnance SWS Shipboard Systems Continuous Production (U.S. Code 10 2218a subsection (i)) of SWS Navigation Subsystem, Inertial Navigation Systems and Gyros and Launcher Subsystem Launch tubes and on-tube components is required to procure critical components to ensure homogeneity and consistent SWS system performance across the SWS unit builds, maintain critical production and fabrication skills, eliminate production re-starts and potential re-qualification risk while gaining manufacturing efficiencies and mitigating SWS Subsystem obsolescence across all COLUMBIA Class hulls during construction. These benefits will reduce schedule risk by ensuring on-time deliveries of SWS GFE to the shipbuilder. Changes since the FY2021 submission include increased funding for Continuous Production of Launch tubes and umbilical housings material and labor for the ship class to reduce risk and support the IEP ship schedules. This line does not fund any efforts related to the development and procurement of the TRIDENT II D5 Life Extension (D5LE) Missile or TRIDENT II D5 Life Extension 2 (D5LE2) Missile. (13) ELECTRONICS: Electronics Equipment AP is required to fund the GFE material that either has a long-lead procurement time, or is required to meet shipbuilding contract early construction delivery requirements. Material includes SONAR hydrophones for the bow array, the Low Cost Conformal Array, the WSQ-9 arrays, and a wide range of components required to outfit the Command and Control System Module (CCSM) where the predominant portion of the C51 system electronics reside. Changes in the ELECTRONICS AP profile reflects adjustments necessary to procure GFE material required to support accelerated shipbuilder need dates, as specified in the Build 1 contract. FY20 SSBN827 (EOQ in support of Multi-Program Procurement) was a Common Submarine Radio Room (CSRR) procurement aligned with VA Class - funds were moved from Full Funding profile in order to achieve EOQ savings. These savings have already been accounted for in the profile. Changes since FY2021 and the addition of funds from the updated 2020 program estimate, this profile has increased with the addition of LLTM procurements for SSBN 827-830. *Note: "When Required" is the number of months required before ship delivery.		

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy										Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 02: Other Warships / BSA 1: Other Warships							P-1 Line Item Number / Title: 2001 / Carrier Replacement Program					
ID Code (A=Service Ready, B=Not Service Ready): A			Program Elements for Code B Items: N/A					Other Related Program Elements: N/A				
Line Item MDAP/MAIS Code: 223												
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (<i>Units in Each</i>)	3	-	-	-	-	-	-	-	-	-	-	-
Gross/Weapon System Cost (<i>\$ in Millions</i>)	37,651.716	0.000	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Less PY Advance Procurement (<i>\$ in Millions</i>)	9,253.301	-	-	-	-	-	-	-	-	-	-	-
Less Cost To Complete (<i>\$ in Millions</i>)	2,072.860	-	-	-	-	-	-	-	-	-	-	-
Less Subsequent Year Full Funding (<i>\$ in Millions</i>)	21,579.937	-	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (<i>\$ in Millions</i>)	4,745.618	0.000	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Plus Subsequent Year Full Funding (<i>\$ in Millions</i>)	13,906.304	1,062.000	958.933	1,068.705	-	1,068.705	-	-	-	-	-	-
Full Funding TOA (<i>\$ in Millions</i>)	18,651.922	1,062.000	958.933	1,068.705	-	1,068.705	-	-	-	-	-	-
Plus CY Advance Procurement (<i>\$ in Millions</i>)	9,253.301	-	-	-	-	-	-	-	-	-	-	-
Plus Cost To Complete (<i>\$ in Millions</i>)	1,394.860	-	71.000	291.000	-	291.000	-	-	-	-	-	-
Total Obligation Authority (<i>\$ in Millions</i>)	29,300.083	1,062.000	1,029.933	1,359.705	0.000	1,359.705	-	-	-	-	-	-
<i>(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)</i>												
Plus Outfitting and Post Delivery (<i>\$ in Millions</i>)	460.366	147.113	117.339	41.981	-	41.981	-	-	-	-	-	-
Total (<i>\$ in Millions</i>)	29,760.449	1,209.113	1,147.272	1,401.686	-	1,401.686	-	-	-	-	-	-
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	12,550.572	-	-	-	-	-	-	-	-	-	-	-
Description: To provide credible, sustainable, independent forward presence during peacetime without access to land bases; operate as the cornerstone of a joint and/or allied maritime expeditionary force in response to crisis; and carry the war to the enemy through joint multi-mission offensive operations. The Navy designed the FORD Class aircraft carrier with the warfighting capability essential for the 21st century, and the flexibility and resilience to rapidly adapt to emerging threats. The Ford class incorporates advances in technology such as a new reactor plant, propulsion system, electric plant, Electromagnetic Aircraft Launch System (EMALS), Advanced Arresting Gear (AAG), machinery control, and integrated warfare systems that increase lethality, while lowering life cycle costs through reductions in maintenance and manning requirements. Increased Service Life Allowances (SLA) for weight and stability enable future modernization and the ability to adapt to new missions over the ship's 50-year life cycle. CVN 80 and CVN 81 were awarded under a two-ship buy. CVN 80 is a separate contract line item under the same contract as CVN 81, which allows discrete hull costs to be captured. The CVN 80/81 construction contract is a Fixed Price Incentive (Firm Target) (FPIF) contract type that limits the Navy's liability and incentivizes the shipyard's best performance. The contract language guarantees a single technical baseline for both ships, which allows the shipyard to re-use engineering rollover products, minimize changes between the two ships and leverages economic order quantities for equipment and material procurement. CVN 81 was moved to BLI 2004 in accordance with the Department of Defense Appropriations Act, 2020. Note:												

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 02: Other Warships / BSA 1: Other Warships			P-1 Line Item Number / Title: 2001 / Carrier Replacement Program	
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: N/A		Other Related Program Elements: N/A
Line Item MDAP/MAIS Code: 223				
The CVN 79 schedule reflects a June 2024 shipbuilder delivery under the single phase Not To Exceed (NTE) contract action which was signed in November 2020. Contract will be definitized in 4th quarter FY 2021.				
Characteristics: Length Overall 1092 ft Beam 134 ft Displacement 97,337 TONS Draft 38.7 ft		Systems: Electronics -SHIP SELF DEFENSE SYSTEM (SSDS) Ordnance -ELECTROMAGNETIC AIRCRAFT LAUNCHING SYSTEM (EMALS) -AN/SPY-6(V)3 ENTERPRISE AIR SURVEILLANCE RADAR (EASR) -ADVANCED ARRESTING GEAR (AAG)		
Production Status: Contract Award Date Jun 2015 Months to Completion a) Award to Delivery 108 months b) Construction Start to Delivery 160 months Delivery Date Jun 2024 Completion Of Fitting Out Feb 2025 Obligation Work Limit Date Jan 2026		CVN 79 ⁽¹⁾ Jun 2015 108 months 160 months Jun 2024 Feb 2025 Jan 2026		
		CVN 80 Jan 2019 110 months 110 months Mar 2028 May 2028 Apr 2029		
Design Schedule		Start / Issue	Complete / Response	Reissue
Issue Date for TLR		Apr 2004	N/A	
Issue Date for TLS		Sep 2006	N/A	
Preliminary Design		Jan 2003	Jul 2008	
Contract Design		May 2004	Apr 2008	
Detail Design		Jan 2004	Sep 2009	
Request for Proposals		Jul 2007	Oct 2007	
Design Agent		Huntington Ingalls Industries		
Classification of Cost Estimate: CLASS C BUDGET ESTIMATE				
Footnotes: ⁽¹⁾ The CVN 79 schedule reflects a June 2024 shipbuilder delivery under the single phase Not To Exceed (NTE) contract action which was signed in November 2020. Contract will be definitized in 4th quarter FY 2021.				

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2001 / Carrier Replacement Program		
Cost Categories ^(†) indicates the presence of a P-8a	FY 2013		FY 2018	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Plan Costs	1	905.299	1	433.200
Basic Construction/Conversion		7,267.030		7,835.507
Change Orders		199.487		193.902
Electronics ^(†)		280.692		280.563
Propulsion Equipment		2,034.582		2,524.461
Hull, Mechanical, and Electrical (HM&E) ^(†)		29.406		22.466
Ordnance ^(†)		1,091.387		1,045.301
Other Cost		121.798		70.143
Total Ship Estimate		11,929.681		12,405.543
Less Advance Procurement FY 2007		52.750		-
Less Advance Procurement FY 2008		123.530		-
Less Advance Procurement FY 2009		1,210.561		-
Less Advance Procurement FY 2010		482.938		-
Less Advance Procurement FY 2011		902.473		-
Less Advance Procurement FY 2012		554.797		-
Less Advance Procurement FY 2016		-		862.358
Less Advance Procurement FY 2017		-		1,370.784
Less Subsequent Full Funding FY 2014		917.553		-
Less Subsequent Full Funding FY 2015		1,219.417		-
Less Subsequent Full Funding FY 2016		1,569.543		-
Less Subsequent Full Funding FY 2017		1,241.783		-
Less Subsequent Full Funding FY 2018		2,556.384		-
Less Subsequent Full Funding FY 2019		-		929.122
Less Subsequent Full Funding FY 2020		-		1,062.000
Less Subsequent Full Funding FY 2021		-		958.933
Less Subsequent Full Funding FY 2022		-		1,068.705
Less Subsequent Full Funding FY 2023		-		1,188.616
Less Subsequent Full Funding FY 2024		-		1,053.526
Less Subsequent Full Funding FY 2025		-		2,341.853
Less Cost to Complete FY 2022		291.000		-
Less Cost to Complete FY 2023		172.900		-
Less Cost to Complete FY 2024		143.100		-
Net P-1 Funding		490.952		1,569.646

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2001 / Carrier Replacement Program
Remarks: CVN 79 Basic Construction: Increase of \$368.4M from the FY 2021 budget submission supports the Government responsible portion of the shipbuilding contract overrun and incorporation of Joint Strike Fighter (JSF) as mandated by the FY 2020 National Defense Authorization Act, and includes CVN 79 single phase delivery, which includes integration and installation of GFE systems previously identified as Phase II systems and time related services. Change Orders: Increase of \$18.3M from the FY 2021 budget submission to support additional requirements due to fact of life changes. Electronics: Increase of \$47.0M from the FY 2021 budget submission supports adding Joint Strike Fighter (JSF) capability as mandated by the FY 2020 National Defense Authorization Act. Other Cost: Increase of \$39.3M from the FY 2021 budget submission due to the addition of the CVN 79 Government Test Team to support single phase delivery and additional Engineering Support. Ordnance: Increase of \$57.9M from the FY 2021 budget submission supports properly pricing Government Furnished Equipment due to fact of life changes. CVN 80 Basic: Increase of \$59.4M from the FY 2021 budget submission due to addition of CVN 78 Class interim spares and adding Classified Storage (JSF) to the technical baseline. Electronics: Net increase of \$29.8M from the FY 2021 budget submission due to addition of Joint Precision Aircraft Landing System (JPALS), Electronic Consolidated Automated Support System (eCASS), Ready Room Modifications and Lithium Ion Battery Handling and Storage, which were added to the technical baseline; and decrease due to Congressional undistributed reduction for hardware ahead of need. Funding will need to be restored in a future budget submission. Other Cost: Decrease of \$3M from the FY 2021 budget submission due to FY 2019 Congressional General reduction (\$1.059M) and Rate/Miscellaneous adjustments (\$2.038M). Ordnance: Net decrease of \$1.9M from the FY 2021 budget submission due to a Congressional undistributed reduction for hardware ahead of need, and due to the addition of Mk-38 Machine Gun System and Joint Strike Fighter Autonomic Logistics Information System (JSF ALIS), which were added to the technical baseline.		

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Exhibit P-27, Ship Production Schedule: PB 2022 Navy

Date: May 2021

Appropriation / Budget Activity / Budget Sub Activity:
1611N / 02 / 1

P-1 Line Item Number / Title:
2001 / Carrier Replacement Program

Ship	Shipbuilder	Fiscal Year	Contract Award	Start of Construction	Delivery Date
CVN 79 ⁽¹⁾	Huntington Ingalls Industries, Newport News Shipbuilding	2013	Jun 2015	Feb 2011	Jun 2024
CVN 80	Huntington Ingalls Industries, Newport News Shipbuilding	2018	Jan 2019	Jan 2019	Mar 2028

Footnotes:
⁽¹⁾ The CVN 79 schedule reflects a June 2024 shipbuilder delivery under the single phase Not To Exceed (NTE) contract action which was signed in November 2020. Contract will be definitized in 4th quarter FY 2021.

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2001 / Carrier Replacement Program		
Electronics	FY 2013		FY 2018	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
P-35 Items				
CONSOLIDATED AFLOAT NETWORK AND ENTERPRISE SERVICES (CANES)	1	19.649	1	8.485
AN/USG-2, COOPERATIVE ENGAGEMENT CAPABILITY (CEC)	1	8.024	1	6.110
DIGITAL MODULAR RADIO (DMR) ULTRA HIGH FREQUENCY/VERY HIGH FREQUENCY LINE OF SIGHT (EHF/VHF LOS) SAT	1	8.859	1	10.904
AN/UPX-29(V), INTERROGATOR FRIEND OR FOE (IFF) W/MK XII	1	7.769	1	6.478
SPN-46, AUTOMATIC CARRIER LANDING SYSTEM	1	12.965	1	6.201
SHIP SELF DEFENSE SYSTEM (SSDS)	1	27.825	1	32.306
AN/SYY-1(V)1 AIR TRAFFIC CONTROL SYSTEM, SHIPBOARD	1	5.797	1	5.513
NAVY MULTI-BAND TERMINAL (NMT)	1	5.492	1	8.646
AN/SLQ-32(V)6, SURFACE ELECTRONIC WARFARE IMPROVEMENT PROGRAM (SEWIP) BLOCK 2	1	14.209	1	1.090
AN/SRQ-6/MCS-21, SHIPS SIGNAL EXPLOITATION EQUIPMENT (SSEE)	1	5.754	1	2.551
AN/SRC-61 (V)X HFDAG	1	5.292	1	6.059
JOINT PRECISION AIRCRAFT LANDING SYSTEMS (JPALS)	1	13.354	1	10.875
ELECTRONIC CONSOLIDATED AUTOMATED SUPPORT SYSTEM (eCASS)	1	36.642	1	39.900
OA-9277 ULTRA HIGH FREQUENCY (UHF) MULTICOUPLER	0	-	1	7.850
P-35 Items Subtotal		171.631		152.968
Major Items				
AN/USQ-155(V)1 TACTICAL VARIANT SWITCH	1	2.277	1	2.743
INFORMATION ASSURANCE (IA)		1.946		2.031
AN/URC-141X(V), MULTI-FUNCTION INFORMATION DISTRIBUTION SYSTEM (MIDS)-ON SHIP (MOS)	1	1.338	1	1.586
AN/SLQ-25E DUAL, SURFACE SHIP TORPEDO DEFENSE SYSTEM, NIXIE	1	7.472	1	5.243
SHIPBOARD AIR TRAFFIC CONTROL COMMUNICATIONS (SATCC)	1	2.131	1	2.343
AN/WSN-7(V)3, RING LASER GYRO NAVIGATOR (RLGN)	1	2.558	1	3.121
DISTRIBUTED SYSTEMS DESIGN INTEGRATION SERVICES	1	7.863	1	19.181
C4I INTEGRATION & COORDINATION		9.245		10.119
DISTRIBUTED COMMON GROUND STATION - NAVY (DCGS-N)	1	1.991	1	2.319
AN/USQ-144K AUTOMATED DIGITAL NETWORK SYSTEM (ADNS)	1	1.257	1	1.315
AN/UYQ-86 COMMON DATA LINK MANAGEMENT SYSTEM (CDLMS) WITH NGC2P	1	1.217	1	1.816
OA-9277 ULTRA HIGH FREQUENCY (UHF) MULTICOUPLER	1	1.862	0	-
ARC-210 CARRIER AIR TRAFFIC CONTROL CENTER (CATCC) - PRIFLY - LANDING SIGNAL OFFICER (LSO) SYSTEM	1	1.427	1	1.668
WARFARE SYSTEM INTEGRATION		14.333		16.627

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2001 / Carrier Replacement Program		
Electronics	FY 2013		FY 2018	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
COMMERCIAL BROADBAND SATELLITE PROGRAM, FORCE LEVEL VARANT (CBSP-FLV)	2	2.367	2	2.465
AN/SSN-6(V)X BLOCK 4, NAVIGATION SENSOR SYSTEM INTERFACE (NAVSSI)	1	2.513	0	-
GLOBAL POSITIONING SYSTEM, NAVIGATION, AND TIMING SERVICE (GPNTS)	0	-	1	1.803
INTEGRATED STRIKE PLANNING & EXECUTION SYSTEMS (ISP&E)	1	5.724	1	8.246
AN/USQ-123(V), COMMUNICATIONS DATA LINK-SYSTEM (CDL-S)	1	3.208	1	2.480
AN/SPN-41 (V), INSTRUMENT LANDING SYSTEM (ILS)	1	4.296	1	3.897
SHIP SIGNAL EXPLOITATION SPACE (SSES/SI) COMMUNICATIONS	1	3.375	1	4.193
TURNKEY RADIO COMMUNICATIONS SYSTEM (RCS)	1	16.573	1	14.884
AN/USQ-T52 TRAINING INTERFACE UNIT (TIU) ADVANCED TRAINING DOMAIN (ATD)	1	1.570	1	2.733
AN/SPS-73(V)18 NEXT GENERATION SURFACE SEARCH RADAR (NGSSR)	0	-	1	1.804
READY ROOM (JSF) MODIFICATIONS		1.674		3.400
LITHIUM ION BATTERY HANDLING & STORAGE		2.100		2.300
Major Items Subtotal		100.317		118.317
Other Cost Elements				
Other ELECTRONICS		8.744		9.278
Other Cost Elements Subtotal		8.744		9.278
Total Electronics		280.692		280.563
Remarks:				
AN/USQ-T52 Training Interface Unit (TIU) Advanced Training Domain (ATD) replaces AN/USQ-T46X(V)X, Battle Force Tactical Training System (BFTT) on CVN 79 and CVN 80.				
Next Generation Surface Search Radar (NGSSR) replaces AN/SPS-73A(V)12-Tech Refresh - Surface Search Radar on the CVN 80, due to SPS-73 no longer being in production.				
AN/SLQ-25E Dual, Surface Ship Torpedo Defense System, NIXIE replaced AN/SLQ-25C Dual, Surface Ship Torpedo Defense System, NIXIE on CVN 79 and CVN 80 due to obsolescence a no longer being available to be refurbished				
Changes from FY 2021 budget submission:				
CVN 79 P-8A Electronics-Major Items:				
- The 2020 National Defense Authorization Act (NDAA) states "before completing the Post-Shakedown Availability of the Ford Class Aircraft Carrier designated CVN -79, the Secretary of the Navy Shall ensure that the Aircraft Carrier is capable of operation and deploying with the F-35C aircraft." Therefore, the JSF suite was added to the technical baseline, comprising Joint Precision Aircraft Landing System (JPALS), Electronic Consolidated Automated Support System (eCASS), and Ready Room Modifications.				
- NIXIE increased due to the replacement of SLQ-25C with SLQ-25E, making this a first of class install for CVN-79. The estimate listed in the FY 2021 budget submission was based on historical costs which is no longer fully applicable for SLQ-25E.				

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2001 / Carrier Replacement Program
- Global Positioning System, Navigation and Timing Service (GPNTS) is the next generation of navigation to address Fleet obsolescence issues. GPNTS replaces AN/SSN-6(V)X Block 4, Navigation Sensor System Interface (NAVSSI) on the CVN 80. - Distributed Systems Design Integration Services was reduced to reflect CANES LAN drops moving to Change Orders. - C4I Integration and Coordination increased due to inclusion of shipyard On-Site Representatives (OSRs) and systems engineering support. - The addition of the Lithium Ion Battery Handling and Storage in support of the JSF suite mandated by the 2020 NDAA. CVN 79 P-8A Electronics-P35 Items: - CANES increase due to Enterprise Licensing and software maintenance fee updates. - CEC increase due to reviewing CVN 78 actual costs, updating current pricing and Integrated Test Program (ITP) requirements. - SEWIP increase due to systems engineering support, program management support, and lessons learned from CVN 78. - IFF increase due to update of software and training interface processors. - SPN 46 increase due to contract pricing updates, which includes upgrades to obsolete hardware and updated PALS certification costs. CVN 80 P-8A Electronics: - Increase of \$56.4M due to the addition of the following systems, which were part of the revised technical baseline: - Joint Precision Aircraft Landing System (JPALS) - Electronic Consolidated Automated Support System (eCASS) - Ready Room Modifications - Lithium Ion Battery Handling and Storage - NMT increase from the FY 2021 budget submission reflects the actual cost of FY 2020 hardware procurements. - Decrease of \$26.6M due to a Congressional undistributed reduction for hardware ahead of need. Funding will need to be restored in a future budget submission. - OA-9277A Ultra High Frequency (UHF) Multicoupler moved from Major Item to P-35 item due to increased hardware costs which met the reporting threshold for moving the budget estimate to a P-35 listed item.		

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy				Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2001 / Carrier Replacement Program		
Hull, Mechanical, and Electrical (HM&E)	FY 2013		FY 2018	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Major Items				
HM&E ENGINEERING SERVICES		24.009		16.494
LIFE RAFTS		1.583		1.900
SUPSHIP MATERIAL AND GFE		0.800		0.620
TRUCKS (FORKLIFTS)		0.747		0.825
Major Items Subtotal		27.139		19.839
Other Cost Elements				
Other HM&E		2.267		2.627
Other Cost Elements Subtotal		2.267		2.627
Total Hull, Mechanical, and Electrical (HM&E)		29.406		22.466

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2001 / Carrier Replacement Program		
Ordnance	FY 2013		FY 2018	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
P-35 Items				
ELECTROMAGNETIC AIRCRAFT LAUNCHING SYSTEM (EMALS)	1	639.177	1	607.873
AN/SPY-6(V)3 ENTERPRISE AIR SURVEILLANCE RADAR (EASR)	1	75.351	1	45.130
ADVANCED ARRESTING GEAR (AAG)	1	243.403	1	251.261
PHALANX BLOCK 1B MK 15 MOD 21 & 22, CLOSE - IN WEAPONS SYSTEM (CIWS)	3	19.643	3	20.959
AN/SQQ-34, CARRIER-TACTICAL SUPPORT CENTER (CV-TSC)	1	5.050	1	4.456
MK-57 NATO SEASPARROW MISSILE SYSTEM (NSSMS)	1	45.853	1	46.523
AVIATION DATA MANAGEMENT AND CONTROL SYSTEM (ADMACS)	1	5.034	1	8.828
MK 49, MOD 5 ROLLING AIRFRAME MISSILE (RAM)	2	16.278	2	16.849
AN/SPQ-9B, ANTI-SHIP MISSILE DEFENSE (ASMD) SURFACE SURVEILLANCE AND TRACKING RADAR	1	8.180	1	13.726
P-35 Items Subtotal		1,057.969		1,015.605
Major Items				
LANDING SIGNAL OFFICER DISPLAY SYSTEM (LSODS)	1	2.601	1	2.112
MORIAH BLOCK 2	1	1.187	1	1.499
LONG RANGE LINEUP SYSTEM (LRLS)	1	0.941	1	0.966
IMPROVED FRESNEL LENS OPTICAL LANDING SYSTEM (IFLOLS)	1	2.112	1	2.272
INTEGRATED LAUNCH AND RECOVERY TELEVISION SYSTEM (ILARTS)	1	4.916	1	5.544
COMPACT SWAGING MACHINE (CSM)	2	2.422	2	3.500
JOINT STRIKE FIGHTER ALIS	1	2.338	1	1.800
MK-38 MOD 3 MACHINE GUN SYSTEM (MGS)	0	-	4	8.300
Major Items Subtotal		16.517		25.993
Other Cost Elements				
DUAL BAND RADAR (DBR) (SPY-3 AND VOLUME SEARCH RADAR (VSR))		10.948		-
Other ORDNANCE		5.953		3.703
Other Cost Elements Subtotal		16.901		3.703
Total Ordnance		1,091.387		1,045.301
Remarks:				
The Enterprise Radar Suite (ERS) consists of AN/SPY-6(V)3 Enterprise Air Surveillance Radar (EASR), AN/SPQ-9B Anti-ship Missile Defense (ASMD) Surface Surveillance and Tracking Radar and MK-9 Target Illuminators (TIs), as part of MK-57 Nato SeaSparrow Missile System (NSSMS). These three systems combine to functionally replace the Dual Band Radar (CVN 78) on CVN 79 and follow. The \$10,948K cost on the CVN 79 represents a sunk cost paid for overruns associated with receiving the Volume Search Radar (VSR) from the DDG 1000 program and was originally planned for installation on CVN 79.				
MK-57 Nato SeaSparrow Missile System (NSSMS) has been updated to become one complete system from MK29 MOD 5, Guided Missile Launching System (GMLS) and MK-9 Target Illuminator.				

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2001 / Carrier Replacement Program
Changes from the FY 2021 budget submission:		
CVN 79 P-8A Ordnance-Major Items: - Joint Strike Fighter Autonomic Logistics Information System (JSF ALIS) is a new requirement in support of the JSF suite mandated by the 2020 NDAA.		
CVN 79 P-8A Ordnance-P35 Items: - EASR increased due to previously immature engineering support plan. - EMALS increase is the result of resolving reliability issues discovered on the first of class CVN 78. - AAG increase is the result from updating Hardware to reflect current contract pricing as a result of incentive payments to General Atomics to accelerate schedule, to implemented Engineering Change Proposals, increase of software testing, and inclusion of refurbishment of Deadloads. Increase also accounts for resolving reliability issues discovered on the first of class CVN 78.		
CVN 79 P-8A Ordnance-Other: - Increase due to refined estimates across multiple systems.		
CVN 80 P-8A Ordnance: - Increase of \$10.1M due to the addition of the following systems, which were part of the revised technical baseline: - MK-38 MOD 3 Machine Gun System - Joint Strike Fighter Autonomic Logistics Information System (JSF ALIS) - Decrease of \$12M due to a Congressional undistributed reduction for hardware ahead of need. Funding will need to be restored in a future budget submission.		

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program			
Equipment Item: CONSOLIDATED AFLOAT NETWORK AND ENTERPRISE SERVICES (CANES)						PARM Code: PMW 160	
P-35 Category	FY 2013		FY 2018				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	11.508	1	3.500			
Spares		0.497		0.474			
System Engineering		1.685		2.365			
Technical Engineering Services		1.061		0.272			
Other Costs		4.898		1.874			
Total	1	19.649	1	8.485			

Description:
CANES will provide the Navy tactical/non-tactical information environment and infrastructure necessary to enable hosting, extended services reach-back and reach-forward, and relay functions. These capabilities will support real time and non-real time tactical/non-tactical edge connected, connectionless, and ad-hoc voice, video and data information exchange requirements. CANES replace and modernize afloat networks with hardware, software and enterprise services infrastructure to enable information warfare from and within the tactical domain. CANES provides complete infrastructure inclusive of hardware, software, processing, storage and end user devices for Unclassified, Coalition, Secret and Sensitive Compartmented Information (SCI) enclaves for all basic network services to Navy surface combatants, submarines and maritime operations centers. The POR CVN 79 is Increment 1.2 and CVN 80 is Increment 2.0.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2013	CVN 79	VARIOUS	C/FFP	Mar 2017		1	11.508
FY 2018	CVN 80	VARIOUS	C/CPFF	Nov 2024		1	3.500

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2013	CVN 79	Jun 2024	64	12	Feb 2018
FY 2018	CVN 80	Mar 2028	28	12	Nov 2024

Competition/Second Source Initiatives:
N/A

Remarks:
The CVN 79 increase from the FY 2021 budget submission is due to Enterprise Licensing and software maintenance fee updates.

The CVN 80 reduction from the FY 2021 budget submission was due to a Congressional undistributed reduction for hardware ahead of need. Funding will need to be restored in a future budget submission.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																									
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program																											
Equipment Item: AN/USG-2, COOPERATIVE ENGAGEMENT CAPABILITY (CEC)						PARM Code: PEO IWS 6.0																									
P-35 Category	FY 2013		FY 2018																												
	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>																											
Major Hardware	1	3.707	1	2.750																											
Spares		0.595		0.470																											
System Engineering		1.320		2.195																											
Technical Engineering Services		0.821		0.197																											
Other Costs		1.581		0.498																											
Total	1	8.024	1	6.110																											
Description: The Cooperative Engagement Capability (CEC) AN/USG-2B system provides real time integration of fire control quality sensor data into a single composite data source, which is used by multiple CEC ships and airborne units for direct and remote missile engagements. CEC significantly improves battle force Anti-Air Warfare (AAW) capability by coordinating all force AAW sensors into a single real time, fire control quality composite track picture.																															
Contract Data: <table border="1" style="width:100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Prime Contractor</th> <th>Contract Method/Type</th> <th>Award Date</th> <th>New/Option</th> <th>Quantity <i>(Each)</i></th> <th>Unit Cost <i>(\$ M)</i></th> </tr> <tr> <td>FY 2013</td> <td>CVN 79</td> <td>DRS</td> <td>C/FFP</td> <td>May 2020</td> <td></td> <td align="center">1</td> <td align="right">3.707</td> </tr> <tr> <td>FY 2018</td> <td>CVN 80</td> <td>DRS</td> <td>C/FFP</td> <td>Mar 2022</td> <td></td> <td align="center">1</td> <td align="right">2.750</td> </tr> </table>								Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity <i>(Each)</i>	Unit Cost <i>(\$ M)</i>	FY 2013	CVN 79	DRS	C/FFP	May 2020		1	3.707	FY 2018	CVN 80	DRS	C/FFP	Mar 2022		1	2.750
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity <i>(Each)</i>	Unit Cost <i>(\$ M)</i>																								
FY 2013	CVN 79	DRS	C/FFP	May 2020		1	3.707																								
FY 2018	CVN 80	DRS	C/FFP	Mar 2022		1	2.750																								
Delivery Date: <table border="1" style="width:100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Earliest Ship Delivery Date</th> <th>Months Required Before Delivery</th> <th>Production Leadtime</th> <th>Required Award Date</th> </tr> <tr> <td>FY 2013</td> <td>CVN 79</td> <td align="center">Jun 2024</td> <td align="center">34</td> <td align="center">18</td> <td align="center">Feb 2020</td> </tr> <tr> <td>FY 2018</td> <td>CVN 80</td> <td align="center">Mar 2028</td> <td align="center">30</td> <td align="center">18</td> <td align="center">Mar 2024</td> </tr> </table>								Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date	FY 2013	CVN 79	Jun 2024	34	18	Feb 2020	FY 2018	CVN 80	Mar 2028	30	18	Mar 2024						
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date																										
FY 2013	CVN 79	Jun 2024	34	18	Feb 2020																										
FY 2018	CVN 80	Mar 2028	30	18	Mar 2024																										
Competition/Second Source Initiatives: N/A																															
Remarks: The CVN 79 increase from the FY 2021 budget submission is due to reviewing CVN 78 actual costs, updating current pricing and Integrated Test Program (ITP) requirements. CVN 80 FY 2022 award date aligns with PARM procurement strategy to achieve program cost goal.																															

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program			
Equipment Item: DIGITAL MODULAR RADIO (DMR) ULTRA HIGH FREQUENCY/VERY HIGH FREQUENCY LINE OF SIGHT (EHF/VHF LOS) SAT						PARM Code: PMW 170	
P-35 Category	FY 2013		FY 2018				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	7.765	1	9.187			
Spares		0.050		0.055			
System Engineering		0.488		0.643			
Technical Engineering Services		0.305		0.565			
Other Costs		0.193		0.380			
Ancillary Equipment		0.058		0.074			
Total	1	8.859	1	10.904			
Description: DMR-VHF/UHF LOS/SATCOM is an open architecture system that allows transmission and reception of UHF and VHF RF signals. The DMR replaces many legacy systems, including some crypto, Line Of Sight (LOS) and Satellite Communications (SATCOM) components.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2013	CVN 79	GENERAL DYNAMICS	C/FFP	Sep 2014		1	7.765
FY 2018	CVN 80	GENERAL DYNAMICS	C/FFP	Feb 2019		1	9.187
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2013	CVN 79	Jun 2024	58	18	Feb 2018		
FY 2018	CVN 80	Mar 2028	38	18	Jul 2023		
Competition/Second Source Initiatives: None							
Remarks: CVN 80 award date updated to reflect earliest hardware procurement.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program			
Equipment Item: AN/UPX-29(V), INTERROGATOR FRIEND OR FOE (IFF) W/MK XII						PARM Code: PMA 213	
P-35 Category	FY 2013		FY 2018				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	5.938	1	5.026			
Ancillary Equipment		0.057		0.102			
Technical Data and Documentation		0.005		-			
Spares		0.249		0.122			
System Engineering		0.145		0.620			
Technical Engineering Services		0.874		0.151			
Other Costs		0.501		0.457			
Total	1	7.769	1	6.478			

Description:
IFF is an approved and fully supported centralized Mark XII/XIIA Interrogator system. It uses one receiver transmitter that synchronizes video with up to four radar sweeps. It supplies synthetic video (symbolology) to, and accepts requests from, as many as 22 remote locations. It provides digital target reporting to the combat systems/weapon systems computer via full scan, sector, and/or pop-up interrogations. It provides instantaneous target reporting at requested range and azimuth through the use of an electronically-steered Antenna Group OE-120/UPX or OE-120B/UPX. It provides electronically evaluated Mode 4 target reporting directly to operators and over the combat systems/weapon system computer interface. It provides full redundancy so identification capabilities are retained in case of main processor, main antenna, or main receiver/transmitter failure.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2013	CVN 79	BAE SYSTEMS	C/FFP	May 2016		1	5.938
FY 2018	CVN 80	NOTHROP GRUMMAN-BAE SYSTEMS	SS/FFP	Mar 2019		1	5.026

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2013	CVN 79	Jun 2024	44	24	Oct 2018
FY 2018	CVN 80	Mar 2028	35	24	Apr 2023

Competition/Second Source Initiatives:
None

Remarks:
The CVN 79 increase from the FY 2021 budget submission is due to the update of software and training interface processors.

CVN 80 award date updated to reflect earliest hardware procurement.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program			
Equipment Item: SPN-46, AUTOMATIC CARRIER LANDING SYSTEM						PARM Code: PMA 213	
P-35 Category	FY 2013		FY 2018				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	8.436	1	2.349			
System Engineering		0.527		1.460			
Technical Engineering Services		0.844		0.340			
Other Costs		3.158		2.052			
Total	1	12.965	1	6.201			
Description: AN/SPN-46 Automatic Carrier Landing System (ACLS) is a precision approach landing system (PALS) which provides electronic guidance to carrier-based aircraft and allows them to land in all-weather conditions with no limitations due to low ceiling or restricted visibility. AN/SPN-46 is a fully automated, all weather approach landing aid for carrier aircraft.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2013	CVN 79	NAWCAD	Various	Dec 2016		1	8.436
FY 2018	CVN 80	NAWCAD	Various	Dec 2020		1	2.349
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2013	CVN 79	Jun 2024	51	24	Mar 2018		
FY 2018	CVN 80	Mar 2028	35	24	Apr 2023		
Competition/Second Source Initiatives: None.							
Remarks: The CVN 79 increased from the FY 2021 budget submission to support current contract pricing updates, which includes upgrades to obsolete hardware and updated PALS certification costs. The CVN 80 reduction from the FY 2021 budget submission was due to a Congressional undistributed reduction for hardware ahead of need. Funding will need to be restored in a future budget submission.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program			
Equipment Item: SHIP SELF DEFENSE SYSTEM (SSDS)						PARM Code: PEO IWS 10.0	
P-35 Category	FY 2013		FY 2018				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	11.795	1	11.900			
Technical Data and Documentation		0.659		1.556			
Spares		1.223		0.644			
System Engineering		4.740		7.467			
Technical Engineering Services		0.191		0.792			
Other Costs		9.217		9.947			
Total	1	27.825	1	32.306			
Description: The SSDS MK 2, Mod (6E) Common C2 system provides capabilities for multi-mission requirements including Ship Protection against air, surface, and subsurface threats using both own-ship and remote data (Joint Composite Track Number (JCTN) and Joint Data Network (JDN)) in support of the Anti-Air Warfare (AAW) Capstone requirements.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2013	CVN 79	LOCKHEED MARTIN	TBD	Apr 2019		1	11.795
FY 2018	CVN 80	LOCKHEED MARTIN	TBD	Jan 2020		1	11.900
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2013	CVN 79	Jun 2024	38	24	Apr 2019		
FY 2018	CVN 80	Mar 2028	31	24	Aug 2023		
Competition/Second Source Initiatives: None							
Remarks: Hardware procurement supports the same CPS (Common Processing System)/CDS (Common Display System) configuration for SSDS, CV-TSC and SEWIP.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program			
Equipment Item: AN/SYY-1(V)1 AIR TRAFFIC CONTROL SYSTEM, SHIPBOARD						PARM Code: PMA 213	
P-35 Category	FY 2013		FY 2018				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	2.893	1	2.404			
Spares		0.267		0.291			
System Engineering		1.012		2.029			
Technical Engineering Services		0.674		0.061			
Other Costs		0.951		0.728			
Total	1	5.797	1	5.513			
Description: AN/SYY-1(V)1 Air Traffic Control System, Shipboard is a processing and display system which correlates and fuses sensor inputs from radar and IFF. The correlated sensor data is displayed to Air Traffic Controllers in a 2D air picture to facilitate the safe and expeditious movement of air traffic operating in the carrier controlled area and launching from/recovering to the ship.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2013	CVN 79	VARIOUS	Various	Jan 2020		1	2.893
FY 2018	CVN 80	VARIOUS	Various	Nov 2021		1	2.404
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2013	CVN 79	Jun 2024	25	24	May 2020		
FY 2018	CVN 80	Mar 2028	34	24	May 2023		
Competition/Second Source Initiatives: N/A							
Remarks: This system replaces AN/TPX-42. AN/SYY-1(V)1 addresses obsolescence, incorporates existing AN/TPX-42A(V) functionality and will add additional capability and interfaces across all platforms. The CVN 80 reduction from the FY 2021 budget submission was due to a Congressional undistributed reduction for hardware ahead of need. Funding will need to be restored in a future budget submission.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																									
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program																											
Equipment Item: NAVY MULTI-BAND TERMINAL (NMT)						PARM Code: PMW 170																									
P-35 Category	FY 2013		FY 2018																												
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																											
Major Hardware	1	5.131	1	7.892																											
Ancillary Equipment		0.005		-																											
Spares		-		0.263																											
System Engineering		0.042		0.075																											
Technical Engineering Services		0.076		0.087																											
Other Costs		0.238		0.329																											
Total	1	5.492	1	8.646																											
Description: The Advanced Extremely High Frequency (AEHF) Navy Multi-band Terminal (NMT) will be used to receive signals from the Advanced EHF satellites which is a follow-on to the DoD's highly secure, highly protected MILSTAR communications satellite system.																															
Contract Data: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Prime Contractor</th> <th>Contract Method/Type</th> <th>Award Date</th> <th>New/Option</th> <th>Quantity (Each)</th> <th>Unit Cost (\$ M)</th> </tr> <tr> <td>FY 2013</td> <td>CVN 79</td> <td>RAYTHEON</td> <td>C/FFP</td> <td>Jun 2014</td> <td></td> <td style="text-align: center;">1</td> <td style="text-align: center;">5.131</td> </tr> <tr> <td>FY 2018</td> <td>CVN 80</td> <td>RAYTHEON</td> <td>C/FFP</td> <td>Jan 2020</td> <td></td> <td style="text-align: center;">1</td> <td style="text-align: center;">7.892</td> </tr> </table>								Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2013	CVN 79	RAYTHEON	C/FFP	Jun 2014		1	5.131	FY 2018	CVN 80	RAYTHEON	C/FFP	Jan 2020		1	7.892
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																								
FY 2013	CVN 79	RAYTHEON	C/FFP	Jun 2014		1	5.131																								
FY 2018	CVN 80	RAYTHEON	C/FFP	Jan 2020		1	7.892																								
Delivery Date: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Earliest Ship Delivery Date</th> <th>Months Required Before Delivery</th> <th>Production Leadtime</th> <th>Required Award Date</th> </tr> <tr> <td>FY 2013</td> <td>CVN 79</td> <td>Jun 2024</td> <td style="text-align: center;">58</td> <td style="text-align: center;">18</td> <td style="text-align: center;">Feb 2018</td> </tr> <tr> <td>FY 2018</td> <td>CVN 80</td> <td>Mar 2028</td> <td style="text-align: center;">31</td> <td style="text-align: center;">18</td> <td style="text-align: center;">Feb 2024</td> </tr> </table>								Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date	FY 2013	CVN 79	Jun 2024	58	18	Feb 2018	FY 2018	CVN 80	Mar 2028	31	18	Feb 2024						
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date																										
FY 2013	CVN 79	Jun 2024	58	18	Feb 2018																										
FY 2018	CVN 80	Mar 2028	31	18	Feb 2024																										
Competition/Second Source Initiatives: None																															
Remarks: CVN 80 NMT increase from the FY 2021 budget submission reflects the actual cost of the hardware procurement in FY 2020.																															

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program			
Equipment Item: AN/SLQ-32(V)6, SURFACE ELECTRONIC WARFARE IMPROVEMENT PROGRAM (SEWIP) BLOCK 2						PARM Code: PEO IWS 2E	

P-35 Category	FY 2013		FY 2018	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Major Hardware	1	10.409	1	0.635
Ancillary Equipment		0.195		0.343
Spares		0.245		-
System Engineering		0.898		0.099
Technical Engineering Services		1.342		-
Other Costs		1.120		0.013
Total	1	14.209	1	1.090

Description:
 SEWIP Block 2 is a scalable Electronic Warfare enterprise suite to provide improved Electromagnetic Interference (EMI) mitigation and Combat System Interface capabilities to select new construction ships as well as upgrade current AN/SLQ-32(V)3 and (V)4 Electronic Warfare (EW) suites on existing ships. It provides enhanced shipboard Electronic Warfare (EW) for early detection, analysis, threat warning and protection from anti-ship missiles. SEWIP Block 2 focused on Electronic Support (ES) capability improvements.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2013	CVN 79	LOCKHEED MARTIN	C/FFP	Jan 2020		1	10.409
FY 2018	CVN 80	LOCKHEED MARTIN	C/FFP	Jan 2020		1	0.635

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2013	CVN 79	Jun 2024	24	18	Dec 2020
FY 2018	CVN 80	Mar 2028	24	18	Sep 2024

Competition/Second Source Initiatives:
 None

Remarks:
 The CVN 79 increase from the FY 2021 budget submission is due to fact of life changes for systems engineering support, program management support, and lessons learned from CVN 78.

 Hardware procurement supports the same CPS (Common Processing System)/CDS (Common Display System) configuration for SSDS, CV-TSC and SEWIP.

 The CVN 80 reduction from the FY 2021 budget submission was due to a Congressional undistributed reduction for hardware ahead of need. Funding will need to be restored in a future budget submission.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program			
Equipment Item: AN/SRQ-6/MCS-21, SHIPS SIGNAL EXPLOITATION EQUIPMENT (SSEE)						PARM Code: PMW 120	
P-35 Category	FY 2013		FY 2018				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	3.801	1	-			
Ancillary Equipment		-		0.085			
Spares		0.192		0.209			
System Engineering		0.654		0.900			
Technical Engineering Services		0.176		0.191			
Other Costs		0.931		1.166			
Total	1	5.754	1	2.551			
Description: SSEE provided for cryptological signal acquisition, recognition, analysis and geo-location. It replaces Maritime Cryptological System (MCS-21) which replaces the Battle Group Passive Horizon Extension System (BGPHEs).							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2013	CVN 79	VARIOUS	Various	Apr 2020		1	3.801
FY 2018	CVN 80	VARIOUS	Various	Oct 2023		1	0.000
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2013	CVN 79	Jun 2024	29	18	Jul 2020		
FY 2018	CVN 80	Mar 2028	35	18	Oct 2023		
Competition/Second Source Initiatives: None							
Remarks: The CVN 80 reduction from the FY 2021 budget submission was due to a Congressional undistributed reduction for hardware ahead of need. Funding will need to be restored in a future budget submission.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program			
Equipment Item: AN/SRC-61 (V)X HFDAG						PARM Code: PMW 170	
P-35 Category	FY 2013		FY 2018				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	4.553	1	4.816			
Ancillary Equipment		-		0.052			
Spares		0.010		0.011			
System Engineering		0.246		0.216			
Technical Engineering Services		0.208		0.527			
Other Costs		0.275		0.437			
Total	1	5.292	1	6.059			
Description: High Frequency (HF) Distributed Amplifier Group (DAG) is the Navy's Program of Record (POR) HF system and is the follow-on replacement of HF Radio Group (HFRG). HFDAG has a modular architecture and utilizes COTS equipment to the maximum extent possible. It provides Line Of Sight (LOS/Beyond Line of Sight (BLOS) voice and data transmission capabilities to USN Ships. The 16-channel CVN variant greatly improves capabilities from HFRG: (1) increases availability (Ao), (2) provides reprogrammable waveforms, (3) increases the number of waveforms available, (4) provides automatic link establishment (ALE).							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2013	CVN 79	GENERAL DYNAMICS	C/FFP	Feb 2019		1	4.553
FY 2018	CVN 80	GENERAL DYNAMICS	C/FFP	Feb 2019		1	4.816
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2013	CVN 79	Jun 2024	48	18	Dec 2018		
FY 2018	CVN 80	Mar 2028	35	18	Oct 2023		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																									
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program																											
Equipment Item: JOINT PRECISION AIRCRAFT LANDING SYSTEMS (JPALS)						PARM Code: PMA 213																									
P-35 Category	FY 2013		FY 2018																												
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																											
Major Hardware	1	8.036	1	6.948																											
Spares		1.197		1.061																											
System Engineering		1.060		0.037																											
Technical Engineering Services		1.761		1.664																											
Other Costs		1.300		1.165																											
Total	1	13.354	1	10.875																											
Description: The Joint Precision Approach and Landing System (JPALS) is the primary precision approach and landing system for CVN ships to support F-35B, F-35C, MQ-25 and future platforms. JPALS ship systems are required to provide a primary precision approach capability during night and instrument flight conditions and coupled approach to the deck (auto-land) capability aboard CVN ships. JPALS also provides over-the-air inertial alignment capability for CVN ships to support aircraft platforms without Link-4A capability, including F-35, MQ-25 and future platforms.																															
Contract Data: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Prime Contractor</th> <th>Contract Method/Type</th> <th>Award Date</th> <th>New/Option</th> <th>Quantity (Each)</th> <th>Unit Cost (\$ M)</th> </tr> <tr> <td>FY 2013</td> <td>CVN 79</td> <td>Raytheon</td> <td>C/FFP</td> <td>Nov 2021</td> <td></td> <td style="text-align: center;">1</td> <td style="text-align: right;">8.036</td> </tr> <tr> <td>FY 2018</td> <td>CVN 80</td> <td>Raytheon</td> <td>C/FFP</td> <td>Sep 2024</td> <td></td> <td style="text-align: center;">1</td> <td style="text-align: right;">6.948</td> </tr> </table>								Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2013	CVN 79	Raytheon	C/FFP	Nov 2021		1	8.036	FY 2018	CVN 80	Raytheon	C/FFP	Sep 2024		1	6.948
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																								
FY 2013	CVN 79	Raytheon	C/FFP	Nov 2021		1	8.036																								
FY 2018	CVN 80	Raytheon	C/FFP	Sep 2024		1	6.948																								
Delivery Date: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Earliest Ship Delivery Date</th> <th>Months Required Before Delivery</th> <th>Production Leadtime</th> <th>Required Award Date</th> </tr> <tr> <td>FY 2013</td> <td>CVN 79</td> <td>Jun 2024</td> <td style="text-align: center;">32</td> <td style="text-align: center;">12</td> <td style="text-align: center;">Oct 2020</td> </tr> <tr> <td>FY 2018</td> <td>CVN 80</td> <td>Mar 2028</td> <td style="text-align: center;">30</td> <td style="text-align: center;">12</td> <td style="text-align: center;">Sep 2024</td> </tr> </table>								Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date	FY 2013	CVN 79	Jun 2024	32	12	Oct 2020	FY 2018	CVN 80	Mar 2028	30	12	Sep 2024						
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date																										
FY 2013	CVN 79	Jun 2024	32	12	Oct 2020																										
FY 2018	CVN 80	Mar 2028	30	12	Sep 2024																										
Competition/Second Source Initiatives: N/A																															
Remarks: The CVN 79 increase from the FY 2021 budget submission is the result of the new requirement in support of the JSF suite mandated by the 2020 NDAA. CVN 79 Contract planned award date of November 2021 is after required award date, with no impact to construction. CVN 80: New requirement added to the technical baseline.																															

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program			
Equipment Item: ELECTRONIC CONSOLIDATED AUTOMATED SUPPORT SYSTEM (eCASS)						PARM Code: PMA 260	
P-35 Category	FY 2013		FY 2018				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	36.372	1	39.504			
Technical Engineering Services		0.270		0.396			
Total	1	36.642	1	39.900			
Description: The electronic Consolidated Automated Support System (eCASS)supports JSF integration on FORD Class. AN/USM-737(V) electronic Consolidated Automated Support System (eCASS) will be the Navy's next generation Automatic Test Equipment (ATE) system, replacing the legacy mainframe AN/USM-636(V) Consolidated Automated Support System (CASS) due to imminent obsolescence and emerging test requirements. eCASS will be the newest member of the CASS family of testers, designated by OPNAVINST 3960.16B as the US Navy's standard ATE used to perform avionics weapons system diagnostics and repair across a multitude of Naval and Marine Corps aircraft platforms and uses joint service coordinated test technologies that will be capable of interoperating with the Future Force. The eCASS system is designed with four configurations, with reserved design space allocations for future growth and mission peculiar requirements. The four eCASS configurations provide the capability to execute the Hybrid (HYB) Mission, Radio Frequency (RF) Mission, High Power (HP) Mission, and the Electro Optical (EO) Mission.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2013	CVN 79	LOCKHEED MARTIN	SS/FFP	Sep 2021		1	36.372
FY 2018	CVN 80	LOCKHEED MARTIN	SS/FFP	Dec 2022		1	39.504
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2013	CVN 79	Jun 2024	15	18	Sep 2021		
FY 2018	CVN 80	Mar 2028	45	18	Dec 2022		
Competition/Second Source Initiatives: N/A Remarks: The CVN 79 increase from the FY 2021 budget submission is the result of the new requirement in support of the JSF suite mandated by the 2020 NDAA. CVN 80 increase from the FY 2021 budget submission is the result of a new requirement added to the technical baseline.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program																			
Equipment Item: OA-9277 ULTRA HIGH FREQUENCY (UHF) MULTICOUPLER						PARM Code: PMW 170																	
P-35 Category	FY 2013		FY 2018																				
	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>																			
Major Hardware	0	-	1	7.660																			
System Engineering		-		0.025																			
Technical Engineering Services		-		0.061																			
Other Costs		-		0.104																			
Total	0	-	1	7.850																			
Description: OA-9277 Description: Ultra High Frequency (UHF) Program supports the exchange of secure Battle Group coordination data, organizational messaging, tactical data and voice over UHF SATCOM via Demand Assigned Multiple Access (DAMA) over 5/25kHz channels. UHF legacy Communication Line of Sight (LOS) Systems also support Anti-Access/Area Denial (A2AD) by providing a redundant or overlapping form of communication in case U.S. satellites are targeted by threat nations.																							
Contract Data: <table border="1" style="width:100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th style="text-align: center;">Program Year</th> <th style="text-align: center;">Hull</th> <th style="text-align: center;">Prime Contractor</th> <th style="text-align: center;">Contract Method/Type</th> <th style="text-align: center;">Award Date</th> <th style="text-align: center;">New/Option</th> <th style="text-align: center;">Quantity <i>(Each)</i></th> <th style="text-align: center;">Unit Cost <i>(\$ M)</i></th> </tr> <tr> <td align="center">FY 2018</td> <td align="center">CVN 80</td> <td align="center">RF PRODUCTS INC</td> <td align="center">C/FFP</td> <td align="center">May 2021</td> <td></td> <td align="center">1</td> <td align="center">7.660</td> </tr> </table>								Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity <i>(Each)</i>	Unit Cost <i>(\$ M)</i>	FY 2018	CVN 80	RF PRODUCTS INC	C/FFP	May 2021		1	7.660
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity <i>(Each)</i>	Unit Cost <i>(\$ M)</i>																
FY 2018	CVN 80	RF PRODUCTS INC	C/FFP	May 2021		1	7.660																
Delivery Date: <table border="1" style="width:100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th style="text-align: center;">Program Year</th> <th style="text-align: center;">Hull</th> <th style="text-align: center;">Earliest Ship Delivery Date</th> <th style="text-align: center;">Months Required Before Delivery</th> <th style="text-align: center;">Production Leadtime</th> <th style="text-align: center;">Required Award Date</th> </tr> <tr> <td align="center">FY 2018</td> <td align="center">CVN 80</td> <td align="center">Mar 2028</td> <td align="center">30</td> <td align="center">24</td> <td align="center">Sep 2023</td> </tr> </table>								Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date	FY 2018	CVN 80	Mar 2028	30	24	Sep 2023				
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date																		
FY 2018	CVN 80	Mar 2028	30	24	Sep 2023																		
Competition/Second Source Initiatives: N/A																							
Remarks: CVN 80 OA-9277A Ultra High Frequency (UHF) Multicoupler moved from Major Item at the FY 2021 budget submission to P-35 item due to increased hardware costs which met the reporting threshold for moving the budget estimate to a P-35 listed item. Hardware cost increase due to lack of available assets for refurbishment and new hardware was required to be procured.																							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program			
Equipment Item: ELECTROMAGNETIC AIRCRAFT LAUNCHING SYSTEM (EMALS)						PARM Code: PMA 251	
P-35 Category	FY 2013		FY 2018				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	573.585	1	542.163			
Spares		9.150		28.050			
System Engineering		21.663		17.507			
Technical Engineering Services		2.082		2.556			
Other Costs		32.697		17.597			
Total	1	639.177	1	607.873			

Description:
EMALS is an advanced technology electrically generated launching system that uses a moving electromagnetic field to propel aircraft to launch speed. EMALS is made up of six primary sub-systems: prime power interface, energy storage, energy distribution, power conversion, launch motor, and launch control subsystem. Benefits over the current C13 steam catapults include reduced weight and volume, greater launching flexibility for future aircraft, improved control, and reduced manning workload requirements.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2013	CVN 79	GENERAL ATOMICS	SS/FFP	May 2014	New	1	573.585
FY 2018	CVN 80	GENERAL ATOMICS	SS/FFP	Jan 2017	Option	1	542.163

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2013	CVN 79	Jun 2024	90	48	Dec 2012
FY 2018	CVN 80	Mar 2028	57	48	Jun 2019

Competition/Second Source Initiatives:
None

Remarks:
The CVN 79 increase from the FY 2021 budget submission is the result of Hardware and System Engineering changes required to resolve reliability issues discovered on the first of class CVN 78. The Other Cost increase is the result of system engineering, software modifications, hardware changes, and ILS documentation updates required to resolve reliability issues also discovered on the first of class CVN 78.

The CVN 80 Spares P-35 category includes \$28.05M for CVN 78 Class Interim Spares.

Long Lead Time Materials Undefinitized Contract Action (UCA) awarded May 2014, Undefinitized Production UCA awarded June 2015 for CVN 79, and Production UCA definitized December 2016 for CVN 79 with option for CVN 80. CVN 80 option exercised January 2017 EMALS and AAG bundled savings on single production contract for CVN 79 and CVN 80 are reflective of contract negotiations.

CVN 79 - Negotiations with the Original Equipment Manufacturer (OEM) were required to negotiate improvements to the contractor's schedule since award was late. Additionally, several changes to the CVN 79 construction contract were required to achieve the ship delivery date.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program			
Equipment Item: AN/SPY-6(V)3 ENTERPRISE AIR SURVEILLANCE RADAR (EASR)						PARM Code: PEO IWS 2.0	
P-35 Category	FY 2013		FY 2018				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	42.588	1	42.130			
Ancillary Equipment		0.175		-			
Spares		1.542		-			
System Engineering		9.055		3.000			
Technical Engineering Services		9.579		-			
Other Costs		12.412		-			
Total	1	75.351	1	45.130			

Description:
The Enterprise Air Surveillance Radar (EASR) suite will be a modern long-range, three-dimensional (3D) radar used to search, detect and provide space-stabilized, three-coordinate (range, bearing, height) data for air intercept control and designation to a weapon system and Air Traffic Control (ATC) system. The Enterprise Radar Suite (ERS), which includes EASR, is intended to replace the functions that Dual Band Radar (DBR) performed on CVN 78, but at a much lower cost.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2013	CVN 79	RAYTHEON	C/CPIF	Dec 2019		1	42.588
FY 2018	CVN 80	RAYTHEON	C/CPIF	Aug 2021		1	42.130

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2013	CVN 79	Jun 2024	27	30	Sep 2019
FY 2018	CVN 80	Mar 2028	30	30	Mar 2023

Competition/Second Source Initiatives:
None

Remarks:
The hardware configuration for the CVN 79, CVN 80, and CVN 81 (non-rotating) is essentially three times that of a rotating configuration, which is currently planned for the big deck amphibious warfare ships. CVN 79, CVN 80, and CVN 81 will have three phased arrays mounted around the island, while the amphibious warfare ships will use one rotating array. Below deck equipment is also provided at a larger scale with the non-rotating variant of EASR.

The Enterprise Radar Suite (ERS) consists of AN/SPY-6(V)2 Enterprise Air Surveillance Radar (EASR), AN/SPQ-9B Anti-Ship Missile Defense (ASMD) Surface Surveillance and Tracking Radar, and MK-9 Target Illuminators (TIs) as part of MK-57 Nato SeaSparrow Missile System (NSSMS).

The CVN 79 overall total requirement has increased from the FY 2021 budget submission. Hardware decreased to reflect current contract pricing. Systems Engineering and other costs increased due to previously immature engineering support plan for CVN 79 which will be the first 3 array configuration installed in the fleet.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1	P-1 Line Item Number / Title: 2001 / Carrier Replacement Program	
Equipment Item: AN/SPY-6(V)3 ENTERPRISE AIR SURVEILLANCE RADAR (EASR)		PARM Code: PEO IWS 2.0
The CVN 80 reduction from the FY 2021 budget submission was due to a Congressional undistributed reduction for hardware ahead of need. Funding will need to be restored in a future budget submission. CVN 80 - Hardware is being procured in FY 2021 for CVN 80 to take advantage of an EOQ buy with other ship classes to achieve the estimated price as reflected in the exhibit.		

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																									
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program																											
Equipment Item: ADVANCED ARRESTING GEAR (AAG)						PARM Code: PMA 251																									
P-35 Category	FY 2013		FY 2018																												
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																											
Major Hardware	1	212.663	1	199.219																											
Spares		0.670		32.497																											
System Engineering		12.450		4.471																											
Technical Engineering Services		0.918		4.771																											
Other Costs		16.702		10.303																											
Total	1	243.403	1	251.261																											
Description: AAG provides an upgraded ability to recover all existing and projected aircraft carrier based air vehicles. The AAG system will replace the Mark 7 arresting gear system found on the NIMITZ class carriers and will be the aircraft recovery system for the CVN 79, CVN 80, and CVN 81. AAG consists of six primary systems: energy absorption subsystem, energy storage subsystem, dynamic control subsystem, thermal management subsystem, cross deck pendant, and the control subsystem.																															
Contract Data: <table border="1" style="width:100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Prime Contractor</th> <th>Contract Method/Type</th> <th>Award Date</th> <th>New/Option</th> <th>Quantity (Each)</th> <th>Unit Cost (\$ M)</th> </tr> <tr> <td>FY 2013</td> <td>CVN 79</td> <td>GENERAL ATOMICS</td> <td>SS/FFP</td> <td>May 2014</td> <td>New</td> <td align="center">1</td> <td align="right">212.663</td> </tr> <tr> <td>FY 2018</td> <td>CVN 80</td> <td>GENERAL ATOMICS</td> <td>SS/FFP</td> <td>Jan 2017</td> <td>Option</td> <td align="center">1</td> <td align="right">199.219</td> </tr> </table>								Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2013	CVN 79	GENERAL ATOMICS	SS/FFP	May 2014	New	1	212.663	FY 2018	CVN 80	GENERAL ATOMICS	SS/FFP	Jan 2017	Option	1	199.219
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																								
FY 2013	CVN 79	GENERAL ATOMICS	SS/FFP	May 2014	New	1	212.663																								
FY 2018	CVN 80	GENERAL ATOMICS	SS/FFP	Jan 2017	Option	1	199.219																								
Delivery Date: <table border="1" style="width:100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Earliest Ship Delivery Date</th> <th>Months Required Before Delivery</th> <th>Production Leadtime</th> <th>Required Award Date</th> </tr> <tr> <td>FY 2013</td> <td>CVN 79</td> <td align="center">Jun 2024</td> <td align="center">88</td> <td align="center">48</td> <td align="center">Feb 2013</td> </tr> <tr> <td>FY 2018</td> <td>CVN 80</td> <td align="center">Mar 2028</td> <td align="center">61</td> <td align="center">48</td> <td align="center">Feb 2019</td> </tr> </table>								Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date	FY 2013	CVN 79	Jun 2024	88	48	Feb 2013	FY 2018	CVN 80	Mar 2028	61	48	Feb 2019						
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date																										
FY 2013	CVN 79	Jun 2024	88	48	Feb 2013																										
FY 2018	CVN 80	Mar 2028	61	48	Feb 2019																										
Competition/Second Source Initiatives: None																															
Remarks: The CVN 79 increase from the FY 2021 budget submission results from updating Hardware to reflect current contract pricing as a result of incentive payments to General Atomics to accelerate schedule, to implemented Engineering Change Proposals, increase of software testing, and inclusion of refurbishment of Deadloads. Increase also accounts for resolving reliability issues discovered on the first of class CVN 78. The CVN 80 Spares P-35 category includes \$30.727M for CVN 78 Class Interim Spares and \$1.770M for Initial Installation & Checkout Spares. Long Lead Time Materials Undefinitized Contract Action (UCA) awarded May 2014, Undefinitized Production UCA awarded June 2015 for CVN 79, and Production UCA definitized December 2016 for CVN 79 with option for CVN 80. CVN 80 option exercised January 2017. EMALS and AAG bundled savings on single production contract for CVN 79 and CVN 80 are reflective of contract negotiations.																															

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program			
Equipment Item: PHALANX BLOCK 1B MK 15 MOD 21 & 22, CLOSE - IN WEAPONS SYSTEM (CIWS)						PARM Code: IWS 11	
P-35 Category	FY 2013		FY 2018				
	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>			
Major Hardware	3	15.427	3	16.297			
Ancillary Equipment		0.091		0.251			
Spares		0.381		0.302			
System Engineering		1.215		2.020			
Technical Engineering Services		1.034		0.683			
Other Costs		1.495		1.406			
Total	3	19.643	3	20.959			
Description: Phalanx is a high fire rate Close-In Weapon System (CIWS) that automatically acquires, tracks and destroys Anti-Ship cruise missiles, Helos, Aircraft, and all types of Surface threats. The installed version will have one MK-15 Mod 21 and two MK-15 Mod 22 CIWS systems.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity <i>(Each)</i>	Unit Cost <i>(\$ M)</i>
FY 2013	CVN 79	RAYTHEON	C/FFP	Mar 2020		3	5.142
FY 2018	CVN 80	RAYTHEON	C/FFP	Jan 2024		3	5.432
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2013	CVN 79	Jun 2024	36	22	Aug 2019		
FY 2018	CVN 80	Mar 2028	28	22	Jan 2024		
Competition/Second Source Initiatives: None							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program			
Equipment Item: AN/SQQ-34, CARRIER-TACTICAL SUPPORT CENTER (CV-TSC)						PARM Code: PEO IWS 5E	
P-35 Category	FY 2013		FY 2018				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	3.150	1	3.199			
Spares		0.262		0.109			
System Engineering		0.351		0.381			
Technical Engineering Services		0.177		0.272			
Other Costs		1.110		0.495			
Total	1	5.050	1	4.456			
Description: CV-TSC provides for carrier organic Anti-submarine Warfare (ASW), Mine Warfare (MIW), Surface Warfare (SUW), and other composite warfare area sensor data processing, tactical command and control, and organic/battle-group aircraft mission support. CV-TSC supports both ship self defense and embarked Destroyer Squadron (DESRON) missions. This system is Open Architecture Computing Environment (OACE), Joint Fires Network (JFN), and FORCENet compliant, and includes redesign to maximize introduction of expected transformational technologies such as Common Processing System (CPS), Common Display System (CDS), sensor processing in support of the MH-60R helicopter, high speed bandwidth network, Excomm systems, net-centric warfare components, etc.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2013	CVN 79	Various	Various	Jan 2020		1	3.150
FY 2018	CVN 80	Various	Various	Jan 2020		1	3.199
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2013	CVN 79	Jun 2024	38	18	Oct 2019		
FY 2018	CVN 80	Mar 2028	35	18	Oct 2023		
Competition/Second Source Initiatives: None							
Remarks: Hardware procurement supports the same CPS (Common Processing System)/CDS (Common Display System) configuration for SSDS, CV-TSC and SEWIP.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program			
Equipment Item: MK-57 NATO SEASPARROW MISSILE SYSTEM (NSSMS)						PARM Code: PEO IWS 12	
P-35 Category	FY 2013		FY 2018				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	39.340	1	40.644			
Ancillary Equipment		0.269		0.435			
Spares		1.423		1.958			
System Engineering		1.358		0.816			
Technical Engineering Services		0.363		0.773			
Other Costs		3.100		1.897			
Total	1	45.853	1	46.523			

Description:
 The NATO SeaSparrow Surface Missile System (NSSMS) is a medium range self-defense missile system capable of defeating near/mid-term air/surface threats. The NSSMS MK 57 Mod 15 is comprised of (4) MK 9 tracker/illuminator radars and (2) MK 29 ESSM Guided Missile Launchers (GML) that connect to and support the SSDS Integrated Combat System (ICS).

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2013	CVN 79	RAYTHEON	C/FFP	Nov 2019		1	39.340
FY 2018	CVN 80	RAYTHEON	C/FFP	Nov 2021		1	40.644

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2013	CVN 79	Jun 2024	28	29	Sep 2019
FY 2018	CVN 80	Mar 2028	47	29	Nov 2021

Competition/Second Source Initiatives:
 None

Remarks:
 Guided Missile Launching System (GMLS) (P-35) and MK-9 Target Illuminator (P-35) have been combined in the budget as one complete system.

The CVN 79 increase from the FY 2021 budget submission is the result of Hardware increases due to the condition of the refurbishment asset (Target Illuminators), which will require more overhaul than originally anticipated. Systems Engineering and Other Costs increased to support contract management, ILS support for a new configuration on CVN 79, and engineering support for obsolescence claims during refurbishment.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																									
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program																											
Equipment Item: AVIATION DATA MANAGEMENT AND CONTROL SYSTEM (ADMACS)						PARM Code: PMA 251																									
P-35 Category	FY 2013		FY 2018																												
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																											
Major Hardware	1	2.467	1	5.141																											
System Engineering		0.468		0.950																											
Technical Engineering Services		0.673		0.592																											
Other Costs		1.426		2.145																											
Total	1	5.034	1	8.828																											
Description: ADMACS is a virtual, seamless, data sharing, knowledge based data system that provides interface for all aviation data systems. It is a tactical real-time information management system maintaining data integrity throughout the ship spaces that manage aircraft launch and recovery operations on board the carrier. ADMACS includes data from launch and recovery equipment, air traffic control, aviation maintenance, landing signaling officer, etc.																															
Contract Data: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th style="text-align: center;">Program Year</th> <th style="text-align: center;">Hull</th> <th style="text-align: center;">Prime Contractor</th> <th style="text-align: center;">Contract Method/Type</th> <th style="text-align: center;">Award Date</th> <th style="text-align: center;">New/Option</th> <th style="text-align: center;">Quantity (Each)</th> <th style="text-align: center;">Unit Cost (\$ M)</th> </tr> <tr> <td style="text-align: center;">FY 2013</td> <td style="text-align: center;">CVN 79</td> <td style="text-align: center;">BOWHEAD</td> <td style="text-align: center;">C/FFP</td> <td style="text-align: center;">Jul 2016</td> <td style="text-align: center;">Option</td> <td style="text-align: center;">1</td> <td style="text-align: right;">2.467</td> </tr> <tr> <td style="text-align: center;">FY 2018</td> <td style="text-align: center;">CVN 80</td> <td style="text-align: center;">Laurel Technologies Partnership/DRS</td> <td style="text-align: center;">TBD</td> <td style="text-align: center;">Feb 2022</td> <td></td> <td style="text-align: center;">1</td> <td style="text-align: right;">5.141</td> </tr> </table>								Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2013	CVN 79	BOWHEAD	C/FFP	Jul 2016	Option	1	2.467	FY 2018	CVN 80	Laurel Technologies Partnership/DRS	TBD	Feb 2022		1	5.141
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																								
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Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date																										
FY 2013	CVN 79	Jun 2024	56	12	Oct 2018																										
FY 2018	CVN 80	Mar 2028	28	12	Nov 2024																										
Competition/Second Source Initiatives: N/A																															
Remarks: Hardware is being procured in FY 2022 for CVN 80 due to the significant amount of obsolescence required to complete the assembly of the ADMACS system. The system that was procured for CVN 79 is no longer procurable (as is) and requires significant refresh (thus causing the spike in CVN 80 hardware requirements).																															

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																									
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program																											
Equipment Item: MK 49, MOD 5 ROLLING AIRFRAME MISSILE (RAM)						PARM Code: PEO IWS 11																									
P-35 Category	FY 2013		FY 2018																												
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																											
Major Hardware	2	11.884	2	7.902																											
Ancillary Equipment		0.263		1.503																											
Technical Data and Documentation		-		0.038																											
Spares		-		0.152																											
System Engineering		2.150		2.383																											
Technical Engineering Services		0.403		0.413																											
Other Costs		1.578		4.458																											
Total	2	16.278	2	16.849																											
Description: The MK 49 Mod 5 Rolling Airframe Missile (RAM) Weapon System is a lightweight, low cost, high power system for anti-ship missile defense against current and evolving threats. The Block 2 upgrade adds the capability of infrared, all-the-way missile guidance while maintaining the original dual-mode (RF/IR) capability. The helos, aircraft, and surface (HAS) upgrade enables the engagement of asymmetric threats.																															
Contract Data: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th style="text-align: center;">Program Year</th> <th style="text-align: center;">Hull</th> <th style="text-align: center;">Prime Contractor</th> <th style="text-align: center;">Contract Method/Type</th> <th style="text-align: center;">Award Date</th> <th style="text-align: center;">New/Option</th> <th style="text-align: center;">Quantity (Each)</th> <th style="text-align: center;">Unit Cost (\$ M)</th> </tr> <tr> <td style="text-align: center;">FY 2013</td> <td style="text-align: center;">CVN 79</td> <td style="text-align: center;">RAYTHEON</td> <td style="text-align: center;">C/FFP</td> <td style="text-align: center;">Jun 2019</td> <td></td> <td style="text-align: center;">2</td> <td style="text-align: right;">5.942</td> </tr> <tr> <td style="text-align: center;">FY 2018</td> <td style="text-align: center;">CVN 80</td> <td style="text-align: center;">RAYTHEON</td> <td style="text-align: center;">C/FFP</td> <td style="text-align: center;">Sep 2023</td> <td></td> <td style="text-align: center;">2</td> <td style="text-align: right;">3.951</td> </tr> </table>								Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2013	CVN 79	RAYTHEON	C/FFP	Jun 2019		2	5.942	FY 2018	CVN 80	RAYTHEON	C/FFP	Sep 2023		2	3.951
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																								
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FY 2018	CVN 80	Mar 2028	33	21	Sep 2023																										
Competition/Second Source Initiatives: None																															
Remarks: The CVN 79 Hardware increase from the FY 2021 budget submission is the result of executing Ancillary Requirements and Spares within the Major Hardware P-35 line. The actual hardware unit has not increased as it is already on contract.																															

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																									
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2001 / Carrier Replacement Program																											
Equipment Item: AN/SPQ-9B, ANTI-SHIP MISSILE DEFENSE (ASMD) SURFACE SURVEILLANCE AND TRACKING RADAR						PARM Code: PEO IWS2B																									
P-35 Category	FY 2013		FY 2018																												
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																											
Major Hardware	1	4.290	1	7.469																											
Spares		0.651		0.490																											
System Engineering		0.980		1.066																											
Technical Engineering Services		0.556		0.655																											
Other Costs		1.703		4.046																											
Total	1	8.180	1	13.726																											
Description: SPQ-9B is a multimode, x-band, narrow beam, pulse Doppler radar that detects and tracks sea-skimming missiles (ASMD) at the horizon in heavy clutter while simultaneously providing detection and tracking of surface targets.																															
Contract Data: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Prime Contractor</th> <th>Contract Method/Type</th> <th>Award Date</th> <th>New/Option</th> <th>Quantity (Each)</th> <th>Unit Cost (\$ M)</th> </tr> <tr> <td>FY 2013</td> <td>CVN 79</td> <td>LAUREL TECHNOLOGIES PARTNERSHIP</td> <td>SS/FFP</td> <td>Mar 2020</td> <td></td> <td style="text-align: center;">1</td> <td style="text-align: right;">4.290</td> </tr> <tr> <td>FY 2018</td> <td>CVN 80</td> <td>NGES</td> <td>SS/FFP</td> <td>Jan 2022</td> <td></td> <td style="text-align: center;">1</td> <td style="text-align: right;">7.469</td> </tr> </table>								Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2013	CVN 79	LAUREL TECHNOLOGIES PARTNERSHIP	SS/FFP	Mar 2020		1	4.290	FY 2018	CVN 80	NGES	SS/FFP	Jan 2022		1	7.469
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FY 2013	CVN 79	Jun 2024	28	18	Aug 2020																										
FY 2018	CVN 80	Mar 2028	35	18	Oct 2023																										
Competition/Second Source Initiatives: None																															
Remarks: The Enterprise Radar Suite (ERS) consists of AN/SPY-6(V)3 Enterprise Air Surveillance Radar (EASR), AN/SPQ-9B Anti-ship Missile Defense (ASMD) Surface Surveillance and Tracking Radar and MK-9 Target Illuminators (TIs) as part of MK-57 Nato SeaSparrow Missile System (NSSMS). Hardware is being procured in FY 2022 for CVN 80 to take advantage of an EOQ buy with CVN 81 and other ship classes to achieve the estimated price as reflected in the exhibit.																															

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 02: Other Warships / BSA 1: Other Warships	P-1 Line Item Number / Title: 2004 / CVN-81
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ID Code (A=Service Ready, B=Not Service Ready): A	Program Elements for Code B Items: N/A	Other Related Program Elements: N/A
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Line Item MDAP/MAIS Code: 223

Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (<i>Units in Each</i>)	-	1	-	-	-	-	-	-	-	-	-	-
Gross/Weapon System Cost (<i>\$ in Millions</i>)	0.000	12,483.595	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Less PY Advance Procurement (<i>\$ in Millions</i>)	-	0.000	-	-	-	-	-	-	-	-	-	-
Less Cost To Complete (<i>\$ in Millions</i>)	-	0.000	-	-	-	-	-	-	-	-	-	-
Less Subsequent Year Full Funding (<i>\$ in Millions</i>)	-	10,626.095	-	-	-	-	-	-	-	-	-	-
Less Funding in SCN Line Item 2001 (<i>\$ in Millions</i>)	-	1,429.682	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (<i>\$ in Millions</i>)	0.000	427.818	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Plus Subsequent Year Full Funding (<i>\$ in Millions</i>)	-	-	1,606.432	1,299.764	-	1,299.764	-	-	-	-	-	-
Full Funding TOA (<i>\$ in Millions</i>)	-	427.818	1,606.432	1,299.764	-	1,299.764	-	-	-	-	-	-
Plus CY Advance Procurement (<i>\$ in Millions</i>)	-	0.000	-	-	-	-	-	-	-	-	-	-
Plus Cost To Complete (<i>\$ in Millions</i>)	-	0.000	-	-	-	-	-	-	-	-	-	-
Plus Funding in SCN Line Item 2001 (<i>\$ in Millions</i>)	643.000	786.682	-	-	-	-	-	-	-	-	-	-
Total Obligation Authority (<i>\$ in Millions</i>)	643.000	1,214.500	1,606.432	1,299.764	0.000	1,299.764	-	-	-	-	-	-
(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)												
Plus Outfitting and Post Delivery (<i>\$ in Millions</i>)	-	0.000	-	-	-	-	-	-	-	-	-	-
Total (<i>\$ in Millions</i>)	643.000	1,214.500	1,606.432	1,299.764	-	1,299.764	-	-	-	-	-	-
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	-	12,483.595	-	-	-	-	-	-	-	-	-	-

Description:

To provide credible, sustainable, independent forward presence during peacetime without access to land bases; operate as the cornerstone of a joint and/or allied maritime expeditionary force in response to crisis; and carry the war to the enemy through joint multi-mission offensive operations.

The Navy designed the FORD Class aircraft carrier with the warfighting capability essential for the 21st century, and the flexibility and resilience to rapidly adapt to emerging threats. The Ford class incorporates advances in technology such as a new reactor plant, propulsion system, electric plant, Electromagnetic Aircraft Launch System (EMALS), Advanced Arresting Gear (AAG), machinery control, and integrated warfare systems that increase lethality, while lowering life cycle costs through reductions in maintenance and manning requirements. Increased Service Life Allowances (SLA) for weight and stability enable future modernization and the ability to adapt to new missions over the ship's 50-year life cycle.

CVN 80 and CVN 81 were awarded under a two-ship buy. CVN 81 is a separate contract line item under the same contract as CVN 80, which allows discrete hull costs to be captured. The CVN 80/81 construction contract is a Fixed Price Incentive (Firm Target) (FPIF) contract type that limits the Navy's liability and incentivizes the shipyard's best performance. The contract language guarantees a single technical baseline for both ships, which allows the shipyard to re-use engineering rollover products, minimize changes between the two ships and leverages economic order quantities for equipment and material procurement.

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy				Date: May 2021																																									
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 02: Other Warships / BSA 1: Other Warships			P-1 Line Item Number / Title: 2004 / CVN-81																																										
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: N/A		Other Related Program Elements: N/A																																									
Line Item MDAP/MAIS Code: 223																																													
<table style="width: 100%; border: none;"> <tr> <td style="width: 33%; vertical-align: top;"> Characteristics: Length Overall 1092 ft Beam 134 ft Displacement 97,337 TONS Draft 38.7 ft </td> <td style="width: 33%; vertical-align: top;"> Systems: Electronics -SHIP SELF DEFENSE SYSTEM (SSDS) </td> <td style="width: 33%; vertical-align: top;"> Ordnance -ELECTROMAGNETIC AIRCRAFT LAUNCHING SYSTEM (EMALS) -AN/SPY-6(V)3 ENTERPRISE AIR SURVEILLANCE RADAR (EASR) -ADVANCED ARRESTING GEAR (AAG) </td> </tr> </table>						Characteristics: Length Overall 1092 ft Beam 134 ft Displacement 97,337 TONS Draft 38.7 ft	Systems: Electronics -SHIP SELF DEFENSE SYSTEM (SSDS)	Ordnance -ELECTROMAGNETIC AIRCRAFT LAUNCHING SYSTEM (EMALS) -AN/SPY-6(V)3 ENTERPRISE AIR SURVEILLANCE RADAR (EASR) -ADVANCED ARRESTING GEAR (AAG)																																					
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<table style="width: 100%; border: none;"> <tr> <td style="width: 20%;">Production Status:</td> <td style="width: 80%;">CVN 81</td> </tr> <tr> <td>Contract Award Date</td> <td>Jan 2019</td> </tr> <tr> <td>Months to Completion</td> <td></td> </tr> <tr> <td> a) Award to Delivery</td> <td>157 months</td> </tr> <tr> <td> b) Construction Start to Delivery</td> <td>157 months</td> </tr> <tr> <td>Delivery Date</td> <td>Feb 2032</td> </tr> <tr> <td>Completion Of Fitting Out</td> <td>Apr 2032</td> </tr> <tr> <td>Obligation Work Limit Date</td> <td>Mar 2033</td> </tr> </table>						Production Status:	CVN 81	Contract Award Date	Jan 2019	Months to Completion		a) Award to Delivery	157 months	b) Construction Start to Delivery	157 months	Delivery Date	Feb 2032	Completion Of Fitting Out	Apr 2032	Obligation Work Limit Date	Mar 2033																								
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Completion Of Fitting Out	Apr 2032																																												
Obligation Work Limit Date	Mar 2033																																												
<table style="width: 100%; border: none;"> <tr> <td style="width: 30%;">Design Schedule</td> <td style="width: 15%;"><u>Start / Issue</u></td> <td style="width: 15%;"><u>Complete / Response</u></td> <td style="width: 15%;"><u>Reissue</u></td> <td style="width: 25%;"><u>Reissue Complete / Response</u></td> </tr> <tr> <td>Issue Date for TLR</td> <td>Apr 2004</td> <td>N/A</td> <td></td> <td></td> </tr> <tr> <td>Issue Date for TLS</td> <td>Sep 2006</td> <td>N/A</td> <td></td> <td></td> </tr> <tr> <td>Preliminary Design</td> <td>Jan 2003</td> <td>Jul 2008</td> <td></td> <td></td> </tr> <tr> <td>Contract Design</td> <td>May 2004</td> <td>Apr 2008</td> <td></td> <td></td> </tr> <tr> <td>Detail Design</td> <td>Jan 2004</td> <td>Sep 2009</td> <td></td> <td></td> </tr> <tr> <td>Request for Proposals</td> <td>Jul 2007</td> <td>Oct 2007</td> <td></td> <td></td> </tr> <tr> <td>Design Agent</td> <td colspan="4">Huntington Ingalls Industries</td> </tr> </table>						Design Schedule	<u>Start / Issue</u>	<u>Complete / Response</u>	<u>Reissue</u>	<u>Reissue Complete / Response</u>	Issue Date for TLR	Apr 2004	N/A			Issue Date for TLS	Sep 2006	N/A			Preliminary Design	Jan 2003	Jul 2008			Contract Design	May 2004	Apr 2008			Detail Design	Jan 2004	Sep 2009			Request for Proposals	Jul 2007	Oct 2007			Design Agent	Huntington Ingalls Industries			
Design Schedule	<u>Start / Issue</u>	<u>Complete / Response</u>	<u>Reissue</u>	<u>Reissue Complete / Response</u>																																									
Issue Date for TLR	Apr 2004	N/A																																											
Issue Date for TLS	Sep 2006	N/A																																											
Preliminary Design	Jan 2003	Jul 2008																																											
Contract Design	May 2004	Apr 2008																																											
Detail Design	Jan 2004	Sep 2009																																											
Request for Proposals	Jul 2007	Oct 2007																																											
Design Agent	Huntington Ingalls Industries																																												
<u>Classification of Cost Estimate:</u> CLASS C BUDGET ESTIMATE																																													

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy		Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2004 / CVN-81	
Cost Categories (†) indicates the presence of a P-8a	FY 2020		
	Qty <i>(Each)</i>		Total Cost <i>(\$ M)</i>
Plan Costs	1		
Basic Construction/Conversion			8,111.485
Change Orders			152.000
Electronics ^(†)			264.700
Propulsion Equipment			2,887.110
Hull, Mechanical, and Electrical (HM&E) ^(†)			13.600
Ordnance ^(†)			1,015.606
Other Cost			39.094
Total Ship Estimate			12,483.595
Less Subsequent Full Funding FY 2021			1,606.432
Less Subsequent Full Funding FY 2022			1,299.764
Less Subsequent Full Funding FY 2023			731.352
Less Subsequent Full Funding FY 2024			637.468
Less Subsequent Full Funding FY 2025			566.367
Less Subsequent Full Funding FY 2026			2,087.200
Less Subsequent Full Funding FY 2027			2,026.828
Less Subsequent Full Funding FY 2028			1,670.684
Less Funding in SCN Line Item 2001 FY 2019			643.000
Less Funding in SCN Line Item 2001 FY 2020			786.682
Net P-1 Funding			427.818
Remarks: Basic increased by \$0.6M from the FY 2021 budget submission due to addition of Classified Storage (JSF) to the technical baseline. Electronics increased by \$22.1M from the FY 2021 budget submission due to the addition of Joint Precision Aircraft Landing System (JPALS), Ready Room Modifications and Lithium Ion Battery Handling and Storage. These additions are part of the revised technical baseline. Ordnance increased by \$10.2M from the FY 2021 budget submission due to the addition of MK-38 MOD 3 Machine Gun System and Joint Strike Fighter Autonomic Logistics System (JSF ALIS), which were added to the technical baseline. \$786.682 million of FY 2020 funding for CVN 81 was executed in SCN line item 2001 during the FY 2020 continuing resolution.			

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy		Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2004 / CVN-81	
Electronics		FY 2020	
		Qty (Each)	Total Cost (\$ M)
P-35 Items			
CONSOLIDATED AFLOAT NETWORK AND ENTERPRISE SERVICES (CANES)		1	15.951
AN/USG-2, COOPERATIVE ENGAGEMENT CAPABILITY (CEC)		1	6.071
DIGITAL MODULAR RADIO (DMR) ULTRA HIGH FREQUENCY/VERY HIGH FREQUENCY LINE OF SIGHT (EHF/VHF LOS) SAT		1	10.835
AN/UPX-29(V), INTERROGATOR FRIEND OR FOE (IFF) W/MK XII		1	6.438
SPN-46, AUTOMATIC CARRIER LANDING SYSTEM		1	9.661
SHIP SELF DEFENSE SYSTEM (SSDS)		1	29.699
AN/SYY-1(V)1 AIR TRAFFIC CONTROL SYSTEM, SHIPBOARD		1	6.219
NAVY MULTI-BAND TERMINAL (NMT)		1	6.259
AN/SLQ-32(V)6, SURFACE ELECTRONIC WARFARE IMPROVEMENT PROGRAM (SEWIP) BLOCK 2		1	10.488
AN/SRQ-6/MCS-21, SHIPS SIGNAL EXPLOITATION EQUIPMENT (SSEE)		1	7.716
AN/SRC-61 (V)X HFDAG		1	6.021
SEA BASED JPALS JOINT PRECISION AIRCRAFT LANDING SYSTEM (JPALS)		1	16.100
P-35 Items Subtotal			131.458
Major Items			
AN/USQ-155(V)1 TACTICAL VARIANT SWITCH		1	2.725
INFORMATION ASSURANCE (IA)			2.018
AN/URC-141X(V), MULTI-FUNCTION INFORMATION DISTRIBUTION SYSTEM (MIDS)-ON SHIP (MOS)		1	1.576
AN/SLQ-25C DUAL, SURFACE SHIP TORPEDO DEFENSE SYSTEM, NIXIE		1	4.627
SHIPBOARD AIR TRAFFIC CONTROL COMMUNICATIONS (SATCC)		1	2.328
AN/WSN-7(V)3, RING LASER GYRO NAVIGATOR (RLGN)		1	3.102
DISTRIBUTED SYSTEMS DESIGN INTEGRATION SERVICES		1	19.037
C4I INTEGRATION & COORDINATION			7.114
DISTRIBUTED COMMON GROUND STATION - NAVY (DCGS-N)		1	2.304
AN/USQ-144K AUTOMATED DIGITAL NETWORK SYSTEM (ADNS)		1	1.307
AN/UYQ-86 COMMON DATA LINK MANAGEMENT SYSTEM (CDLMS) WITH NGC2P		1	1.805
OA-9277 ULTRA HIGH FREQUENCY (UHF) MULTICOUPLER		1	1.954
ARC-210 CARRIER AIR TRAFFIC CONTROL CENTER (CATCC) - PRIFLY - LANDING SIGNAL OFFICER (LSO) SYSTEM		1	1.657
WARFARE SYSTEM INTEGRATION			19.995
COMMERCIAL BROADBAND SATELLITE PROGRAM, FORCE LEVEL VARANT (CBSP-FLV)		2	2.450
GLOBAL POSITIONING SYSTEM, NAVIGATION, AND TIMING SERVICE (GPNTS)		1	1.719
INTEGRATED STRIKE PLANNING & EXECUTION SYSTEMS (ISP&E)		1	8.194
AN/USQ-123(V), COMMUNICATIONS DATA LINK-SYSTEM (CDL-S)		1	2.464

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy		Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2004 / CVN-81	
Electronics	FY 2020		
	Qty (Each)	Total Cost (\$ M)	
AN/SPN-41 (V), INSTRUMENT LANDING SYSTEM (ILS)	1	3.873	
SHIP SIGNAL EXPLOITATION SPACE (SSES/SI) COMMUNICATIONS	1	4.044	
TURNKEY RADIO COMMUNICATIONS SYSTEM (RCS)	1	14.789	
AN/USQ-T52 TRAINING INTERFACE UNIT (TIU) ADVANCED TRAINING DOMAIN (ATD)	1	1.828	
AN/SPS-73(V)18 NEXT GENERATION SURFACE SEARCH RADAR (NGSSR)		2.341	
READY ROOM (JSF)	1	3.600	
LITHIUM ION BATTERY HANDLING & STORAGE		2.400	
Major Items Subtotal		119.251	
Other Cost Elements			
Other ELECTRONICS		13.991	
Other Cost Elements Subtotal		13.991	
Total Electronics		264.700	
Remarks: P-8A Electronics: AN/SYY-1(V)1 Air Traffic Control System Shipboard replaces AN/TPX-42A(V)14, Carrier Air Traffic Control Center - Direct Altitude and Identify Readout (CATCC-DAIR). AN/SYY-1(V)1 addresses obsolescence, incorporates existing AN/TPX-42A(V) functionality and will add additional capability and interfaces across all platforms. Global Positioning System, Navigation and Timing Service (GPNTS) is the next generation of navigation to address Fleet obsolescence issues. GPNTS replaces AN/SSN-6(V)X Block 4, Navigation Sensor System Interface (NAVSSI). AN/USQ-T52 Training Interface Unit (TIU) Advanced Training Domain (ATD) replaces AN/USQ-T46X(V)X, Battle Force Tactical Training System (BFTT). Official nomenclature for Next Generation Surface Search Radar (NGSSR) has been updated to AN/SPS-73(V)18, Next Generation Surface Search Radar (NGSSR). AN/SLQ-25E Dual, Surface Ship Torpedo Defense System, NIXIE replaces AN/SLQ-25C Dual, Surface Ship Torpedo Defense System, NIXIE due to obsolescence and no longer being available to procure or be refurbished. P-8A Electronics increased by \$22.1M from the FY 2021 budget submission due to the addition of the following systems, which were part of the revised technical baseline: - Joint Precision Aircraft Landing System (JPALS) - Ready Room Modifications - Lithium Ion Battery Handling and Storage			

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy		Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2004 / CVN-81	
Ordnance	FY 2020		
	Qty (Each)	Total Cost (\$ M)	
P-35 Items			
ELECTROMAGNETIC AIRCRAFT LAUNCHING SYSTEM (EMALS)	1	593.314	
AN/SPY-6(V)3 ENTERPRISE AIR SURVEILLANCE RADAR (EASR)	1	57.854	
ADVANCED ARRESTING GEAR (AAG)	1	229.279	
PHALANX BLOCK 1B MK 15 MOD 21 & 22, CLOSE - IN WEAPONS SYSTEM (CIWS)	3	19.286	
AN/SQQ-34, CARRIER-TACTICAL SUPPORT CENTER (CV-TSC)	1	4.100	
MK-57 NATO SEASPARROW MISSILE SYSTEM (NSSMS)	1	45.849	
AVIATION DATA MANAGEMENT AND CONTROL SYSTEM (ADMACS)	1	8.772	
MK 49, MOD 3 ROLLING AIRFRAME MISSILE (RAM)	2	15.788	
AN/SPQ-9B, ANTI-SHIP MISSILE DEFENSE (ASMD) SURFACE SURVEILLANCE AND TRACKING RADAR	1	13.718	
P-35 Items Subtotal		987.960	
Major Items			
LANDING SIGNAL OFFICER DISPLAY SYSTEM (LSODS)	1	2.098	
MORIAH BLOCK 2	1	1.490	
LONG RANGE LINEUP SYSTEM (LRLS)	1	0.960	
IMPROVED FRESNEL LENS OPTICAL LANDING SYSTEM (IFLOLS)	1	2.257	
INTEGRATED LAUNCH AND RECOVERY TELEVISION SYSTEM (ILARTS)	1	5.509	
COMPACT SWAGING MACHINE (CSM)	2	2.500	
JOINT STRIKE FIGHTER ALIS	1	1.900	
MK-38 MOD 3 Machine Gun System (MGS)	4	8.300	
Major Items Subtotal		25.014	
Other Cost Elements			
Other ORDNANCE		2.632	
Other Cost Elements Subtotal		2.632	
Total Ordnance		1,015.606	
Remarks: P-8A Ordnance: - The Enterprise Air Surveillance Radar (EASR) replaces Dual Band Radar (DBR). - Official nomenclature for Enterprise Air Surveillance Radar (EASR) has been updated to AN/SPY-6(V)3, Enterprise Air Surveillance Radar (EASR). - MK-57 NATO SeaSparrow Missile System (NSSMS) has been updated to become one complete system from MK29 MOD 5 Guided Missile Launching System (GMLS) (P-35) and MK-9 Target Illuminator (P-35).			

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2004 / CVN-81
- P-8A Ordnance increased by \$10.2M from the FY 2021 budget submission due to the addition of the following systems, which were part of the revised technical baseline: - MK-38 MOD 3 Machine Gun System - Joint Strike Fighter Autonomic Logistics System (JSF ALIS)		

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2004 / CVN-81			
Equipment Item: CONSOLIDATED AFLOAT NETWORK AND ENTERPRISE SERVICES (CANES)						PARM Code: PMW 160	
P-35 Category				FY 2020			
				Qty (Each)	Total Cost (\$ M)		
Major Hardware				1	10.998		
Spares					0.471		
System Engineering					2.350		
Technical Engineering Services					0.270		
Other Costs					1.862		
Total				1	15.951		
Description: CANES will provide the Navy tactical/non-tactical information environment and infrastructure necessary to enable hosting, extended services reach-back and reach-forward, and relay functions. These capabilities will support real time and non-real time tactical/non-tactical edge connected, connectionless, and ad-hoc voice, video and data information exchange requirements. CANES replace and modernize afloat networks with hardware, software and enterprise services infrastructure to enable information warfare from and within the tactical domain. CANES provides complete infrastructure inclusive of hardware, software, processing, storage and end user devices for Unclassified, Coalition, Secret and Sensitive Compartmented Information (SCI) enclaves for all basic network services to Navy surface combatants, submarines and maritime operations centers. The POR is Increment 2.0.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	CVN 81	VARIOUS	C/FFP	Oct 2028		1	10.998
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 81	Feb 2032	28	12	Oct 2028		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2004 / CVN-81			
Equipment Item: AN/USG-2, COOPERATIVE ENGAGEMENT CAPABILITY (CEC)						PARM Code: PEO IWS 6.0	
P-35 Category				FY 2020			
				Qty (Each)	Total Cost (\$ M)		
Major Hardware				1	2.733		
Spares					0.467		
System Engineering					2.181		
Technical Engineering Services					0.196		
Other Costs					0.494		
Total				1	6.071		
Description: The Cooperative Engagement Capability (CEC) AN/USG-2B system provides real time integration of fire control quality sensor data into a single composite data source, which is used by multiple CEC ships and airborne units for direct and remote missile engagements. CEC significantly improves battle force Anti-Air Warfare (AAW) capability by coordinating all force AAW sensors into a single real time, fire control quality composite track picture.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	CVN 81	TBD	TBD	Feb 2028		1	2.733
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 81	Feb 2032	30	18	Feb 2028		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2004 / CVN-81			
Equipment Item: DIGITAL MODULAR RADIO (DMR) ULTRA HIGH FREQUENCY/VERY HIGH FREQUENCY LINE OF SIGHT (EHF/VHF LOS) SAT						PARM Code: PMW 170	
P-35 Category				FY 2020			
				Qty (Each)	Total Cost (\$ M)		
Major Hardware				1	9.129		
Spares					0.055		
System Engineering					0.639		
Technical Engineering Services					0.561		
Other Costs					0.378		
Ancillary Equipment					0.073		
Total				1	10.835		
Description: DMR-VHF/UHF LOS/SATCOM is an open architecture system that allows transmission and reception of UHF and VHF RF signals. The DMR replaces many legacy systems, including some crypto, Line Of Sight (LOS) and Satellite Communications (SATCOM) components.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	CVN 81	TBD	TBD	May 2027		1	9.129
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 81	Feb 2032	39	18	May 2027		
Competition/Second Source Initiatives: None							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2004 / CVN-81																			
Equipment Item: AN/UPX-29(V), INTERROGATOR FRIEND OR FOE (IFF) W/MK XII						PARM Code: PMA 213																	
P-35 Category				FY 2020																			
				Qty (Each)	Total Cost (\$ M)																		
Major Hardware				1	4.994																		
Ancillary Equipment					0.101																		
Spares					0.121																		
System Engineering					0.616																		
Technical Engineering Services					0.150																		
Other Costs					0.456																		
Total				1	6.438																		
Description: IFF is an approved and fully supported centralized Mark XII/XIIA Interrogator system. It uses one receiver transmitter that synchronizes video with up to four radar sweeps. It supplies synthetic video (symbolology) to, and accepts requests from, as many as 22 remote locations. It provides digital target reporting to the combat systems/weapon systems computer via full scan, sector, and/or pop-up interrogations. It provides instantaneous target reporting at requested range and azimuth through the use of an electronically-steered Antenna Group OE-120/UPX or OE-120B/UPX. It provides electronically evaluated Mode 4 target reporting directly to operators and over the combat systems/weapon system computer interface. It provides full redundancy so identification capabilities are retained in case of main processor, main antenna, or main receiver/transmitter failure.																							
Contract Data: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Prime Contractor</th> <th>Contract Method/Type</th> <th>Award Date</th> <th>New/Option</th> <th>Quantity (Each)</th> <th>Unit Cost (\$ M)</th> </tr> <tr> <td>FY 2020</td> <td>CVN 81</td> <td>NOTHROP GRUMMAN-BAE SYSTEMS</td> <td>SS/FFP</td> <td>Mar 2022</td> <td></td> <td style="text-align: center;">1</td> <td style="text-align: right;">4.994</td> </tr> </table>								Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2020	CVN 81	NOTHROP GRUMMAN-BAE SYSTEMS	SS/FFP	Mar 2022		1	4.994
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																
FY 2020	CVN 81	NOTHROP GRUMMAN-BAE SYSTEMS	SS/FFP	Mar 2022		1	4.994																
Delivery Date: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Earliest Ship Delivery Date</th> <th>Months Required Before Delivery</th> <th>Production Leadtime</th> <th>Required Award Date</th> </tr> <tr> <td>FY 2020</td> <td>CVN 81</td> <td>Feb 2032</td> <td style="text-align: center;">35</td> <td style="text-align: center;">24</td> <td style="text-align: center;">Mar 2027</td> </tr> </table>								Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date	FY 2020	CVN 81	Feb 2032	35	24	Mar 2027				
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date																		
FY 2020	CVN 81	Feb 2032	35	24	Mar 2027																		
Competition/Second Source Initiatives: None																							
Remarks: The award date has changed from March 2023 as submitted in the FY 2021 budget submission to March 2022 in order to take advantage of current contract pricing. The current IFF contract (with EOQ pricing) is available through FY 2022. The future contract award will have an increased unit cost, and the configuration is likely to change (due to obsolescence, etc.).																							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2004 / CVN-81			
Equipment Item: SPN-46, AUTOMATIC CARRIER LANDING SYSTEM						PARM Code: PMA 213	
P-35 Category				FY 2020			
				Qty (Each)		Total Cost (\$ M)	
Major Hardware				1		5.833	
System Engineering						1.451	
Technical Engineering Services						0.338	
Other Costs						2.039	
Total				1		9.661	
Description: AN/SPN-46 Automatic Carrier Landing System (ACLS) is a precision approach landing system (PALS) which provides electronic guidance to carrier-based aircraft and allows them to land in all-weather conditions with no limitations due to low ceiling or restricted visibility. AN/SPN-46 is a fully automated, all weather approach landing aid for carrier aircraft.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	CVN 81	NAWCAD	Various	Mar 2022		1	5.833
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 81	Feb 2032	35	24	Mar 2027		
Competition/Second Source Initiatives: None.							
Remarks: Hardware procurement moved from FY 2021 to FY 2022 with no change in ability to achieve an EOQ buy with other ship classes and achieve the estimated price as reflected in the exhibit.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2004 / CVN-81			
Equipment Item: SHIP SELF DEFENSE SYSTEM (SSDS)						PARM Code: PEO IWS 10.0	
P-35 Category				FY 2020			
				Qty (Each)	Total Cost (\$ M)		
Major Hardware				1	10.940		
Technical Data and Documentation					1.430		
Spares					0.592		
System Engineering					6.865		
Technical Engineering Services					0.728		
Other Costs					9.144		
Total				1	29.699		
Description: The SSDS MK 2, Mod (6E) Common C2 system provides capabilities for multi-mission requirements including Ship Protection against air, surface, and subsurface threats using both own-ship and remote data (Joint Composite Track Number (JCTN) and Joint Data Network (JDN)) in support of the Anti-Air Warfare (AAW) Capstone requirements.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	CVN 81	Lockheed Martin	TBD	Jul 2027		1	10.940
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 81	Feb 2032	31	24	Jul 2027		
Competition/Second Source Initiatives: None							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2004 / CVN-81			
Equipment Item: AN/SYY-1(V)1 AIR TRAFFIC CONTROL SYSTEM, SHIPBOARD						PARM Code: PMA 213	
P-35 Category				FY 2020			
				Qty (Each)	Total Cost (\$ M)		
Major Hardware				1	3.176		
Spares					0.285		
System Engineering					1.986		
Technical Engineering Services					0.060		
Other Costs					0.712		
Total				1	6.219		
Description: AN/SYY-1(V)1 Air Traffic Control System, Shipboard" is a processing and display system which correlates and fuses sensor inputs from radar and IFF. The correlated sensor data is displayed to Air Traffic Controllers in a 2D air picture to facilitate the safe and expeditious movement of air traffic operating in the carrier controlled area and launching from/recovering to the ship.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	CVN 81	TBD	TBD	Apr 2022		1	3.176
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 81	Feb 2032	34	24	Apr 2027		
Competition/Second Source Initiatives: N/A							
Remarks: This system replaces AN/TPX-42. AN/SYY-1(V)1 addresses obsolescence, incorporates existing AN/TPX-42A(V) functionality, and adds additional capability and interfaces across all platforms. Hardware is being procured in FY 2022 via an EOQ buy to achieve the estimated price as reflected in the exhibit. The system is being procured in FY 2022 to make use of an existing contract already in place.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2004 / CVN-81			
Equipment Item: NAVY MULTI-BAND TERMINAL (NMT)						PARM Code: PMW 170	
P-35 Category				FY 2020			
				Qty (Each)	Total Cost (\$ M)		
Major Hardware				1	5.646		
Ancillary Equipment					0.052		
System Engineering					0.097		
Technical Engineering Services					0.097		
Other Costs					0.367		
Total				1	6.259		
Description: The Advanced Extremely High Frequency (AEHF) Navy Multi-band Terminal (NMT) will be used to receive signals from the Advanced EHF satellites which is a follow-on to the DoD's highly secure, highly protected MILSTAR communications satellite system.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	CVN 81	TBD	TBD	Jan 2028		1	5.646
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 81	Feb 2032	31	18	Jan 2028		
Competition/Second Source Initiatives: None							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2004 / CVN-81			
Equipment Item: AN/SLQ-32(V)6, SURFACE ELECTRONIC WARFARE IMPROVEMENT PROGRAM (SEWIP) BLOCK 2						PARM Code: PEO IWS 2E	
P-35 Category				FY 2020			
				Qty (Each)		Total Cost (\$ M)	
Major Hardware				1		10.036	
Ancillary Equipment						0.341	
System Engineering						0.098	
Other Costs						0.013	
Total				1		10.488	
Description: SEWIP Block 2 is a scalable Electronic Warfare enterprise suite to provide improved Electromagnetic Interference (EMI) mitigation and Combat System Interface capabilities to select new construction ships as well as upgrade current AN/SLQ-32(V)3 and (V)4 Electronic Warfare (EW) suites on existing ships. It provides enhanced shipboard Electronic Warfare (EW) for early detection, analysis, threat warning and protection from anti-ship missiles. SEWIP Block 2 focused on Electronic Support (ES) capability improvements.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	CVN 81	TBD	TBD	Aug 2028		1	10.036
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 81	Feb 2032	24	18	Aug 2028		
Competition/Second Source Initiatives: None							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2004 / CVN-81			
Equipment Item: AN/SRQ-6/MCS-21, SHIPS SIGNAL EXPLOITATION EQUIPMENT (SSEE)						PARM Code: PMW 120	
P-35 Category				FY 2020			
				Qty (Each)	Total Cost (\$ M)		
Major Hardware				1	5.181		
Ancillary Equipment					0.084		
Spares					0.208		
System Engineering					0.894		
Technical Engineering Services					0.190		
Other Costs					1.159		
Total				1	7.716		
Description: SSEE provided for cryptological signal acquisition, recognition, analysis and geo-location. It replaces Maritime Cryptological System (MCS-21) which replaces the Battle Group Passive Horizon Extension System (BGPHEs).							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	CVN 81	TBD	TBD	Sep 2027		1	5.181
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 81	Feb 2032	35	18	Sep 2027		
Competition/Second Source Initiatives: None							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2004 / CVN-81			
Equipment Item: AN/SRC-61 (V)X HFDAG						PARM Code: PMW 170	
P-35 Category				FY 2020			
				Qty (Each)	Total Cost (\$ M)		
Major Hardware				1	4.786		
Ancillary Equipment					0.052		
Spares					0.011		
System Engineering					0.215		
Technical Engineering Services					0.524		
Other Costs					0.433		
Total				1	6.021		
Description: High Frequency (HF) Distributed Amplifier Group (DAG) is the Navy's Program of Record (POR) HF system and is the follow-on replacement of HF Radio Group (HFRG). HFDAG has a modular architecture and utilizes COTS equipment to the maximum extent possible. It provides Line Of Sight (LOS/Beyond Line of Sight (BLOS) voice and data transmission capabilities to USN Ships. The 16-channel CVN variant greatly improves capabilities from HFRG: (1) increases availability (Ao), (2) provides reprogrammable waveforms, (3) increases the number of waveforms available, (4) provides automatic link establishment (ALE).							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	CVN 81	TBD	TBD	Sep 2027		1	4.786
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 81	Feb 2032	35	18	Sep 2027		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2004 / CVN-81			
Equipment Item: SEA BASED JPALS JOINT PRECISION AIRCRAFT LANDING SYSTEM (JPALS)						PARM Code: PMA 213	
P-35 Category				FY 2020			
				Qty (Each)	Total Cost (\$ M)		
Major Hardware				1	10.465		
Spares					1.066		
System Engineering					2.073		
Technical Engineering Services					0.436		
Other Costs					2.060		
Total				1	16.100		
Description: The Joint Precision Approach and Landing System (JPALS) is the primary precision approach and landing system for CVN ships to support F-35B, F-35C, MQ-25 and future platforms. JPALS ship systems are required to provide a primary precision approach capability during night and instrument flight conditions and coupled approach to the deck (auto-land) capability aboard CVN ships. JPALS also provides over-the-air inertial alignment capability for CVN ships to support aircraft platforms without Link-4A capability, including F-35, MQ-25 and future platforms.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	CVN 81	RAYTHEON	C/FFP	Aug 2028		1	10.465
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 81	Feb 2032	30	12	Aug 2028		
Competition/Second Source Initiatives: N/A							
Remarks: JPALS required to support JSF integration on FORD Class.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2004 / CVN-81																			
Equipment Item: ELECTROMAGNETIC AIRCRAFT LAUNCHING SYSTEM (EMALS)						PARM Code: PMA 251																	
P-35 Category				FY 2020																			
				Qty (Each)		Total Cost (\$ M)																	
Major Hardware				1	557.474																		
Technical Data and Documentation																							
Spares																							
System Engineering					18.001																		
Technical Engineering Services					2.628																		
Other Costs					15.211																		
Total				1	593.314																		
Description: EMALS is an advanced technology electrically generated launching system that uses a moving electromagnetic field to propel aircraft to launch speed. EMALS is made up of six primary sub-systems: prime power interface, energy storage, energy distribution, power conversion, launch motor, and launch control subsystem. Benefits over the current C13 steam catapults include reduced weight and volume, greater launching flexibility for future aircraft, improved control, and reduced manning workload requirements.																							
Contract Data: <table border="1" style="width:100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Prime Contractor</th> <th>Contract Method/Type</th> <th>Award Date</th> <th>New/Option</th> <th>Quantity (Each)</th> <th>Unit Cost (\$ M)</th> </tr> <tr> <td align="center">FY 2020</td> <td align="center">CVN 81</td> <td align="center">GENERAL ATOMICS</td> <td align="center">SS/FFP</td> <td align="center">Aug 2021</td> <td align="center">Option</td> <td align="center">1</td> <td align="center">557.474</td> </tr> </table>								Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2020	CVN 81	GENERAL ATOMICS	SS/FFP	Aug 2021	Option	1	557.474
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																
FY 2020	CVN 81	GENERAL ATOMICS	SS/FFP	Aug 2021	Option	1	557.474																
Delivery Date: <table border="1" style="width:100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Earliest Ship Delivery Date</th> <th>Months Required Before Delivery</th> <th>Production Leadtime</th> <th>Required Award Date</th> </tr> <tr> <td align="center">FY 2020</td> <td align="center">CVN 81</td> <td align="center">Feb 2032</td> <td align="center">57</td> <td align="center">48</td> <td align="center">May 2023</td> </tr> </table>								Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date	FY 2020	CVN 81	Feb 2032	57	48	May 2023				
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date																		
FY 2020	CVN 81	Feb 2032	57	48	May 2023																		
Competition/Second Source Initiatives: None																							
Remarks: Motor Generator Long Lead Time Material (LLTM) hardware is being procured in FY 2021 to lessen the production gap and reduce lost learning on units completed early in production on CVN 80.																							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2004 / CVN-81																			
Equipment Item: AN/SPY-6(V)3 ENTERPRISE AIR SURVEILLANCE RADAR (EASR)						PARM Code: PEO IWS 2.0																	
P-35 Category				FY 2020																			
				Qty (Each)	Total Cost (\$ M)																		
Major Hardware				1	53.365																		
Ancillary Equipment																							
Spares																							
System Engineering					4.489																		
Technical Engineering Services																							
Other Costs																							
Total				1	57.854																		
Description: The Enterprise Air Surveillance Radar (EASR) suite will be a modern long-range, three-dimensional (3D) radar used to search, detect and provide space-stabilized, three-coordinate (range, bearing, height) data for air intercept control and designation to a weapon system and Air Traffic Control (ATC) system. The Enterprise Radar Suite (ERS), which includes EASR, is intended to replace the functions that Dual Band Radar (DBR) performed on CVN 78, but at a much lower cost.																							
Contract Data: <table border="1" style="width:100%; border-collapse: collapse;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Prime Contractor</th> <th>Contract Method/Type</th> <th>Award Date</th> <th>New/Option</th> <th>Quantity (Each)</th> <th>Unit Cost (\$ M)</th> </tr> <tr> <td>FY 2020</td> <td>CVN 81</td> <td>RAYTHEON</td> <td>C/CPIF</td> <td>Feb 2027</td> <td></td> <td style="text-align: center;">1</td> <td style="text-align: right;">53.365</td> </tr> </table>								Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2020	CVN 81	RAYTHEON	C/CPIF	Feb 2027		1	53.365
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																
FY 2020	CVN 81	RAYTHEON	C/CPIF	Feb 2027		1	53.365																
Delivery Date: <table border="1" style="width:100%; border-collapse: collapse;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Earliest Ship Delivery Date</th> <th>Months Required Before Delivery</th> <th>Production Leadtime</th> <th>Required Award Date</th> </tr> <tr> <td>FY 2020</td> <td>CVN 81</td> <td>Feb 2032</td> <td style="text-align: center;">30</td> <td style="text-align: center;">30</td> <td style="text-align: center;">Feb 2027</td> </tr> </table>								Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date	FY 2020	CVN 81	Feb 2032	30	30	Feb 2027				
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date																		
FY 2020	CVN 81	Feb 2032	30	30	Feb 2027																		
Competition/Second Source Initiatives: None																							
Remarks: The hardware configuration (non-rotating) is essentially three times that of a rotating configuration, which is currently planned for the big deck amphibious warfare ships. EASR will have three phased arrays mounted around the island, while the amphibious warfare ships will use one rotating array. Below deck equipment is also provided at a larger scale with the non-rotating variant of EASR. The Enterprise Radar Suite (ERS) consists of AN/SPY-6(V)3 Enterprise Air Surveillance Radar (EASR), AN/SPQ-9B Anti-ship Missile Defense (ASMD) Surface Surveillance and Tracking Radar, and MK-9 Target Illuminators (TIs) (now part of MK-57 NATO SEASPARROW MISSILE SYSTEM (NSSMS)).																							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2004 / CVN-81			
Equipment Item: ADVANCED ARRESTING GEAR (AAG)						PARM Code: PMA 251	
P-35 Category				FY 2020			
				Qty (Each)	Total Cost (\$ M)		
Major Hardware				1	204.845		
Spares					1.827		
System Engineering					4.597		
Technical Engineering Services					4.906		
Other Costs					13.104		
Total				1	229.279		
Description: AAG provides an upgraded ability to recover all existing and projected aircraft carrier based air vehicles. The AAG system will replace the Mark 7 arresting gear system found on the NIMITZ Class carriers for the FORD Class. AAG consists of six primary systems: energy absorption subsystem, energy storage subsystem, dynamic control subsystem, thermal management subsystem, cross deck pendant, and the control subsystem.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	CVN 81	GENERAL ATOMICS	SS/FFP	Aug 2021	Option	1	204.845
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 81	Feb 2032	64	48	Oct 2022		
Competition/Second Source Initiatives: None							
Remarks: Motor Generator Long Lead Time Material (LLTM) hardware is being procured in FY 2021 to lessen the production gap and reduce lost learning on units completed early in production on CVN 80.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2004 / CVN-81			
Equipment Item: PHALANX BLOCK 1B MK 15 MOD 21 & 22, CLOSE - IN WEAPONS SYSTEM (CIWS)						PARM Code: IWS 11	
P-35 Category				FY 2020			
				Qty (Each)	Total Cost (\$ M)		
Major Hardware				3	14.996		
Ancillary Equipment					0.231		
Spares					0.278		
System Engineering					1.858		
Technical Engineering Services					0.629		
Other Costs					1.294		
Total				3	19.286		
Description: Phalanx is a high fire rate Close-In Weapon System (CIWS) that automatically acquires, tracks, and destroys Anti-Ship cruise missiles, Helos, Aircraft, and all types of Surface threats. The installed version will have one MK-15 Mod 21 and two MK-15 Mod 22 CIWS systems.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	CVN 81	RAYTHEON	C/FFP	Dec 2027		3	4.999
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 81	Feb 2032	28	22	Dec 2027		
Competition/Second Source Initiatives: None							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2004 / CVN-81			
Equipment Item: AN/SQQ-34, CARRIER-TACTICAL SUPPORT CENTER (CV-TSC)						PARM Code: PEO IWS 5E	
P-35 Category				FY 2020			
				Qty (Each)	Total Cost (\$ M)		
Major Hardware				1	3.621		
Spares					0.245		
System Engineering					0.029		
Technical Engineering Services					0.035		
Other Costs					0.170		
Total				1	4.100		
Description: CV-TSC provides for carrier organic Anti-submarine Warfare (ASW), Mine Warfare (MIW), Surface Warfare (SUW), and other composite warfare area sensor data processing, tactical command and control, and organic/battle-group aircraft mission support. CV-TSC supports both ship self-defense and embarked Destroyer Squadron (DESRON) missions. This system is Open Architecture Computing Environment (OACE), Joint Fires Network (JFN), and FORCENet compliant. It includes redesign to maximize introduction of expected transformational technologies such as Common Processing System (CPS), Common Display System (CDS), sensor processing in support of the MH-60R helicopter, high speed bandwidth network, Excomm systems, net-centric warfare components, etc.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	CVN 81	TBD	TBD	Sep 2027		1	3.621
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 81	Feb 2032	35	18	Sep 2027		
Competition/Second Source Initiatives: None							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2004 / CVN-81			
Equipment Item: MK-57 NATO SEASPARROW MISSILE SYSTEM (NSSMS)						PARM Code: PEO IWS 12	
P-35 Category				FY 2020			
				Qty (Each)	Total Cost (\$ M)		
Major Hardware				1	40.165		
Ancillary Equipment					0.432		
Spares					1.788		
System Engineering					0.811		
Technical Engineering Services					0.768		
Other Costs					1.885		
Total				1	45.849		
Description: The NATO SeaSparrow Surface Missile System (NSSMS) is a medium range self-defense missile system capable of defeating near/mid-term air/surface threats. The NSSMS MK 57 Mod 15 is comprised of (4) MK 9 tracker/illuminator radars and (2) MK 29 ESSM Guided Missile Launchers (GML) that connect to and support the SSDS Integrated Combat System (ICS).							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	CVN 81	TBD	TBD	Oct 2025		2	3.513
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 81	Feb 2032	47	29	Oct 2025		
Competition/Second Source Initiatives: None							
Remarks: Starting in the FY 2022 budget submission, MK 29 MOD 5, Guided Missile Launching System (GMLS) and MK-9 Target Illuminator are now combined as one complete system.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2004 / CVN-81			
Equipment Item: AVIATION DATA MANAGEMENT AND CONTROL SYSTEM (ADMACS)						PARM Code: PMA 251	
P-35 Category				FY 2020			
				Qty (Each)	Total Cost (\$ M)		
Major Hardware				1	5.108		
System Engineering					0.944		
Technical Engineering Services					0.588		
Other Costs					2.132		
Total				1	8.772		
Description: ADMACS is a virtual, seamless, data sharing, knowledge based data system that provides interface for all aviation data systems. It is a tactical real-time information management system maintaining data integrity throughout the ship spaces that manage aircraft launch and recovery operations on board the carrier. ADMACS includes data from launch and recovery equipment, air traffic control, aviation maintenance, landing signaling officer, etc.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	CVN 81	TBD	TBD	Oct 2028		1	5.108
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 81	Feb 2032	28	12	Oct 2028		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2004 / CVN-81			
Equipment Item: MK 49, MOD 3 ROLLING AIRFRAME MISSILE (RAM)						PARM Code: PEO IWS 11	
P-35 Category				FY 2020			
				Qty (Each)	Total Cost (\$ M)		
Major Hardware				2	7.404		
Ancillary Equipment					1.408		
Technical Data and Documentation					0.036		
Spares					0.142		
System Engineering					2.233		
Technical Engineering Services					0.387		
Other Costs					4.178		
Total				2	15.788		
Description: The MK 49 Mod 5 Rolling Airframe Missile (RAM) Weapon System is a lightweight, low cost, high power system for anti-ship missile defense against current and evolving threats. The Block 2 upgrade adds the capability of infrared, all-the-way missile guidance while maintaining the original dual-mode (RF/IR) capability. The helos, aircraft, and surface (HAS) upgrade enables the engagement of asymmetric threats.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	CVN 81	TBD	TBD	Aug 2027		2	3.702
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 81	Feb 2032	33	21	Aug 2027		
Competition/Second Source Initiatives: None							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2004 / CVN-81			
Equipment Item: AN/SPQ-9B, ANTI-SHIP MISSILE DEFENSE (ASMD) SURFACE SURVEILLANCE AND TRACKING RADAR						PARM Code: PEO IWS2B	
P-35 Category				FY 2020			
				Qty (Each)	Total Cost (\$ M)		
Major Hardware				1	7.576		
Spares					0.497		
System Engineering					1.081		
Technical Engineering Services					0.664		
Other Costs					3.900		
Total				1	13.718		
Description: SPQ-9B is a multimode, x-band, narrow beam, pulse Doppler radar that detects and tracks sea-skimming missiles (ASMD) at the horizon in heavy clutter while simultaneously providing detection and tracking of surface targets.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	CVN 81	Laurel Technologies Partnership/DRS	SS/FFP	Jan 2022		1	7.576
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 81	Feb 2032	35	18	Sep 2027		
Competition/Second Source Initiatives: None							
Remarks: The Enterprise Radar Suite (ERS) consists of AN/SPY-6(V)3 Enterprise Air Surveillance Radar (EASR), AN/SPQ-9B Anti-ship Missile Defense (ASMD) Surface Surveillance and Tracking Radar and MK-9 Target Illuminators (TIs) (now part of MK-57 NATO SEASPARROW MISSILE SYSTEM (NSSMS)). The award date has changed to January 2022 in order to take advantage of current contract pricing. The current SPQ-9B contract has pricing and options through FY 2022. The Prime Contractor advised the Government that unit cost will go up significantly in the next contract award due to the overall amount of systems projected to be procured during that period of time.							

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy										Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 02: Other Warships / BSA 1: Other Warships						P-1 Line Item Number / Title: 2013 / Virginia Class Submarine						
ID Code (A=Service Ready, B=Not Service Ready): A			Program Elements for Code B Items: N/A				Other Related Program Elements: 0604558N, 0604580N, 0204281N					
Line Item MDAP/MAIS Code: 516												
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (<i>Units in Each</i>)	30	2	2	2	-	2	-	-	-	-	-	-
Gross/Weapon System Cost (<i>\$ in Millions</i>)	79,782.762	7,483.677	7,215.253	6,915.757	0.000	6,915.757	-	-	-	-	-	-
Less PY Advance Procurement (<i>\$ in Millions</i>)	22,914.124	1,756.902	1,840.679	1,888.328	-	1,888.328	-	-	-	-	-	-
Less Cost To Complete (<i>\$ in Millions</i>)	1,844.685	-	200.000	-	-	-	-	-	-	-	-	-
Less Economic Order Quantity (<i>\$ in Millions</i>)	3,820.647	361.594	571.361	778.189	-	778.189	-	-	-	-	-	-
Net Procurement (P-1) (<i>\$ in Millions</i>)	51,203.306	5,365.181	4,603.213	4,249.240	0.000	4,249.240	-	-	-	-	-	-
Plus CY Advance Procurement (<i>\$ in Millions</i>)	25,893.064	2,087.588	1,745.767	2,120.407	-	2,120.407	-	-	-	-	-	-
Plus Cost To Complete (<i>\$ in Millions</i>)	1,844.685	-	-	-	-	-	-	-	-	-	-	-
Plus Economic Order Quantity (<i>\$ in Millions</i>)	5,000.496	881.964	427.420	-	-	-	-	-	-	-	-	-
Total Obligation Authority (<i>\$ in Millions</i>)	83,941.551	8,334.733	6,776.400	6,369.647	0.000	6,369.647	-	-	-	-	-	-
<i>(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)</i>												
Plus Outfitting and Post Delivery (<i>\$ in Millions</i>)	1,293.822	141.436	142.629	71.454	-	71.454	-	-	-	-	-	-
Total (<i>\$ in Millions</i>)	85,235.373	8,476.169	6,919.029	6,441.101	-	6,441.101	-	-	-	-	-	-
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	2,659.425	3,741.839	3,607.627	3,457.879	-	3,457.879	-	-	-	-	-	-

Description:

MISSION: To seek out and destroy enemy ships across a wide spectrum of tactical scenarios, working both independently and in concert with a battle group/other ships, providing Joint Commanders with early, accurate knowledge of the battlefield on which power may be projected from sea; covert striking power against targets ashore; the capability to establish covertly an expeditionary force on land; and the maritime strength to destroy enemy naval forces and interdict seaborne commerce.

NOTE: The program is currently executing the fourth Multi-Year Procurement (MYP) (Block V) contract which was awarded on 2 December 2019 incorporating Acoustic Superiority (AS) modifications on all SSNs. VIRGINIA Payload Module (VPM) is included from SSN 803 and subsequent SSNs. All Block V and future SSNs will include modifications to provide enhanced capability and improved performance based upon efforts performed via RDT&E (PE 0604558N/Prj 1947). VPM is an 84-foot hull section with four additional payload tubes, each capable of carrying seven Tomahawk cruise missiles or various other payloads. VPM helps mitigate the loss of undersea strike capability with the retirement of the Service's four guided missile submarines (SSGNs) in the late-2020s. These changes to the baseline require additional funding which is shown in more detail on subsequent pages of these exhibits.

The Block V tenth SSN option was awarded on 19 March 2021. All FY 2021 values have been updated to reflect the procurement of this additional submarine. Completion of Prior Year Shipbuilding Programs (cost to complete) funds are for the completion of mission essential activities and procurement of Electronics, HM&E, and Other Cost items for the FY 2021 (SSN 807) Block V SSN (\$200.0M).

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy						Date: May 2021																																																																	
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 02: Other Warships / BSA 1: Other Warships				P-1 Line Item Number / Title: 2013 / Virginia Class Submarine																																																																			
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: N/A		Other Related Program Elements: 0604558N, 0604580N, 0204281N																																																																			
Line Item MDAP/MAIS Code: 516																																																																							
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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy				Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 02: Other Warships / BSA 1: Other Warships			P-1 Line Item Number / Title: 2013 / Virginia Class Submarine		
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: N/A		Other Related Program Elements: 0604558N, 0604580N, 0204281N	
Line Item MDAP/MAIS Code: 516					
Design Schedule		Start / Issue	Complete / Response	Reissue	Reissue Complete / Response
Request for Proposals		N/A	N/A		
Design Agent		Electric Boat			
Classification of Cost Estimate: C					
Justification: The Block V tenth SSN option was awarded on 19 March 2021. All FY 2021 values have been updated to reflect the procurement of this additional submarine. Cost to complete funds are for the completion of mission essential activities and procurement of Electronics, HM&E, and Other Cost items for the FY 2021 (SSN 807) Block V SSN (\$200.0M). FY 2022 funds two VIRGINIA Class Submarines (VCS), Advance Procurement (AP) for future SSNs, as well as \$50 million of submarine industrial base investment (funded in Basic). The Program Total Cost reflects the VIRGINIA Class 48 SSN Acquisition Program Baseline (APB) established in 2017. Update of baseline estimates to support current force structure projections are in process. Footnotes: ⁽¹⁾ Applies to all SSNs shown - P-27 Exhibit Construction Start to Delivery duration is based on a 1st of month to 1st of month convention. Therefore, computed construction spans may be one month short of actual for a projected delivery at the end of the month. ⁽²⁾ These VIRGINIA Class exhibits reflect a Block V FY 2019-FY 2023; incorporating changes for AS on all SSNs and VPM on SSN 803 and subsequent SSNs. ⁽³⁾ The Navy conducts Construction Readiness Reviews (CRR) with the shipbuilders every six months to determine at an enterprise level if the shipbuilders are ready to begin construction of the next VCS Block V SSN. The provided dates for SSN 805 and subsequent SSNs are the current Program Manager estimates and may be adjusted based on the outcomes of future CRRs.					

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy **Date:** May 2021

Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1 **P-1 Line Item Number / Title:** 2013 / Virginia Class Submarine

Cost Categories (†) indicates the presence of a P-8a	FY 2014		FY 2015		FY 2016		FY 2017		FY 2018		FY 2019		FY 2020		FY 2021	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Plan Costs	2	167.937	2	177.095	2	183.078	2	180.184	2	187.778	2	472.813	2	394.106	2	303.171
Basic Construction/Conversion		3,492.087		3,335.501		3,384.290		3,430.573		3,508.117		4,397.589		5,186.180		4,988.260
Change Orders		104.021		89.481		91.459		73.043		74.536		87.952		103.891		100.610
Electronics (†)		503.718		504.701		514.795		515.852		525.653		482.551		546.354		529.936
Technology Insertion		73.500		28.835		13.535		12.501		18.000		8.500		15.096		14.800
Propulsion Equipment		910.157		970.000		1,025.000		1,032.500		1,051.100		871.675		1,047.000		1,083.600
Hull, Mechanical, and Electrical (HM&E) (†)		105.248		106.822		109.920		110.190		112.394		119.028		125.882		128.400
Other Cost		52.658		53.233		54.777		54.058		55.140		60.070		65.168		66.476
Total Ship Estimate		5,409.326		5,265.668		5,376.854		5,408.901		5,532.718		6,500.178		7,483.677		7,215.253
Less Advance Procurement FY 2012		988.246		-		-		-		-		-		-		-
Less Advance Procurement FY 2013		540.376		1,110.000		-		-		-		-		-		-
Less Advance Procurement FY 2014		-		467.014		1,145.000		-		-		-		-		-
Less Advance Procurement FY 2015		-		-		468.536		1,152.500		-		-		-		-
Less Advance Procurement FY 2016		-		-		-		470.788		1,171.100		-		-		-
Less Advance Procurement FY 2017		-		-		-		-		475.940		1,376.294		-		-
Less Advance Procurement FY 2018		-		-		-		-		-		752.597		1,167.999		-
Less Advance Procurement FY 2019		-		-		-		-		-		-		588.903		1,222.038
Less Advance Procurement FY 2020		-		-		-		-		-		-		-		618.641
Less Advance Procurement FY 2021		-		-		-		-		-		-		-		-
Less Cost to Complete FY 2023		-		-		-		-		-		-		-		200.000
Less EOQ FY 2014		-		158.400		219.380		194.909		169.909		-		-		-
Less EOQ FY 2015		-		-		197.568		251.603		231.618		-		-		-
Less EOQ FY 2016		-		-		-		151.116		178.836		-		-		-
Less EOQ FY 2018		-		-		-		-		-		30.611		115.229		31.008
Less EOQ FY 2019		-		-		-		-		-		-		246.365		246.365

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy

Date: May 2021

Appropriation / Budget Activity / Budget Sub Activity:
1611N / 02 / 1

P-1 Line Item Number / Title:
2013 / Virginia Class Submarine

Cost Categories ^(†) indicates the presence of a P-8a	FY 2014		FY 2015		FY 2016		FY 2017		FY 2018		FY 2019		FY 2020		FY 2021	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Less EOQ FY 2020		-		-		-		-		-		-		-		293.988
Less EOQ FY 2021		-		-		-		-		-		-		-		-
Net P-1 Funding		3,880.704		3,530.254		3,346.370		3,187.985		3,305.315		4,340.676		5,365.181		4,603.213

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy		Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2013 / Virginia Class Submarine	
Cost Categories (†) indicates the presence of a P-8a	FY 2022		Total Cost (\$ M)
	Qty (Each)		
Plan Costs	2		252.399
Basic Construction/Conversion			4,758.283
Change Orders			95.212
Electronics (†)			540.536
Technology Insertion			15.398
Propulsion Equipment			1,046.830
Hull, Mechanical, and Electrical (HM&E) (†)			139.293
Other Cost			67.806
Total Ship Estimate			6,915.757
Less Advance Procurement FY 2012			-
Less Advance Procurement FY 2013			-
Less Advance Procurement FY 2014			-
Less Advance Procurement FY 2015			-
Less Advance Procurement FY 2016			-
Less Advance Procurement FY 2017			-
Less Advance Procurement FY 2018			-
Less Advance Procurement FY 2019			-
Less Advance Procurement FY 2020			1,268.947
Less Advance Procurement FY 2021			619.381
Less Cost to Complete FY 2023			-
Less EOQ FY 2014			-
Less EOQ FY 2015			-
Less EOQ FY 2016			-
Less EOQ FY 2018			24.126
Less EOQ FY 2019			246.365
Less EOQ FY 2020			293.988
Less EOQ FY 2021			213.710
Net P-1 Funding			4,249.240
Remarks: These VIRGINIA Class exhibits reflect a Block V FY 2019-FY 2023 incorporating changes for AS on all SSNs and VPM on SSN 803 and all subsequent SSNs. The program identified cost efficiencies in the Electronics category to support the award of the Block V MYP contract. The overall cost efficiencies are carried through all future program years. Additionally, the Electronics profile fluctuation (FY 19 - FY 22) reflects improved alignment of non-recurring engineering support to physically and electronically integrate the pre-planned NPES TI-24 Tech Insertion configuration and various phased Undersea Dominance Payloads.			

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1	P-1 Line Item Number / Title: 2013 / Virginia Class Submarine	

The Block V tenth SSN option was awarded on 19 March 2021. All FY 2021 values have been updated to reflect the procurement of this additional submarine. Cost to complete funds are for the completion of mission essential activities and procurement of Electronics, HM&E, and Other Cost items for the FY 2021 (SSN 807) Block V SSN (\$200.0M).

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Exhibit P-27, Ship Production Schedule: PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1	P-1 Line Item Number / Title: 2013 / Virginia Class Submarine
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Ship	Shipbuilder	Fiscal Year	Contract Award	Start of Construction	Delivery Date
SSN 793 ⁽¹⁾	EB/HII-NNS	2014	Apr 2014	Sep 2014	Oct 2021
SSN 794	EB/HII-NNS	2015	Apr 2014	Apr 2015	Nov 2021
SSN 795	EB/HII-NNS	2015	Apr 2014	Sep 2015	Apr 2022
SSN 796	EB/HII-NNS	2016	Apr 2014	Mar 2016	Dec 2022
SSN 797	EB/HII-NNS	2016	Apr 2014	Sep 2016	Apr 2023
SSN 798	EB/HII-NNS	2017	Apr 2014	Mar 2017	Nov 2023
SSN 799	EB/HII-NNS	2017	Apr 2014	Sep 2017	Mar 2024
SSN 800	EB/HII-NNS	2018	Apr 2014	Mar 2018	Oct 2024
SSN 801	EB/HII-NNS	2018	Apr 2014	Sep 2018	Feb 2025
SSN 802 ⁽²⁾	EB/HII-NNS	2019	Dec 2019	Sep 2019	Jun 2025
SSN 803	EB/HII-NNS	2019	Dec 2019	Mar 2020	Apr 2026
SSN 804	EB/HII-NNS	2020	Dec 2019	Sep 2020	Oct 2026
SSN 805 ⁽³⁾	EB/HII-NNS	2020	Dec 2019	Jul 2021	May 2027
SSN 806	EB/HII-NNS	2021	Dec 2019	Dec 2021	Nov 2027
SSN 807	EB/HII-NNS	2021	Dec 2019	May 2022	May 2028
SSN 808	EB/HII-NNS	2022	Dec 2019	Oct 2022	Nov 2028
SSN 809	EB/HII-NNS	2022	Dec 2019	Mar 2023	May 2029

Footnotes:

⁽¹⁾ Applies to all SSNs shown - P-27 Exhibit Construction Start to Delivery duration is based on a 1st of month to 1st of month convention. Therefore, computed construction spans may be one month short of actual for a projected delivery at the end of the month.

⁽²⁾ These VIRGINIA Class exhibits reflect a Block V FY 2019-FY 2023; incorporating changes for AS on all SSNs and VPM on SSN 803 and subsequent SSNs.

⁽³⁾ The Navy conducts Construction Readiness Reviews (CRR) with the shipbuilders every six months to determine at an enterprise level if the shipbuilders are ready to begin construction of the next VCS Block V SSN. The provided dates for SSN 805 and subsequent SSNs are the current Program Manager estimates and may be adjusted based on the outcomes of future CRRs.

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy					Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1			P-1 Line Item Number / Title: 2013 / Virginia Class Submarine			
Electronics	FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
P-35 Items						
Sonar, Combat Control & Architecture	2	224.315	2	217.572	2	221.924
Electronic Support Measures (ESM)	2	61.277	2	59.436	2	60.623
Photonics Masts	2	41.210	2	39.972	2	40.772
Universal Modular Mast (UMM)	2	23.504	2	22.796	2	23.252
Exterior Communications System (ECS) Recurring	2	55.595	2	53.922	2	55.001
P-35 Items Subtotal		405.901		393.698		401.572
Major Items						
System Level Activities	2	41.255	2	40.018	2	40.816
AN/BPS-16	2	6.325	2	6.138	2	6.261
Navigation	2	7.173	2	6.958	2	7.097
CWITT	2	46.652	2	45.254	2	46.159
Non-Propulsion Electronics System, Systems Engineering and Integration (NPES SE&I)	2	36.515	2	35.416	2	36.125
Major Items Subtotal		137.920		133.784		136.458
Other Cost Elements						
Misc Electronics	0	2.533	0	2.454	0	2.506
Other Cost Elements Subtotal		2.533		2.454		2.506
Total Electronics		546.354		529.936		540.536

Remarks:

The program identified cost efficiencies in the Electronics category to support the award of the Block V MYP contract. The overall cost efficiencies are carried through all future program years. Additionally, the Electronics profile fluctuation (FY 2019 - FY 2022) reflects improved alignment of non-recurring engineering support to physically and electronically integrate the pre-planned NPES TI-24 Tech Insertion configuration and various phased Undersea Dominance Payloads.

The Block V tenth SSN option was awarded on 19 March 2021. All FY 2021 values have been updated to reflect the procurement of this additional submarine. Cost to complete funds are for the completion of mission essential activities and procurement of Electronics, HM&E, and Other Cost items for the FY 2021 (SSN 807) Block V SSN (\$200.0M).

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy					Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1			P-1 Line Item Number / Title: 2013 / Virginia Class Submarine			
Hull, Mechanical, and Electrical (HM&E)	FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
P-35 Items						
Propulsor	2	81.022	2	82.642	2	92.621
P-35 Items Subtotal		81.022		82.642		92.621
Major Items						
CSA MK2		3.432		3.500		3.570
Major Items Subtotal		3.432		3.500		3.570
Other Cost Elements						
HM&E Installation and testing		20.290		20.696		21.110
T&E		18.838		19.216		19.600
SUPSHIP responsible material		2.300		2.346		2.392
Other Cost Elements Subtotal		41.428		42.258		43.102
Total Hull, Mechanical, and Electrical (HM&E)		125.882		128.400		139.293

Remarks:

VCS includes additional hardware funding to insert Improved Advanced Hybrid (IAH) Propulsors beginning with SSNs authorized in FY 2022. Construction of these Propulsors is within existing foundry capacity and has been successfully proven during at-sea demonstration. The IAH Propulsor provides significantly enhanced capability, reduced maintenance complexity, improved performance, and meets the objectives of the Tactical Submarine Evolution Plan (TSEP). Insertion of IAH Propulsors accounts for the increase to Propulsor hardware material costs above inflation.

The Block V tenth SSN option was awarded on 19 March 2021. All FY 2021 values have been updated to reflect the procurement of this additional submarine. Cost to complete funds are for the completion of mission essential activities and procurement of Electronics, HM&E, and Other Cost items for the FY 2021 (SSN 807) Block V SSN (\$200.0M).

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2013 / Virginia Class Submarine			
Equipment Item: Sonar, Combat Control & Architecture						PARM Code: N/A	
P-35 Category	FY 2020		FY 2021		FY 2022		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
Major Hardware	2	183.095	2	186.940	2	190.679	
Technical Engineering Services		3.278		3.346		3.413	
Other Costs		37.942		27.286		27.832	
Total	2	224.315	2	217.572	2	221.924	
Description: The VIRGINIA Class Command, Control, Communications and Intelligence (C3I) System is the electronics suite which will provide required operational and warfighting capability for the Navy's newest attack submarine. The C3I System includes 15 subsystems (23 if all electronically interfaced subsystems are included) integrated by an overarching Architecture Subsystem. This P-35 covers the procurement requirements for the following: C3I Prime Contractor Furnished Equipment (Sonar, Combat Control and Architecture subsystems) and associated Government Furnished Equipment; technical data documentation; spares; technical engineering services; design engineering services; field engineering services; management support services; and shipboard certification efforts.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	SSN 804	Lockheed Martin	C/CPIF	Jan 2020	Option	2	50.867
FY 2021	SSN 806	Lockheed Martin	C/CPIF	Jan 2021	Option	2	51.884
FY 2022	SSN 808	Lockheed Martin	C/CPIF	Jan 2022	Option	2	52.922
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	SSN 804	Oct 2026	26	32	Dec 2021		
FY 2021	SSN 806	Aug 2027	26	32	Oct 2022		
FY 2022	SSN 808	Aug 2028	26	32	Oct 2023		
Competition/Second Source Initiatives: N/A							
Remarks: The Sonar, Combat Control & Architecture P-35 exhibit includes funding for multiple subsystems, contracts and field activities. However, the unit cost in the contract data section reflects only the major contract, Sonar acoustics processing. Other Costs - The program identified cost efficiencies in the Electronics category to support the award of the Block V MYP contract. The overall cost efficiencies are carried through all future program years. Additionally, the Electronics profile fluctuation (FY 2019 - FY 2022) reflects improved alignment of non-recurring engineering support to physically and electronically integrate the pre-planned NPES TI-24 Tech Insertion configuration and various phased Undersea Dominance Payloads. The Block V tenth SSN option was awarded on 19 March 2021. All FY 2021 values have been updated to reflect the procurement of this additional submarine. Cost to complete funds are for the completion of mission essential activities and procurement of Electronics, HM&E, and Other Cost items for the FY 2021 (SSN 807) Block V SSN (\$200.0M).							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2013 / Virginia Class Submarine			
Equipment Item: Electronic Support Measures (ESM)						PARM Code: N/A	
P-35 Category	FY 2020		FY 2021		FY 2022		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
Major Hardware	2	46.519	2	47.496	2	48.444	
Technical Engineering Services		2.605		2.660		2.713	
Other Costs		12.153		9.280		9.466	
Total	2	61.277	2	59.436	2	60.623	

Description:
The VIRGINIA Class Command, Control, Communications and Intelligence (C3I) System is the electronics suite which will provide required operational and warfighting capability for the Navy's newest attack submarine. The C3I System includes 15 subsystems (23 if all electronically interfaced subsystems are included) integrated by an overarching Architecture Subsystem. This P-35 covers the procurement requirements for the following: Electronic Support Measures subsystem Prime Contractor Furnished Equipment, and associated Government Furnished Equipment; technical data documentation; spares; systems engineering; technical engineering services; computer program support; system test & evaluation; field engineering services; management support services; shipboard certification efforts; quality assurance and reliability/maintainability assurance; maintenance of technical data; and contractor support services efforts. This system provides the capability to process a variety of electromagnetic signal types over a wide frequency range in support of all applicable submarine mission areas.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	SSN 804	Lockheed Martin	C/FPIF	Jun 2020	Option	2	23.260
FY 2021	SSN 806	Lockheed Martin	C/FPIF	Jun 2021	Option	2	23.748
FY 2022	SSN 808	Lockheed Martin	C/FPIF	Jun 2022	Option	2	24.222

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	SSN 804	Oct 2026	26	24	Aug 2022
FY 2021	SSN 806	Aug 2027	26	24	Jun 2023
FY 2022	SSN 808	Aug 2028	26	24	Jun 2024

Competition/Second Source Initiatives:
Multi-Functional Modular Mast: Full and open competition for SSN 802 through SSN 811.

Remarks:
Other Costs - The program identified cost efficiencies in the Electronics category to support the award of the Block V MYP contract. The overall cost efficiencies are carried through all future program years. Additionally, the Electronics profile fluctuation (FY 2019 - FY 2022) reflects improved alignment of non-recurring engineering support to physically and electronically integrate the pre-planned NPES TI-24 Tech Insertion configuration and various phased Undersea Dominance Payloads.

The Block V tenth SSN option was awarded on 19 March 2021. All FY 2021 values have been updated to reflect the procurement of this additional submarine. Cost to complete funds are for the completion of mission essential activities and procurement of Electronics, HM&E, and Other Cost items for the FY 2021 (SSN 807) Block V SSN (\$200.0M).

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1	P-1 Line Item Number / Title: 2013 / Virginia Class Submarine
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Equipment Item: Photonics Masts	PARM Code: N/A
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P-35 Category	FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Major Hardware	2	28.517	2	29.116	2	29.699
Technical Engineering Services		1.281		1.308		1.334
Other Costs		11.412		9.548		9.739
Total	2	41.210	2	39.972	2	40.772

Description:

The VIRGINIA Class Command, Control, Communications and Intelligence (C3I) System is the electronics suite which will provide required operational and warfighting capability for the Navy's newest attack submarine. The C3I System includes 15 subsystems (23 if all electronically interfaced subsystems are included) integrated by an overarching Architecture Subsystem. This P-35 covers the procurement requirements for the following: Photonics subsystem Prime Contractor Furnished Equipment; spares; systems engineering; technical engineering services; computer program support; field engineering services; management support services; shipboard certification; maintenance of technical data; and contractor support services efforts. This system consists of two outboard mast/antenna/camera assemblies and the associated inboard processing and display equipment. This system supports visual and infrared (IR) imaging, RF signal communications, early warning and contact direction finding capability.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	SSN 804	Lockheed Martin	C/FFP	Sep 2020	Option	2	14.259
FY 2021	SSN 806	Lockheed Martin	C/FFP	Sep 2021	Option	2	14.558
FY 2022	SSN 808	Lockheed Martin	C/FFP	Sep 2022	Option	2	14.850

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	SSN 804	Oct 2026	26	24	Aug 2022
FY 2021	SSN 806	Aug 2027	26	24	Jun 2023
FY 2022	SSN 808	Aug 2028	26	24	Jun 2024

Competition/Second Source Initiatives:

AN/BVY-1 Integrated Submarine Imaging System (ISIS): Full and open competition for SSN 802 through SSN 811.

Remarks:

Other Costs - The program identified cost efficiencies in the Electronics category to support the award of the Block V MYP contract. The overall cost efficiencies are carried through all future program years. Additionally, the Electronics profile fluctuation (FY 2019 - FY 2022) reflects improved alignment of non-recurring engineering support to physically and electronically integrate the pre-planned NPES TI-24 Tech Insertion configuration and various phased Undersea Dominance Payloads.

The Block V tenth SSN option was awarded on 19 March 2021. All FY 2021 values have been updated to reflect the procurement of this additional submarine. Cost to complete funds are for the completion of mission essential activities and procurement of Electronics, HM&E, and Other Cost items for the FY 2021 (SSN 807) Block V SSN (\$200.0M).

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy					Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1			P-1 Line Item Number / Title: 2013 / Virginia Class Submarine			
Equipment Item: Universal Modular Mast (UMM)					PARM Code: N/A	
P-35 Category	FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Major Hardware	2	17.534	2	17.902	2	18.260
Technical Engineering Services		2.908		2.970		3.029
Other Costs		3.062		1.924		1.963
Total	2	23.504	2	22.796	2	23.252

Description:
 The VIRGINIA Class Command, Control, Communications and Intelligence (C3I) System is the electronics suite which will provide required operational and warfighting capability for the Navy's newest attack submarine. The C3I System includes 15 subsystems (23 if all electronically interfaced subsystems are included) integrated by an overarching Architecture Subsystem. This P-35 covers the procurement requirements for the following: Modular Mast Prime Contractor Furnished Equipment; technical data documentation; spares; systems engineering; technical engineering services; management support services; shipboard certification; and maintenance of technical data efforts. This system consists of eight common masts for purposes of housing, raising and lowering antenna and other sensor units.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	SSN 804	L3 KEO	SS/FP	Apr 2020	Option	2	8.767
FY 2021	SSN 806	L3 KEO	SS/FP	Apr 2021	New	2	8.951
FY 2022	SSN 808	L3 KEO	SS/FP	Apr 2022	Option	2	9.130

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	SSN 804	Oct 2026	37	21	Dec 2021
FY 2021	SSN 806	Aug 2027	37	21	Oct 2022
FY 2022	SSN 808	Aug 2028	37	21	Oct 2023

Competition/Second Source Initiatives:
 N/A

Remarks:
 Other Costs - The program identified cost efficiencies in the Electronics category to support the award of the Block V MYP contract. The overall cost efficiencies are carried through all future program years. Additionally, the Electronics profile fluctuation (FY 2019 - FY 2022) reflects improved alignment of non-recurring engineering support to physically and electronically integrate the pre-planned NPES TI-24 Tech Insertion configuration and various phased Undersea Dominance Payloads.

The Block V tenth SSN option was awarded on 19 March 2021. All FY 2021 values have been updated to reflect the procurement of this additional submarine.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy					Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1			P-1 Line Item Number / Title: 2013 / Virginia Class Submarine			
Equipment Item: Exterior Communications System (ECS) Recurring					PARM Code: N/A	

P-35 Category	FY 2020		FY 2021		FY 2022	
	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>
Major Hardware	2	37.329	2	38.114	2	38.877
Technical Engineering Services		6.328		6.460		6.589
Other Costs		11.938		9.348		9.535
Total	2	55.595	2	53.922	2	55.001

Description:
The VIRGINIA Class Command, Control, Communications and Intelligence (C3I) System is the electronics suite which will provide required operational and warfighting capability for the Navy's newest attack submarine. The C3I System includes 15 subsystems (23 if all electronically interfaced subsystems are included) integrated by an overarching Architecture Subsystem. Exterior Communications Systems (ECS) is an integration effort with multiple Government-Off-The-Shelf (GOTS) components providing the core ECS capability. The GOTS components of ECS will be provided using existing contracts. This P-35 covers the procurement requirements for the following: ECS GOTS equipment; fabrication/production; systems engineering; system test & evaluation; training; data; technical engineering services; spares and repair parts; and program management. This system provides the capability for seamless, transparent, secure connectivity for information exchange between submarine users and the Global Command and Communications System (GCCS).

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity <i>(Each)</i>	Unit Cost <i>(\$ M)</i>
FY 2020	SSN 804	SAIC	C/IDIQ	May 2022	Option	2	18.665
FY 2021	SSN 806	SAIC	C/IDIQ	May 2023	Option	2	19.057
FY 2022	SSN 808	SAIC	C/IDIQ	May 2024	Option	2	19.438

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	SSN 804	Oct 2026	24	9	Jan 2024
FY 2021	SSN 806	Aug 2027	24	9	Nov 2024
FY 2022	SSN 808	Aug 2028	24	9	Nov 2025

Competition/Second Source Initiatives:
SAIC is the prime contractor for fabrication and production.

Remarks:
Other Costs - The program identified cost efficiencies in the Electronics category to support the award of the Block V MYP contract. The overall cost efficiencies are carried through all future program years. Additionally, the Electronics profile fluctuation (FY 2019 - FY 2022) reflects improved alignment of non-recurring engineering support to physically and electronically integrate the pre-planned NPES TI-24 Tech Insertion configuration and various phased Undersea Dominance Payloads.

The Block V tenth SSN option was awarded on 19 March 2021. All FY 2021 values have been updated to reflect the procurement of this additional submarine. Cost-To-Complete funds are for the completion of mission essential activities and procurement of Electronics, HM&E, and Other Cost items for the FY 2021 (SSN 807) Block V SSN (\$200.0M).

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2013 / Virginia Class Submarine			
Equipment Item: Propulsor						PARM Code: N/A	
P-35 Category	FY 2020		FY 2021		FY 2022		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
Major Hardware	2	70.077	2	71.478	2	81.234	
TECH ENGINEERING SERVICES		10.945		11.164		11.387	
Total	2	81.022	2	82.642	2	92.621	

Description:
The propulsor consists of Ni-Al-bronze blades and a large steel and inconel fabrication piece. The purpose of the propulsor is to generate proper thrust to propel the ship at a rated speed within the approved limits of torque and shaft RPM, while at the same time meeting acoustic and structural requirements. This design is unique to the VIRGINIA Class. The propulsor consists of a large quantity of government supplied material and a contract for the fixed portion construction and assembly.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	SSN 804	BAE Systems	C/FFP	Mar 2020	Option	2	27.426
FY 2021	SSN 806	BAE Systems	C/FFP	Mar 2021	Option	2	27.974
FY 2022	SSN 808	BAE Systems	C/FFP	Mar 2022	Option	2	32.696

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	SSN 804	Oct 2026	40	30	Dec 2020
FY 2021	SSN 806	Aug 2027	40	30	Oct 2021
FY 2022	SSN 808	Aug 2028	40	30	Oct 2022

Competition/Second Source Initiatives:
N/A

Remarks:
VCS includes additional hardware funding to insert Improved Advanced Hybrid (IAH) Propulsors beginning with SSNs authorized in FY 2022. Construction of these Propulsors is within existing foundry capacity and has been successfully proven effective during at-sea demonstration. The IAH Propulsor provides significantly enhanced capability, reduced maintenance complexity, improved performance, and meet the objectives of the Tactical Submarine Evolution Plan (TSEP). Insertion of IAH Propulsors accounts for the increase to Propulsor hardware material costs above inflation.

The Block V tenth SSN option was awarded on 19 March 2021. All FY 2021 values have been updated to reflect the procurement of this additional submarine. Cost to complete funds are for the completion of mission essential activities and procurement of Electronics, HM&E, and Other Cost items for the FY 2021 (SSN 807) Block V SSN (\$200.0M).

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Exhibit P-10, Advance Procurement Requirements Analysis (page 1 - Budget Funding Justification): PB 2022 Navy							Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2013 / Virginia Class Submarine					
First System (2022) Award Date:		First System (2022) Completion Date:			Interval Between Systems: 0 Months				
Cost Elements	Production Leadtime (Months)	When Required* (Months)	FY 2020 (\$ M)	FY 2021 (\$ M)	FY 2022 (\$ M)	FY 2023 (\$ M)	FY 2024 (\$ M)	FY 2025 (\$ M)	FY 2026 (\$ M)
Advance Procurement									
Nuclear Propulsion Plant Equipment ⁽¹⁾	30-72	Various	1,246.830	883.115	1,121.470	-	-	-	-
Electronics Equipment ⁽²⁾	37-43	Various	29.940	30.540	31.150	-	-	-	-
NON-Nuclear Propulsion Plant Equipment - Propulsor ⁽³⁾	36	Various	45.738	46.653	47.214	-	-	-	-
Long Lead-Time CFE One Year AP ⁽⁴⁾	24-42	Various	542.963	542.188	533.556	-	-	-	-
Long Lead-Time CFE Two Year AP (4) ⁽⁵⁾	24-42	Various	222.117	243.271	387.017	-	-	-	-
<i>Total: Advance Procurement</i>			2,087.588	1,745.767	2,120.407	-	-	-	-
Economic Order of Quantity									
EOQ for FY20 SSNs ⁽⁶⁾	-	Various	-	-	0.000	-	-	-	-
EOQ for FY21 SSNs	-	Various	293.988	-	0.000	-	-	-	-
EOQ for FY22 SSNs	-	Various	293.988	213.710	0.000	-	-	-	-
EOQ for FY23 SSNs	-	Various	293.988	213.710	0.000	-	-	-	-
EOQ for FY25 SSNs	-	Various	-	-	0.000	-	-	-	-
EOQ for FY26 SSNs	-	Various	-	-	0.000	-	-	-	-
EOQ for FY27 SSNs	-	Various	-	-	0.000	-	-	-	-
EOQ for FY28 SSNs	-	Various	-	-	0.000	-	-	-	-
<i>Total: Economic Order of Quantity</i>			881.964	427.420	-	-	-	-	-
Total Advance Procurement/Obligation Authority			2,969.552	2,173.187	2,120.407	-	-	-	-
*Note: "When Required" is the number of months required before ship delivery.									

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Exhibit P-10, Advance Procurement Requirements Analysis (page 2 - Budget Funding Justification): PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1	P-1 Line Item Number / Title: 2013 / Virginia Class Submarine
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Cost Elements	FY 2022						
	Production Leadtime (Months)	When Required* (Months)	Unit Cost (\$ M)	Contract Forecast Date	2022 Qty (Each)	For FY	Total Cost Request (\$ M)
Advance Procurement							
Nuclear Propulsion Plant Equipment ⁽¹⁾	30-72	Various	-	Oct 2021	-	2024	1,121.470
Electronics Equipment ⁽²⁾	37-43	Various	-	Dec 2021	-	2023	31.150
NON-Nuclear Propulsion Plant Equipment - Propulsor ⁽³⁾	36	Various	-	Dec 2021	-	2023	47.214
Long Lead-Time CFE One Year AP ⁽⁴⁾	24-42	Various	-	Jan 2022	-	2023	533.556
Long Lead-Time CFE Two Year AP (4) ⁽⁵⁾	24-42	Various	-	Jan 2022	-	2024	387.017
<i>Total: Advance Procurement</i>							<i>2,120.407</i>
Economic Order of Quantity							
EOQ for FY20 SSNs ⁽⁶⁾	-	Various	-		-		0.000
EOQ for FY21 SSNs	-	Various	-		-		0.000
EOQ for FY22 SSNs	-	Various	-		-		0.000
EOQ for FY23 SSNs	-	Various	-		-		0.000
EOQ for FY25 SSNs	-	Various	-		-		0.000
EOQ for FY26 SSNs	-	Various	-		-		0.000
EOQ for FY27 SSNs	-	Various	-		-		0.000
EOQ for FY28 SSNs	-	Various	-		-		0.000
<i>Total: Economic Order of Quantity</i>							<i>-</i>
Total Advance Procurement/Obligation Authority							2,120.407

Description:
In FY2020 funding, Program realigned \$200M of Long Lead CFE One Year AP for the FY21 ships to Nuclear Propulsion Plant Equipment (Now \$1,247M) to fully fund 2 ship sets of Propulsion Equipment for the FY23 Ships. When added to the \$883M of FY21 AP for Nuclear Propulsion Plant Equipment, the total cost for the FY23 ship Propulsion Equipment cost category is \$1,083M. The \$200M of FY21 of Long Lead CFE One Year AP will now be financed with FY21 full funding. The Navy has programmed cost to complete funding to fully finance the FY 2021 ships.

*Note: "When Required" is the number of months required before ship delivery.

Footnotes:
⁽¹⁾ Reactor Plant Equipment Advance Procurement is required to fund long-lead time propulsion plant equipment, which is the longest lead-time equipment required for construction of nuclear-powered attack submarines, and ensure production capability that supports projected production quantities. To support the VIRGINIA Class' innovative and more efficient modular construction method, reactor plant components must be delivered earlier in the construction process than previous submarine classes. Under the new method, the VIRGINIA Class reactor plant is assembled and tested before being mounted and installed in the hull. In FY2020 funding, Program realigned \$200M of Long Lead CFE One Year AP for the FY2021 ships to Nuclear Propulsion Plant Equipment (Now \$1,247M) to fully fund 2 ship sets of Propulsion Equipment for the FY2023 Ships. When added to the \$883M of FY2021 AP for Nuclear Propulsion Plant Equipment, the total cost for the FY2023 ship Propulsion Equipment cost category is \$1,083M.

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Exhibit P-10, Advance Procurement Requirements Analysis (page 2 - Budget Funding Justification): PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2013 / Virginia Class Submarine
(2) Electronics Equipment AP is required to fund the long-lead time material for the Command and Control System Module (CCSM). AP for the CCSM plays a critical role in early system installation and test in order to keep the CCSM out of the critical path to ship delivery and minimize risk to ship construction. AP is required to procure selected electronics and associated pre-cable kits, cabling, connector plates and mechanical structures to be installed in this module in accordance with Shipyard Required in Yard Dates (RIYD). Pre-cable kits allow the shipyard to establish cable runs and checkout platform interfaces prior to electronics installation. Mechanical structures establish footprint unique packaging to allow electronics to install efficiently. Additionally, this 1 YR AP is for long lead items such as metal fabrication parts (mechanical structures, chassis, drawer slides, mounting hardware), power supplies and cable connectors, subcontract items (Aft Sonar Receive Unit), and acoustic hull sensors (such as DT-574 LAB Hydrophones). (3) Non-Nuclear Propulsion Plant Equipment Propulsor AP is required to satisfy in-yard need dates for ship delivery. Other prior year non-nuclear propulsion plant equipment has been negotiated as CFE in the Construction Contract. (4) Long Lead-Time CFE AP is required to fund long lead time contractor furnished material including the Weapons Handling and Reactor Plant Modules and the Main Propulsion Unit (MPU)/Ship Service Turbine Generator (SSTG). Additionally, VPM LLTM CFE required to support the increased material procurement (i.e. electrical, valves, flanges, fittings, pipe, fabricated parts, hardware, and tools, etc.) associated with the increased VPM workload and to maintain anticipated ship construction schedules is included. These and other components are required early in the construction phase to meet the delivery schedule. The one and two year CFE includes essential funding to support the Block V LLTM that has been negotiated into the MYP Construction Contract (not inflation driven due to VPM fluctuation and cadence of vendor base procurements) in combination with the EOQ profile to maximize cost savings and meet in yard need dates of materials. (5) FY 2022 long-lead time CFE AP reflects initial material procurements supporting SSNs anticipated to be authorized in FY 2024. As this is the first instance of LLTM procurement for FY 2024 SSNs, two-year AP values do not have the benefit of prior year economic order quantity (EOQ) procurements and are higher than prior years. In FY 2021, two-year AP values reflected lower amounts due to the funding benefits associated with prior year EOQ procurements. (6) EOQ is for Economic Order Quantity for large lot procurements of shipbuilder material and major Government Furnished Equipment to achieve MYP savings. Examples of shipbuilder large lot procurements include items such as Electrical (cable, wire, fittings, switches, instrumentation, connectors, resistors, etc.); Valves, flanges and fittings, piping; Fabricated Parts (bearings, sound isolation mounts, pipe hanged assemblies, machined parts); Hardware and Tools (fasteners, marine fittings, locks, latches, small tools). Examples of GFE large lot procurements include items such as: Sonar - Large Aperture Bow (LAB) Arrays and associated bottles, Light Weight Wide Aperture Array (LWWAA) Receivers & electronic components (network servers, switches) ECS - High Data Rate (HDR) Antennas, Digital Modular Radios (DMRs) & associated power amplifiers, Navy Multiband Terminals (NMTs), and Multifunction Masts (MFMs) OE-538. ESM - Photonics ESM Performance Improvement (PEPI)-3 systems and Multifunctional Modular Masts (MMMs) Photonics Masts - outboard equipment only, such as Diploops along with complex electronic & mechanical components that are required to manufacture the Photonics masts.		

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy										Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 02: Other Warships / BSA 1: Other Warships							P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls					
ID Code (A=Service Ready, B=Not Service Ready): A				Program Elements for Code B Items: N/A				Other Related Program Elements: N/A				
Line Item MDAP/MAIS Code: N/A												
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (<i>Units in Each</i>)	6	1	-	-	-	-	-	-	-	-	-	-
Gross/Weapon System Cost (<i>\$ in Millions</i>)	23,049.658	5,853.456	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Less PY Advance Procurement (<i>\$ in Millions</i>)	5,275.153	749.870	-	-	-	-	-	-	-	-	-	-
Less Cost To Complete (<i>\$ in Millions</i>)	525.598	-	-	-	-	-	-	-	-	-	-	-
Less Subsequent Year Full Funding (<i>\$ in Millions</i>)	10,085.568	3,987.171	-	-	-	-	-	-	-	-	-	-
Less Transfer (<i>\$ in Millions</i>)	128.131	-	-	-	-	-	-	-	-	-	-	-
Less Previously Appropriated OPN (<i>\$ in Millions</i>)	-	481.789	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (<i>\$ in Millions</i>)	7,035.208	634.626	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Plus Subsequent Year Full Funding (<i>\$ in Millions</i>)	10,085.568	-	1,531.153	2,456.018	-	2,456.018	-	-	-	-	-	-
Full Funding TOA (<i>\$ in Millions</i>)	17,120.776	634.626	1,531.153	2,456.018	-	2,456.018	-	-	-	-	-	-
Plus CY Advance Procurement (<i>\$ in Millions</i>)	6,025.023	16.900	17.384	66.262	-	66.262	-	-	-	-	-	-
Plus Cost To Complete (<i>\$ in Millions</i>)	180.598	-	186.200	158.800	-	158.800	-	-	-	-	-	-
Plus Transfer (<i>\$ in Millions</i>)	128.131	-	-	-	-	-	-	-	-	-	-	-
Plus Previously Appropriated OPN (<i>\$ in Millions</i>)	1,588.582	-	-	-	-	-	-	-	-	-	-	-
Total Obligation Authority (<i>\$ in Millions</i>)	25,043.110	651.526	1,734.737	2,681.080	0.000	2,681.080	-	-	-	-	-	-
<i>(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)</i>												
Plus Outfitting and Post Delivery (<i>\$ in Millions</i>)	125.240	21.661	16.897	39.122	-	39.122	-	-	-	-	-	-
Total (<i>\$ in Millions</i>)	25,168.350	673.187	1,751.634	2,720.202	-	2,720.202	-	-	-	-	-	-
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	3,841.610	5,853.456	-	-	-	-	-	-	-	-	-	-

Description:

To support and operate aircraft to engage in attacks on targets afloat and ashore which threaten our use of the sea and to engage in sustained operations in support of other forces. The refueling of the reactors and repair and upgrade of the main propulsion equipment will provide for reliable operations during its remaining 23 plus years of ship life using only the normal maintenance cycle.

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy				Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 02: Other Warships / BSA 1: Other Warships			P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls		
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: N/A		Other Related Program Elements: N/A	
Line Item MDAP/MAIS Code: N/A					
Characteristics:	CVN 73	CVN 74	Systems:		
Length Overall	1092 ft	1092 ft	Electronics	Hull, Mechanical, and Electrical (HM&E)	Ordnance
Beam	252 ft	252 ft	-C4ISR	-AIRCRAFT ELECTRICAL SERVICE STATION (AESS) INSTALL	-AVIATION EQUIPMENT & SUPPORT SYSTEM (NSSMS)
Displacement	101,200 LT	100,700 LT	-INTEGRATED COMMUNICATION NETWORK (ICAN / DDCN & IVCN)	-FURNITURE (NON PROPULSION PLANT)	-AN/SPS-48G - 3D AIR SEARCH RADAR
Draft	39.96 ft	39.81 ft	-SHIP SELF DEFENSE SYSTEM (SSDS) MK2 MOD 1E	-LOW PRESSURE AIR PLANT (LPAP)	-AN/SPS-49(V)1 OVERHAUL/REFURBISHMENT
			-ELECTRONIC CONSOLIDATED AUTOMATED SUPPORT SYSTEM (ECASS)	-AUTOMATIC VOLTAGE REGULATOR (AVR)	-AN/SPQ-14 - ADVANCED SENSOR DISTRIBUTION SYSTEM (ASDS)
			-JOINT PRECISION APPROACH AND LANDING SYSTEM (JPALS)	-VSA O2 GENERATOR	-CIWS/RAM DEFENSE CAPABILITY (CRDC) BLOCK 1
			-AN/SPN-46 OVERHAUL/UPGRADE	-NODE ROOM RIPOUT/INSTALL	-MK 38 MOD 3 GUN SYSTEM
			-MQ-25 - UNMANNED AVIATION WARFARE CENTER (UAWC)	-LAUNDRY DRYERS	-ENTERPRISE AIR SURVEILLANCE RADAR (EASR)
			-AN/USG-2B - COOPERATIVE ENGAGEMENT CAPABILITY (CEC)	-COMBAT SYSTEMS SUPPORT CENTER (CSSC) RIPOUT/INSTALL	-SURFACE ELECTRONIC WARFARE IMPROVEMENT PROGRAM (SEWIP) BLOCK 2
			-AN/SLQ-59 - ELECTRONIC WARFARE (EW) SYSTEM	-AFT CREW MESS	-COMBAT DIRECTION CENTER (CDC)/FLAG RIPOUT/INSTALL
			-AN/SPN-41 REFURBISHMENT	-PASSIVE COUNTER MEASURE SYSTEM (PCMS)	-AN/SPQ-9B - ANTI-SHIP CRUISE MISSILE DEFENSE RADAR
			-AN/SPN-50(V)1 - AIR TRAFFIC CONTROL RADAR	-COMBI-OVENS	-AN/SQQ-34C - CARRIER TACTICAL SUPPORT CENTER
			-AN/SLQ-32A(V)4 - ELECTRONIC WARFARE SUITE	-CARRIER INTELLIGENCE CENTER (CVIC) RIPOUT/INSTALL	-RAM GUIDED MISSILE LAUNCHING SYSTEM
			-BATTLE FORCE TACTICAL TRAINING (BFTT) SYSTEM	-MODULAR REFRIGERATION UNIT (MRU)	-AN/SPS-73 NEXT GENERATION SURFACE SEARCH RADAR (NGSSR)
			-AN/SPN-43C REFURBISHMENT	-MEDICAL AND DENTAL SUITE	-MK 53 DECOY LAUNCHING SYSTEM (DLS)
			-AN/UPX-29 - IDENTIFICATION FRIEND OR FOE (IFF) INTERROGATOR SET	-ACE PLC CONTROL SYSTEM UPGRADE	-AN/SPA-25 INDICATOR GROUP
			-NAVAL STRIKE WARFARE PLANNING CENTER (NSWPC)	-OPERATOR BALLISTIC PROTECTION FOR CREW SERVED WEAPONS STATIONS	
			-JOINT STRIKE FIGHTER AUTONOMIC LOGISTICS INFORMATION SYSTEM (JSF-ALIS)		
			-AN/TPX-42(V)15 UPGRADE		
			-AN/SYY-1 - AIR TRAFFIC CONTROL SYSTEM		
Production Status:	CVN 73	CVN 74			
Contract Award Date	Aug 2017	Jan 2021			
Months to Completion					
a) Award to Delivery	60 months	55 months			
b) Construction Start to Delivery	60 months	51 months			
Delivery Date	Aug 2022	Aug 2025			
Completion Of Fitting Out	Oct 2022	Dec 2025			
Obligation Work Limit Date	Sep 2023	Nov 2026			

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy				Date: May 2021																																														
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<table border="0" style="width: 100%;"> <tr> <td style="width: 33%;"><u>Design Schedule</u></td> <td style="width: 16.5%;"><u>Start / Issue</u></td> <td style="width: 16.5%;"><u>Complete / Response</u></td> <td style="width: 16.5%;"><u>Reissue</u></td> <td style="width: 16.5%;"><u>Reissue Complete / Response</u></td> </tr> <tr> <td>Issue Date for TLR</td> <td>Jan 2000</td> <td>Feb 2000</td> <td>Mar 2000</td> <td>Apr 2000</td> </tr> <tr> <td>Issue Date for TLS</td> <td>Jan 2001</td> <td>Feb 2001</td> <td>Mar 2001</td> <td>Apr 2001</td> </tr> <tr> <td>Preliminary Design</td> <td>Jan 2002</td> <td>Feb 2002</td> <td>N/A</td> <td>N/A</td> </tr> <tr> <td>Contract Design</td> <td>Jan 2003</td> <td>Feb 2003</td> <td>N/A</td> <td>N/A</td> </tr> <tr> <td>Detail Design</td> <td>Jan 2004</td> <td>Feb 2004</td> <td>N/A</td> <td>Apr 2004</td> </tr> <tr> <td>Request for Proposals</td> <td>Jan 2005</td> <td>Feb 2005</td> <td>Mar 2005</td> <td>N/A</td> </tr> <tr> <td>Design Agent</td> <td colspan="4">[Design Agent]</td> </tr> <tr> <td colspan="5"><u>Classification of Cost Estimate:</u> [cost estimate]</td> </tr> </table>						<u>Design Schedule</u>	<u>Start / Issue</u>	<u>Complete / Response</u>	<u>Reissue</u>	<u>Reissue Complete / Response</u>	Issue Date for TLR	Jan 2000	Feb 2000	Mar 2000	Apr 2000	Issue Date for TLS	Jan 2001	Feb 2001	Mar 2001	Apr 2001	Preliminary Design	Jan 2002	Feb 2002	N/A	N/A	Contract Design	Jan 2003	Feb 2003	N/A	N/A	Detail Design	Jan 2004	Feb 2004	N/A	Apr 2004	Request for Proposals	Jan 2005	Feb 2005	Mar 2005	N/A	Design Agent	[Design Agent]				<u>Classification of Cost Estimate:</u> [cost estimate]				
<u>Design Schedule</u>	<u>Start / Issue</u>	<u>Complete / Response</u>	<u>Reissue</u>	<u>Reissue Complete / Response</u>																																														
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Design Agent	[Design Agent]																																																	
<u>Classification of Cost Estimate:</u> [cost estimate]																																																		
Justification: The FY 2021 enacted includes congressional rescissions of \$13.1M to CVN 74 full funding. However, these rescissions are not reflected on the P-40. FY 2022 is the third of three years of full funding for CVN 74. FY 2022 includes a \$571.3M increase for CVN 74 and is primarily the result of \$347.3M for the prime contractor re-phased from FY 2021 to FY 2022 and \$190.6M in cost increases in Basic Construction. The remaining net change of \$33.4M is detailed in subsequent exhibits. This increase is driven by a higher than budgeted Execution Contract award value that incorporated CVN 74 specific repair items into the base work package. CVN 74 specific repair items were finalized with the prime contractor in 2020 and the Execution Contract was awarded with the Prime Contractor with period of performance starting January 2021. FY 2022 is the third of five years of advance procurement for CVN 75. The CVN 75 RCOH starts in FY 2025 with five years of advance procurement from FY 2020 to FY 2024. Previously appropriated OPN represents reactor power units procured for CVN 74-76 RCOHs. Funding was included with the end cost per Section 1018 of the 2019 National Defense Authorization Act. FY 2022 includes \$158.8M of completion of prior year shipbuilding programs funding for CVN 73. The funding is required to finance the Government responsible portion of the shipbuilding construction contract overrun for CVN 73 RCOH. The FY 2022 requirement has increased from the FY 2021 budget request to account for the re-delivery date change from December 2021 to August 2022. These estimates do not include COVID-19 impacts.																																																		

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls		
Cost Categories ^(†) indicates the presence of a P-8a	FY 2016		FY 2020	
	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>
Plan Costs	1	65.548	1	68.794
Basic Construction/Conversion		4,104.091		4,274.946
Electronics ^(†)		355.096		376.505
Propulsion Equipment		157.088		615.386
Hull, Mechanical, and Electrical (HM&E) ^(†)		168.067		179.894
Ordnance ^(†)		126.701		243.163
Other Cost		88.105		94.768
Total Ship Estimate		5,064.696		5,853.456
Less Advance Procurement FY 2012		14.008		-
Less Advance Procurement FY 2013		69.918		-
Less Advance Procurement FY 2014		245.793		-
Less Advance Procurement FY 2015		483.600		-
Less Advance Procurement FY 2016		-		14.951
Less Advance Procurement FY 2017		-		233.149
Less Advance Procurement FY 2018		-		75.897
Less Advance Procurement FY 2019		-		425.873
Less Subsequent Full Funding FY 2017		1,699.120		-
Less Subsequent Full Funding FY 2018		1,569.669		-
Less Subsequent Full Funding FY 2021		-		1,531.153
Less Subsequent Full Funding FY 2022		-		2,456.018
Less Cost to Complete FY 2021		186.200		-
Less Cost to Complete FY 2022		158.800		-
Less Previously Appropriated OPN FY 2004		-		234.028
Less Previously Appropriated OPN FY 2005		-		247.761
Net P-1 Funding		637.588		634.626
Remarks: The FY 2021 enacted includes congressional rescissions of \$13.1M to CVN 74 full funding. However, these rescissions are not reflected on the P-5c. The \$3.942M decrease for CVN 74 in Plans from the FY 2021 budget request reflects savings from engineering resource efficiencies and contract cost reductions. The \$84.528M increase for CVN 73 in Basic Construction from the FY 2021 budget request is due to prime contractor performance, non-contract funds cost overrun (customer contracted teams and alteration installation teams efforts), and eight-month schedule shift for a August 2022 re-delivery. These estimates do not include COVID-19 impacts. The \$190.605M increase for CVN 74 in Basic Construction from the FY 2021 budget request is due to the higher than budgeted Execution Contract award value and updated cost estimates received by the prime contractor and other repair activities to include identified CVN 74-specific repair items into the base work package. The base work package was finalized with the prime contractor in 2020.				

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls
The \$1.200M increase for CVN 73 in Propulsion Equipment from the FY 2021 budget request is due to the extension of the CVN 73 RCOH schedule. The \$12.007M increase for CVN 74 in Propulsion Equipment from the FY 2021 budget request is due to design work for obsolescence, material degradation beyond original scoping, and realized vendor costs higher than expected. Non-nuclear government furnished equipment (GFE) is captured in Electronics, HM&E, and Ordnance. The GFE Governance Program was initiated to target soft costs by challenging performers to achieve savings in program management, system engineering, and logistics support. Procurement details are recorded in NAVSEA Form 7300s and reviewed annually to refine and adjust estimates for required products and services to the most current data available. CVN 74 increases of \$12.934M in Electronics and \$13.056M in Ordnance are the result of updated cost estimates based on more accurate scope and CVN 73 RCOH actual costs. See further details in respective P-8a and P-35 exhibits. The \$10.191M decrease for CVN 73 in Other Cost from the FY 2021 budget request is the result of descoping contractor support services. The \$0.302M decrease for CVN 74 in Other Cost from the FY 2021 budget request is the result of lower than anticipated costs for integrated logistics products.		

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LI 2086 - CVN Refueling Overhauls

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy				Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1			P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls		
Electronics	FY 2016		FY 2020		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
P-35 Items					
C4ISR	1	128.710	1	120.132	
INTEGRATED COMMUNICATION NETWORK (ICAN / DDCN & IVCN)	1	53.058	1	75.810	
SHIP SELF DEFENSE SYSTEM (SSDS) MK2 MOD 1E	1	44.509	1	29.916	
ELECTRONIC CONSOLIDATED AUTOMATED SUPPORT SYSTEM (ECASS)	1	36.791	1	37.038	
JOINT PRECISION APPROACH AND LANDING SYSTEM (JPALS)	1	14.131	1	14.372	
AN/SPN-46 OVERHAUL/UPGRADE	1	12.842	1	12.820	
MQ-25 - UNMANNED AVIATION WARFARE CENTER (UAWC)	1	11.597	1	24.159	
AN/USG-2B - COOPERATIVE ENGAGEMENT CAPABILITY (CEC)	1	10.026	1	12.351	
AN/SLQ-59 - ELECTRONIC WARFARE (EW) SYSTEM	1	6.692	0	-	
AN/SPN-41 REFURBISHMENT	1	5.718	1	5.821	
AN/SPN-50(V)1 - AIR TRAFFIC CONTROL RADAR	0	-	1	16.879	
P-35 Items Subtotal		324.074		349.298	
Major Items					
AN/SLQ-32A(V)4 - ELECTRONIC WARFARE SUITE	1	6.752	0	-	
BATTLE FORCE TACTICAL TRAINING (BFTT) SYSTEM	1	6.408	1	4.118	
AN/SPN-43C REFURBISHMENT	1	3.923	0	-	
AN/UPX-29 - IDENTIFICATION FRIEND OR FOE (IFF) INTERROGATOR SET	1	3.064	1	4.207	
NAVAL STRIKE WARFARE PLANNING CENTER (NSWPC)	1	3.028	1	5.265	
JOINT STRIKE FIGHTER AUTONOMIC LOGISTICS INFORMATION SYSTEM (JSF-ALIS)	1	1.876	1	1.826	
AN/TPX-42(V)15 UPGRADE	1	1.534	0	-	
AN/SYY-1 - AIR TRAFFIC CONTROL SYSTEM	0	-	1	4.091	
Major Items Subtotal		26.585		19.507	
Other Cost Elements					
TEST & CERTIFICATIONS, MISC.		4.437		7.700	
Other Cost Elements Subtotal		4.437		7.700	
Total Electronics		355.096		376.505	
Remarks: AN/SLQ-32A(V)4 - ELECTRONIC WARFARE SUITE: CVN 73 overall reduction (\$0.706M) from the FY 2021 budget request is a result of realized savings from the GFE Governance Program to reduce program management, system engineering, and logistics support. Replaced by Surface Electronic Warfare Improvement Program (SEWIP) Block 2 onboard CVN 74 (listed under Ordnance P-35 items). BATTLE FORCE TACTICAL TRAINING (BFTT) SYSTEM: CVN 73 cost reduction (\$1.176M) from the FY 2021 budget request due to a reduction in Combat Systems Integration Part II funding requirements. CVN 74 is utilizing the same hardware & software configuration as CVN 73, resulting in significantly less non-recurring engineering (NRE).					

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls
AN/SPN-43C REFURBISHMENT: CVN 73 overall reduction (\$0.114M) from the FY 2021 budget request is a result of realized savings from the GFE Governance Program to reduce program management, system engineering, and logistics support. Replaced by AN/SPN-50(V)1 Air Traffic Control Radar onboard CVN 74. AN/UPX-29 - IDENTIFICATION FRIEND OR FOE (IFF) INTERROGATOR SET: CVN 73 cost increase (\$0.118M) from the FY 2021 budget request due to growth work (material and labor) to procure and install system cabling. Original cabling tagged for re-use was found damaged and required replacement. CVN 74 cost increase (\$1.143M) from CVN 73 is due to CVN 74 requiring new hardware, whereas hardware on CVN 73 was able to be refurbished. NAVAL STRIKE WARFARE PLANNING CENTER (NSWPC): CVN 74 cost increase (\$2.237M) from CVN 73 is due to incorporation of additional Ready Room modernization and collaborative conferencing equipment technology refresh not required for CVN 73. AN/TPX-42(V)15 UPGRADE: CVN 73 cost increase (\$0.084M) from the FY 2021 budget request due to growth work (material and labor) to procure and install system cabling. Original cabling tagged for re-use was found damaged and required replacement. Replaced by AN/SYY-1 - Air Traffic Control System onboard CVN 74 vice AN/TPX-42(V)15 refurbishment. AN/SYY-1 - AIR TRAFFIC CONTROL SYSTEM: New display system that replaces AN/TPX-42(V)15 onboard CVN 74. CVN 73 RCOH Other Cost Elements includes eight unlisted systems, each under \$1 million in major hardware requirement. CVN 74 RCOH Other Cost Elements includes eleven unlisted systems, each under \$1 million in major hardware requirement. CVN 73 cost reduction (\$3.036M) from the FY 2021 budget request is due to descope of GFE for MK-38 Chain Guns and CV-TSC, which will need to be installed during a post-RCOH opportunity prior to deployment. CVN 74 RCOH increase (\$0.541M) from the FY 2021 budget request is due to the refinement of scope for AN/SLQ-59 and Aircraft Inertial Alignment System (AIAS) requirements.		

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls		
Hull, Mechanical, and Electrical (HM&E)	FY 2016		FY 2020	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
P-35 Items				
AIRCRAFT ELECTRICAL SERVICE STATION (AESS) INSTALL	1	14.480	0	-
FURNITURE (NON PROPULSION PLANT)	1	13.853	1	15.975
LOW PRESSURE AIR PLANT (LPAP)	1	6.126	1	6.527
AUTOMATIC VOLTAGE REGULATOR (AVR)	1	4.340	0	-
VSA O2 GENERATOR	0	-	1	5.363
P-35 Items Subtotal		38.799		27.865
Major Items				
NODE ROOM RIPOUT/INSTALL	1	6.831	1	7.036
LAUNDRY DRYERS	1	6.720	0	-
COMBAT SYSTEMS SUPPORT CENTER (CSSC) RIPOUT/INSTALL	1	5.970	1	6.376
AFT CREW MESS	1	5.608	1	6.390
PASSIVE COUNTER MEASURE SYSTEM (PCMS)	1	5.565	1	6.065
COMBI-OVENS	1	5.128	1	2.130
CARRIER INTELLIGENCE CENTER (CVIC) RIPOUT/INSTALL	1	3.074	1	3.166
MODULAR REFRIGERATION UNIT (MRU)	1	2.106	1	2.826
MEDICAL AND DENTAL SUITE	1	1.965	1	2.203
ACE PLC CONTROL SYSTEM UPGRADE	1	1.785	0	-
OPERATOR BALLISTIC PROTECTION FOR CREW SERVED WEAPONS STATIONS	0	-	1	20.000
Major Items Subtotal		44.752		56.192
Other Cost Elements				
ENGINEERING, TEST & CERTIFICATION		60.528		63.179
MISCELLANEOUS GOVERNMENT FURNISHED EQUIPMENT (GFE)		23.988		32.658
Other Cost Elements Subtotal		84.516		95.837
Total Hull, Mechanical, and Electrical (HM&E)		168.067		179.894
Remarks:				
The FY 2021 enacted includes congressional rescissions of \$13.1M to CVN 74 for Operator Ballistic Protection for Crew Served Weapons Stations. However, these rescissions are not reflected on the P-8a.				
LAUNDRY DRYERS: CVN 73 cost increase (\$2.440M) from the FY 2021 budget request is due to growth work (material and labor) to replace inoperable steam presses, faulty steam valves, and lagging replacement in the overhead. Growth work is required to ensure fully functioning shipboard laundry services. Laundry Dryers (SCD 3186) was descoped from the CVN 74 RCOH as the existing equipment was inspected and determined to fulfill the requirement.				
COMBAT SYSTEMS SUPPORT CENTER (CSSC) RIPOUT/INSTALL: CVN 74 cost reduction (\$0.031M) from the FY 2021 budget request is a result of realized savings from the GFE Governance Program to reduce program management, system engineering, and logistics support.				

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls
AFT CREW MESS: CVN 73 cost increase (\$1.902M) from the FY 2021 budget request is due to increase labor (added second shift) in order to meet critical Crew Move Aboard (CMA) key event. This work is required to move the crew back onto the ship supporting the CMA key event. CVN 74 cost increase (\$2.646M) from the FY 2021 budget request is due to additional fees associated with the labor contract and AIT install. PASSIVE COUNTER MEASURE SYSTEM (PCMS): CVN 73 cost increase (\$0.017M) from the FY 2021 budget request is due to an increase in contract field services required to complete ship checks and drawing development. COMBI-OVENS: Not previously scoped for CVN 74 but shipcheck revealed that a set (four double stacks) is required in one space, and the existing 11 combi-ovens require groom, repair and testing efforts. This differs from CVN 73, which is installing all new combi-ovens in five spaces. CARRIER INTELLIGENCE CENTER (CVIC) RIPOUT/INSTALL: CVN 74 cost reduction (\$0.250M) from the FY 2021 budget request is a result of realized savings from the GFE Governance Program to reduce program management, system engineering, and logistics support. MODULAR REFRIGERATION UNIT (MRU): CVN 74 cost increase (\$0.240M) from the FY 2021 budget request is due to updated material cost. CVN 74 increase from CVN 73 is due to increased GFE cost under new contract. MEDICAL AND DENTAL SUITE: CVN 73 (\$0.744M) and CVN 74 (\$0.629M) cost reductions from the FY 2021 budget request is a result of realized savings from the GFE Governance Program to reduce program management, system engineering, and logistics support. CVN 74 cost increase (\$0.238M) from CVN 73 is due to additional GFE procurements based on condition reports from the CVN 73 RCOH. ACE PLC CONTROL SYSTEM UPGRADE: Descoped from the CVN 74 RCOH as the existing equipment was inspected and determined to fulfill the requirement. OPERATOR BALLISTIC PROTECTION FOR CREW SERVED WEAPONS STATIONS: New modernization added to CVN 74 at the direction of the Department of Defense Appropriations Act, 2020. The modernization will provide ballistic shielding protection around exposed common served weapons stations. The FY 2021 enacted includes congressional rescissions of \$13.1M but these rescissions are not reflected on this submission. ENGINEERING, TEST & CERTIFICATION: Engineering funds a lean team of subject matter expert engineers to diagnose issues and provide solutions at a lower cost point to mitigate risk of cost and scope creep at the shipyard. CVN 73 decrease of \$3.595M from the FY 2021 budget request due to descope. CVN 74 increase of \$0.061M is to support a full test & certification program required to re-deliver the ship. MISCELLANEOUS GOVERNMENT FURNISHED EQUIPMENT (GFE): The CVN 73 RCOH includes twenty-seven unlisted systems, each under \$1 million in major hardware requirement. The CVN 74 RCOH includes nineteen unlisted systems, each under \$1 million in major hardware requirement. CVN 73 decrease of \$0.213M from the FY 2021 budget request is a result of realized savings from the GFE Governance Program to reduce program management, system engineering, and logistics support. CVN 74 decrease of \$2.700M from the FY 2021 budget request is primarily the result of re-categorizing Shock Hardened Breaker (\$1.228M) to Basic Construction and cost efficiencies realized through the GFE Governance Program.		

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls		
Ordnance	FY 2016		FY 2020	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
P-35 Items				
AVIATION EQUIPMENT & SUPPORT	1	42.849	1	54.417
NATO SEASPARROW SURFACE MISSILE SYSTEM (NSSMS)	1	18.589	1	29.517
AN/SPS-48G - 3D AIR SEARCH RADAR	1	13.772	0	-
AN/SPS-49(V)1 OVERHAUL/REFURBISHMENT	1	7.915	0	-
AN/SPQ-14 - ADVANCED SENSOR DISTRIBUTION SYSTEM (ASDS)	1	3.425	0	-
CIWS/RAM DEFENSE CAPABILITY (CRDC) BLOCK 1	0	2.296	1	24.466
MK 38 MOD 3 GUN SYSTEM	1	1.969	1	7.543
ENTERPRISE AIR SURVEILLANCE RADAR (EASR)	0	-	1	55.266
SURFACE ELECTRONIC WARFARE IMPROVEMENT PROGRAM (SEWIP) BLOCK 2	0	-	1	18.329
P-35 Items Subtotal		90.815		189.538
Major Items				
COMBAT DIRECTION CENTER (CDC)/FLAG RIPOUT/INSTALL	1	23.116	1	25.808
AN/SPQ-9B - ANTI-SHIP CRUISE MISSILE DEFENSE RADAR	1	3.646	1	4.790
AN/SQQ-34C - CARRIER TACTICAL SUPPORT CENTER	1	1.248	1	0.286
RAM GUIDED MISSILE LAUNCHING SYSTEM	1	1.156	1	2.542
AN/SPS-73 NEXT GENERATION SURFACE SEARCH RADAR (NGSSR)	0	-	1	3.362
MK 53 DECOY LAUNCHING SYSTEM (DLS)	1	0.775	1	4.885
AN/SPA-25 INDICATOR GROUP	0	-	1	2.696
Major Items Subtotal		29.941		44.369
Other Cost Elements				
TEST & CERTIFICATIONS, MISC		5.945		9.256
Other Cost Elements Subtotal		5.945		9.256
Total Ordnance		126.701		243.163
Remarks:				
CVN 74 RCOH Ordnance increase from CVN 73 RCOH includes new significant capabilities in Enterprise Air Surveillance Radar (EASR), CIWS/RAM Defense Capability (CRDC) Block 1, and Surface Electronic Warfare Improvement Program (SEWIP) Block 2. EASR replaces AN/SPS-48 and AN/SPS-49 Air Search Radars (listed under Ordnance P-35 items). CIWS/RAM Defense Capability (CRDC) Block 1 is a full GFE procurement and installation on CVN 74. SEWIP replaces AN/SLQ-32A(V)4 Electronic Warfare Suite (listed under Electronics major items).				
COMBAT DIRECTION CENTER (CDC)/FLAG RIPOUT/INSTALL: CVN 73 cost increase (\$2.156M) from the FY 2021 budget request is due to growth work (design, material, labor) associated with HVAC installation (2) in the Computer Room and Flag Briefing and Analysis spaces. CVN 74 cost increase (\$3.171M) from the FY 2021 budget request is due to incorporation of CVN 73 RCOH actual costs and additional work for the electronic countermeasure room and AN/SQQ-34C Carrier Tactical Support Center (CV-TSC) antenna with cabling, which is scope that was not included in the CVN 73 RCOH.				

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls
AN/SPQ-9B - ANTI-SHIP CRUISE MISSILE DEFENSE RADAR: CVN 73 cost increase (\$0.094M) from the FY 2021 budget request is due to growth work associated with the manufacture and installation of (material and labor) additional, required waveguide hangers. Both in quantity and type. CVN 74 cost increase (\$0.731M) from the FY 2021 budget request is due to increased hardware cost. AN/SQQ-34C - CARRIER TACTICAL SUPPORT CENTER: CVN 73 cost increase (\$0.088M) from the FY 2021 budget request is due to Combat Systems Integration Part II On Site Tech Assist cost required to complete overarching Stage 5 (system to system) testing. CVN 74 cost reduction (\$2.199M) from the FY 2021 budget request is due to reduction in scope to remove hardware only. Installation of upgraded system being planned for post-ROH opportunity. RAM GUIDED MISSILE LAUNCHING SYSTEM: CVN 73 cost reduction (\$0.043M) from the FY 2021 budget request is a result of realized savings from the GFE Governance Program to reduce program management, system engineering, and logistics support. CVN 74 cost increase (\$1.386M) from CVN 73 due to procurement of new hardware. CVN 73 hardware was obtained from decommissioned assets and refurbished. Additional decommissioned hardware assets are not available for CVN 74 and all new hardware is required. AN/SPS-73 NEXT GENERATION SURFACE SEARCH RADAR (NGSSR): Previously accounted for under Other Cost Elements. CVN 74 will install NGSSR vice original plan to remove, groom, test, and reinstall both AN/SPS-73(V)12 and AN/SPS-67(V)1 radars. MK 53 DECOY LAUNCHING SYSTEM (DLS): CVN 73 cost increase (\$0.332M) from the FY 2021 budget request is due to an updated Alteration Installation Team (AIT) cost based on the final delivered drawing package. CVN 74 increase from CVN 73 is because CVN 74 requires procurement of all new equipment whereas CVN 73 is refurbishing the existing equipment. AN/SPA-25 INDICATOR GROUP: Replaces AN/SPQ-14 - Advanced Sensor Distribution System (ASDS) onboard CVN 74. CVN 73 RCOH Other Cost Elements includes four unlisted systems, each under \$1 million in major hardware requirement. CVN 74 RCOH Other Cost Elements includes four unlisted systems, each under \$1 million in major hardware requirement. CVN 74 cost increase (\$3.690M) from the FY 2021 budget request is due to the addition of Combat System Integration efforts and new test and certification requirements for first-in-class installation of EASR, SEWIP, and SPN-50. These increases were partially offset by the re-categorization of AN/SPS-73 NGSSR to Major Items, as the major equipment requirement is above \$1 million.		

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls			
Equipment Item: C4ISR						PARM Code: NAVWAR PMW 750	
P-35 Category	FY 2016		FY 2020				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	50.828	1	60.640			
Ancillary Equipment		2.383		2.813			
Technical Data and Documentation		0.193		0.667			
Spares		1.034		2.732			
System Engineering		9.697		10.861			
Technical Engineering Services		53.148		30.047			
Other Costs		11.427		12.372			
Total	1	128.710	1	120.132			

Description:
Provides an integrated communications infrastructure to support both tactical and non-tactical applications in all warfare and support areas, an improved shipboard RF distribution system and multiband antennas, and capabilities for the control and monitoring of RF assets introducing network automation and provide interoperable communications for joint operations. It will interconnect forces of the Battle Group (BG)/ Amphibious Readiness Group (ARG) and connects the BG/ARG with expeditionary forces and the Commander-in-Chief Command Complex (CCC) ashore crossing all available media including Ultra High Frequency (UHF), Super High Frequency (SHF), Extremely High Frequency (EHF), commercial satellite links, and new medium-to-high data rate HF and UHF line of sight (LOS) links. C4ISR includes RCS, weather, navigational, signal exploitation, and command and control equipment.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2016	CVN 73	Various	Various	Various	Various	1	50.828
FY 2020	CVN 74	Various	Various	Various	Various	1	60.640

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2016	CVN 73	Aug 2022	0	0	Aug 2022
FY 2020	CVN 74	Aug 2025	0	0	Aug 2025

Competition/Second Source Initiatives:
N/A

Remarks:
CVN 73 RCOH - Comprised of 36 discretely funded line items. Overall increase due to growth work associate with to CANES installation. Growth work included foundation re-work (design issues) required to complete the C4ISR Install (\$3.535M). CVN 73 is the first hull to significantly increase the use of government installation teams for GFE vice HII-NNS installation teams during RCOH. Technical engineering services costs in P-35 exhibits appear higher than previous hulls because these costs were categorized as Basic Construction in the P-5 exhibit when the work was performed by the shipbuilder, HII-NNS. Government installation teams are approximately 25% less costly and are an overall cost savings to the program.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls
Equipment Item: C4ISR		PARM Code: NAVWAR PMW 750
CVN 74 RCOH - Comprised of 34 discretely funded line items (down from 35, with the descope of ARC-210 Radio System). CVN 74 increase (\$5.422M) is due primarily to hardware increases to Distributed Systems (\$5.167M), NTCDL (\$4.400M), RCS integration (\$1.459M), and other miscellaneous small value hardware increases (totaling \$2.241M). The hardware increases were partially offset by decreases in other categories.		

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls			
Equipment Item: INTEGRATED COMMUNICATION NETWORK (ICAN / DDCN & IVCN)						PARM Code: NAVSEA 05H3, NSWC Philadelphia	
P-35 Category	FY 2016		FY 2020				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	23.924	1	24.649			
Ancillary Equipment		0.039		1.042			
Technical Data and Documentation		1.186		1.400			
Spares		0.519		0.288			
System Engineering		8.546		9.835			
Technical Engineering Services		13.115		29.364			
Other Costs		5.729		9.232			
Total	1	53.058	1	75.810			
Description: The Integrated Communication Network consists of the following systems: An Integrated Communications System (ICS) that provides the ship's Internal Command and Control Communications. In addition, ICS provides connectivity to other onboard systems such as Announcing Systems, Sound Powered Circuits, Secure / Non Secure off-ship Communications, Shipboard Air Traffic Control Communications (SATCC) and Hierarchical Yet Dynamically Reprogrammable Architecture (HYDRA). The Machinery Control Monitoring System (MCMS) controls and monitors approximately 3500 machinery signals for various HM&E auxiliary systems (e.g. JP5, firemain, IC/SM panels) for aircraft carriers. It utilizes the Machinery Control Network (MCN) for signals. The MCN is the core network that provides communication services and transport for the MCMS system and part of the backbone that rides over the Fiber Optic Cable Plant (FOCP). It consists of five network switches, associated racks, and cabling. The Navigation Critical Distribution System (NAVCRT) is a switched network providing communication services and transport for the NAV Standard Message, which is originated in the Naval Sensor System Interface (NAVSSI) system. The NAVCRIT Distribution consists of three backbone switches and eight I/O controllers to convert digital NAV data for analog outputs. It will use the FOCP to the maximum extent for connectivity. The Ship Control System (SCS) provides control and display of rudder position, Engine and Propeller Order Telegraph functions. SCS provides data for heading, speed, and rudder angles through NAVCRIT Network from NAVSSI. The SCS interfaces to an Electronic Chart Display Information System. Shipboard Multipurpose Copiers includes the acquisition and installation of Class III Copier/Printer (B&W), Class III Color Copier/Printer, Class IV Copier/Printer (B&W) and Class IV Color Copier/Printer. The related equipment is for use on surface vessels in the US Navy as part of the Shipboard Multipurpose Copier Program.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2016	CVN 73	Various	Various	Various	Various	1	23.924
FY 2020	CVN 74	Various	Various	Various	Various	1	24.649
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2016	CVN 73	Aug 2022	0	0	Aug 2022		
FY 2020	CVN 74	Aug 2025	0	0	Aug 2025		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1	P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls	
Equipment Item: INTEGRATED COMMUNICATION NETWORK (ICAN / DDCN & IVCN)		PARM Code: NAVSEA 05H3, NSWC Philadelphia
Remarks: CVN 73 RCOH - Comprised of 24 discretely funded line items. Overall decrease (\$1.496M) due to a reduction in Navigation Critical Distribution System (NCDS) hardware (Network Management System) and associated technical engineering services (shipboard install costs). CVN 74 RCOH - Comprised of 30 discretely funded line items. CVN 74 increase (\$10.903M) due primarily to the addition of several new cybersecurity systems (\$2.520M), as well as increases to SBS Suite for technical engineering services related to shock requirements and IA efforts (\$2.602M); to NAVCRIT for hardware, software and licensing (\$1.061M); to IVN for technical engineering services (\$1.963M); and other miscellaneous small value increases. CVN 74 increases from CVN 73 are driven by several new system requirements including Cyber Security Upgrades (\$7.890M), Afloat Multi-Media Production System (AMMPS) (\$2.791M), and Advanced Damage Control Systems (ADCS) (\$2.431M); as well as the increases to SBS Suite, NAVCRIT, IVN and other small value increases discussed above.		

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																									
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls																											
Equipment Item: SHIP SELF DEFENSE SYSTEM (SSDS) MK2 MOD 1E						PARM Code: NAVSEA PEO IWS 10																									
P-35 Category	FY 2016		FY 2020																												
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																											
Major Hardware	1	12.864	1	12.757																											
Technical Data and Documentation		0.770		0.475																											
Spares		0.899		0.940																											
System Engineering		7.028		4.633																											
Technical Engineering Services		4.445		4.825																											
Other Costs		18.503		6.286																											
Total	1	44.509	1	29.916																											
Description: The Ship Self Defense System (SSDS) MK2 provides primary support for force/own ship combat systems control and enhanced self-defense capabilities. SSDS is the heart of the Combat System integrating sensors, weapons systems, data links, and command and control elements into a unified Combat System.																															
Contract Data: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Prime Contractor</th> <th>Contract Method/Type</th> <th>Award Date</th> <th>New/Option</th> <th>Quantity (Each)</th> <th>Unit Cost (\$ M)</th> </tr> <tr> <td>FY 2016</td> <td>CVN 73</td> <td>Raytheon/Lockheed Martin</td> <td>C/CPFF</td> <td>Apr 2018</td> <td>Option</td> <td style="text-align: center;">1</td> <td style="text-align: right;">12.864</td> </tr> <tr> <td>FY 2020</td> <td>CVN 74</td> <td>Lockheed Martin</td> <td>Various</td> <td>Sep 2020</td> <td>Various</td> <td style="text-align: center;">1</td> <td style="text-align: right;">12.757</td> </tr> </table>								Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2016	CVN 73	Raytheon/Lockheed Martin	C/CPFF	Apr 2018	Option	1	12.864	FY 2020	CVN 74	Lockheed Martin	Various	Sep 2020	Various	1	12.757
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																								
FY 2016	CVN 73	Raytheon/Lockheed Martin	C/CPFF	Apr 2018	Option	1	12.864																								
FY 2020	CVN 74	Lockheed Martin	Various	Sep 2020	Various	1	12.757																								
Delivery Date: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Earliest Ship Delivery Date</th> <th>Months Required Before Delivery</th> <th>Production Leadtime</th> <th>Required Award Date</th> </tr> <tr> <td>FY 2016</td> <td>CVN 73</td> <td>Aug 2022</td> <td style="text-align: center;">26</td> <td style="text-align: center;">18</td> <td style="text-align: center;">Dec 2018</td> </tr> <tr> <td>FY 2020</td> <td>CVN 74</td> <td>Aug 2025</td> <td style="text-align: center;">28</td> <td style="text-align: center;">24</td> <td style="text-align: center;">Apr 2021</td> </tr> </table>								Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date	FY 2016	CVN 73	Aug 2022	26	18	Dec 2018	FY 2020	CVN 74	Aug 2025	28	24	Apr 2021						
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date																										
FY 2016	CVN 73	Aug 2022	26	18	Dec 2018																										
FY 2020	CVN 74	Aug 2025	28	24	Apr 2021																										
Competition/Second Source Initiatives: N/A																															
Remarks: CVN 73 RCOH - Overall decrease (\$0.917M) is a result of realized savings from the GFE Governance Program to reduce program management, system engineering, and logistics support. CVN 74 RCOH - Overall decrease (\$7.261M) primarily due to cost sharing initiative for TI-16 software development, licenses and computer support across all TI-16 platforms (aircraft carriers and amphibious ships). CVN 74 decrease from CVN 73 due to CVN 74 utilizing the same CVN 73 TI-16 hardware configuration, thus having significantly less non-recurring engineering (NRE).																															

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls			
Equipment Item: ELECTRONIC CONSOLIDATED AUTOMATED SUPPORT SYSTEM (ECASS)						PARM Code: NAVAIR PMA 260	
P-35 Category	FY 2016		FY 2020				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	34.953	1	35.017			
Technical Engineering Services		1.638		1.606			
Other Costs		0.200		0.415			
Total	1	36.791	1	37.038			
Description: The Electronic Consolidated Automated Support System (eCASS) provides repair capability for aircraft instruments, components ("black boxes"), subcomponents (e.g. circuit cards), avionics and missile systems for all current deployable aircraft, F/A-18 ATFLIR (Forward Looking Infrared Receiver) and ALQ-99 (electronic jamming) systems, as well as new and future aircraft such as E-2D and F-35C. The eCASS replaces the obsolete Consolidated Automated Support System (CASS) that formerly provided this support. The eCASS suite provides expeditious, on-site repair capability for more than 1,100 different components, without which parts support for the ship's AIRWING (which routinely operates at great distances from logistics supply points) would be degraded to the point that it would result in increased support costs and negatively affect mission accomplishment, combat readiness, and required sortie generation rates.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2016	CVN 73	Lockheed Martin	C/CPFF	Oct 2018	New	1	34.953
FY 2020	CVN 74	Lockheed Martin	SS/FFP	Nov 2021	Option	1	35.017
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2016	CVN 73	Aug 2022	27	12	May 2019		
FY 2020	CVN 74	Aug 2025	25	12	Jul 2022		
Competition/Second Source Initiatives: N/A							
Remarks: CVN 73 RCOH - Overall decrease (\$0.178M) is a result of realized savings from the GFE Governance Program to reduce program management, system engineering, and logistics support. CVN 74 RCOH - Overall increase (\$0.057M) due to updated AIT cost estimate previously based on CVN 73 actuals.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls			
Equipment Item: JOINT PRECISION APPROACH AND LANDING SYSTEM (JPALS)						PARM Code: NAVAIR PMA 213	
P-35 Category	FY 2016		FY 2020				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	11.353	1	9.921			
Spares		0.637		0.946			
System Engineering		0.178		0.762			
Technical Engineering Services		1.377		1.435			
Other Costs		0.586		1.308			
Total	1	14.131	1	14.372			
Description: The Joint Precision Approach and Landing System (JPALS) is the future precision approach and landing system which will be the primary landing system for the Joint Strike Fighter (F-35B/F-35C), Unmanned Carrier Aviation Air System (MQ-25A), and future aircraft platforms onboard CVNs and LHA/LHD type ships. JPALS is the Navy certified sea-based system to have the capabilities necessary to provide ship range/bearing for JPALS-equipped aircraft operating within 200NM; provide air traffic control surveillance of JPALS-equipped aircraft via secure, two-way data link with the ship; and support auto-land functionality for the F-35C, MQ-25A, and future platforms to CVNs. JPALS is critical for MQ-25A as currently no secondary landing system exists for MQ-25A operations at sea.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2016	CVN 73	Raytheon	C/FPIF	May 2019	New	1	11.353
FY 2020	CVN 74	Raytheon	SS/FFP	Oct 2020	Option	1	9.921
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2016	CVN 73	Aug 2022	19	12	Jan 2020		
FY 2020	CVN 74	Aug 2025	34	17	May 2021		
Competition/Second Source Initiatives: N/A							
Remarks: No change from the FY 2021 budget request.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls			
Equipment Item: AN/SPN-46 OVERHAUL/UPGRADE						PARM Code: NAVAIR PMA 213	
P-35 Category	FY 2016		FY 2020				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	6.651	1	5.768			
System Engineering		0.527		0.638			
Technical Engineering Services		4.070		3.392			
Other Costs		1.594		3.022			
Total	1	12.842	1	12.820			
Description: The AN/SPN-46 Automatic Carrier Landing System (ACLS) is a precision approach landing system (PALS) which provides electronic guidance to carrier-based aircraft and allows them to land in all-weather conditions with no limitations due to low ceiling or restricted visibility. AN/SPN-46 is a fully automated, all-weather approach landing aid for carrier aircraft that enhances safety of flight during recovery, enables the execution of all-weather air combat operations, and is required to achieve full air traffic control certification following RCOH. AN/SPN-46 is required to be removed during the RCOH to prevent damage and allow for major infrastructure recapitalization and reconfiguration of the ship's island, mast, and tower.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2016	CVN 73	NAWC Aircraft Division	WR	Mar 2016		1	6.651
FY 2020	CVN 74	NAWC Aircraft Division	WR	Oct 2020		1	5.768
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2016	CVN 73	Aug 2022	28	41	Nov 2016		
FY 2020	CVN 74	Aug 2025	13	38	May 2021		
Competition/Second Source Initiatives: N/A							
Remarks: CVN 73 RCOH - Overall decrease (\$0.009M) is a result of realized savings from the GFE Governance Program to reduce program management, system engineering, and logistics support. CVN 74 RCOH - Overall decrease (\$0.210M) is a result of realized savings from the GFE Governance Program to reduce program management, system engineering, and logistics support.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																									
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls																											
Equipment Item: MQ-25 - UNMANNED AVIATION WARFARE CENTER (UAWC)						PARM Code: NAVAIR PMA 268																									
P-35 Category	FY 2016		FY 2020																												
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																											
Major Hardware	1	2.687	1	12.010																											
Technical Data and Documentation		0.182		0.096																											
Technical Engineering Services		8.597		11.256																											
Other Costs		0.131		0.797																											
Total	1	11.597	1	24.159																											
Description: MQ-25 - Unmanned Aviation Warfare Center (UAWC) will incorporate a family of systems providing a carrier-based unmanned aircraft system that supports long-endurance, proven Intelligence, Surveillance, Reconnaissance, and Targeting (ISR&T) and precision strike capability to Joint and Naval Warfare Commanders.																															
Contract Data: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Prime Contractor</th> <th>Contract Method/Type</th> <th>Award Date</th> <th>New/Option</th> <th>Quantity (Each)</th> <th>Unit Cost (\$ M)</th> </tr> <tr> <td>FY 2016</td> <td>CVN 73</td> <td>Leonardo DRS/Lockheed Martin</td> <td>Various</td> <td>May 2020</td> <td>Various</td> <td>1</td> <td>2.687</td> </tr> <tr> <td>FY 2020</td> <td>CVN 74</td> <td>Leonardo DRS/GTS</td> <td>Various</td> <td>Jan 2021</td> <td>Various</td> <td>1</td> <td>12.010</td> </tr> </table>								Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2016	CVN 73	Leonardo DRS/Lockheed Martin	Various	May 2020	Various	1	2.687	FY 2020	CVN 74	Leonardo DRS/GTS	Various	Jan 2021	Various	1	12.010
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																								
FY 2016	CVN 73	Leonardo DRS/Lockheed Martin	Various	May 2020	Various	1	2.687																								
FY 2020	CVN 74	Leonardo DRS/GTS	Various	Jan 2021	Various	1	12.010																								
Delivery Date: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Earliest Ship Delivery Date</th> <th>Months Required Before Delivery</th> <th>Production Leadtime</th> <th>Required Award Date</th> </tr> <tr> <td>FY 2016</td> <td>CVN 73</td> <td>Aug 2022</td> <td>7</td> <td>14</td> <td>Nov 2020</td> </tr> <tr> <td>FY 2020</td> <td>CVN 74</td> <td>Aug 2025</td> <td>21</td> <td>27</td> <td>Aug 2021</td> </tr> </table>								Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date	FY 2016	CVN 73	Aug 2022	7	14	Nov 2020	FY 2020	CVN 74	Aug 2025	21	27	Aug 2021						
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date																										
FY 2016	CVN 73	Aug 2022	7	14	Nov 2020																										
FY 2020	CVN 74	Aug 2025	21	27	Aug 2021																										
Competition/Second Source Initiatives: N/A																															
Remarks: MQ-25 infrastructure installation restored to the CVN 73 work package for the FY 2021 budget request as mandated by the National Defense Authorization Act for Fiscal Year 2019. CVN 74 is installing the full system with GFE hardware. CVN 73 RCOH - Overall decrease (\$0.226M) is a result of realized savings from the GFE Governance Program to reduce program management, system engineering, and logistics support. CVN 74 RCOH - Overall increase (\$2.659M) due to added capabilities for Digital Modular Radio (DMR) cabling solutions (necessary for narrow band SATCOM capabilities), data transfer storage (required product to load mission data), and TPX-42/SYY-1 interface (existing hardware/software product to bring the airspace radar picture into UAWC for additional situational awareness). CVN 74 increase from CVN 73 is due to CVN 74 installing the full system with GFE hardware, whereas CVN 73 is installing the infrastructure only.																															

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																									
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls																											
Equipment Item: AN/USG-2B - COOPERATIVE ENGAGEMENT CAPABILITY (CEC)						PARM Code: NAVSEA PEO IWS 6.0																									
P-35 Category	FY 2016		FY 2020																												
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																											
Major Hardware	1	3.448	1	5.048																											
Technical Data and Documentation		0.115		-																											
Spares		0.338		0.476																											
System Engineering		2.520		3.123																											
Technical Engineering Services		2.569		1.674																											
Other Costs		1.036		2.030																											
Total	1	10.026	1	12.351																											
Description: AN/USG-2B - Cooperative Engagement Capability (CEC) provides Battle Force Anti-Air Warfare (AAW) capability by coordinating all force AAW sensors into a single real time, fire control quality composite track picture. CEC distributes sensor measurement data from each Cooperating Unit (CU) to all other CUs. Each CU has a Data Distribution System (DDS) and a Cooperative Engagement Processor (CEP). The DDS encodes and distributes ownship sensor and engagement data to other CUs, and receives and decodes other CU's data. The CEP processes ownship data and DDS supplied remote sensor and weapon data needed to provide the common air picture.																															
Contract Data: <table border="1" style="width:100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Prime Contractor</th> <th>Contract Method/Type</th> <th>Award Date</th> <th>New/Option</th> <th>Quantity (Each)</th> <th>Unit Cost (\$ M)</th> </tr> <tr> <td>FY 2016</td> <td>CVN 73</td> <td>Raytheon/Sechan Electronics</td> <td>C/FFP</td> <td>Jan 2016</td> <td>Option</td> <td align="center">1</td> <td align="right">3.448</td> </tr> <tr> <td>FY 2020</td> <td>CVN 74</td> <td>Leonardo DRS</td> <td>C/FFP</td> <td>May 2020</td> <td>Option</td> <td align="center">1</td> <td align="right">5.048</td> </tr> </table>								Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2016	CVN 73	Raytheon/Sechan Electronics	C/FFP	Jan 2016	Option	1	3.448	FY 2020	CVN 74	Leonardo DRS	C/FFP	May 2020	Option	1	5.048
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																								
FY 2016	CVN 73	Raytheon/Sechan Electronics	C/FFP	Jan 2016	Option	1	3.448																								
FY 2020	CVN 74	Leonardo DRS	C/FFP	May 2020	Option	1	5.048																								
Delivery Date: <table border="1" style="width:100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Earliest Ship Delivery Date</th> <th>Months Required Before Delivery</th> <th>Production Leadtime</th> <th>Required Award Date</th> </tr> <tr> <td>FY 2016</td> <td>CVN 73</td> <td align="center">Aug 2022</td> <td align="center">38</td> <td align="center">18</td> <td align="center">Dec 2017</td> </tr> <tr> <td>FY 2020</td> <td>CVN 74</td> <td align="center">Aug 2025</td> <td align="center">38</td> <td align="center">18</td> <td align="center">Dec 2020</td> </tr> </table>								Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date	FY 2016	CVN 73	Aug 2022	38	18	Dec 2017	FY 2020	CVN 74	Aug 2025	38	18	Dec 2020						
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date																										
FY 2016	CVN 73	Aug 2022	38	18	Dec 2017																										
FY 2020	CVN 74	Aug 2025	38	18	Dec 2020																										
Competition/Second Source Initiatives: N/A																															
Remarks: CVN 73 RCOH - Overall decrease (\$1.254M) due to re-phasing Combat Systems Integration Part II efforts for CSSQT to the Post Delivery period. CVN 74 RCOH - Increases from CVN 73 in hardware and other costs are due to upgrade of antennas from PAO-cooled (polyalphaolefin, a liquid coolant) to air-cooled aboard CVN 74 to address obsolescence and maintainability issues.																															

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls			
Equipment Item: AN/SLQ-59 - ELECTRONIC WARFARE (EW) SYSTEM						PARM Code: NAVSEA PEO IWS 2E	
P-35 Category	FY 2016		FY 2020				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	1.499	0	-			
Ancillary Equipment		0.100		-			
Spares		0.045		-			
System Engineering		0.291		-			
Technical Engineering Services		4.495		-			
Other Costs		0.262		-			
Total	1	6.692	0	-			
Description: AN/SLQ-59 - ELECTRONIC WARFARE (EW) SYSTEM is a CNO designated program in response to PACFLT/CJTF519 Urgent Operational Needs Statement (UONS) designed to enhance existing shipboard Surface Electronic Warfare Systems. AN/SLQ-59 provides enhanced shipboard Electronic Support (ES) and Electronic Attack (EA) capabilities.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2016	CVN 73	Electro Impulse Laboratory	SS/IDIQ	Nov 2018	Option	1	1.499
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2016	CVN 73	Aug 2022	22	12	Oct 2019		
Competition/Second Source Initiatives: N/A							
Remarks: CVN 73 RCOH - Overall increase (\$0.389M) due to an updated Alteration Installation Team (AIT) cost based on the final delivered drawing package. CVN 74 RCOH - AN/SLQ-59 was installed onboard CVN 74 prior to the RCOH.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls			
Equipment Item: AN/SPN-41 REFURBISHMENT						PARM Code: NAVAIR PMA 213	
P-35 Category	FY 2016		FY 2020				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	3.569	1	3.796			
System Engineering		0.393		0.399			
Technical Engineering Services		1.550		1.384			
Other Costs		0.206		0.242			
Total	1	5.718	1	5.821			
Description: The AN/SPN-41 Aircraft Approach Control Transmitting Set provides all-weather instrument approach guidance from the ship to the aircraft. It is used as the ship's Instrument Landing System (ILS) and Monitor to provide azimuth and elevation alignment information to landing aircraft on final approach to the deck. It also serves as an independent monitor of other shipboard landing systems for the pilot as well as providing a backup landing guidance option. AN/SPN-41 enhances safety of flight during recovery, enables the execution of all-weather air combat operations, and is required to achieve full air traffic control certification following RCOH. AN/SPN-41 is required to be removed during the RCOH to prevent damage and allow for major infrastructure recapitalization and reconfiguration of the ship's island, mast, and tower.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2016	CVN 73	NAWC Aircraft Division	WR	Jul 2017		1	3.569
FY 2020	CVN 74	NAWC Aircraft Division	WR	Dec 2020		1	3.796
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2016	CVN 73	Aug 2022	29	24	Mar 2018		
FY 2020	CVN 74	Aug 2025	25	24	Jul 2021		
Competition/Second Source Initiatives: N/A							
Remarks: No significant change from the FY 2021 budget request.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls																			
Equipment Item: AN/SPN-50(V)1 - AIR TRAFFIC CONTROL RADAR						PARM Code: NAVAIR PMA 213																	
P-35 Category	FY 2016		FY 2020																				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																			
Major Hardware	0	-	1	12.432																			
Technical Data and Documentation		-		0.130																			
Spares		-		1.034																			
System Engineering		-		0.593																			
Technical Engineering Services		-		2.192																			
Other Costs		-		0.498																			
Total	0	-	1	16.879																			
Description: Digital radar system capable of modern radar processing to improve target detection and track in the presence of competing clutter. C-band radar alleviating operational congestion of new shipboard S-band radars and restrictions caused by increasing spectrum encroachment and National Broadband Plan.																							
Contract Data: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th style="text-align: center;">Program Year</th> <th style="text-align: center;">Hull</th> <th style="text-align: center;">Prime Contractor</th> <th style="text-align: center;">Contract Method/Type</th> <th style="text-align: center;">Award Date</th> <th style="text-align: center;">New/Option</th> <th style="text-align: center;">Quantity (Each)</th> <th style="text-align: center;">Unit Cost (\$ M)</th> </tr> <tr> <td style="text-align: center;">FY 2020</td> <td style="text-align: center;">CVN 74</td> <td style="text-align: center;">SAAB</td> <td style="text-align: center;">C/FPIF</td> <td style="text-align: center;">Jul 2020</td> <td style="text-align: center;">New</td> <td style="text-align: center;">1</td> <td style="text-align: center;">12.432</td> </tr> </table>								Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2020	CVN 74	SAAB	C/FPIF	Jul 2020	New	1	12.432
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																
FY 2020	CVN 74	SAAB	C/FPIF	Jul 2020	New	1	12.432																
Delivery Date: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th style="text-align: center;">Program Year</th> <th style="text-align: center;">Hull</th> <th style="text-align: center;">Earliest Ship Delivery Date</th> <th style="text-align: center;">Months Required Before Delivery</th> <th style="text-align: center;">Production Leadtime</th> <th style="text-align: center;">Required Award Date</th> </tr> <tr> <td style="text-align: center;">FY 2020</td> <td style="text-align: center;">CVN 74</td> <td style="text-align: center;">Aug 2025</td> <td style="text-align: center;">24</td> <td style="text-align: center;">22</td> <td style="text-align: center;">Oct 2021</td> </tr> </table>								Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date	FY 2020	CVN 74	Aug 2025	24	22	Oct 2021				
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date																		
FY 2020	CVN 74	Aug 2025	24	22	Oct 2021																		
Competition/Second Source Initiatives: N/A																							
Remarks: AN/SPN-50(V)1 is a new equipment installation for the CVN 74 RCOH program of record, replacing AN/SPN-43C. CVN 73 AN/SPN-43C refurbishment is an Electronics major item. CVN 74 increase from the FY 2021 request of \$1.586M is due to improved estimates based on results of hardware negotiations and necessary software and licensing.																							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls			
Equipment Item: AIRCRAFT ELECTRICAL SERVICE STATION (AESS) INSTALL						PARM Code: NSWC Philadelphia	
P-35 Category	FY 2016		FY 2020				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	6.668	0	-			
System Engineering		0.244		-			
Technical Engineering Services		7.335		-			
Other Costs		0.233		-			
Total	1	14.480	0	-			
Description: Aircraft Electrical Service Station (AESS) supports the F-35 Joint Strike Fighter (JSF) and JUCAS electrical requirements. They require 270VDC electrical power for maintenance and pre-flight operations. This type of power is not currently available on CVN 68 class aircraft carriers. This ship alteration will equip CVN 68 class ships with a dual purpose AESS station providing either: 90KVA of 115VAC, 400Hz, power for the Advanced Hawkeye (E-2D), and all legacy aircraft, or 70KW of 270VDC electrical power for the JSF (F-35) and JUCAS. This upgrade will replace the obsolete components now part of the AESS with a quiet, lightweight, low-cost, dual-purpose system that decreases maintenance costs and improves habitability.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2016	CVN 73	Various	C/FFP	Jan 2017	New	1	6.668
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2016	CVN 73	Aug 2022	55	12	Jan 2017		
Competition/Second Source Initiatives: N/A							
Remarks: Aircraft Electrical Service Station (AESS) was installed onboard CVN 74 prior to the RCOH. No further AESS work is required during the RCOH.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																									
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls																											
Equipment Item: FURNITURE (NON PROPULSION PLANT)						PARM Code: NSWC Philadelphia																									
P-35 Category	FY 2016		FY 2020																												
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																											
Major Hardware	1	4.647	1	4.973																											
Ancillary Equipment		0.069		0.975																											
System Engineering		0.377		-																											
Technical Engineering Services		8.703		10.027																											
Other Costs		0.057		-																											
Total	1	13.853	1	15.975																											
Description: Replaces damaged/worn furniture for 250 spaces. During the RCOH, all furniture is offloaded and evaluated for reuse. Most furniture is stored and reinstalled. However, damaged furniture must be replaced as it cannot be properly secured and poses a personnel risk during high sea states.																															
Contract Data: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th style="text-align: center;">Program Year</th> <th style="text-align: center;">Hull</th> <th style="text-align: center;">Prime Contractor</th> <th style="text-align: center;">Contract Method/Type</th> <th style="text-align: center;">Award Date</th> <th style="text-align: center;">New/Option</th> <th style="text-align: center;">Quantity (Each)</th> <th style="text-align: center;">Unit Cost (\$ M)</th> </tr> <tr> <td style="text-align: center;">FY 2016</td> <td style="text-align: center;">CVN 73</td> <td style="text-align: center;">Tecnico</td> <td style="text-align: center;">C/CPFF</td> <td style="text-align: center;">Dec 2016</td> <td style="text-align: center;">New</td> <td style="text-align: center;">1</td> <td style="text-align: right;">4.647</td> </tr> <tr> <td style="text-align: center;">FY 2020</td> <td style="text-align: center;">CVN 74</td> <td style="text-align: center;">TBD</td> <td style="text-align: center;">C/CPFF</td> <td style="text-align: center;">Aug 2021</td> <td style="text-align: center;">New</td> <td style="text-align: center;">1</td> <td style="text-align: right;">4.973</td> </tr> </table>								Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2016	CVN 73	Tecnico	C/CPFF	Dec 2016	New	1	4.647	FY 2020	CVN 74	TBD	C/CPFF	Aug 2021	New	1	4.973
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																								
FY 2016	CVN 73	Tecnico	C/CPFF	Dec 2016	New	1	4.647																								
FY 2020	CVN 74	TBD	C/CPFF	Aug 2021	New	1	4.973																								
Delivery Date: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th style="text-align: center;">Program Year</th> <th style="text-align: center;">Hull</th> <th style="text-align: center;">Earliest Ship Delivery Date</th> <th style="text-align: center;">Months Required Before Delivery</th> <th style="text-align: center;">Production Leadtime</th> <th style="text-align: center;">Required Award Date</th> </tr> <tr> <td style="text-align: center;">FY 2016</td> <td style="text-align: center;">CVN 73</td> <td style="text-align: center;">Aug 2022</td> <td style="text-align: center;">43</td> <td style="text-align: center;">6</td> <td style="text-align: center;">Jul 2018</td> </tr> <tr> <td style="text-align: center;">FY 2020</td> <td style="text-align: center;">CVN 74</td> <td style="text-align: center;">Aug 2025</td> <td style="text-align: center;">39</td> <td style="text-align: center;">6</td> <td style="text-align: center;">Nov 2021</td> </tr> </table>								Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date	FY 2016	CVN 73	Aug 2022	43	6	Jul 2018	FY 2020	CVN 74	Aug 2025	39	6	Nov 2021						
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date																										
FY 2016	CVN 73	Aug 2022	43	6	Jul 2018																										
FY 2020	CVN 74	Aug 2025	39	6	Nov 2021																										
Competition/Second Source Initiatives: N/A																															
Remarks: CVN 74 increase (\$3.236M) is due to updated estimates based on actuals from CVN 73 and updated labor contract expenses.																															

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls			
Equipment Item: LOW PRESSURE AIR PLANT (LPAP)						PARM Code: NSWC Philadelphia	
P-35 Category	FY 2016		FY 2020				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	5.990	1	6.242			
System Engineering		0.064		0.061			
Technical Engineering Services		0.021		0.093			
Other Costs		0.051		0.131			
Total	1	6.126	1	6.527			
Description: Remove three Ship Service Air Compressors (SSAC), four Control Air Compressors, and associated dryers from two machinery rooms and two reactor rooms. Install nine MARC 350 Low Pressure Air Plants (LPAPs) to serve both ship service air and control air systems.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2016	CVN 73	RIX Industries	SS/IDIQ	Feb 2015	Option	1	5.990
FY 2020	CVN 74	RIX Industries	SS/FFP	Sep 2019	Option	1	6.242
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2016	CVN 73	Aug 2022	55	12	Jan 2017		
FY 2020	CVN 74	Aug 2025	48	16	Apr 2020		
Competition/Second Source Initiatives: N/A							
Remarks: No change from the FY 2021 budget request.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls			
Equipment Item: AUTOMATIC VOLTAGE REGULATOR (AVR)						PARM Code: NAVSEA PMS 312	
P-35 Category	FY 2016		FY 2020				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	4.340	0	-			
Total	1	4.340	0	-			
Description: The Turbine Generator Automatic Voltage Regulator is a digital upgrade to the legacy voltage regulator. Its purpose is to regulate output voltage from shipboard turbine generators to meet electrical requirements for all ship systems. This upgrade is required to improve operational safety of the turbine generators and eliminate material shortages due to obsolescence of legacy voltage regulator system components.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2016	CVN 73	Northrop Grumman Power/Control Systems	C/FFP	Jun 2015	Option	1	4.340
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2016	CVN 73	Aug 2022	51	26	Mar 2016		
Competition/Second Source Initiatives: N/A							
Remarks: No change from the FY 2021 budget request. Automatic Voltage Regulator was installed onboard CVN 74 prior to the RCOH. No further AVR work is required during the RCOH.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls																			
Equipment Item: VSA O2 GENERATOR						PARM Code: NSWC Philadelphia																	
P-35 Category	FY 2016		FY 2020																				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																			
Major Hardware	0	-	1	3.849																			
Spares		-		0.170																			
System Engineering		-		0.752																			
Technical Engineering Services		-		0.294																			
Other Costs		-		0.298																			
Total	0	-	1	5.363																			
Description: One liquid oxygen generating and storage plant with associated support equipment; one gaseous nitrogen generator with associated storage flasks.																							
Contract Data: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th style="text-align: center;">Program Year</th> <th style="text-align: center;">Hull</th> <th style="text-align: center;">Prime Contractor</th> <th style="text-align: center;">Contract Method/Type</th> <th style="text-align: center;">Award Date</th> <th style="text-align: center;">New/Option</th> <th style="text-align: center;">Quantity (Each)</th> <th style="text-align: center;">Unit Cost (\$ M)</th> </tr> <tr> <td style="text-align: center;">FY 2020</td> <td style="text-align: center;">CVN 74</td> <td style="text-align: center;">Pacific Consolidated Industries</td> <td style="text-align: center;">SS/FFP</td> <td style="text-align: center;">Jan 2020</td> <td style="text-align: center;">Option</td> <td style="text-align: center;">1</td> <td style="text-align: center;">3.849</td> </tr> </table>								Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2020	CVN 74	Pacific Consolidated Industries	SS/FFP	Jan 2020	Option	1	3.849
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																
FY 2020	CVN 74	Pacific Consolidated Industries	SS/FFP	Jan 2020	Option	1	3.849																
Delivery Date: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th style="text-align: center;">Program Year</th> <th style="text-align: center;">Hull</th> <th style="text-align: center;">Earliest Ship Delivery Date</th> <th style="text-align: center;">Months Required Before Delivery</th> <th style="text-align: center;">Production Leadtime</th> <th style="text-align: center;">Required Award Date</th> </tr> <tr> <td style="text-align: center;">FY 2020</td> <td style="text-align: center;">CVN 74</td> <td style="text-align: center;">Aug 2025</td> <td style="text-align: center;">36</td> <td style="text-align: center;">24</td> <td style="text-align: center;">Aug 2020</td> </tr> </table>								Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date	FY 2020	CVN 74	Aug 2025	36	24	Aug 2020				
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date																		
FY 2020	CVN 74	Aug 2025	36	24	Aug 2020																		
Competition/Second Source Initiatives: N/A																							
Remarks: No change from the FY 2021 budget request.																							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls			
Equipment Item: AVIATION EQUIPMENT & SUPPORT						PARM Code: NAVAIR PMA 251	
P-35 Category	FY 2016		FY 2020				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	25.019	1	37.020			
Ancillary Equipment		0.015		0.135			
Technical Data and Documentation		-		0.199			
Spares		0.118		0.332			
System Engineering		3.691		3.268			
Technical Engineering Services		8.680		8.096			
Other Costs		5.326		5.367			
Total	1	42.849	1	54.417			
Description: Provides procurement, engineering and logistics support for launch and recovery equipment (includes overhaul/replacement of catapult launch valves and arresting gear engines), ADMACS (Aviation Data Management and Control System Phase II upgrade; includes Cyber Security requirement and future aircraft ready), Moriah Wind System, ILARTS (Integrated Launch and Recovery TV Surveillance System; includes Technical Refresh Service Change to mitigate obsolescence issues), mission pods, Jet Blast Deflectors (includes Service Change to provide side panel cooling to meet JSF requirements), aviation maintenance facility, weapons compatibility, aircraft spotting, aviation servicing facilities, Landing Signal Officer Display System (LSODS; includes ADMACS interfacing and Cyber Security updates), Long Range Lineup System (LRLS), Improved Fresnel Lens Optical Landing System (IFLOLS; includes Phase IV upgrade), Manually Operated Visual Landing Aid System (MOVLAS) and Flight Deck Lighting and Marking and Lighting. All of these systems are required to be repaired, updated, overhauled as required and tested during RCOH to attain final Flight Deck Certification authorizing launch and recovery of USN aircraft.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2016	CVN 73	Various	Various	Various	Various	1	25.019
FY 2020	CVN 74	Various	Various	Various	Various	1	37.020
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2016	CVN 73	Aug 2022	0	0	Aug 2022		
FY 2020	CVN 74	Aug 2025	0	0	Aug 2025		
Competition/Second Source Initiatives: N/A							
Remarks: CVN 73 RCOH - Comprised of 18 discretely funded line items. Overall decrease (\$3.039M) is a result of a cost reduction for the low loss launch valve refurbishment cost. CVN 74 RCOH - Comprised of 18 discretely funded line items. Increase from CVN 73 is due to CVN 74 requiring overhaul of additional crosshead and fixed sheaves and an additional engine new-build, which are not required on the CVN 73.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																									
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls																											
Equipment Item: NATO SEASPARROW SURFACE MISSILE SYSTEM (NSSMS)						PARM Code: NAVSEA PEO IWS 12																									
P-35 Category	FY 2016		FY 2020																												
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																											
Major Hardware	1	6.287	1	15.709																											
Ancillary Equipment		-		0.050																											
Spares		0.913		1.260																											
System Engineering		0.518		0.861																											
Technical Engineering Services		9.268		9.346																											
Other Costs		1.603		2.291																											
Total	1	18.589	1	29.517																											
Description: The NATO Seasparrow Surface Missile System (NSSMS) is a medium range self defense missile system capable of defeating near/mid-term air/surface threats.																															
Contract Data: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th style="text-align: center;">Program Year</th> <th style="text-align: center;">Hull</th> <th style="text-align: center;">Prime Contractor</th> <th style="text-align: center;">Contract Method/Type</th> <th style="text-align: center;">Award Date</th> <th style="text-align: center;">New/Option</th> <th style="text-align: center;">Quantity (Each)</th> <th style="text-align: center;">Unit Cost (\$ M)</th> </tr> <tr> <td style="text-align: center;">FY 2016</td> <td style="text-align: center;">CVN 73</td> <td style="text-align: center;">Raytheon</td> <td style="text-align: center;">SS/FFP</td> <td style="text-align: center;">Apr 2017</td> <td style="text-align: center;">New</td> <td style="text-align: center;">1</td> <td style="text-align: right;">6.287</td> </tr> <tr> <td style="text-align: center;">FY 2020</td> <td style="text-align: center;">CVN 74</td> <td style="text-align: center;">Raytheon</td> <td style="text-align: center;">SS/FFP</td> <td style="text-align: center;">Nov 2020</td> <td style="text-align: center;">Option</td> <td style="text-align: center;">1</td> <td style="text-align: right;">15.709</td> </tr> </table>								Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2016	CVN 73	Raytheon	SS/FFP	Apr 2017	New	1	6.287	FY 2020	CVN 74	Raytheon	SS/FFP	Nov 2020	Option	1	15.709
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																								
FY 2016	CVN 73	Raytheon	SS/FFP	Apr 2017	New	1	6.287																								
FY 2020	CVN 74	Raytheon	SS/FFP	Nov 2020	Option	1	15.709																								
Delivery Date: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th style="text-align: center;">Program Year</th> <th style="text-align: center;">Hull</th> <th style="text-align: center;">Earliest Ship Delivery Date</th> <th style="text-align: center;">Months Required Before Delivery</th> <th style="text-align: center;">Production Leadtime</th> <th style="text-align: center;">Required Award Date</th> </tr> <tr> <td style="text-align: center;">FY 2016</td> <td style="text-align: center;">CVN 73</td> <td style="text-align: center;">Aug 2022</td> <td style="text-align: center;">32</td> <td style="text-align: center;">30</td> <td style="text-align: center;">Jun 2017</td> </tr> <tr> <td style="text-align: center;">FY 2020</td> <td style="text-align: center;">CVN 74</td> <td style="text-align: center;">Aug 2025</td> <td style="text-align: center;">19</td> <td style="text-align: center;">31</td> <td style="text-align: center;">Jun 2021</td> </tr> </table>								Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date	FY 2016	CVN 73	Aug 2022	32	30	Jun 2017	FY 2020	CVN 74	Aug 2025	19	31	Jun 2021						
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date																										
FY 2016	CVN 73	Aug 2022	32	30	Jun 2017																										
FY 2020	CVN 74	Aug 2025	19	31	Jun 2021																										
Competition/Second Source Initiatives: N/A																															
Remarks: CVN 73 RCOH - Overall increase (\$0.655M) to accommodate growth work associate with Launcher (2) refurbishment. Material condition was worse than originally scoped. CVN 74 RCOH - Overall decrease (\$1.353M) due to configuration update to connect NSSMS radars and launchers directly to SSDS (removes NATO Q70 consoles). CVN 74 increase from CVN 73 is due to the requirement to buy new hardware to replace high wear items that have significantly degraded due to the age of the system. CVN 73 was limited to refurbishment of the removed hardware. Spare components are not available to support refurbishment of the CVN 74 removed hardware.																															

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls			
Equipment Item: AN/SPS-48G - 3D AIR SEARCH RADAR						PARM Code: NAVSEA PEO IWS 2RI	
P-35 Category	FY 2016		FY 2020				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	8.260	0	-			
Technical Data and Documentation		0.017		-			
Spares		0.328		-			
System Engineering		2.186		-			
Technical Engineering Services		2.453		-			
Other Costs		0.528		-			
Total	1	13.772	0	-			
Description: AN/SPS-48G (V)1 is a long range three dimensional (3D) radar used to search, detect and provide space-stabilized, three-coordinate (range, bearing, height) data to track airborne contacts.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2016	CVN 73	Harris	SS/FPIF	Sep 2016	Option	1	8.260
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2016	CVN 73	Aug 2022	32	18	Jun 2018		
Competition/Second Source Initiatives: N/A							
Remarks: CVN 73 RCOH - Overall increase (\$0.442M) due to an updated Alteration Installation Team (AIT) cost based on the final delivered drawing package. CVN 74 RCOH - AN/SPS-48G will be replaced by Enterprise Air Surveillance Radar (EASR) on CVN 74, enhancing air surveillance radar capability, increasing system availability, and re-baselining sustainment.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls																			
Equipment Item: AN/SPS-49(V)1 OVERHAUL/REFURBISHMENT						PARM Code: NAVSEA PEO IWS 2RI																	
P-35 Category	FY 2016		FY 2020																				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																			
Major Hardware	1	3.210	0	-																			
Ancillary Equipment		0.030		-																			
Spares		0.275		-																			
System Engineering		0.423		-																			
Technical Engineering Services		3.495		-																			
Other Costs		0.482		-																			
Total	1	7.915	0	-																			
Description: The AN/SPS-49A(V)1 Radar is a narrow beam, very long range, two dimensional air search radar. This is the primary air search radar for the ship providing early detection of airborne contacts (range, bearing, and altitude).																							
Contract Data: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th style="text-align: center;">Program Year</th> <th style="text-align: center;">Hull</th> <th style="text-align: center;">Prime Contractor</th> <th style="text-align: center;">Contract Method/Type</th> <th style="text-align: center;">Award Date</th> <th style="text-align: center;">New/Option</th> <th style="text-align: center;">Quantity (Each)</th> <th style="text-align: center;">Unit Cost (\$ M)</th> </tr> <tr> <td style="text-align: center;">FY 2016</td> <td style="text-align: center;">CVN 73</td> <td style="text-align: center;">NSWC Crane</td> <td style="text-align: center;">WR</td> <td style="text-align: center;">Apr 2017</td> <td></td> <td style="text-align: center;">1</td> <td style="text-align: right;">3.210</td> </tr> </table>								Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2016	CVN 73	NSWC Crane	WR	Apr 2017		1	3.210
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																
FY 2016	CVN 73	NSWC Crane	WR	Apr 2017		1	3.210																
Delivery Date: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th style="text-align: center;">Program Year</th> <th style="text-align: center;">Hull</th> <th style="text-align: center;">Earliest Ship Delivery Date</th> <th style="text-align: center;">Months Required Before Delivery</th> <th style="text-align: center;">Production Leadtime</th> <th style="text-align: center;">Required Award Date</th> </tr> <tr> <td style="text-align: center;">FY 2016</td> <td style="text-align: center;">CVN 73</td> <td style="text-align: center;">Aug 2022</td> <td style="text-align: center;">26</td> <td style="text-align: center;">30</td> <td style="text-align: center;">Dec 2017</td> </tr> </table>								Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date	FY 2016	CVN 73	Aug 2022	26	30	Dec 2017				
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date																		
FY 2016	CVN 73	Aug 2022	26	30	Dec 2017																		
Competition/Second Source Initiatives: N/A																							
Remarks: CVN 73 RCOH - Overall decrease (\$0.366M) is a result of realized savings from the GFE Governance Program to reduce program management, system engineering, and logistics support. CVN 74 RCOH - AN/SPS-49A(V)1 will be replaced by Enterprise Air Surveillance Radar (EASR) on CVN 74, enhancing air surveillance radar capability, increasing system availability, and re-baselining sustainment.																							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls			
Equipment Item: AN/SPQ-14 - ADVANCED SENSOR DISTRIBUTION SYSTEM (ASDS)						PARM Code: NAVSEA PEO IWS 1.0	
P-35 Category	FY 2016		FY 2020				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	0.893	0	-			
Ancillary Equipment		0.010		-			
Spares		0.045		-			
System Engineering		1.070		-			
Technical Engineering Services		1.094		-			
Other Costs		0.313		-			
Total	1	3.425	0	-			
Description: AN/SPQ-14(V) - Advanced Sensor Distribution System (ASDS) provides the distribution of radar sensor data and video to radar displays on board the ship.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2016	CVN 73	Lockheed Martin/DRS Technologies	C/FFP	Jan 2018	Option	1	0.893
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2016	CVN 73	Aug 2022	26	18	Dec 2018		
Competition/Second Source Initiatives: N/A							
Remarks: CVN 73 RCOH - Overall decrease (\$0.365M) is a result of realized savings from the GFE Governance Program to reduce program management, system engineering, and logistics support. CVN 74 RCOH - Replaced by AN/SPA-25 Indicator Group (listed under Ordnance major items).							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls			
Equipment Item: CIWS/RAM DEFENSE CAPABILITY (CRDC) BLOCK 1						PARM Code: NAVSEA PEO IWS 3G	
P-35 Category	FY 2016		FY 2020				
	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>			
Major Hardware	0	-	1	20.200			
Ancillary Equipment		-		0.500			
Technical Engineering Services		2.271		3.671			
Other Costs		0.025		0.095			
Total	0	2.296	1	24.466			
Description: Close-In Weapon System (CIWS)/Rolling Airframe Missile (RAM) Defense Capability (CRDC) is a high fire rate gun weapon system that automatically acquires, tracks, and destroys threats. Active radars, supporting platforms, below deck equipment, and interfaces them to CIWS for Quick Reaction Capability (QRC).							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity <i>(Each)</i>	Unit Cost <i>(\$ M)</i>
FY 2020	CVN 74	Raytheon	C/FFP	Dec 2020	New	1	20.200
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 74	Aug 2025	24	24	Aug 2021		
Competition/Second Source Initiatives: N/A							
Remarks: CIWS/RAM Defense Capability (CRDC) Block 1 is a significant enhancement of Phalanx MK 15 MOD 22 (CIWS). On CVN 73, infrastructure-only efforts (no GFE hardware) have been added since the FY 2020 budget request. On CVN 74, the increase (\$3.196M) from the FY 2021 budget request is due to increased cost in HW for sensors and enclosures. CVN 74 increase from CVN 73 is because CVN 74 will receive the full installation (with GFE) whereas CVN 73 will install system infrastructure only during RCOH.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls			
Equipment Item: MK 38 MOD 3 GUN SYSTEM						PARM Code: NAVSEA PEO IWS 3C	
P-35 Category	FY 2016		FY 2020				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	0	-	1	5.002			
Spares		-		0.097			
Technical Engineering Services		1.850		2.056			
Other Costs		0.119		0.388			
Total	0	1.969	1	7.543			
Description: MK 38 Mod 3 Gun System is a 25mm stabilized gun with auto tracking and day/night capability that is also capable of interfacing with 7.62 mm MK 52 MOD 0 Coaxial gun.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	CVN 74	BAE Systems	SS/FP	Jun 2020	New	1	5.002
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 74	Aug 2025	37	18	Jan 2021		
Competition/Second Source Initiatives: N/A							
Remarks: CVN 73 RCOH - Overall decrease (\$0.061M) is a result of realized savings from the GFE Governance Program to reduce program management, system engineering, and logistics support. CVN 74 RCOH - Overall decrease (\$1.528M) due to a reduction in hardware costs. Original estimate was to procure, install, and test a five gun configuration; only four are required. CVN 74 increase from CVN 73 is because CVN 74 will receive the full installation (with GFE) whereas CVN 73 will install system infrastructure (cable, foundations) only during RCOH.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls			
Equipment Item: ENTERPRISE AIR SURVEILLANCE RADAR (EASR)						PARM Code: NAVSEA PEO IWS 2RI	
P-35 Category	FY 2016		FY 2020				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	0	-	1	27.646			
Ancillary Equipment		-		0.010			
Spares		-		1.401			
System Engineering		-		5.312			
Technical Engineering Services		-		14.550			
Other Costs		-		6.347			
Total	0	-	1	55.266			
Description: Enterprise Air Surveillance Radar (EASR) is the next generation S-band air search radar with 3-D search capability supporting: Anti-Air Warfare, Anti-Surface Warfare, Air Traffic Control.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	CVN 74	Raytheon	C/CPIF	Jul 2020	Option	1	27.646
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 74	Aug 2025	30	24	Feb 2021		
Competition/Second Source Initiatives: N/A							
Remarks: Enterprise Air Surveillance Radar (EASR) is a replacement for the legacy AN/SPS-48 and AN/SPS-49 Air Search Radars that are no longer in production and are experiencing increasing service life obsolescence issues. CVN 74 overall increase (\$6.355M) is due to additional costs associated with efforts to address unique system configuration complexity, as CVN 74 is the only CVN receiving EASR Variant 1, single face rotating array.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls			
Equipment Item: SURFACE ELECTRONIC WARFARE IMPROVEMENT PROGRAM (SEWIP) BLOCK 2						PARM Code: NAVSEA PEO IWS 2E	
P-35 Category	FY 2016		FY 2020				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	0	-	1	10.402			
Technical Data and Documentation		-		0.808			
Spares		-		0.391			
System Engineering		-		0.828			
Technical Engineering Services		-		4.815			
Other Costs		-		1.085			
Total	0	-	1	18.329			
Description: Surface Electronic Warfare Improvement Program (SEWIP) Block 2 detects and classifies special signals not processed by other SLQ-32 receivers. SEWIP Block 2 provides enhanced Electronic Support (ES) capability with improved detection and accuracy to pace evolving threats through an upgraded ES antenna & receiver along with an open combat system interface.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	CVN 74	Lockheed Martin	C/FFP	Mar 2020	Option	1	10.402
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	CVN 74	Aug 2025	29	29	Oct 2020		
Competition/Second Source Initiatives: N/A							
Remarks: CVN 73 is receiving AN/SLQ-32A(V)4 Electronic Warfare Suite, which is being replaced by Surface Electronic Warfare Improvement Program (SEWIP) Block 2 on CVN 74.							

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Exhibit P-10, Advance Procurement Requirements Analysis (page 1 - Budget Funding Justification): PB 2022 Navy						Date: May 2021				
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls						
First System (2022) Award Date: June 2025		First System (2022) Completion Date: March 2029			Interval Between Systems: 41 Months					
Cost Elements		Production Leadtime (Months)	When Required* (Months)	FY 2020 (\$ M)	FY 2021 (\$ M)	FY 2022 (\$ M)	FY 2023 (\$ M)	FY 2024 (\$ M)	FY 2025 (\$ M)	FY 2026 (\$ M)
Advance Procurement										
Plans ⁽⁷⁾	-	Various	1.270	1.524	6.896	-	-	-	-	
Basic	-	Various	0.056	11.037	9.316	-	-	-	-	
Other	-	Various	2.469	3.863	0.000	-	-	-	-	
Propulsion Equipment	-	Various	13.105	-	47.355	-	-	-	-	
HM&E	-	Various	-	0.960	0.000	-	-	-	-	
Electronics	-	Various	-	-	2.628	-	-	-	-	
Ordnance	-	Various	-	-	0.067	-	-	-	-	
Total: Advance Procurement			16.900	17.384	66.262	-	-	-	-	
Total Advance Procurement/Obligation Authority			16.900	17.384	66.262					

*Note: "When Required" is the number of months required before ship delivery.

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Exhibit P-10, Advance Procurement Requirements Analysis (page 2 - Budget Funding Justification): PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1	P-1 Line Item Number / Title: 2086 / CVN Refueling Overhauls
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Cost Elements	FY 2022						
	Production Leadtime (Months)	When Required* (Months)	Unit Cost (\$ M)	Contract Forecast Date	2022 Qty (Each)	For FY	Total Cost Request (\$ M)
Advance Procurement							
Plans ⁽⁷⁾	-	Various	-	Dec 2021	-	2025	6.896
Basic	-	Various	-	Apr 2022	-	2025	9.316
Other	-	Various	-		-		0.000
Propulsion Equipment	-	Various	-	Dec 2021	-	2025	47.355
HM&E	-	Various	-		-		0.000
Electronics	-	Various	-		-		2.628
Ordnance	-	Various	-		-		0.067
<i>Total: Advance Procurement</i>							66.262
Total Advance Procurement/Obligation Authority							66.262

Description:

FY 2022 is the third of five years of advance procurement for CVN 75 RCOH. FY 2022 resumes the required advance procurement funding profile to support a FY 2025 start date.

Plans: Efforts include advance planning engineering support; authorized work package (AWP) development; shipchecks and shipcheck oversight; government furnished information (GFI) development; and technical oversight and authority.

Basic: Efforts include prime contractor advance planning; integration of the AWP into the execution integrated master schedule; Ship's Force work package material procurement; and technical support.

Propulsion Equipment: Nuclear component procurement and technical support services.

Electronics: GFI/GFE and technical support services for combat systems, interior communications, and C4I.

Ordnance: GFI/GFE and technical support services for radars and weapons systems.

*Note: "When Required" is the number of months required before ship delivery.

Footnotes:

⁽⁷⁾ FY 2020 and FY 2021 funding re-aligned by P-5 categories to reflect actual and planned obligations.

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy										Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 02: Other Warships / BSA 1: Other Warships							P-1 Line Item Number / Title: 2119 / DDG 1000					
ID Code (A=Service Ready, B=Not Service Ready): A			Program Elements for Code B Items: N/A					Other Related Program Elements: N/A				
Line Item MDAP/MAIS Code: N/A												
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (<i>Units in Each</i>)	3	-	-	-	-	-	-	-	-	-	-	-
Gross/Weapon System Cost (<i>\$ in Millions</i>)	13,305.936	0.000	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Less PY Advance Procurement (<i>\$ in Millions</i>)	1,160.113	-	-	-	-	-	-	-	-	-	-	-
Less Subsequent Year Full Funding (<i>\$ in Millions</i>)	8,054.049	-	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (<i>\$ in Millions</i>)	4,091.774	0.000	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Plus Subsequent Year Full Funding (<i>\$ in Millions</i>)	7,734.925	155.944	78.205	56.597	-	56.597	-	-	-	-	-	-
Full Funding TOA (<i>\$ in Millions</i>)	11,826.699	155.944	78.205	56.597	-	56.597	-	-	-	-	-	-
Plus CY Advance Procurement (<i>\$ in Millions</i>)	1,160.113	-	-	-	-	-	-	-	-	-	-	-
Total Obligation Authority (<i>\$ in Millions</i>)	12,986.812	155.944	78.205	56.597	0.000	56.597	-	-	-	-	-	-
<i>(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)</i>												
Plus Outfitting and Post Delivery (<i>\$ in Millions</i>)	312.634	62.193	35.483	40.420	-	40.420	-	-	-	-	-	-
Total (<i>\$ in Millions</i>)	13,299.446	218.137	113.688	97.017	-	97.017	-	-	-	-	-	-
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	4,435.312	-	-	-	-	-	-	-	-	-	-	-

Description:
 DDG 1000, a multi-mission surface combatant will serve as a versatile asset in the context of future Naval Strategy. Armed with an array of weapons, DDG 1000 will provide the Joint Force Commander with precision strike and volume fires. Designed with sustainable payload, multi-spectral stealth and optimal manning, DDG 1000 will take the fight to the enemy with unprecedented striking power, sustainability, survivability and information dominance. FY22 funding will support DDG 1001 mission systems activation and DDG 1002 Combat Availability.

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy				Date: May 2021																																																	
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 02: Other Warships / BSA 1: Other Warships			P-1 Line Item Number / Title: 2119 / DDG 1000																																																		
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<u>Classification of Cost Estimate:</u> CLASS C BUDGET ESTIMATE																																																					
Footnotes: ⁽¹⁾ USS Michael Monsoor (DDG 1001) completed Combat System Availability in March 2020 and is activating weapons, sensors and communication systems. Delay in SPY-3 parts has pushed the mission system activation completion to March 2022. ⁽²⁾ Construction of Lyndon B. Johnson (DDG 1002), the final ship of the class, is 98% complete. Acceptance from BIW is forecast for November 2021. Final Delivery is scheduled for April 2024.																																																					

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2119 / DDG 1000		
Cost Categories ^(†) indicates the presence of a P-8a	FY 2007		FY 2009	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Plan Costs	2	1,553.687	1	563.435
Basic Construction/Conversion		3,645.347		1,303.350
Change Orders		320.320		80.328
Electronics ^(†)		2,805.702		1,434.813
Hull, Mechanical, and Electrical (HM&E) ^(†)		242.156		76.848
Ordnance ^(†)		515.811		264.891
Other Cost		367.807		131.441
Total Ship Estimate		9,450.830		3,855.106
Less Advance Procurement FY 2005		304.046		-
Less Advance Procurement FY 2006		706.240		-
Less Advance Procurement FY 2008		-		149.827
Less Subsequent Full Funding FY 2008		3,009.929		-
Less Subsequent Full Funding FY 2010		313.025		1,065.507
Less Subsequent Full Funding FY 2011		107.020		140.062
Less Subsequent Full Funding FY 2012		437.938		70.789
Less Subsequent Full Funding FY 2013		536.503		138.020
Less Subsequent Full Funding FY 2014		230.944		31.349
Less Subsequent Full Funding FY 2015		374.916		85.933
Less Subsequent Full Funding FY 2016		277.658		155.643
Less Subsequent Full Funding FY 2017		198.180		73.576
Less Subsequent Full Funding FY 2018		150.006		66.962
Less Subsequent Full Funding FY 2019		81.518		189.447
Less Subsequent Full Funding FY 2020		93.714		62.230
Less Subsequent Full Funding FY 2021		26.675		51.530
Less Subsequent Full Funding FY 2022		14.950		41.647
Less Subsequent Full Funding FY 2023		-		28.378
Net P-1 Funding		2,587.568		1,504.206
Remarks: DDG 1001 increases in FY21 \$26.675M and FY22 \$14.95M due to the extended mission system activation period pushing final delivery from September 2021 to March 2022. FY21 funding programmed for DDG 1002 Combat Availability was used to finance this increased cost. DDG 1001 increase in Basic Construction necessary to execute work items and trial cards at execution yard and correct deficiencies identified during PDA. DDG 1001 increase in Other Cost to fund government management and oversight of waterfront activities during planning and execution of DDG 1001 work packages. DDG 1002 decreases in FY21 and FY22 to finance the increased costs on DDG 1001 mission system activation.				

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Exhibit P-27, Ship Production Schedule: PB 2022 Navy				Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1			P-1 Line Item Number / Title: 2119 / DDG 1000		
Ship	Shipbuilder	Fiscal Year	Contract Award	Start of Construction	Delivery Date
DDG 1001 ⁽¹⁾	BIW	2007	Sep 2011	Mar 2010	Mar 2022
DDG 1002 ⁽²⁾	BIW	2009	Sep 2011	Apr 2012	Apr 2024
Footnotes: ⁽¹⁾ USS Michael Monsoor (DDG 1001) completed Combat System Availability in March 2020 and is activating weapons, sensors and communication systems. Delay in SPY-3 parts has pushed the mission system activation completion to March 2022. ⁽²⁾ Construction of Lyndon B. Johnson (DDG 1002), the final ship of the class, is 98% complete. Acceptance from BIW is forecast for November 2021. Final Delivery is scheduled for April 2024.					

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy				Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1			P-1 Line Item Number / Title: 2119 / DDG 1000		
Electronics	FY 2007		FY 2009		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
P-35 Items					
EXTERIOR COMMUNICATIONS (EXCOMMS)	2	470.348	1	79.962	
INTEGRATED UNDERSEA WARFARE (IUSW) SYSTEM	2	216.263	1	105.136	
MULTI FUNCTION RADAR (MFR)	2	519.609	1	297.999	
COMMON ARRAY POWER SYSTEM (CAPS)	2	97.017	1	16.409	
TOTAL SHIP COMPUTING ENVIRONMENT (TSCE)	2	382.591	1	259.191	
ELECTRO-OPTICAL / INFRARED (EO/IR)	2	94.411	1	31.452	
IDENTIFICATION FRIEND OR FOE (IFF)	2	35.532	1	28.138	
COMMON ARRAY COOLING SYSTEM (CACS)	2	20.065	1	0.965	
SHIP CONTROL SYSTEM (SCS)	2	111.527	1	117.229	
COOPERATIVE ENGAGEMENT CAPABILITY (CEC)	2	16.025	1	7.800	
SURFACE ELECTRONIC WARFARE IMPROVEMENT PROGRAM (SEWIP)	2	40.242	1	17.682	
VERTICAL LAUNCHING SYSTEM (VLS) MK 57 4-CELL MODULES	40	276.782	20	302.815	
P-35 Items Subtotal		2,280.412		1,264.778	
Other Cost Elements					
MISSION SYSTEM ENGR INTEGR & TEST (MSEIT)		322.274		132.510	
MISSION SYSTEM ACTIVATION		193.016		32.025	
SPARES		10.000		5.500	
Other Cost Elements Subtotal		525.290		170.035	
Total Electronics		2,805.702		1,434.813	
Remarks: DDG 1001 increases of \$37.225M in mission systems activation is due to the extended mission system activation period pushing final delivery from September 2021 to March 2022. FY21 funding programmed for DDG 1002 Combat Availability was used to finance this increased cost.					

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity:		P-1 Line Item Number / Title:		
1611N / 02 / 1		2119 / DDG 1000		
Hull, Mechanical, and Electrical (HM&E)	FY 2007		FY 2009	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
P-35 Items				
MAIN TURBINE GENERATOR (MTG)	4	84.797	2	39.412
P-35 Items Subtotal		84.797		39.412
Major Items				
BATTLE SPARES (MTG)		32.168		-
RIGID HULL INFLATABLE BOAT (RHIB)	4	2.100	2	1.100
Major Items Subtotal		34.268		1.100
Other Cost Elements				
HM&E (NGVLA, Moriah Wind Measurement System (WMS), Aviation Integration)		68.492		12.446
MISSION SYSTEM ACTIVATION		18.781		16.390
INTERIM SPARES		35.818		7.500
Other Cost Elements Subtotal		123.091		36.336
Total Hull, Mechanical, and Electrical (HM&E)		242.156		76.848

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy				Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1			P-1 Line Item Number / Title: 2119 / DDG 1000		
Ordnance	FY 2007		FY 2009		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
P-35 Items					
ADVANCED GUN SYSTEM (AGS)	4	458.711	2	248.596	
CLOSE-IN GUN SYSTEM (CIGS)	4	36.151	2	13.795	
P-35 Items Subtotal		494.862		262.391	
Major Items					
BATTLE SPARES (AGS)		18.449		-	
Major Items Subtotal		18.449		-	
Other Cost Elements					
MISSION SYSTEM ACTIVATION		2.500		2.500	
New Cost Element					
Other Cost Elements Subtotal		2.500		2.500	
Total Ordnance		515.811		264.891	

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2119 / DDG 1000			
Equipment Item: EXTERIOR COMMUNICATIONS (EXCOMMS)						PARM Code: PEOC4I	
P-35 Category	FY 2007		FY 2009				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	2	195.953	1	20.600			
Technical Support Services		33.947		6.585			
Other / NRE		240.448		52.777			
Total	2	470.348	1	79.962			
Description: EXCOMMs are part of the DDG 1000 C3I Segment and consists of a set of seven (7) external communications elements. The EXCOMM Elements support the DDG 1000 system in achieving its mission by providing communications between DDG 1000 and other land, air, and sea based platforms as well as pier-side communications. These EXCOMM elements provide the voice, data, and video communications between DDG 1000 and the external world at sea as well as when in port. The 7 elements are: Satellite Communications (SATCOMs), Line of Sight (LOS), Common Data Link-Navy (CDL-N), Information Security (INFOSEC), Common Array Element (CAE), Cooperative Engagement Capability (CEC) and Integrated Communications Controller Software (ICCS). Government legacy systems include: Distributed Common Ground System, Navy (DCGS-N), Cooperative Engagement Capability (CEC), Communication Terminals, AN/WSC-6(V)9 Shipboard Terminal, Common Link Integrated Processor (CLIP), Automated Digital Network System (ADNS), Global Broadcast Service (GBS), Communications Data Link System (CDLS), & Naval Modular Automated Communications System (NAVMACS).							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2007	DDG 1001	Raytheon	C/CPIF	May 2008		2	97.977
FY 2009	DDG 1002	Raytheon	C/CPIF	May 2012		1	20.600
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2007	DDG 1001	Mar 2022	43	26	Jun 2016		
FY 2009	DDG 1002	Apr 2024	43	26	Jul 2018		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2119 / DDG 1000			
Equipment Item: INTEGRATED UNDERSEA WARFARE (IUSW) SYSTEM						PARM Code: IWS 5.0 XR	
P-35 Category	FY 2007		FY 2009				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	2	95.829	1	54.300			
Technical Support Services		11.293		5.639			
Other / NRE		109.141		45.197			
Total	2	216.263	1	105.136			
Description: The IUSW suite supports DDG 1000 in achieving Undersea and Surface Dominance with the capability to detect and track hostile surface vessels, submarines, and moored volume mines. It supports the Sensor Systems Segment in accomplishing its Integrated Air and Surface Dominance (IASD) and Integrated Undersea Dominance (IUSD) objectives by providing the capability to conduct Anti-Submarine Warfare (ASW), Torpedo Defense (TD) and Mine Warfare (MIW) missions. Military Operations Other than War (MOOTW) objectives, such as Search and Rescue (SAR) (locating downed aircraft and vessels in the ocean) are also supported. There are four major subcomponents: Bow Array Component, Towed Array Component, Towed Torpedo Countermeasures Component, as well as software.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2007	DDG 1001	Raytheon	C/CPIF	May 2008		2	47.915
FY 2009	DDG 1002	Raytheon	C/CPIF	Oct 2012		1	54.300
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2007	DDG 1001	Mar 2022	47	18	Oct 2016		
FY 2009	DDG 1002	Apr 2024	46	18	Dec 2018		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2119 / DDG 1000			
Equipment Item: MULTI FUNCTION RADAR (MFR)						PARM Code: IWS 2.0 SQ	
P-35 Category	FY 2007		FY 2009				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	2	314.313	1	189.573			
Technical Support Services		21.993		11.145			
Other / NRE		183.303		97.281			
Total	2	519.609	1	297.999			
Description: The Multi Function Radar (MFR) element supports the DDG 1000 system in achieving Integrated Air and Surface Dominance with the capability to neutralize hostile surface vessels and aircraft at short ranges. The MFR is comprised of X-Band (AN/SPY-3) arrays integrated through a common signal data processor offering surface and horizon search capabilities and 3-D air search radar capabilities. The X-Band portion also has two navigation modes (high power and lower power) for use in piloting and marine navigation.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2007	DDG 1001	Raytheon	C/CPIF	Mar 2008		2	157.157
FY 2009	DDG 1002	Raytheon	C/CPIF	Oct 2012		1	189.573
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2007	DDG 1001	Mar 2022	45	28	Feb 2016		
FY 2009	DDG 1002	Apr 2024	36	28	Dec 2018		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2119 / DDG 1000			
Equipment Item: COMMON ARRAY POWER SYSTEM (CAPS)						PARM Code: IWS 2.0 SQ	
P-35 Category	FY 2007		FY 2009				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	2	56.185	1	12.624			
Battle Spares		1.000		-			
Technical Support Services		4.490		0.420			
Other / NRE		35.342		3.365			
Total	2	97.017	1	16.409			
Description: The Common Array Power System (CAPS) provides electrical power for the Multi Function Radar (MFR), Identification of Friend or Foe (IFF), EW/Cryptology and External Communications (EXCOMMs) Elements. The CAPS is a distributed power system designed to operate from the ship-supplied medium voltage distribution Integrated Power System's (IPS) 13.8 kV AC power source. The CAPS consists of two Power Distribution Units (PDUs) and four Power Conversion Units (PCUs).							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2007	DDG 1001	Raytheon	C/CPIF	Mar 2008		2	28.093
FY 2009	DDG 1002	Raytheon	C/CPIF	Nov 2012		1	12.624
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2007	DDG 1001	Mar 2022	48	28	Nov 2015		
FY 2009	DDG 1002	Apr 2024	35	28	Jan 2019		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2119 / DDG 1000			
Equipment Item: TOTAL SHIP COMPUTING ENVIRONMENT (TSCE)						PARM Code: IWS 9.0 XV	
P-35 Category	FY 2007		FY 2009				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	2	196.450	1	147.453			
Technical Support Services		21.834		10.499			
Other / NRE		164.307		101.239			
Total	2	382.591	1	259.191			
Description: The Total Ship Computing Environment (TSCE) Segment provides all computing resources and associated software to the DDG 1000 System. It is a single computing environment for Ship, Combat and Support Systems. The TSCE provides a common middleware platform upon which all application/functional software can build and execute. The segment applications software, combined with TSCE hardware and software infrastructure represent the majority of the computing resources and associated software for the DDG 1000 System.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2007	DDG 1001	Raytheon	C/CPIF	May 2008		2	98.225
FY 2009	DDG 1002	Raytheon	C/CPIF	Oct 2012		1	147.453
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2007	DDG 1001	Mar 2022	48	21	Jun 2016		
FY 2009	DDG 1002	Apr 2024	43	21	Dec 2018		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2119 / DDG 1000			
Equipment Item: ELECTRO-OPTICAL / INFRARED (EO/IR)						PARM Code: IWS 2.0 SJ	
P-35 Category	FY 2007		FY 2009				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	2	33.368	1	12.973			
Technical Support Services		6.900		1.551			
Other / NRE		54.143		16.928			
Total	2	94.411	1	31.452			
Description: The Electro-Optical / Infrared (EO/IR) Sensor Suite Element is composed of both the hardware and software components required to detect and range on specified targets and report track data to C2. The EO/IR sensor suite consists of five (5) gimballed EO sensors located on the cardinal faces of the deckhouse and associated electronics in Electronic Modular Enclosures (EMEs). Also included are Detect and Tracking Software components that provide embedded control and generate tracks for the C2 system and Mine Like Object (MLO) detection algorithm.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2007	DDG 1001	Raytheon	C/CPIF	May 2008		2	16.684
FY 2009	DDG 1002	Raytheon	C/CPIF	Nov 2012		1	12.973
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2007	DDG 1001	Mar 2022	47	22	Jun 2016		
FY 2009	DDG 1002	Apr 2024	41	22	Jan 2019		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2119 / DDG 1000			
Equipment Item: IDENTIFICATION FRIEND OR FOE (IFF)						PARM Code: NAVAIR	
P-35 Category	FY 2007		FY 2009				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	2	16.018	1	8.640			
Technical Support Services		2.186		2.163			
Other / NRE		17.328		17.335			
Total	2	35.532	1	28.138			
Description: Identification Friend or Foe (IFF) sensor element supports the DDG 1000 Ship System segment in accomplishing Anti-Air Warfare (AAW) and Anti-Surface Warfare (ASUW) missions. The IFF Sensor Element is a cooperative "challenge and reply" system that assists in the rapid identification, tracking and control of friendly platforms. IFF is comprised of three hardware components to include the Interrogator component, the Transponder component and the Electronically Scanned Antenna (ESA) component, as well as software.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2007	DDG 1001	Raytheon	C/CPIF	May 2008		2	8.009
FY 2009	DDG 1002	Raytheon	C/CPIF	Dec 2012		1	8.640
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2007	DDG 1001	Mar 2022	40	29	Jun 2016		
FY 2009	DDG 1002	Apr 2024	33	29	Feb 2019		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2119 / DDG 1000			
Equipment Item: COMMON ARRAY COOLING SYSTEM (CACS)						PARM Code: IWS 2.0 SQ	
P-35 Category	FY 2007		FY 2009				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	2	11.766	1	-			
Battle Spares		1.000		-			
Technical Support Services		0.824		0.107			
Other / NRE		6.475		0.858			
Total	2	20.065	1	0.965			
Description: The Common Array Cooling System (CACS) provides liquid cooling for the Multi Function Radar (MFR) and External Communications (EXCOMMs) arrays. CACS is a distributed cooling system consisting of three Cooling Equipment Units (CEUs). Each CEU operates an independent coolant loop used to transport, monitor and control coolant flow to the DBR and EXCOMMs Equipment. CEUs consist of redundant pumps, a heat exchanger and filtration system. It is designed to provide liquid coolant to the MFR and EXCOMM equipment and dissipate heat to the ship-supplied chilled water.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2007	DDG 1001	Raytheon	C/CPIF	May 2008		2	5.883
FY 2009	DDG 1002	Raytheon	C/CPIF	Nov 2012		1	0.000
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2007	DDG 1001	Mar 2022	49	28	Oct 2015		
FY 2009	DDG 1002	Apr 2024	35	28	Jan 2019		
Competition/Second Source Initiatives: N/A							
Remarks: CACS Technical Services are incorporated into DBR Technical Services. DDG 1002 CACS costs are included in the DDG 1002 MFR value.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2119 / DDG 1000			
Equipment Item: SHIP CONTROL SYSTEM (SCS)						PARM Code: SPAWAR	
P-35 Category	FY 2007		FY 2009				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	2	58.000	1	42.801			
Technical Support Services		6.031		8.256			
Other / NRE		47.496		66.172			
Total	2	111.527	1	117.229			
Description: The Flight 1 Ship Control System (SCS) element is a system of hardware and software items that provide hierarchical and integrated ship control by the DDG 1000 crew. The SCS software architecture allows for various levels of automation for monitoring, control, reporting and configuration of SCS equipment and operations to support mission and low manning concepts. From workstation positions on the ship bridge or in the ship mission centers, the SCS coordinates, controls and monitors the navigation, hull, electric plant, machinery plant and damage control functions on the DDG 1000.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2007	DDG 1001	Raytheon	C/CPIF	May 2008		2	29.000
FY 2009	DDG 1002	Raytheon	C/CPIF	May 2012		1	42.801
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2007	DDG 1001	Mar 2022	38	31	Jun 2016		
FY 2009	DDG 1002	Apr 2024	38	31	Jul 2018		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2119 / DDG 1000			
Equipment Item: COOPERATIVE ENGAGEMENT CAPABILITY (CEC)						PARM Code: IWS 6.0 XN	
P-35 Category	FY 2007		FY 2009				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	2	12.000	1	6.800			
Technical Support Services		4.025		1.000			
Total	2	16.025	1	7.800			
Description: Cooperative Engagement Capability (CEC) is a sensor network with Integrated Fire Control capability that significantly improves Battle Force air and missile defense capabilities by coordinating measurement data from Battle Force air search sensors on CEC-equipped units into a single, real-time, composite cooperating unit (CU), to all other CUs in the Battle Force through a real-time, line of sight, high data rate sensor and engagement data distribution network. CEC is highly resistant to jamming and provides accurate grid locking (relative spatial positioning) between CUs. Each CU independently employs high capacity, parallel processing and advanced algorithms to combine all distributed sensor data into a high quality track picture that is the same for all CUs. CEC data is presented as a superset of the best air and missile defense sensor capabilities from each CU, all of which are integrated into a single input to each CU's combat weapon system. CEC significantly improves Battle Force defense in depth, including both local and area defense capabilities against current and future air missile threats.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2007	DDG 1001	Raytheon	C/FPIF	Feb 2007		2	6.000
FY 2009	DDG 1002	Raytheon	C/FPIF	Oct 2013		1	6.800
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2007	DDG 1001	Mar 2022	34	18	Nov 2017		
FY 2009	DDG 1002	Apr 2024	34	18	Dec 2019		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2119 / DDG 1000			
Equipment Item: SURFACE ELECTRONIC WARFARE IMPROVEMENT PROGRAM (SEWIP)						PARM Code: IWS 2.0 SJ	
P-35 Category	FY 2007		FY 2009				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	2	36.214	1	15.906			
Technical Support Services		2.406		0.935			
Other / NRE		1.622		0.841			
Total	2	40.242	1	17.682			
Description: SEWIP provides enhanced Electronic Warfare (EW) capabilities to improve anti-ship missile defense, counter-targeting and counter surveillance capabilities, as well as improved situational awareness to pace the threat, improving detection, accuracy, and mitigation of EMI. The SEWIP Block 2 is an upgraded antenna, receiver and combat system interface for AN/SLQ-32.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2007	DDG 1001	Lockheed Martin	C/FPIF	Jul 2012		2	18.107
FY 2009	DDG 1002	Lockheed Martin	C/FPIF	Jan 2015		1	15.906
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2007	DDG 1001	Mar 2022	2	19	Jun 2020		
FY 2009	DDG 1002	Apr 2024	2	16	Oct 2022		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2119 / DDG 1000			
Equipment Item: VERTICAL LAUNCHING SYSTEM (VLS) MK 57 4-CELL MODULES						PARM Code: IWS 3L S8	
P-35 Category	FY 2007		FY 2009				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	40	180.987	20	234.670			
Technical Support Services		9.029		4.231			
Other / NRE		86.766		63.914			
Total	40	276.782	20	302.815			

Description:
 The MK 57 VLS is a general purpose, operationally unmanned launching system capable of stowing, preparing, and launching missiles in support of DDG-1000 mission areas including: land attack warfare, integrated air and surface dominance, and integrated undersea dominance. The MK57 VLS provides the capability for rapid launch of missiles into a 360-degree hemispherical volume above and about the ship. The canistered missiles are stowed within the launching systems below-deck cells. DDG-1000 will have 80 total cells grouped into 20 four-cell modules. Flight 1 missiles to be carried include: Enhanced Sea Sparrow Missile (ESSM), Standard Missile-2 (SM-2) Blk III, Tomahawk Land Attack Missile (TLAM) Blk III/IV, and Vertical Launch Anti-Submarine Rocket (VLA).

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2007	DDG 1001	Raytheon	C/CPIF	May 2008		40	4.525
FY 2009	DDG 1002	Raytheon	C/CPIF	Oct 2012		20	11.734

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2007	DDG 1001	Mar 2022	40	24	Nov 2016
FY 2009	DDG 1002	Apr 2024	40	24	Dec 2018

Competition/Second Source Initiatives:
 N/A

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2119 / DDG 1000			
Equipment Item: MAIN TURBINE GENERATOR (MTG)						PARM Code: PMS 500 WA	
P-35 Category	FY 2007		FY 2009				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	4	73.262	2	39.412			
Technical Support Services		6.126		-			
Other / NRE		5.409		-			
Total	4	84.797	2	39.412			
Description: The Main Turbine Generator Set (MTG) shall be capable of being utilized as the prime power source on the DDG 1000 Destroyer for electrical power applications (propulsion, ship services, and combat systems loads). The DDG 1000 baseline includes two MTGs. The minimum output power from each MTG shall be 35.25 MWe. The engine utilizes a Full Authority Digital Control Local Operating Panel (FADC LOCOP) and electric start system. The generator contains redundant automatic voltage regulators (AVR) with automatic changeover.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2007	DDG 1001	Rolls-Royce	C/FFP	Mar 2007	New	4	18.316
FY 2009	DDG 1002	Rolls-Royce	C/FFP	Jan 2008	Option	2	19.706
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2007	DDG 1001	Mar 2022	33	24	Jun 2017		
FY 2009	DDG 1002	Apr 2024	33	24	Jul 2019		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2119 / DDG 1000			
Equipment Item: ADVANCED GUN SYSTEM (AGS)						PARM Code: IWS 3C YF	
P-35 Category	FY 2007		FY 2009				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	4	298.654	2	206.747			
Technical Support Services		14.500		3.860			
Other / NRE		145.557		37.989			
Total	4	458.711	2	248.596			
Description: The Advanced Gun System is a fully automated, single barrel, 155mm, vertically loaded, stabilized gun mount that is capable of storing, initializing/programming, loading and firing projectiles and propelling charges. Its primary mission is Land Attack Warfare in support of ground and expeditionary forces beyond the Line of Sight in the DDG 1000 system's littoral engagement area where precise, rapid-response, high-volume, long-range fire support is required. Each DDG 1000 will carry two complete AGS systems - Mount 61 and 62. The above deck configurations are identical but each has a slightly different below deck configuration.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2007	DDG 1001	BAE	C/CPIF	Apr 2008		4	74.664
FY 2009	DDG 1002	BAE	C/CPIF	Apr 2012		2	103.374
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2007	DDG 1001	Mar 2022	31	39	May 2016		
FY 2009	DDG 1002	Apr 2024	31	39	Jun 2018		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy				Date: May 2021			
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1			P-1 Line Item Number / Title: 2119 / DDG 1000				
Equipment Item: CLOSE-IN GUN SYSTEM (CIGS)				PARM Code: IWS 3C YF			
P-35 Category	FY 2007		FY 2009				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	4	16.034	2	7.534			
Technical Support Services		7.177		3.381			
Other / NRE		12.940		2.880			
Total	4	36.151	2	13.795			
Description: The Close-In Gun System (CIGS) supports the DDG 1000 system in achieving Integrated Air and Surface Dominance with the capability to neutralize hostile surface vessels and aircraft at short ranges. CIGS also supports the Military Operations Other than War (MOOTW) missions, such as performing maritime interdiction, conducting maritime law enforcement, and supporting hostage rescue. Two (2) CIGS will be mounted on the aft end of the hanger. The CIGS MK 46 MOD 2 GWS is composed of a turret assembly that houses the MK 44 MOD 2 cannon and an advanced Fire Control System that includes a ballistic solution computer, an electro-optical sensor package, and an eye-safe laser range finder. The system uses a forward-looking infrared sensor, a low-light television camera, and eye safe laser range finder with a closed-loop tracking system to optimize accuracy against small, high-speed surface targets. The system can be operated locally from the gun control station inside the turret, remotely from the MK 46 MOD 2 GWS Remote Gun Station Operator (RGSO) panel in the Combat Information Center (CIC), or manually using hand cranks from inside the turret. The 30mm cannon, MK 44 MOD 2, is a single barrel, open bolt, dual feed, electrically powered, chain-driven automatic cannon. The system has a magazine capacity of 424 rounds, a dual-feed capability with a firing rate of 200 rounds per minute, and is capable of selectively switching between ammunition types and firing modes.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2007	DDG 1001	General Dynamics Land Systems	C/FFP	Jan 2015		4	4.008
FY 2009	DDG 1002	General Dynamics Land Systems	C/FFP	Mar 2016		2	3.767
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2007	DDG 1001	Mar 2022	6	22	Nov 2019		
FY 2009	DDG 1002	Apr 2024	6	18	Apr 2022		
Competition/Second Source Initiatives: N/A							

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy									Date: May 2021				
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 02: Other Warships / BSA 1: Other Warships						P-1 Line Item Number / Title: 2122 / DDG-51							
ID Code (A=Service Ready, B=Not Service Ready): A			Program Elements for Code B Items: N/A					Other Related Program Elements: N/A					
Line Item MDAP/MAIS Code: N/A													
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total	
Procurement Quantity (<i>Units in Each</i>)	82	3	2	1	-	1	-	-	-	-	-	-	
Gross/Weapon System Cost (<i>\$ in Millions</i>)	89,488.691	5,428.955	4,016.478	2,401.719	0.000	2,401.719	-	-	-	-	-	-	
Less PY Advance Procurement (<i>\$ in Millions</i>)	2,910.850	-	-	-	-	-	-	-	-	-	-	-	
Less Cost To Complete (<i>\$ in Millions</i>)	1,677.033	-	-	-	-	-	-	-	-	-	-	-	
Less Subsequent Year Full Funding (<i>\$ in Millions</i>)	433.000	-	-	-	-	-	-	-	-	-	-	-	
Less Hurricane (<i>\$ in Millions</i>)	227.100	-	-	-	-	-	-	-	-	-	-	-	
Less EOQ (<i>\$ in Millions</i>)	460.946	363.660	796.635	384.932	-	384.932	-	-	-	-	-	-	
Less Escalation (<i>\$ in Millions</i>)	48.200	-	-	-	-	-	-	-	-	-	-	-	
Less Transfer (<i>\$ in Millions</i>)	218.500	-	-	-	-	-	-	-	-	-	-	-	
Net Procurement (P-1) (<i>\$ in Millions</i>)	83,513.062	5,065.295	3,219.843	2,016.787	0.000	2,016.787	-	-	-	-	-	-	
Plus Subsequent Year Full Funding (<i>\$ in Millions</i>)	433.000	-	-	-	-	-	-	-	-	-	-	-	
Full Funding TOA (<i>\$ in Millions</i>)	83,946.062	5,065.295	3,219.843	2,016.787	-	2,016.787	-	-	-	-	-	-	
Plus CY Advance Procurement (<i>\$ in Millions</i>)	3,332.434	-	-	-	-	-	-	-	-	-	-	-	
Plus Cost To Complete (<i>\$ in Millions</i>)	1,139.452	-	9.634	45.753	-	45.753	-	-	-	-	-	-	
Plus EOQ (<i>\$ in Millions</i>)	681.264	744.028	159.297	-	-	-	-	-	-	-	-	-	
Plus Escalation (<i>\$ in Millions</i>)	48.200	-	-	-	-	-	-	-	-	-	-	-	
Plus Transfer (<i>\$ in Millions</i>)	218.500	-	-	-	-	-	-	-	-	-	-	-	
Plus Hurricane (<i>\$ in Millions</i>)	227.100	-	-	-	-	-	-	-	-	-	-	-	
Total Obligation Authority (<i>\$ in Millions</i>)	89,593.012	5,809.323	3,388.774	2,062.540	0.000	2,062.540	-	-	-	-	-	-	
(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)													
Plus Outfitting and Post Delivery (<i>\$ in Millions</i>)	2,504.678	78.053	132.311	129.580	-	129.580	-	-	-	-	-	-	
Total (<i>\$ in Millions</i>)	92,097.690	5,887.376	3,521.085	2,192.120	-	2,192.120	-	-	-	-	-	-	
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	1,091.326	1,809.652	2,008.239	2,401.719	-	2,401.719	-	-	-	-	-	-	

Description:

DDG 51 will be able to operate offensively and defensively, independently or as units of Carrier Strike Groups and Surface Action Groups, in support of Marine Amphibious Task Forces in multi-threat environments that include air, surface and subsurface threats. These ships will respond to Low Intensity Conflict/Coastal and Littoral Offshore Warfare (LIC/CALOW) scenarios as well as open ocean conflict providing or augmenting power projection and forward presence requirements, and escort operations at sea. Beginning in FY17, DDG 51 Flight III with the Air and Missile Defense Radar (SPY-6) will significantly enhance Integrated Air and Missile Defense capability against current and future threats. Starting in FY21, Bridge System Upgrades will be incorporated for improved navigation capability. Starting in FY22, DDG 51 Flight III AN/SLQ-32 system will be upgraded from AN/SLQ-32(V)6 to AN/SLQ-32(V)7, with the addition of SEWIP Block 3, which provides advanced Electronic Attack (EA) capability to keep pace with the evolving Anti-Ship Missile (ASM) threat and counter-targeting.

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy					Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 02: Other Warships / BSA 1: Other Warships				P-1 Line Item Number / Title: 2122 / DDG-51			
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: N/A			Other Related Program Elements: N/A		
Line Item MDAP/MAIS Code: N/A							
Note: (1) The FY18-22 acquisition strategy is a 10 ship Multi-year Procurement (MYP) with options. The FY 2022 President's Budget request reflects savings for the ships in FY18-FY22 associated with EOQ procurement and a MYP strategy. Additional ships in FY19 and FY20 also reflect quantity savings over annual ship prices.							
Characteristics:	FLIGHT IIA	FLIGHT III					
Length Overall	509 ft	509 ft					
Beam	59 ft	59 ft					
Displacement	9217 TONS	9650 TONS					
Draft	-						
Production Status:	DDG 120	DDG 121	DDG 122	DDG 123	DDG 124	DDG 127	DDG 125
Contract Award Date	Mar 2014	Jun 2013	Jun 2013	Jun 2013	Jun 2013	Sep 2017	Jun 2013
Months to Completion							
a) Award to Delivery	96 months	101 months	115 months	109 months	123 months	78 months	118 months
b) Construction Start to Delivery	66 months	67 months	64 months	66 months	62 months	59 months	59 months
Delivery Date	Mar 2022	Nov 2021	Jan 2023	Jul 2022	Sep 2023	Mar 2024	Apr 2023
Completion Of Fitting Out	Jul 2022	Mar 2022	May 2023	Nov 2022	Jan 2024	Jul 2024	Aug 2023
Obligation Work Limit Date	Jun 2023	Feb 2023	Apr 2024	Oct 2023	Dec 2024	Jun 2025	Jul 2024
Production Status:	DDG 126	DDG 128	DDG 129	DDG 130 ⁽¹⁾	DDG 131 ⁽²⁾	DDG 132	DDG 134
Contract Award Date	Jun 2013	Sep 2018	Sep 2018	Sep 2018	Sep 2018	Dec 2018	Sep 2018
Months to Completion							
a) Award to Delivery	132 months	73 months	82 months	82 months	91 months	89 months	98 months
b) Construction Start to Delivery	51 months	54 months	54 months	55 months	49 months	59 months	55 months
Delivery Date	Jun 2024	Oct 2024	Jul 2025	Jul 2025	Apr 2026	May 2026	Nov 2026
Completion Of Fitting Out	Oct 2024	Feb 2025	Nov 2025	Nov 2025	Aug 2026	Sep 2026	Mar 2027
Obligation Work Limit Date	Sep 2025	Jan 2026	Oct 2026	Oct 2026	Jul 2027	Aug 2027	Feb 2028
Production Status:	DDG 133	DDG 135	DDG 136	DDG 137	DDG 138 ⁽³⁾		
Contract Award Date	Sep 2018	Jun 2020	Sep 2018	Sep 2018	Sep 2018		
Months to Completion							
a) Award to Delivery	98 months	84 months	105 months	110 months	115 months		
b) Construction Start to Delivery	49 months	49 months	53 months	49 months	50 months		
Delivery Date	Nov 2026	Jun 2027	Jun 2027	Nov 2027	Apr 2028		
Completion Of Fitting Out	Mar 2027	Oct 2027	Oct 2027	Mar 2028	Aug 2028		
Obligation Work Limit Date	Feb 2028	Sep 2028	Sep 2028	Feb 2029	Oct 2029		
Design Schedule	Start / Issue		Complete / Response		Reissue	Reissue Complete / Response	
Issue Date for TLR	Jun 1983		N/A				
Issue Date for TLS	N/A		N/A				
Preliminary Design	Mar 1982		Dec 1982				

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy				Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 02: Other Warships / BSA 1: Other Warships			P-1 Line Item Number / Title: 2122 / DDG-51		
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: N/A		Other Related Program Elements: N/A	
Line Item MDAP/MAIS Code: N/A					
<u>Design Schedule</u>		<u>Start / Issue</u>	<u>Complete / Response</u>	<u>Reissue</u>	<u>Reissue Complete / Response</u>
Contract Design		May 1983	Jun 1984		
Detail Design		N/A	N/A		
Request for Proposals		N/A	N/A		
Design Agent		BIW			
<u>Classification of Cost Estimate:</u> CLASS C BUDGET ESTIMATE					
Footnotes: ⁽¹⁾ DDG 118 - DDG 130 reflects actual and contract milestone dates based on ship production progress. ⁽²⁾ DDG 131 - DDG 137 reflects contract milestone dates. ⁽³⁾ DDG 138 dates are notional based on latest allowable delivery date in the FY18-FY22 shipbuilding contracts.					

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1	P-1 Line Item Number / Title: 2122 / DDG-51
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Cost Categories ^(†) indicates the presence of a P-8a	FY 2013		FY 2015		FY 2016		FY 2017		FY 2018		FY 2019		FY 2020		FY 2021	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Plan Costs	3	67.450	2	68.814	3	204.160	2	81.454	2	72.811	3	74.195	3	75.713	2	77.117
Basic Construction/Conversion		2,209.492		1,454.083		2,303.947		1,741.806		1,751.705		2,806.468		2,894.944		2,215.906
Change Orders		49.091		81.010		158.144		122.866		69.321		122.992		85.974		56.334
Electronics ^(†)		531.642		349.746		517.100		346.044		379.000		576.510		597.388		413.024
Hull, Mechanical, and Electrical (HM&E) ^(†)		185.425		141.283		219.752		153.668		164.571		228.757		226.260		161.970
Ordnance ^(†)		1,168.622		838.247		1,462.352		1,085.987		902.890		1,536.147		1,467.208		1,014.534
Other Cost		81.240		77.775		73.033		73.971		75.846		77.216		81.468		77.593
Total Ship Estimate		4,292.962		3,010.958		4,938.488		3,605.796		3,416.144		5,422.285		5,428.955		4,016.478
Less Advance Procurement FY 2012		92.454		-		-		-		-		-		-		-
Less Advance Procurement FY 2015		-		-		134.039		-		-		-		-		-
Less Subsequent Full Funding FY 2017		-		-		433.000		-		-		-		-		-
Less Cost to Complete FY 2014		100.000		-		-		-		-		-		-		-
Less Cost to Complete FY 2018		31.941		-		-		-		-		-		-		-
Less Cost to Complete FY 2019		37.266		-		-		-		-		-		-		-
Less Cost to Complete FY 2021		-		9.634		-		-		-		-		-		-
Less Cost to Complete FY 2022		-		44.577		1.176		-		-		-		-		-
Less Cost to Complete FY 2023		-		-		39.238		111.961		-		-		-		-
Less Cost to Complete FY 2024		-		-		9.390		121.827		-		-		-		-
Less Cost to Complete FY 2025		-		-		-		7.627		31.603		-		-		-
Less Cost to Complete FY 2026		-		-		-		-		27.462		133.086		-		-
Less EOQ FY 2013		-		224.851		108.345		13.677		-		-		-		-
Less EOQ FY 2014		-		69.989		130.650		168.912		-		-		-		-
Less EOQ FY 2018		-		-		-		-		-		39.362		25.940		12.517
Less EOQ FY 2019		-		-		-		-		-		-		337.720		152.104

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy

Date: May 2021

Appropriation / Budget Activity / Budget Sub Activity:
1611N / 02 / 1

P-1 Line Item Number / Title:
2122 / DDG-51

Cost Categories ^(†) indicates the presence of a P-8a	FY 2013		FY 2015		FY 2016		FY 2017		FY 2018		FY 2019		FY 2020		FY 2021	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Less EOQ FY 2020		-		-		-		-		-		-		-		632.014
Less EOQ FY 2021		-		-		-		-		-		-		-		-
Net P-1 Funding		4,031.301		2,661.907		4,082.650		3,181.792		3,357.079		5,249.837		5,065.295		3,219.843

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy		Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2122 / DDG-51	
Cost Categories (†) indicates the presence of a P-8a	FY 2022		
	Qty (Each)		Total Cost (\$ M)
Plan Costs	1		86.682
Basic Construction/Conversion			1,281.992
Change Orders			40.381
Electronics ^(†)			277.188
Hull, Mechanical, and Electrical (HM&E) ^(†)			82.389
Ordnance ^(†)			556.102
Other Cost			76.985
Total Ship Estimate			2,401.719
Less Advance Procurement FY 2012			-
Less Advance Procurement FY 2015			-
Less Subsequent Full Funding FY 2017			-
Less Cost to Complete FY 2014			-
Less Cost to Complete FY 2018			-
Less Cost to Complete FY 2019			-
Less Cost to Complete FY 2021			-
Less Cost to Complete FY 2022			-
Less Cost to Complete FY 2023			-
Less Cost to Complete FY 2024			-
Less Cost to Complete FY 2025			-
Less Cost to Complete FY 2026			-
Less EOQ FY 2013			-
Less EOQ FY 2014			-
Less EOQ FY 2018			12.517
Less EOQ FY 2019			101.104
Less EOQ FY 2020			112.014
Less EOQ FY 2021			159.297
Net P-1 Funding			2,016.787
Remarks: Total Ship Estimates include incorporation of Bridge System Navigation Modifications on FY15 through FY19 ships, adjustments for government responsible portion of ship contract performance and Integrated Power Node Center (IPNC) resolution for FY15 - FY19 ships, and realized cost adjustments for GFE systems in ORDNANCE for FY17 - FY19. Total Ship Estimates for FY21 include 1) \$130M of FY20 congressionally added advance procurement for the Surface Combatant Supplier Base, and 2) \$215M of Congressionally added full funding for Surface Combatant Shipyard Infrastructure. These amounts are shown in the Basic Construction/Conversion cost category.			

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Exhibit P-27, Ship Production Schedule: PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1	P-1 Line Item Number / Title: 2122 / DDG-51
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Ship	Shipbuilder	Fiscal Year	Contract Award	Start of Construction	Delivery Date
DDG 120	Bath Iron Works	2013	Mar 2014	Sep 2016	Mar 2022
DDG 121	Huntington Ingalls Industries	2015	Jun 2013	Apr 2016	Nov 2021
DDG 122	Bath Iron Works	2015	Jun 2013	Sep 2017	Jan 2023
DDG 123	Huntington Ingalls Industries	2016	Jun 2013	Jan 2017	Jul 2022
DDG 124	Bath Iron Works	2016	Jun 2013	Jul 2018	Sep 2023
DDG 127	Bath Iron Works	2016	Sep 2017	Apr 2019	Mar 2024
DDG 125	Huntington Ingalls Industries	2017	Jun 2013	May 2018	Apr 2023
DDG 126	Bath Iron Works	2017	Jun 2013	Mar 2020	Jun 2024
DDG 128	Huntington Ingalls Industries	2018	Sep 2018	Apr 2020	Oct 2024
DDG 129	Huntington Ingalls Industries	2018	Sep 2018	Jan 2021	Jul 2025
DDG 130 ⁽¹⁾	Bath Iron Works	2019	Sep 2018	Dec 2020	Jul 2025
DDG 131 ⁽²⁾	Huntington Ingalls Industries	2019	Sep 2018	Mar 2022	Apr 2026
DDG 132	Bath Iron Works	2019	Dec 2018	Jun 2021	May 2026
DDG 134	Bath Iron Works	2020	Sep 2018	Apr 2022	Nov 2026
DDG 133	Huntington Ingalls Industries	2020	Sep 2018	Oct 2022	Nov 2026
DDG 135	Huntington Ingalls Industries	2020	Jun 2020	May 2023	Jun 2027
DDG 136	Bath Iron Works	2021	Sep 2018	Jan 2023	Jun 2027
DDG 137	Huntington Ingalls Industries	2021	Sep 2018	Oct 2023	Nov 2027
DDG 138 ⁽³⁾	TBD	2022	Sep 2018	Feb 2024	Apr 2028

Footnotes:

- ⁽¹⁾ DDG 118 - DDG 130 reflects actual and contract milestone dates based on ship production progress.
- ⁽²⁾ DDG 131 - DDG 137 reflects contract milestone dates.
- ⁽³⁾ DDG 138 dates are notional based on latest allowable delivery date in the FY18-FY22 shipbuilding contracts.

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1	P-1 Line Item Number / Title: 2122 / DDG-51
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Electronics	FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
P-35 Items						
SQQ 89 ASW	3	148.191	2	100.968	1	51.114
AN/SLQ-32 SEWIP V(6)	3	48.855	2	33.286	0	-
AN/SLQ-32 SEWIP V(7)	0	-	0	-	1	73.388
USQ 82(V) GEDMS	3	44.060	2	30.020	1	15.280
EXCOMM	3	156.413	2	106.570	1	58.815
AN/UPX 29(V) IFF and TACAN	3	22.710	2	15.474	1	7.876
CEC	3	17.284	2	11.776	1	5.995
P-35 Items Subtotal		437.513		298.094		212.468
Major Items						
NAVIGATION SYSTEM	3	12.274	2	8.362	1	4.257
SLQ 25 NIXIE	3	5.070	2	3.454	1	1.759
SRQ 4 LAMPS III	3	13.575	2	9.250	1	6.384
SSEE/SPECTRAL	3	51.000	2	34.748	1	17.687
MIDS	3	10.563	2	7.196	1	3.663
MK 53 NULKA	3	6.926	2	4.718	1	2.400
TSA ANTENNA	3	5.515	2	3.758	1	1.913
Major Items Subtotal		104.923		71.486		38.063
Other Cost Elements						
MISC. ELECTRONICS	3	54.952	2	43.444	1	26.657
Other Cost Elements Subtotal		54.952		43.444		26.657
Total Electronics		597.388		413.024		277.188

Remarks:

FY22 reflects the upgrade of the Surface Electronic Warfare Improvement Program (SEWIP) System from (V)6 to (V)7 which provides Electronic Attack (EA) capability improvements as part of in-line construction of new DDG-51 class ships.

FY22 EXCOMM system costs increased due to \$4.5M for NMT hardware procured using FY21 advanced procurement funding for a second FY22 ship.

FY22 SRQ-4 system costs increased due to \$1.7M of FY21 advanced procurement funding for a second FY22 ship.

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity:

1611N / 02 / 1

P-1 Line Item Number / Title:

2122 / DDG-51

Hull, Mechanical, and Electrical (HM&E)	FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
P-35 Items						
STC 3 IVCS	3	23.765	2	16.160	1	8.242
Main Reduction Gear	3	116.451	2	79.188	1	40.386
P-35 Items Subtotal		140.216		95.348		48.628
Major Items						
Machinery Control System	3	16.576	2	11.272	1	5.749
Integrated Bridge Navigation System	3	14.301	2	12.694	1	6.475
CYBER BDC	3	9.000	2	6.200	1	3.500
Major Items Subtotal		39.877		30.166		15.724
Other Cost Elements						
MISC. HM&E	3	46.167	2	36.456	1	18.037
Other Cost Elements Subtotal		46.167		36.456		18.037
Total Hull, Mechanical, and Electrical (HM&E)		226.260		161.970		82.389

Remarks:

FY21 and future years include the introduction of Bridge System Modifications to improve navigation capabilities on new construction DDG 51 ships.

Cyber Boundary Defense Capability (BDC) cost growth above inflation in FY22 from FY21 is due to the planned ECPs updating the SABER rack Solid State Drive (SSD) from Hard Disk Drive (HDD) and the Central Processing Unit (CPU) processor and a generic Connector Plate for the BDC.

FY22 MISC HM&E costs increased due to \$.4M for DVSS procured using FY21 advanced procurement funding for a second FY22 ship.

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity:
1611N / 02 / 1

P-1 Line Item Number / Title:
2122 / DDG-51

Ordnance	FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
P-35 Items						
AEGIS WEAPON SYSTEM (MK-7)	3	398.163	2	297.298	1	151.622
AN/SPY-6 (AMDR)	3	497.700	2	338.436	1	182.057
VLS MK 41	3	166.454	2	113.188	1	88.239
MK 45 Light Weight Gun (LWG)	3	82.526	2	56.118	1	28.620
MK 37 TOMAHAWK	3	40.580	2	27.594	1	14.073
PHALANX (CIWS)	3	26.376	2	17.936	1	9.148
SPQ-9B Radar	3	29.816	2	20.276	1	10.341
P-35 Items Subtotal		1,241.615		870.846		484.100
Major Items						
MK 32 Surface Vessel Torpedo Tubes (SVTT)	3	9.520	2	6.474	1	3.302
ELECTRO-OPTICAL SYSTEM	3	10.075	2	6.852	1	3.494
MK 160 Gun Fire Control System (GFCS)	3	10.477	2	7.124	1	3.634
Major Items Subtotal		30.072		20.450		10.430
Other Cost Elements						
MISC. ORDNANCE	3	195.521	2	123.238	1	61.572
Other Cost Elements Subtotal		195.521		123.238		61.572
Total Ordnance		1,467.208		1,014.534		556.102

Remarks:

FY22 AMDR system costs updated in accordance with planned contract cost requirements.

FY22 VLS systems costs include 1) updated pricing in accordance with contract awarded to Lockheed Martin in FEB 2020, and 2) \$13.5M of FY19 advance procurement funding for hardware for the second ship in FY22.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2122 / DDG-51			
Equipment Item: SQQ 89 ASW						PARM Code: N/A	
P-35 Category	FY 2020		FY 2021		FY 2022		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
Major Hardware	3	94.629	2	64.300	1	32.793	
Spares		1.573		1.068		0.545	
System Engineering		13.203		8.972		4.576	
Technical Engineering Services		7.668		5.210		2.657	
Other Costs		31.118		21.418		10.543	
Total	3	148.191	2	100.968	1	51.114	

Description:
Detect, classify, localize and track submerged submarines under all environmental conditions at long range from ASW ships, using bottom reflected and convergence zone acoustic paths. Starting in FY20, the Multi-Function Towed Array (MFTA) sensor along with the Handling and Stowage Gear (H&SG) is included as part of in-line construction of new DDG-51 class ships.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	DDG 133	LOCKHEED MARTIN	C/FFP	Mar 2020	Option	3	31.543
FY 2021	DDG 136	LOCKHEED MARTIN	C/FFP	Mar 2021	Option	2	32.150
FY 2022	DDG 138	LOCKHEED MARTIN	C/FFP	Mar 2022	Option	1	32.793

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	DDG 133	Nov 2026	41	24	Jun 2021
FY 2021	DDG 136	Jun 2027	41	24	Jan 2022
FY 2022	DDG 138	Apr 2028	41	24	Nov 2022

Competition/Second Source Initiatives:
Competitive

Remarks:
Note: Months Required Before Delivery reflect timeline for earliest ship delivery date of each Program Year. Additional deliveries for each program year may have a higher number of Months Required Before Delivery based on shipyard required delivery dates.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2122 / DDG-51			
Equipment Item: AN/SLQ-32 SEWIP V(6)						PARM Code: N/A	

P-35 Category	FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Major Hardware	3	41.529	2	28.296	0	-
Spares		1.182		0.806		-
System Engineering		2.865		1.952		-
Technical Engineering Services		0.417		0.284		-
Other Costs		2.862		1.948		-
Total	3	48.855	2	33.286	0	-

Description:
AN/SLQ-32 Surface Electronic Warfare Improvement Program (SEWIP) V(6) provides the DDG 51 Class Destroyers with the electronic warfare capability of automatically detecting, sorting, classifying, tracking, engaging and continually displaying emitter and platform densities.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	DDG 133	LOCKHEED MARTIN SYRACUSE	C/FFP	Jan 2021	New	3	13.843
FY 2021	DDG 136	LOCKHEED MARTIN SYRACUSE	C/FFP	Jun 2021	Option	2	14.148

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	DDG 133	Nov 2026	43	23	May 2021
FY 2021	DDG 136	Jun 2027	43	23	Dec 2021

Competition/Second Source Initiatives:
Competitive

Remarks:
Note: Months Required Before Delivery reflect timeline for earliest ship delivery date of each Program Year. Additional deliveries for each program year may have a higher number of Months Required Before Delivery based on shipyard required delivery dates.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy					Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1			P-1 Line Item Number / Title: 2122 / DDG-51			
Equipment Item: AN/SLQ-32 SEWIP V(7)					PARM Code: N/A	

P-35 Category	FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Major Hardware	0	-	0	-	1	67.648
Spares		-		-		1.416
System Engineering		-		-		2.341
Technical Engineering Services		-		-		0.112
Other Costs		-		-		1.871
Total	0	-	0	-	1	73.388

Description:
 AN/SLQ-32 Surface Electronic Warfare Improvement Program (SEWIP)V(7) will begin in FY2022. This upgrade will include all the capabilities of the AN/SLQ-32(V)6 providing the DDG 51 Class Destroyers with the electronic warfare capability of automatically detecting, sorting, classifying, tracking, engaging and continually displaying emitter and platform densities. The AN/SLQ-32(V)7 will also provide the addition of SEWIP Block 3, which provides advanced Electronic Attack (EA) capability to keep pace with the evolving Anti-Ship Missile (ASM) threat and counter-targeting.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2022	DDG 138	NORTHROP GRUMMAN	C/FFP	Mar 2022	Option	1	66.041

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2022	DDG 138	Apr 2028	34	30	Dec 2022

Competition/Second Source Initiatives:
 Competitive

Remarks:
 Note: Months Required Before Delivery reflect timeline for earliest ship delivery date of each Program Year. Additional deliveries for each program year may have a higher number of Months Required Before Delivery based on shipyard required delivery dates.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2122 / DDG-51			
Equipment Item: USQ 82(V) GEDMS						PARM Code: N/A	

P-35 Category	FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Major Hardware	3	23.567	2	16.058	1	8.190
Technical Data and Documentation		2.096		1.428		0.728
System Engineering		5.081		3.462		1.766
Technical Engineering Services		0.856		0.584		0.298
Other Costs		12.460		8.488		4.298
Total	3	44.060	2	30.020	1	15.280

Description:
 Gigabit Ethernet Data Multiplex System (GEDMS) is the mission critical ship-wide network that transfers data associated with Machinery, Steering, Navigation, Combat, Alarms & Indicating, and Damage Control Systems. It is a general purpose modular data transfer system that provides high speed, reliable and survivable data from source systems to user systems automatically or on demand.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	DDG 133	BOEING/DRS	C/FFP	May 2020	Option	3	7.856
FY 2021	DDG 136	BOEING/DRS	C/FFP	Mar 2022	Option	2	8.029
FY 2022	DDG 138	BOEING/DRS	C/FFP	Mar 2022	Option	1	8.190

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	DDG 133	Nov 2026	40	16	Mar 2022
FY 2021	DDG 136	Jun 2027	40	16	Oct 2022
FY 2022	DDG 138	Apr 2028	40	16	Aug 2023

Competition/Second Source Initiatives:
 Competitive

Remarks:
 Note: Months Required Before Delivery reflect timeline for earliest ship delivery date of each Program Year. Additional deliveries for each program year may have a higher number of Months Required Before Delivery based on shipyard required delivery dates.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2122 / DDG-51			
Equipment Item: EXCOMM						PARM Code: N/A	

P-35 Category	FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Major Hardware	3	93.508	2	63.846	1	37.019
Technical Data and Documentation		0.375		0.256		0.131
Spares		0.863		0.590		0.301
System Engineering		9.732		6.646		3.389
Technical Engineering Services		5.679		3.878		1.978
Other Costs		18.389		12.326		6.293
Assembly & Integration		27.867		19.028		9.704
Total	3	156.413	2	106.570	1	58.815

Description:
 The Exterior Communication System (EXCOMM) provides voice, data, teletypewriter (TTY), continuous wave (CW), and other communication services on designated frequencies from very low frequency (VLF) to ultra high frequency (UHF) for tactical and record requirements. It includes all external radio communication devices aboard the ship.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	DDG 133	VARIOUS	Various	Various	Various	3	31.169
FY 2021	DDG 136	VARIOUS	Various	Various	Various	2	31.923
FY 2022	DDG 138	VARIOUS	Various	Various	Various	1	37.019

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	DDG 133	Nov 2026	31	24	Apr 2022
FY 2021	DDG 136	Jun 2027	31	24	Nov 2022
FY 2022	DDG 138	Apr 2028	31	24	Sep 2023

Competition/Second Source Initiatives:
 Numerous contract arrangements (sole source/competitive)

Remarks:
 There are numerous components and contracts resulting in various award dates.

FY22 EXCOMM system costs increased due to \$4.5M for NMT hardware procured using FY21 advanced procurement funding for a second FY22 ship.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2122 / DDG-51			
Equipment Item: AN/UPX 29(V) IFF and TACAN						PARM Code: N/A	
P-35 Category	FY 2020		FY 2021		FY 2022		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
Major Hardware	3	19.522	2	13.300	1	6.783	
Spares		0.244		0.166		0.085	
System Engineering		1.184		0.808		0.412	
Technical Engineering Services		0.498		0.338		0.172	
Other Costs		1.262		0.862		0.424	
Total	3	22.710	2	15.474	1	7.876	

Description:
 The UPX-29 Interrogator System is a centralized Mark XIIA interrogator and target processor. It employs a cooperative challenge and reply technique to positively identify friendly platforms. The system is capable of interrogating Mark XII, Mark XIIA, International Civil Aviation Organization (ICAO), or Federal Aviation Administration (FAA)-compliant IFF transponders using a standard shipboard interrogator set, a target processor, and an Electronically Steerable Antenna (ESA) system. TACAN is a navigational beacon system that provides azimuth, slant range, and station identification information to TACAN equipped aircraft, permitting 24/7, all weather landing operations.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	DDG 133	BAE	SS/FFP	Mar 2020	Option	3	6.507
FY 2021	DDG 136	BAE	SS/FFP	Mar 2021	Option	2	6.650
FY 2022	DDG 138	BAE	SS/FFP	Mar 2022	Option	1	6.783

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	DDG 133	Nov 2026	29	24	Jun 2022
FY 2021	DDG 136	Jun 2027	29	24	Jan 2023
FY 2022	DDG 138	Apr 2028	29	24	Nov 2023

Competition/Second Source Initiatives:
N/A

Remarks:
 Note: Months Required Before Delivery reflect timeline for earliest ship delivery date of each Program Year. Additional deliveries for each program year may have a higher number of Months Required Before Delivery based on shipyard required delivery dates.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2122 / DDG-51			
Equipment Item: CEC						PARM Code: N/A	

P-35 Category	FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Major Hardware	3	14.802	2	10.086	1	5.144
System Engineering		0.759		0.518		0.264
Technical Engineering Services		0.543		0.370		0.189
Other Costs		1.180		0.802		0.398
Total	3	17.284	2	11.776	1	5.995

Description:
Cooperative Engagement Capability (CEC) is a sensor netting system which distributes sensor data from each CEC equipped ship, aircraft, and/or Cooperating Unit (CU), to all other CUs in the battle force through a real-time, line of sight, high data rate sensor and engagement data distribution network.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	DDG 133	DRS/ RAYTHEON	C/FFP	May 2021	Option	3	4.934
FY 2021	DDG 136	DRS/ RAYTHEON	C/FFP	May 2022	Option	2	5.043
FY 2022	DDG 138	COMPETITIVE	C/FFP	Feb 2023	New	1	5.144

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	DDG 133	Nov 2026	40	18	Jan 2022
FY 2021	DDG 136	Jun 2027	40	18	Aug 2022
FY 2022	DDG 138	Apr 2028	40	18	Jun 2023

Competition/Second Source Initiatives:
Competitive

Remarks:
Note: Months Required Before Delivery reflect timeline for earliest ship delivery date of each Program Year. Additional deliveries for each program year may have a higher number of Months Required Before Delivery based on shipyard required delivery dates.

Contract Data Notes for FY20 and FY21:
USG-2B CEC Hardware - Contractor: DRS
PAAA Antenna - Contractor: Raytheon

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																																
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2122 / DDG-51																																		
Equipment Item: STC 3 IVCS						PARM Code: N/A																																
P-35 Category	FY 2020		FY 2021		FY 2022																																	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																																
Major Hardware	3	15.292	2	10.398	1	5.303																																
Spares		0.825		0.562		0.287																																
System Engineering		2.874		1.954		0.997																																
Technical Engineering Services		0.734		0.500		0.255																																
Other Costs		4.040		2.746		1.400																																
Total	3	23.765	2	16.160	1	8.242																																
Description: A solid state integrated voice communication system (IVCS) for application with the AEGIS combat system.																																						
Contract Data: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th style="text-align: center;">Program Year</th> <th style="text-align: center;">Hull</th> <th style="text-align: center;">Prime Contractor</th> <th style="text-align: center;">Contract Method/Type</th> <th style="text-align: center;">Award Date</th> <th style="text-align: center;">New/Option</th> <th style="text-align: center;">Quantity (Each)</th> <th style="text-align: center;">Unit Cost (\$ M)</th> </tr> <tr> <td style="text-align: center;">FY 2020</td> <td style="text-align: center;">DDG 133</td> <td style="text-align: center;">DRS</td> <td style="text-align: center;">C/FFP</td> <td style="text-align: center;">Aug 2020</td> <td style="text-align: center;">Option</td> <td style="text-align: center;">3</td> <td style="text-align: right;">5.097</td> </tr> <tr> <td style="text-align: center;">FY 2021</td> <td style="text-align: center;">DDG 136</td> <td style="text-align: center;">DRS</td> <td style="text-align: center;">C/FFP</td> <td style="text-align: center;">Jul 2021</td> <td style="text-align: center;">Option</td> <td style="text-align: center;">2</td> <td style="text-align: right;">5.199</td> </tr> <tr> <td style="text-align: center;">FY 2022</td> <td style="text-align: center;">DDG 138</td> <td style="text-align: center;">DRS</td> <td style="text-align: center;">C/FFP</td> <td style="text-align: center;">Jul 2022</td> <td style="text-align: center;">Option</td> <td style="text-align: center;">1</td> <td style="text-align: right;">5.303</td> </tr> </table>							Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2020	DDG 133	DRS	C/FFP	Aug 2020	Option	3	5.097	FY 2021	DDG 136	DRS	C/FFP	Jul 2021	Option	2	5.199	FY 2022	DDG 138	DRS	C/FFP	Jul 2022	Option	1	5.303
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																															
FY 2020	DDG 133	DRS	C/FFP	Aug 2020	Option	3	5.097																															
FY 2021	DDG 136	DRS	C/FFP	Jul 2021	Option	2	5.199																															
FY 2022	DDG 138	DRS	C/FFP	Jul 2022	Option	1	5.303																															
Delivery Date: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th style="text-align: center;">Program Year</th> <th style="text-align: center;">Hull</th> <th style="text-align: center;">Earliest Ship Delivery Date</th> <th style="text-align: center;">Months Required Before Delivery</th> <th style="text-align: center;">Production Leadtime</th> <th style="text-align: center;">Required Award Date</th> </tr> <tr> <td style="text-align: center;">FY 2020</td> <td style="text-align: center;">DDG 133</td> <td style="text-align: center;">Nov 2026</td> <td style="text-align: center;">46</td> <td style="text-align: center;">12</td> <td style="text-align: center;">Jan 2022</td> </tr> <tr> <td style="text-align: center;">FY 2021</td> <td style="text-align: center;">DDG 136</td> <td style="text-align: center;">Jun 2027</td> <td style="text-align: center;">46</td> <td style="text-align: center;">12</td> <td style="text-align: center;">Aug 2022</td> </tr> <tr> <td style="text-align: center;">FY 2022</td> <td style="text-align: center;">DDG 138</td> <td style="text-align: center;">Apr 2028</td> <td style="text-align: center;">46</td> <td style="text-align: center;">12</td> <td style="text-align: center;">Jun 2023</td> </tr> </table>							Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date	FY 2020	DDG 133	Nov 2026	46	12	Jan 2022	FY 2021	DDG 136	Jun 2027	46	12	Aug 2022	FY 2022	DDG 138	Apr 2028	46	12	Jun 2023								
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date																																	
FY 2020	DDG 133	Nov 2026	46	12	Jan 2022																																	
FY 2021	DDG 136	Jun 2027	46	12	Aug 2022																																	
FY 2022	DDG 138	Apr 2028	46	12	Jun 2023																																	
Competition/Second Source Initiatives: Competitive																																						
Remarks: Note: Months Required Before Delivery reflect timeline for earliest ship delivery date of each Program Year. Additional deliveries for each program year may have a higher number of Months Required Before Delivery based on shipyard required delivery dates.																																						

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																																
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2122 / DDG-51																																		
Equipment Item: Main Reduction Gear						PARM Code: N/A																																
P-35 Category	FY 2020		FY 2021		FY 2022																																	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																																
Major Hardware	3	112.177	2	77.332	1	38.934																																
System Engineering		-		-		-																																
Technical Engineering Services		4.274		1.856		1.452																																
Total	3	116.451	2	79.188	1	40.386																																
Description: The contractor will engineer, manufacture, test and deliver a fully operational DDG 51 Main Reduction Gear (MRG). A DDG 51 Class MRG shipset consists of two gear assemblies. Each reduction gear combines the input of two LM2500 engines to convert the high speed, low torque of the engine to low speed, high torque output suitable to drive the propulsion shafting, and the related support systems and equipment.																																						
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FY 2022	DDG 138	Apr 2028	38	32	Jun 2022																																	
Competition/Second Source Initiatives: Competitive																																						
Remarks: Note: Months Required Before Delivery reflect timeline for earliest ship delivery date of each Program Year. Additional deliveries for each program year may have a higher number of Months Required Before Delivery based on shipyard required delivery dates.																																						

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2122 / DDG-51			
Equipment Item: AEGIS WEAPON SYSTEM (MK-7)						PARM Code: N/A	
P-35 Category	FY 2020		FY 2021		FY 2022		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
Major Hardware	3	232.412	2	166.358	1	84.843	
System Engineering		2.759		1.876		0.957	
Technical Engineering Services		2.597		1.766		0.900	
Other Costs		13.402		27.342		13.944	
Logistics Support		31.546		21.452		10.941	
Combat System Integration		115.447		78.504		40.037	
Total	3	398.163	2	297.298	1	151.622	

Description:
AEGIS is a fast reaction, high firepower, all weather weapon system incorporating a high degree of system availability and effectiveness. It consists of a multi-function phase/plane array radar, high powered illuminators, advanced missile guidance and fully digitalized and integrated combat ship control for radar, weapons and command and decision. An Operational Readiness Test System performs continuous on-line assessment and fault detection.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	DDG 133	LM/ RTN/ GD	Various	Jan 2021	Option	3	77.471
FY 2021	DDG 136	LM/ RTN/ GD	Various	Jan 2022	Option	2	83.179
FY 2022	DDG 138	LM/ RTN/ GD	Various	Jul 2022	Option	1	84.843

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	DDG 133	Nov 2026	29	36	Jun 2021
FY 2021	DDG 136	Jun 2027	29	36	Jan 2022
FY 2022	DDG 138	Apr 2028	29	36	Nov 2022

Competition/Second Source Initiatives:
Multiple contract arrangements (sole source/competitive)

Remarks:
Note: Months Required Before Delivery reflect timeline for earliest ship delivery date of each Program Year. Additional deliveries for each program year may have a higher number of Months Required Before Delivery based on shipyard required delivery dates.

Contract Data Notes:
AWS Antenna and Signal Processors - Contractor: Lockheed Martin
Fire Control System Transmitter - Contractor: Raytheon

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2122 / DDG-51
Equipment Item: AEGIS WEAPON SYSTEM (MK-7) AWS Director/Director Controller - Contractor: General Dynamics		PARM Code: N/A

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2122 / DDG-51			
Equipment Item: AN/SPY-6 (AMDR)						PARM Code: N/A	

P-35 Category	FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Major Hardware	3	401.643	2	273.118	1	148.744
System Engineering		30.888		21.004		10.713
Technical Engineering Services		24.089		16.380		8.354
Other Costs		25.884		10.334		5.270
Logistics		15.196		17.600		8.976
Total	3	497.700	2	338.436	1	182.057

Description:
 The AN/SPY-6 Air and Missile Defense Radar (AMDR) suite consists of an S-Band radar (AMDR-S), an X-band radar (via SPQ-9B starting with DDG 119), and a Radar Suite Controller (RSC). AMDR will provide multi-mission capabilities, simultaneously supporting both long range, exoatmospheric detection, tracking and discrimination of ballistic missiles, as well as Area and Self Defense against air and surface threats.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	DDG 133	RAYTHEON	C/FPIF	Dec 2019	Option	3	133.881
FY 2021	DDG 136	COMPETITIVE	C/FPIF	Aug 2021	New	2	136.559
FY 2022	DDG 138	COMPETITIVE	C/FPIF	Feb 2022	Option	1	148.744

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	DDG 133	Nov 2026	40	36	Jul 2020
FY 2021	DDG 136	Jun 2027	40	36	Feb 2021
FY 2022	DDG 138	Apr 2028	38	36	Feb 2022

Competition/Second Source Initiatives:
 Competitive

Remarks:
 AMDR is only applicable to Flight III ships.

FY22 AMDR system costs updated in accordance with planned contract cost requirements.

Note: Months Required Before Delivery reflect timeline for earliest ship delivery date of each Program Year. Additional deliveries for each program year may have a higher number of Months Required Before Delivery based on shipyard required delivery dates.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2122 / DDG-51			
Equipment Item: VLS MK 41						PARM Code: N/A	

P-35 Category	FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Major Hardware	3	107.010	2	72.766	1	67.624
Ancillary Equipment		4.976		3.384		1.726
Technical Data and Documentation		0.879		0.598		0.305
System Engineering		22.101		15.028		7.664
Technical Engineering Services		20.381		13.860		7.069
Other Costs		11.107		7.552		3.851
Total	3	166.454	2	113.188	1	88.239

Description:
 The VLS is a Missile Launching System which provides Surface Combatants with a launcher to carry, prepare for launch and fire, Anti-Air Warfare, Strike/Surface Warfare, and Anti-Submarine Warfare weapons. The MK-41 VLS Launchers consist of twelve modules comprised of eight cells each.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	DDG 133	LOCKHEED MARTIN	C/FFP	Feb 2020	Option	3	35.670
FY 2021	DDG 136	LOCKHEED MARTIN	C/FFP	Jun 2021	Option	2	36.383
FY 2022	DDG 138	LOCKHEED MARTIN	C/FFP	Jan 2022	Option	1	67.624

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	DDG 133	Nov 2026	34	24	Jan 2022
FY 2021	DDG 136	Jun 2027	34	24	Aug 2022
FY 2022	DDG 138	Apr 2028	34	24	Jun 2023

Competition/Second Source Initiatives:
 Competitive

Remarks:
 VLS equipment procurements are partially funded with Advance Procurement funds.

FY22 VLS systems costs include 1) updated pricing in accordance with contract awarded to Lockheed Martin in FEB 2020, and 2) \$13.5M of FY19 advance procurement funding for hardware for the second ship in FY22.

Note: Months Required Before Delivery reflect timeline for earliest ship delivery date of each Program Year. Additional deliveries for each program year may have a higher number of Months Required Before Delivery based on shipyard required delivery dates.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2122 / DDG-51			
Equipment Item: MK 45 Light Weight Gun (LWG)						PARM Code: N/A	
P-35 Category	FY 2020		FY 2021		FY 2022		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
Major Hardware	3	61.095	2	41.544	1	21.187	
Spares		0.521		0.354		0.181	
System Engineering		7.383		5.020		2.560	
Technical Engineering Services		3.965		2.696		1.375	
Other Costs		9.562		6.504		3.317	
Total	3	82.526	2	56.118	1	28.620	

Description:
The 5" 62 caliber MK 45 Mod 4 Gun is a digitized high energy system with the capability to automatically select, load and fire different types of 5"/62 caliber projectiles.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	DDG 133	BAE AD/MCNALLY	Various	Sep 2020	Option	3	20.365
FY 2021	DDG 136	BAE AD/MCNALLY	Various	Oct 2020	Option	2	20.772
FY 2022	DDG 138	BAE AD/MCNALLY	Various	Aug 2022	New	1	21.187

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	DDG 133	Nov 2026	40	24	Jul 2021
FY 2021	DDG 136	Jun 2027	40	24	Feb 2022
FY 2022	DDG 138	Apr 2028	40	24	Dec 2022

Competition/Second Source Initiatives:
Sole Source

Remarks:
Gun Mount Equipment for FY21 ships procured using FY20 AP funds.

Contract Data notes:
Gun Mount contract: BAE Armament Division - Sole Source
Lower Hoist contract: McNally - Sole Source

Note: Months Required Before Delivery reflect timeline for earliest ship delivery date of each Program Year. Additional deliveries for each program year may have a higher number of Months Required Before Delivery based on shipyard required delivery dates.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2122 / DDG-51			
Equipment Item: MK 37 TOMAHAWK						PARM Code: N/A	

P-35 Category	FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Major Hardware	3	13.950	2	9.486	1	4.838
Spares		2.270		1.544		0.787
System Engineering		7.059		4.800		2.448
Technical Engineering Services		6.705		4.560		2.326
Other Costs		10.596		7.204		3.674
Total	3	40.580	2	27.594	1	14.073

Description:
The Tactical Tomahawk Weapon Control System (TTWCS) is an open system architecture of work stations, processors, printers, fiber optic Local Area Network (LAN) and the Navy Standard Mass Measurement storage device which provides target data management, engagement planning, weapon selection and initiation and launch functions for the TOMAHAWK cruise missile.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	DDG 133	NSWC PT HUENEME	WR	Jun 2021	Various	3	4.650
FY 2021	DDG 136	NSWC PT HUENEME	WR	Apr 2022	Various	2	4.743
FY 2022	DDG 138	NSWC PT HUENEME	WR	Apr 2023	Various	1	4.838

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	DDG 133	Nov 2026	24	12	Nov 2023
FY 2021	DDG 136	Jun 2027	24	12	Jun 2024
FY 2022	DDG 138	Apr 2028	24	12	Apr 2025

Competition/Second Source Initiatives:
Navy construction

Remarks:
Note: Months Required Before Delivery reflect timeline for earliest ship delivery date of each Program Year. Additional deliveries for each program year may have a higher number of Months Required Before Delivery based on shipyard required delivery dates.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																																
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2122 / DDG-51																																		
Equipment Item: PHALANX (CIWS)						PARM Code: N/A																																
P-35 Category	FY 2020		FY 2021		FY 2022																																	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																																
Major Hardware	3	19.547	2	13.292	1	6.779																																
System Engineering		1.321		0.898		0.458																																
Technical Engineering Services		2.366		1.608		0.820																																
Other Costs		3.142		2.138		1.091																																
Total	3	26.376	2	17.936	1	9.148																																
Description: Phalanx Close-In Weapon System (CIWS) provides fast reaction terminal defense against anti-ship missiles, aircraft, helicopters, low-slow flyers (e.g. unmanned aerial vehicles) and surface threats. The system is an automatic, self-contained unit consisting of search/track radar, threat evaluation and fire control subsystem, and a 20 mm M61A1 Gatling gun subsystem all mounted in a single structure requiring a minimum of integration with other ship systems.																																						
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Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																															
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Competition/Second Source Initiatives: Sole Source																																						
Remarks: Note: Months Required Before Delivery reflect timeline for earliest ship delivery date of each Program Year. Additional deliveries for each program year may have a higher number of Months Required Before Delivery based on shipyard required delivery dates.																																						

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2122 / DDG-51			
Equipment Item: SPQ-9B Radar						PARM Code: N/A	
P-35 Category	FY 2020		FY 2021		FY 2022		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
Major Hardware	3	25.223	2	17.152	1	8.748	
Spares		0.332		0.226		0.115	
System Engineering		1.315		0.894		0.456	
Technical Engineering Services		1.398		0.950		0.485	
Other Costs		1.548		1.054		0.537	
Total	3	29.816	2	20.276	1	10.341	

Description:
The AN/SPQ-9B Radar detects and tracks low flying Anti-Ship Missile targets in heavy clutter. The mission of the AN/SPQ-9B includes the capability to detect and classify periscopes with the completion and incorporation of a Periscope Detection and Discrimination (PDD) capability designed to operate concurrently with the AN/SPY-6 capability.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	DDG 133	DRS	C/FFP	Mar 2020	Option	3	8.408
FY 2021	DDG 136	DRS	C/FFP	Mar 2021	Option	2	8.576
FY 2022	DDG 138	DRS	C/FFP	Jan 2022	Option	1	8.748

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	DDG 133	Nov 2026	41	30	Dec 2020
FY 2021	DDG 136	Jun 2027	41	30	Jul 2021
FY 2022	DDG 138	Apr 2028	41	30	May 2022

Competition/Second Source Initiatives:
Competitive

Remarks:
Note: Months Required Before Delivery reflect timeline for earliest ship delivery date of each Program Year. Additional deliveries for each program year may have a higher number of Months Required Before Delivery based on shipyard required delivery dates.

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Exhibit P-10, Advance Procurement Requirements Analysis (page 1 - Budget Funding Justification): PB 2022 Navy						Date: May 2021				
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2122 / DDG-51						
First System (2022) Award Date: January 2018		First System (2022) Completion Date: July 2024			Interval Between Systems: 12 Months					
Cost Elements		Production Leadtime (Months)	When Required* (Months)	FY 2020 (\$ M)	FY 2021 (\$ M)	FY 2022 (\$ M)	FY 2023 (\$ M)	FY 2024 (\$ M)	FY 2025 (\$ M)	FY 2026 (\$ M)
SHIP CONSTRUCTION EOQ										
SHIP Construction EOQ FY21 Ships		-	-	375.623	-	0.000	-	-	-	-
SHIP Construction EOQ FY22 Ships		-	-	112.014	146.280	0.000	-	-	-	-
SHIP Construction EOQ FY23 Ships		-	-	-	-	0.000	-	-	-	-
SHIP Construction EOQ FY24 Ships		-	-	-	-	0.000	-	-	-	-
SHIP Construction EOQ FY25 Ships		-	-	-	-	0.000	-	-	-	-
SHIP Construction EOQ FY26 Ships		-	-	-	-	0.000	-	-	-	-
SHIP Construction EOQ FY27 Ships		-	-	-	-	0.000	-	-	-	-
Total: SHIP CONSTRUCTION EOQ				487.637	146.280	-	-	-	-	-
VLS Advanced Procurement										
VLS EOQ FY22 Ships		-	-	-	-	0.000	-	-	-	-
VLS EOQ FY23 Ships		-	-	-	-	0.000	-	-	-	-
VLS EOQ FY24 Ships		-	-	-	-	0.000	-	-	-	-
VLS EOQ FY25 Ships		-	-	-	-	0.000	-	-	-	-
VLS EOQ FY26 Ships		-	-	-	-	0.000	-	-	-	-
VLS EOQ FY27 Ships		-	-	-	-	0.000	-	-	-	-
Total: VLS Advanced Procurement				-	-	-	-	-	-	-
Other GFE										
AWS AP FY21 Ships		-	-	59.443	-	0.000	-	-	-	-
MK 45 AP FY21 Ships		-	-	34.000	-	0.000	-	-	-	-
MK 160-CDP/NPS AP FY21 Ships		-	-	2.964	-	0.000	-	-	-	-
Aviation Support (AVSUP) Recovery, Assist, Securing & Traversing (RAST) AP FY21 Ships ⁽⁸⁾		-	-	11.796	-	0.000	-	-	-	-
Environmental Protection Systems (ENVIRON) Solid Waste Equipment (SWE) AP FY21 Ships ⁽⁹⁾		-	-	1.407	-	0.000	-	-	-	-
NIXIE AP FY21 Ships		-	-	0.700	-	0.000	-	-	-	-
Consoles, Displays & Peripherals/ Network Processing & Storage (CDP/ NPS) AP FY21 Ships ⁽¹⁰⁾		-	-	16.081	-	0.000	-	-	-	-
EXCOMM AP FY22 Ships		-	-	-	8.916	0.000	-	-	-	-

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Exhibit P-10, Advance Procurement Requirements Analysis (page 1 - Budget Funding Justification): PB 2022 Navy							Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2122 / DDG-51					
First System (2022) Award Date: January 2018		First System (2022) Completion Date: July 2024			Interval Between Systems: 12 Months				
Cost Elements	Production Leadtime (Months)	When Required* (Months)	FY 2020 (\$ M)	FY 2021 (\$ M)	FY 2022 (\$ M)	FY 2023 (\$ M)	FY 2024 (\$ M)	FY 2025 (\$ M)	FY 2026 (\$ M)
AN/SRQ-4 AP FY22 Ships ⁽¹¹⁾	-	-	-	3.352	0.000	-	-	-	-
DVSS AP FY22 Ships ⁽¹²⁾	-	-	-	0.749	0.000	-	-	-	-
<i>Total: Other GFE</i>			<i>126.391</i>	<i>13.017</i>	-	-	-	-	-
Industrial Supplier Base									
Surface Combatant Supplier Base	-	-	130.000	-	0.000	-	-	-	-
<i>Total: Industrial Supplier Base</i>			<i>130.000</i>	-	-	-	-	-	-
Total Advance Procurement/Obligation Authority			744.028	159.297	-	-	-	-	-
*Note: "When Required" is the number of months required before ship delivery.									

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Exhibit P-10, Advance Procurement Requirements Analysis (page 2 - Budget Funding Justification): PB 2022 Navy				Date: May 2021			
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2122 / DDG-51			
Cost Elements	FY 2022						
	Production Leadtime (Months)	When Required* (Months)	Unit Cost (\$ M)	Contract Forecast Date	2022 Qty (Each)	For FY	Total Cost Request (\$ M)
SHIP CONSTRUCTION EOQ							
SHIP Construction EOQ FY21 Ships	-	-	-		-		0.000
SHIP Construction EOQ FY22 Ships	-	-	-		-		0.000
SHIP Construction EOQ FY23 Ships	-	-	-		-		0.000
SHIP Construction EOQ FY24 Ships	-	-	-		-		0.000
SHIP Construction EOQ FY25 Ships	-	-	-		-		0.000
SHIP Construction EOQ FY26 Ships	-	-	-		-		0.000
SHIP Construction EOQ FY27 Ships	-	-	-		-		0.000
Total: SHIP CONSTRUCTION EOQ							-
VLS Advanced Procurement							
VLS EOQ FY22 Ships	-	-	-		-		0.000
VLS EOQ FY23 Ships	-	-	-		-		0.000
VLS EOQ FY24 Ships	-	-	-		-		0.000
VLS EOQ FY25 Ships	-	-	-		-		0.000
VLS EOQ FY26 Ships	-	-	-		-		0.000
VLS EOQ FY27 Ships	-	-	-		-		0.000
Total: VLS Advanced Procurement							-
Other GFE							
EXCOMM AP FY22 Ships	-	-	-		-		0.000
AN/SRQ-4 AP FY22 Ships ⁽¹¹⁾	-	-	-		-		0.000
DVSS AP FY22 Ships ⁽¹²⁾	-	-	-		-		0.000
Total: Other GFE							-
Industrial Supplier Base							
Surface Combatant Supplier Base	-	-	-		-		0.000
Total: Industrial Supplier Base							-
Total Advance Procurement/Obligation Authority							-
Description: AP is required for shipbuilder and VLS Economic Order Quantity procurements for material items to achieve savings under the FY18-22 MYP. FY20 Congressional Addition of \$390M for long lead time material (LLTM) for FY21 ship(s)reflected in both Shipbuilder EOQ and GFE Line Items.							

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Exhibit P-10, Advance Procurement Requirements Analysis (page 2 - Budget Funding Justification): PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2122 / DDG-51
Reflects FY20 Congressional Addition of \$130M for the Surface Combatant Supplier Base. FY21 AP includes Congressional Addition of \$130M LLTM for third ship in FY22 reflected in Shipbuilder EOQ line Item. *Note: "When Required" is the number of months required before ship delivery. Footnotes: ⁽⁸⁾ Aviation Support (AVSUP) Recovery, Assist, Securing and Traversing (RAST): Shipboard installed system designed to assist LAMPS MK III and SH-60B helicopters to land on the ship flight deck and then secures the aircraft during the traversing phase to the hangar. ⁽⁹⁾ Environmental Protection Systems - Solid Waste Equipment (SWE): Provides pulpers, shredders and Compressed Melt Units (CMU) onboard ships. Equipment is used to shred metal cans and glass, and to processes paper, cardboard, and food wastes into a solution consisting of 2% solids that is discharged overboard. ⁽¹⁰⁾ Consoles, Displays & Peripherals/ Network Processing & Storage (CDP/NPS) are combat system hardware components for consoles and cabinets. CDP consoles provide the shipboard displays for control of various combat system elements including the Gun Weapon System (GWS) and Aegis Weapon System (AWS). NPS cabinets provide the computing infrastructure and information processing for the CDP consoles operations. ⁽¹¹⁾ The AN/SRQ-4 is the shipboard element of a situational awareness system that links the MH-60R helicopter with surface warships in the area. It provides command and control (C2), sensor data transfer, data link operation, and built-in-test. ⁽¹²⁾ Digital Video Surveillance System (DVSS) utilizes a high definition networked video platform in combination with embedded video analytics to provide proactive monitoring, situational awareness and response mechanisms aboard Navy ships.		

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy										Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 02: Other Warships / BSA 1: Other Warships							P-1 Line Item Number / Title: 2127 / Littoral Combat Ship (LCS)					
ID Code (A=Service Ready, B=Not Service Ready): A				Program Elements for Code B Items: N/A				Other Related Program Elements: N/A				
Line Item MDAP/MAIS Code: N/A												
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (<i>Units in Each</i>)	33	-	-	-	-	-	-	-	-	-	-	-
Gross/Weapon System Cost (<i>\$ in Millions</i>)	16,796.457	0.000	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Less PY Advance Procurement (<i>\$ in Millions</i>)	158.893	-	-	-	-	-	-	-	-	-	-	-
Less Cost To Complete (<i>\$ in Millions</i>)	455.383	-	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (<i>\$ in Millions</i>)	16,182.181	0.000	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Full Funding TOA (<i>\$ in Millions</i>)	16,182.181	-	-	-	-	-	-	-	-	-	-	-
Plus CY Advance Procurement (<i>\$ in Millions</i>)	158.893	-	-	-	-	-	-	-	-	-	-	-
Plus Cost To Complete (<i>\$ in Millions</i>)	374.993	14.000	-	44.860	-	44.860	-	-	-	-	-	-
Total Obligation Authority (<i>\$ in Millions</i>)	16,716.067	14.000	0.000	44.860	0.000	44.860	-	-	-	-	-	-
<i>(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)</i>												
Plus Outfitting and Post Delivery (<i>\$ in Millions</i>)	874.380	154.139	176.498	126.342	-	126.342	-	-	-	-	-	-
Total (<i>\$ in Millions</i>)	17,590.447	168.139	176.498	171.202	-	171.202	-	-	-	-	-	-
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	508.984	-	-	-	-	-	-	-	-	-	-	-

Description:
Provides for the design, construction, integration, and testing of the Littoral Combat Ship (LCS) including ordnance, government furnished equipment (GFE), plans and change order costs.

LCS: Operates with focused-mission packages that deploy manned and unmanned vehicles to execute a variety of missions, including anti-submarine warfare (ASW), surface warfare (SUW), and mine countermeasures (MCM). LCS also possesses inherent capabilities, regardless of the mission package installed, including intelligence, surveillance, and reconnaissance (ISR), anti-terrorism/force protection (AT/FP), air warfare self-defense, joint littoral mobility, and logistic support for movement of personnel and supplies. LCS SUW ships will also include maritime interdiction/interception operations (MIO). This relatively small, shallow-draft, high-speed surface combatant complements the U.S. Navy's Surface Fleet by operating in environments where it is impossible or undesirable to employ larger deeper-draft, multi-mission ships. LCS can deploy independently to overseas littoral regions or remain on station for extended periods of time either with a battle group or through a forward-basing arrangement. LCS will operate with Carrier Strike Groups, Surface Action Groups, or independently as dictated by the mission and environment. Additionally, LCS can operate cooperatively with the U.S. Coast Guard and Allies.

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy						Date: May 2021																																																																																																																																																																																																																	
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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1	P-1 Line Item Number / Title: 2127 / Littoral Combat Ship (LCS)
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Cost Categories	FY 2015		FY 2016		FY 2017		FY 2018		FY 2019	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Plan Costs	3	117.073	3	85.490	3	17.967	3	8.026	3	0.100
Basic Construction/Conversion		1,178.115		1,173.962		1,336.563		1,346.630		1,329.750
Change Orders		35.223		33.998		34.500		36.770		36.596
Electronics		43.431		43.812		46.183		47.014		47.954
Hull, Mechanical, and Electrical (HM&E)		11.041		11.228		11.419		11.625		11.858
Ordnance		25.401		27.270		27.290		27.304		27.514
Other Cost		100.451		74.684		131.300		89.602		117.472
Total Ship Estimate		1,510.735		1,450.444		1,605.222		1,566.971		1,571.244
Less Advance Procurement FY 2015		-		80.000		-		-		-
Less Cost to Complete FY 2019		83.686		-		-		-		-
Less Cost to Complete FY 2020		-		14.000		-		-		-
Less Cost to Complete FY 2022		-		24.860		20.000		-		-
Less Cost to Complete FY 2023		-		-		21.530		-		-
Net P-1 Funding		1,427.049		1,331.584		1,563.692		1,566.971		1,571.244

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Exhibit P-27, Ship Production Schedule: PB 2022 Navy				Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1			P-1 Line Item Number / Title: 2127 / Littoral Combat Ship (LCS)		
Ship	Shipbuilder	Fiscal Year	Contract Award	Start of Construction	Delivery Date
LCS 21	LOCKHEED MARTIN	2015	Mar 2015	Feb 2017	Oct 2021
LCS 23	LOCKHEED MARTIN	2016	Nov 2015	Sep 2017	Dec 2021
LCS 25	LOCKHEED MARTIN	2016	Mar 2016	Feb 2018	May 2022
LCS 28	AUSTAL	2017	Jun 2017	Oct 2018	Jun 2021
LCS 27	LOCKHEED MARTIN	2017	Oct 2017	Nov 2018	Oct 2022
LCS 30	AUSTAL	2017	Oct 2017	May 2019	Dec 2021
LCS 29	LOCKHEED MARTIN	2018	Sep 2018	Jun 2019	May 2023
LCS 32	AUSTAL	2018	Sep 2018	Dec 2019	Jul 2022
LCS 34	AUSTAL	2018	Sep 2018	Jun 2020	Feb 2023
LCS 36	AUSTAL	2019	Dec 2018	Jan 2021	Sep 2023
LCS 38	AUSTAL	2019	Dec 2018	Jul 2021	Apr 2024
LCS 31	LOCKHEED MARTIN	2019	Jan 2019	Jun 2020	Apr 2024

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Exhibit P-40, Budget Line Item Justification: PB 2021 Navy										Date: May 2021			
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 02: Other Warships / BSA 1: Other Warships							P-1 Line Item Number / Title: 2128 / FFG-Frigate						
ID Code (A=Service Ready, B=Not Service Ready): A				Program Elements for Code B Items: N/A				Other Related Program Elements: N/A					
Line Item MDAP/MAIS Code: N/A													
Resource Summary		Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (Units in Each)		-	1	1	1	-	1	-	-	-	-	-	-
Gross/Weapon System Cost (\$ in Millions)		0.000	1,281.177	1,053.123	1,087.900	0.000	1,087.900	-	-	-	-	-	-
Less PY Advance Procurement (\$ in Millions)		-	0.000	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (\$ in Millions)		0.000	1,281.177	1,053.123	1,087.900	0.000	1,087.900	-	-	-	-	-	-
Plus CY Advance Procurement (\$ in Millions)		-	-	-	69.100	-	69.100	-	-	-	-	-	-
Total Obligation Authority (\$ in Millions)		0.000	1,281.177	1,053.123	1,157.000	0.000	1,157.000	-	-	-	-	-	-
(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)													
Plus Outfitting and Post Delivery (\$ in Millions)		-	0.000	-	-	-	-	-	-	-	-	-	-
Total (\$ in Millions)		-	1,281.177	1,053.123	1,157.000	-	1,157.000	-	-	-	-	-	-
Gross/Weapon System Unit Cost (\$ in Millions)		-	1,281.177	1,053.123	1,087.900	-	1,087.900	-	-	-	-	-	-
Description:													
The Guided Missile Frigate (FFG 62 Class) is a more lethal and survivable multi-mission small surface combatant. With the FFG 62 Class, the Navy will maximize the small surface combatant capabilities in the Anti-Surface Warfare (SUW), Anti-Submarine Warfare (ASW), Electronic Warfare/Information Operations (EW/IO), Air Warfare (AW) mission areas, and survivability while keeping the ship affordable and as a part of a "high-low" mix of surface ships. The FFG 62 Class Capability Development Document was JROC approved in Feb 2019 and completed Milestone B, an informed Independent Cost Estimate, and award of the Detail Design & Construction contract in April 2020. As of October 2020, the previously known FFG(X) program was designated the FFG 62 Class.													
Characteristics:													
-													
Length Overall 496 ft													
Beam 65 ft													
Displacement 7406 tons													
Draft 18 ft													
Production Status:													
FFG 62		FFG 63		FFG 64									
Contract Award Date		Apr 2020		May 2021		Apr 2022							
Months to Completion													
a) Award to Delivery		72 months		68 months		74 months							
b) Construction Start to Delivery		48 months		51 months		62 months							
Delivery Date		Apr 2026		Jan 2027		Jun 2028							
Completion Of Fitting Out		Oct 2026		Mar 2027		Aug 2028							
Obligation Work Limit Date		Sep 2027		Feb 2028		Jul 2029							
Design Schedule				Start / Issue		Complete / Response		Reissue		Reissue Complete / Response			
Issue Date for TLR				Feb 2017		Oct 2017							
Issue Date for TLS				N/A		N/A							

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy				Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 02: Other Warships / BSA 1: Other Warships			P-1 Line Item Number / Title: 2128 / FFG-Frigate		
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: N/A		Other Related Program Elements: N/A	
Line Item MDAP/MAIS Code: N/A					
<u>Design Schedule</u>		<u>Start / Issue</u>	<u>Complete / Response</u>	<u>Reissue</u>	<u>Reissue Complete / Response</u>
Preliminary Design		N/A	N/A		
Contract Design		Feb 2018	May 2019		
Detail Design		Apr 2020	N/A		
Request for Proposals		Jun 2019	Sep 2019		
Design Agent					
<u>Classification of Cost Estimate:</u> Class C Estimate					

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy					Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1			P-1 Line Item Number / Title: 2128 / FFG-Frigate			
Cost Categories ^(†) indicates the presence of a P-8a	FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Plan Costs	1	206.706	1	38.744	1	131.461
Basic Construction/Conversion		651.080		598.880		584.122
Change Orders		31.650		24.679		23.021
Electronics ^(†)		266.830		264.281		230.783
Hull, Mechanical, and Electrical (HM&E) ^(†)		26.930		22.531		17.389
Ordnance ^(†)		49.981		55.445		52.274
Other Cost		48.000		48.563		48.850
Total Ship Estimate		1,281.177		1,053.123		1,087.900
Net P-1 Funding		1,281.177		1,053.123		1,087.900
Remarks: The changes from FY 2021 to the FY 2022 request are due to the completed Component Cost Position and the Detail Design and Construction contract award. Plan Costs: Changes from FY21 to FY22 account for program Component Cost Position and establishment of an FFG 62 Land Based Engineering Site (LBES). Change Orders: FY22 increased to account for technical baseline changes pursuant to Buy American.						

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Exhibit P-27, Ship Production Schedule: PB 2022 Navy				Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1			P-1 Line Item Number / Title: 2128 / FFG-Frigate		
Ship	Shipbuilder	Fiscal Year	Contract Award	Start of Construction	Delivery Date
FFG 62	Fincantieri Marinette Marine	2020	Apr 2020	Apr 2022	Apr 2026
FFG 63	Fincantieri Marinette Marine	2021	May 2021	Oct 2022	Jan 2027
FFG 64	Fincantieri Marinette Marine	2022	Apr 2022	Apr 2023	Jun 2028

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy				Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1			P-1 Line Item Number / Title: 2128 / FFG-Frigate			
Electronics	FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
P-35 Items						
Enterprise Air Surveillance Radar (EASR)	1	65.995	1	59.165	1	55.307
AEGIS Weapon System (AWS)	1	44.454	1	45.091	1	36.965
Anti-Submarine Warfare (ASW) Combat Suite	1	45.787	1	37.535	1	33.221
Surface Electronic Warfare Improvement Program (SEWIP) BLK II (SLQ-32(V)6)	1	13.989	1	12.573	1	12.520
Tactical COMINT System- Spectral	1	11.709	1	14.641	1	13.480
Cooperative Engagement Capability (CEC)	1	7.582	1	10.080	1	6.512
Identification Friend or Foe (IFF) UPX-29	1	8.043	1	5.890	1	5.715
External Communication Suite	1	7.420	1	7.413	1	10.193
Navy Advanced Extremely High Frequency (AEHF) Multiband Terminal (NMT)(AN/WSC-9(V)1)	1	6.375	1	9.150	1	6.207
Consolidated Afloat Network Enterprise System (CANES)	1	6.672	1	9.904	1	6.771
Network Tactical Common Data Link (NTCDL) Variant B	1	6.105	1	7.155	1	6.094
Inertial Navigation System (INS) AN/WSN-7(V)1	1	5.816	1	5.813	1	5.824
Situation Awareness EO/IR	1	4.570	1	3.696	1	3.753
NIXIE AN/SLQ-25E	1	3.340	1	2.502	1	2.552
P-35 Items Subtotal		237.857		230.608		205.114
Major Items						
Next Generation Surface Search Radar (NGSSR)	2	3.249	2	3.314	2	3.047
OA-9277A	1	2.066	1	2.064	0	-
Moriah Wind System (MWS) (AN/SMQ-13)	1	1.525	1	1.513	1	1.235
Tactical Variant Switch (TVS) (AN/USQ-155(V))	1	1.704	1	1.737	1	1.772
Major Items Subtotal		8.544		8.628		6.054
Other Cost Elements						
Other Electronics	1	20.429	1	25.045	1	19.615
Other Cost Elements Subtotal		20.429		25.045		19.615
Total Electronics		266.830		264.281		230.783
Remarks: Cost increase in FY21 AWS hardware due to resolution of obsolescence centered on the Common Display Systems and Common Processing Systems. FY21 costs increases for CEC, and CANES due to updated non-recurring scope related to detail design, integration, test, and cybersecurity. FY21 NTCDL cost increase based on estimate of quantity buy pricing on base year of hardware production contract. Tactical COMINT - Spectral costs have increased in FY21 due to on-going design maturation. FY21 NMT costs have increased due to loss of quantity buy opportunity. Beginning in FY22, the OA-9772A will be included in the External Communications Suite costs.						

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1		P-1 Line Item Number / Title: 2128 / FFG-Frigate
Other Electronics include C4I System Innovation Facility (C-SIF) / Technical Integration Facility (TIF), Advanced Training Domain (ATD), Navy Electronic Chart Display Information System (ECDIS), Global Positioning System (GPS) Based Positioning Navigation and Timing Service (GPNTS), Antennas, and various other systems.		

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy **Date:** May 2021

Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1 **P-1 Line Item Number / Title:** 2128 / FFG-Frigate

Hull, Mechanical, and Electrical (HM&E)	FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
P-35 Items						
Aircraft Ship Integrated Securing and Traversing System (ASIST)	1	10.106	1	10.306	1	6.071
Internal Communication (IC) Voice	1	3.422	1	3.440	1	3.078
P-35 Items Subtotal		13.528		13.746		9.149
Major Items						
Advanced Flight Deck Lighting System (AFDLS)	1	2.413	1	2.237	1	2.220
Unclassified Video System (UVS)	1	2.248	1	1.650	1	1.360
Interior Wireless Communication System (IWCS)	1	2.500	1	1.580	1	1.402
Major Items Subtotal		7.161		5.467		4.982
Other Cost Elements						
Other HM&E	1	6.241	1	3.318	1	3.258
Other Cost Elements Subtotal		6.241		3.318		3.258
Total Hull, Mechanical, and Electrical (HM&E)		26.930		22.531		17.389

Remarks:

FY20 and FY21 ASIST cost increase due to continuation of non-recurring integration/test scope and subsequent update to documentation. Increase in FY20 Other HM&E costs due to nonrecurring costs previously unidentified for multiple systems including the Advanced Stabilized Glide Slope Indicator (ASGSI), Horizon Reference Set (HRS), and Advance Damage Control System (ADCS).

Other HM&E includes 7M Rigid Hull Inflatable Boat (RHIB), Horizon Reference Set (HRS), Enhanced Maritime Biological Detection (EMBD), Plastic Shredder, and other systems.

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy **Date:** May 2021

Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1 **P-1 Line Item Number / Title:** 2128 / FFG-Frigate

Ordnance	FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
P-35 Items						
MK 41 Vertical Launch System (VLS)- 32 Cell	1	19.681	1	26.040	1	22.700
MK 48 Gun Weapon System (GWS)	1	14.838	1	13.622	1	12.785
Rolling Airframe Missile (RAM) Guided Missile Launching System- 21 Cell	1	12.254	1	12.453	1	13.205
P-35 Items Subtotal		46.773		52.115		48.690
Major Items						
MK 53 Decoy Launching System (DLS) (Nulka)- 4 Launcher	1	1.359	1	1.396	1	1.402
Major Items Subtotal		1.359		1.396		1.402
Other Cost Elements						
Other Ordnance	1	1.849	1	1.934	1	2.182
Other Cost Elements Subtotal		1.849		1.934		2.182
Total Ordnance		49.981		55.445		52.274

Remarks:

FY20 and FY21 VLS costs have increased as the system is currently being procured under a Unilateral Contract Agreement until contract negotiations are completed. FY20 RAM costs have increased based on timing of procurements with other programs. FY22 RAM cost increase over inflation from FY21 due to changes in ship profile and loss of quantity buy opportunity to meet ship construction schedule.

Other Ordnance includes Over the Horizon (OTH) Missile Launcher, Portable Ordnance Handling Equipment (POHE) and Machine Gun Mounts.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2128 / FFG-Frigate			
Equipment Item: Enterprise Air Surveillance Radar (EASR)						PARM Code: PEO IWS 2A	
P-35 Category	FY 2020		FY 2021		FY 2022		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
Major Hardware	1	48.557	1	43.129	1	42.148	
System Engineering		3.048		3.083		2.375	
Technical Engineering Services		2.360		2.173		2.168	
Other Costs		12.030		10.780		8.616	
Total	1	65.995	1	59.165	1	55.307	

Description:
Enterprise Air Surveillance Radar (EASR) is the next generation S-band air search radar with 3-D search capability supporting Air Warfare (AW) for self- and local area defense; Surface Warfare (SUW) for Fast Attack Craft (FAC), Fast Inshore Attack Craft (FIAC); Asymmetric / Force Protection; Electromagnetic Maneuver Warfare. The EASR consists of arrays, below deck gear support equipment computer program for radar processing and radar control.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	FFG 62	Raytheon	Various	Aug 2020	Option	1	48.557
FY 2021	FFG 63	Raytheon	Various	Aug 2021	Option	1	43.129
FY 2022	FFG 64	Raytheon	Various	Aug 2024	Option	1	42.148

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	FFG 62	Apr 2026	40	30	Jun 2020
FY 2021	FFG 63	Jan 2027	40	29	Apr 2021
FY 2022	FFG 64	Jun 2028	40	29	Sep 2022

Competition/Second Source Initiatives:
N/A

Remarks:
Both FFG 62 Class and CVN (starting at CVN 79) will have fixed faced 9-RMA EASRs, however the software integration to the Aegis Weapon System (AWS) will be different. FFG will integrate EASR into the AEGIS Combat System FFG(X) Baseline.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2128 / FFG-Frigate			
Equipment Item: AEGIS Weapon System (AWS)						PARM Code: PEO IWS 8.0	
P-35 Category	FY 2020		FY 2021		FY 2022		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
Major Hardware	1	14.413	1	14.646	1	14.481	
System Engineering		1.993		1.594		0.690	
Technical Engineering Services		1.763		1.477		1.143	
Other Costs		26.285		27.374		20.651	
Total	1	44.454	1	45.091	1	36.965	

Description:
 The AEGIS Weapon System (AWS, formerly Frigate Weapon System (FWS)) is an integrated combat management system that provides doctrine, management, control, and display functionality for FFG Combat System (FCS) missions as well as command and control functionality for all warfare areas. The AWS comprises: Command and Decision (C&D) System, Combat System Computing Infrastructure (CSCI), Shipboard Gridlock System/Automatic Correlation (SGS/AC), AEGIS Display System (ADS), Weapons Control System (WCS), Operational Readiness Training System (ORTS), Mission Planner (MP), Vehicle Control Domain (VCD), and resource management for the Enterprise Air Surveillance Radar.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	FFG 62	Various	C/FFP	Mar 2021	New	1	14.413
FY 2021	FFG 63	TBD	C/FFP	TBD		1	14.646
FY 2022	FFG 64	TBD	C/CPFF	TBD		1	14.481

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	FFG 62	Apr 2026	37	24	Mar 2021
FY 2021	FFG 63	Jan 2027	37	24	Dec 2021
FY 2022	FFG 64	Jun 2028	37	24	May 2023

Competition/Second Source Initiatives:
 AWS will be procured under the competitive combat system contract run by IWS 1.0 with award in FY 2021.

Remarks:
 FY21 AWS cost increase due to required resolution of obsolescence centered on the Common Display System and Common Processing System.

Other costs represent the system integration and test for the entire combat system (not just the AWS). The change from FWS to AEGIS CS Frigate baseline occurred to align with AEGIS Baseline 10 (BL 10). The Frigate AWS is an AEGIS BL 10 derivative.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy					Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1			P-1 Line Item Number / Title: 2128 / FFG-Frigate			
Equipment Item: Anti-Submarine Warfare (ASW) Combat Suite					PARM Code: PEO IWS 5.0	
P-35 Category	FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Major Hardware	1	30.379	1	24.118	1	22.565
System Engineering		6.292		5.747		3.731
Technical Engineering Services		2.990		2.914		2.379
Other Costs		6.126		4.756		4.546
Total	1	45.787	1	37.535	1	33.221

Description:
The Anti-Submarine Warfare Combat Suite provides surface warships with an integrated undersea/anti-submarine warfare detection, localization, classification, and targeting ability. It includes AN/SQQ-89 (V)16, Undersea Warfare Decision Support System (USW-DSS), Multi Function Towed Array (MFTA), expendable bathythermograph (XBT) launcher LM-48, and the fathometer. Variable Depth Sonar was the selected sensor per the awarded Detail Design & Construction contract.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	FFG 62	Various	Various	Various		1	30.379
FY 2021	FFG 63	Various	Various	Various		1	24.118
FY 2022	FFG 64	Various	Various	Various		1	22.565

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	FFG 62	Apr 2026	40	24	Dec 2020
FY 2021	FFG 63	Jan 2027	40	24	Sep 2021
FY 2022	FFG 64	Jun 2028	40	24	Feb 2023

Competition/Second Source Initiatives:
N/A

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2128 / FFG-Frigate			
Equipment Item: Surface Electronic Warfare Improvement Program (SEWIP) BLK II (SLQ-32(V)6)						PARM Code: PEO IWS 2E	
P-35 Category	FY 2020		FY 2021		FY 2022		
	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>	
Major Hardware	1	10.409	1	10.311	1	10.977	
System Engineering		0.966		0.969		0.389	
Technical Engineering Services		0.108		0.111		0.114	
Other Costs		2.506		1.182		1.040	
Total	1	13.989	1	12.573	1	12.520	

Description:
The AN/SLQ-32 SEWIP Block II is a scalable Electronic Warfare enterprise suite that interfaces to the AWS. It provides enhanced shipboard Electronic Warfare (EW) for early detection, analysis, threat warning and soft kill protection from anti-ship missiles.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity <i>(Each)</i>	Unit Cost <i>(\$ M)</i>
FY 2020	FFG 62	Various	Various	Mar 2021	Various	1	10.409
FY 2021	FFG 63	Various	Various	Mar 2021	Various	1	10.311
FY 2022	FFG 64	Various	Various	Mar 2021	Various	1	10.977

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	FFG 62	Apr 2026	40	29	Jul 2020
FY 2021	FFG 63	Jan 2027	40	29	Apr 2021
FY 2022	FFG 64	Jun 2028	40	29	Sep 2022

Competition/Second Source Initiatives:
A phased delivery of equipment will be used for FFG 62 to mitigate risk due to award later than required.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2128 / FFG-Frigate			
Equipment Item: Tactical COMINT System- Spectral						PARM Code: PMW 120	
P-35 Category	FY 2020		FY 2021		FY 2022		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
Major Hardware	1	9.885	1	12.363	1	12.363	
System Engineering		0.477		0.450		0.360	
Technical Engineering Services		0.678		0.880		0.262	
Other Costs		0.669		0.948		0.495	
Total	1	11.709	1	14.641	1	13.480	

Description:
Spectral is the next generation information warfare weapons system, enhancing Ship Signal Exploitation System (SSES) capabilities that detects, classifies, and tracks SOI (Signals of Interest) used in the Tasking, Collection, Processing, Exploitation and Dissemination (TCPED) process. Spectral is scalable, mission configurable, modular, and remotable, rapidly responding to new threats and/or capabilities. A integrated capability enables data sharing with other systems to support Electromagnetic Maneuver Warfare (EMW/Integrated Fires (IF)).

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	FFG 62	TBD	TBD	TBD		1	9.885
FY 2021	FFG 63	TBD	TBD	TBD		1	12.363
FY 2022	FFG 64	TBD	TBD	TBD		1	12.363

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	FFG 62	Apr 2026	40	16	Aug 2021
FY 2021	FFG 63	Jan 2027	40	16	May 2022
FY 2022	FFG 64	Jun 2028	40	16	Oct 2023

Competition/Second Source Initiatives:
N/A

Remarks:
FY21 cost estimate increased due to design iterations leading to higher production and integration costs.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																																
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2128 / FFG-Frigate																																		
Equipment Item: Cooperative Engagement Capability (CEC)						PARM Code: PEO IWS 6.0																																
P-35 Category	FY 2020		FY 2021		FY 2022																																	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																																
Major Hardware	1	5.110	1	6.503	1	4.680																																
System Engineering		0.740		0.990		0.401																																
Technical Engineering Services		0.514		1.091		0.536																																
Other Costs		1.218		1.496		0.895																																
Total	1	7.582	1	10.080	1	6.512																																
Description: The AN/USG-2B Cooperative Engagement Capability (CEC) system provides real time integration of fire control quality sensor data into a single composite data source, which can be used by multiple CEC ships and airborne units for direct and remote missile engagements. CEC significantly improves battle force Air Warfare (AW) capability by coordinating all force AW sensors into a single real time, fire control quality composite track picture. FFG(X) technical baseline for CEC was reverted to legacy phased array antenna as schedule risk mitigation, incurring additional costs in FY21 for engineering and integration.																																						
Contract Data: <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Prime Contractor</th> <th>Contract Method/Type</th> <th>Award Date</th> <th>New/Option</th> <th>Quantity (Each)</th> <th>Unit Cost (\$ M)</th> </tr> <tr> <td>FY 2020</td> <td>FFG 62</td> <td>DRS Technologies & Raytheon</td> <td>Various</td> <td>Various</td> <td>Option</td> <td>1</td> <td>5.110</td> </tr> <tr> <td>FY 2021</td> <td>FFG 63</td> <td>DRS Technologies & Raytheon</td> <td>Various</td> <td>Various</td> <td>Option</td> <td>1</td> <td>6.503</td> </tr> <tr> <td>FY 2022</td> <td>FFG 64</td> <td>DRS Technologies & Raytheon</td> <td>Various</td> <td>Various</td> <td>Option</td> <td>1</td> <td>4.680</td> </tr> </table>							Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2020	FFG 62	DRS Technologies & Raytheon	Various	Various	Option	1	5.110	FY 2021	FFG 63	DRS Technologies & Raytheon	Various	Various	Option	1	6.503	FY 2022	FFG 64	DRS Technologies & Raytheon	Various	Various	Option	1	4.680
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																															
FY 2020	FFG 62	DRS Technologies & Raytheon	Various	Various	Option	1	5.110																															
FY 2021	FFG 63	DRS Technologies & Raytheon	Various	Various	Option	1	6.503																															
FY 2022	FFG 64	DRS Technologies & Raytheon	Various	Various	Option	1	4.680																															
Delivery Date: <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Earliest Ship Delivery Date</th> <th>Months Required Before Delivery</th> <th>Production Leadtime</th> <th>Required Award Date</th> </tr> <tr> <td>FY 2020</td> <td>FFG 62</td> <td>Apr 2026</td> <td>21</td> <td>18</td> <td>Jan 2023</td> </tr> <tr> <td>FY 2021</td> <td>FFG 63</td> <td>Jan 2027</td> <td>21</td> <td>18</td> <td>Oct 2023</td> </tr> <tr> <td>FY 2022</td> <td>FFG 64</td> <td>Jun 2028</td> <td>21</td> <td>18</td> <td>Mar 2025</td> </tr> </table>							Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date	FY 2020	FFG 62	Apr 2026	21	18	Jan 2023	FY 2021	FFG 63	Jan 2027	21	18	Oct 2023	FY 2022	FFG 64	Jun 2028	21	18	Mar 2025								
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date																																	
FY 2020	FFG 62	Apr 2026	21	18	Jan 2023																																	
FY 2021	FFG 63	Jan 2027	21	18	Oct 2023																																	
FY 2022	FFG 64	Jun 2028	21	18	Mar 2025																																	
Competition/Second Source Initiatives: N/A																																						
Remarks: Initial cost decrease in FY20 due to FFG 62 Class CEC technical baseline reversion to legacy phased array antenna. Cost increase in FY21 for additional engineering and integration efforts required for the legacy phased array antenna.																																						

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1	P-1 Line Item Number / Title: 2128 / FFG-Frigate
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Equipment Item: Identification Friend or Foe (IFF) UPX-29	PARM Code: PMA 213
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P-35 Category	FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Major Hardware	1	6.443	1	5.243	1	5.280
System Engineering		0.572		0.387		0.217
Technical Engineering Services		0.845		0.210		0.168
Other Costs		0.183		0.050		0.050
Total	1	8.043	1	5.890	1	5.715

Description:

The AN/UPX-29(V) provides a centralized identification system that operates independently to transmit interrogations, detect transponder replies, and process the resulting information for use by a ship's operators and combat weapons system computers. The system provides Modes 1, 2, 3A, C and provides secure, positive ID of Mode 4, Mode S and Mode 5 targets.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	FFG 62	TBD	Various	Various		1	6.443
FY 2021	FFG 63	TBD	Various	Various		1	5.243
FY 2022	FFG 64	TBD	Various	Various		1	5.280

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	FFG 62	Apr 2026	40	24	Dec 2020
FY 2021	FFG 63	Jan 2027	40	24	Sep 2021
FY 2022	FFG 64	Jun 2028	40	24	Feb 2023

Competition/Second Source Initiatives:

N/A

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2128 / FFG-Frigate			
Equipment Item: External Communication Suite						PARM Code: PMW 760	
P-35 Category	FY 2020		FY 2021		FY 2022		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
Major Hardware	1	6.335	1	6.333	1	9.113	
System Engineering		0.268		0.266		0.231	
Technical Engineering Services		0.478		0.476		0.511	
Other Costs		0.339		0.338		0.338	
Total	1	7.420	1	7.413	1	10.193	
Description: External Communication Suite (EXCOMM) is a software programmable tactical radio (with embedded Information Security [INFOSEC]) that provides interoperable Line of Sight/Beyond Line of Sight (LOS/BLOS) C4I capabilities to the fleet. For FY22, the OA-9227A antenna now included in the baseline EXCOMM DMR HFDAG system.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	FFG 62	Various	C/FFP	TBD		1	6.335
FY 2021	FFG 63	Various	C/FFP	TBD		1	6.333
FY 2022	FFG 64	Various	C/FFP	TBD		1	9.113
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	FFG 62	Apr 2026	40	24	Dec 2020		
FY 2021	FFG 63	Jan 2027	40	24	Sep 2021		
FY 2022	FFG 64	Jun 2028	40	24	Feb 2023		
Competition/Second Source Initiatives: N/A							
Remarks: Cost increase in FY22 due to the OA-9227A antenna (a major electronics item in FY20 and FY21) now included in the baseline EXCOMM DMR HFDAG system.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2128 / FFG-Frigate			
Equipment Item: Navy Advanced Extremely High Frequency (AEHF) Multiband Terminal (NMT)(AN/WSC-9(V)1))						PARM Code: PMW 170	
P-35 Category	FY 2020		FY 2021		FY 2022		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
Major Hardware	1	5.448	1	8.239	1	5.522	
System Engineering		0.052		0.052		0.053	
Technical Engineering Services		0.243		0.231		0.240	
Other Costs		0.632		0.628		0.392	
Total	1	6.375	1	9.150	1	6.207	

Description:
The Navy Multiband Terminal (NMT) is the Navy's next generation ground terminal for military protected and wideband satellite communications, providing voice, video and data communications. NMT supports protected strategic/tactical warfare communications down to housekeeping and humanitarian (quality of life) communications for the Sailor and maintains backwards compatibility with legacy waveforms and systems while ensuring access to new and future MILSATCOM capabilities (Advanced Extremely High Frequency [AEHF], Wideband Global Satellite [WGS], and Enhanced Polar Satellite [EPS]).

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	FFG 62	Raytheon	C/FFP	Dec 2020	Option	1	5.448
FY 2021	FFG 63	Raytheon	C/FFP	Dec 2021	Option	1	8.239
FY 2022	FFG 64	Raytheon		Dec 2022	Option	1	5.522

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	FFG 62	Apr 2026	21	24	Jul 2022
FY 2021	FFG 63	Jan 2027	21	24	Apr 2023
FY 2022	FFG 64	Jun 2028	21	24	Sep 2024

Competition/Second Source Initiatives:
N/A

Remarks:
FY21 costs have increased due to a loss of quantity buy opportunity resulting from timing of NMT contract award date and production lead time to meet ship construction schedule. FY20 and FY22 costs reflect quantity buy pricing.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2128 / FFG-Frigate			
Equipment Item: Consolidated Afloat Network Enterprise System (CANES)						PARM Code: PMW 160	
P-35 Category	FY 2020		FY 2021		FY 2022		
	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>	
Major Hardware	1	4.332	1	4.332	1	4.338	
System Engineering		1.785		3.367		2.049	
Technical Engineering Services		0.426		1.258		0.264	
Other Costs		0.129		0.947		0.120	
Total	1	6.672	1	9.904	1	6.771	
Description: CANES is a single, highly scalable network, computing, and services infrastructure supporting mission area application hosting and service delivery across a diverse set of Communities of Interest (COIs). It provides Navy tactical/nontactical information environment and infrastructure necessary to enable hosting, extended services reach back & reach forward and relay functions.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity <i>(Each)</i>	Unit Cost <i>(\$ M)</i>
FY 2020	FFG 62	TBD	TBD	TBD		1	4.332
FY 2021	FFG 63	TBD	TBD	TBD		1	4.332
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	FFG 62	Apr 2026	18	15	Jul 2023		
FY 2021	FFG 63	Jan 2027	18	15	Apr 2024		
Competition/Second Source Initiatives: N/A							
Remarks: FY21 cost increase due to update on non-recurring engineering and cybersecurity support as result of Frigate Architecture Framework solution for boundary defense.							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2128 / FFG-Frigate			
Equipment Item: Network Tactical Common Data Link (NTCDL) Variant B						PARM Code: PMW 170	
P-35 Category	FY 2020		FY 2021		FY 2022		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
Major Hardware	1	5.117	1	6.314	1	5.421	
System Engineering		0.297		0.285		0.228	
Technical Engineering Services		0.255		0.215		0.172	
Other Costs		0.436		0.341		0.273	
Total	1	6.105	1	7.155	1	6.094	

Description:
NTCDL provides the ability to transmit/receive real-time ISR data simultaneously from multiple sources (air, surface, subsurface, and man-portable) and exchange command and control information (voice, data, imagery, and full-motion video) across dissimilar joint, service, coalition, and civil networks. NTCDL provides warfighters the capability to support multiple, simultaneous, networked operations with in-service CDL equipped aircraft (e.g., F/A-18, P-3, and MH-60R) in addition to next-generation manned and unmanned platforms (e.g., Fire Scout). The NTCDL Variant planned for the FFG(X) is comprised of 4 Transmit and 4 Receive Phased Array Antennas (PAAs).

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	FFG 62	TBD	TBD	TBD		1	5.117
FY 2021	FFG 63	TBD	TBD	TBD		1	5.117
FY 2022	FFG 64	TBD	TBD	TBD		1	5.421

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	FFG 62	Apr 2026	21	12	Jul 2023
FY 2021	FFG 63	Jan 2027	21	12	Apr 2024
FY 2022	FFG 64	Jun 2028	21	12	Sep 2025

Competition/Second Source Initiatives:
N/A

Remarks:
FY21 cost increase based on estimate of quantity buy pricing on base year of hardware production contract.

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2128 / FFG-Frigate			
Equipment Item: Inertial Navigation System (INS) AN/WSN-7(V)1						PARM Code: PEO IWS 6.0	
P-35 Category	FY 2020		FY 2021		FY 2022		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
Major Hardware	1	5.035	1	5.035	1	5.035	
System Engineering		0.088		0.104		0.120	
Technical Engineering Services		0.534		0.534		0.430	
Other Costs		0.159		0.140		0.239	
Total	1	5.816	1	5.813	1	5.824	
Description: The AN/WSN-7 Ring Laser Gyro Navigation (RLGN) System calculates and disseminates own ship's position, velocity and attitude (heading, roll and pitch) data outputs. The AN/WSN-7 RLGN System provides real time navigation data to use by Navigation & combat systems.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	FFG 62	Sperry Marine	C/CPFF	Sep 2020	New	1	5.035
FY 2021	FFG 63	Sperry Marine	TBD	TBD		1	5.035
FY 2022	FFG 64	Sperry Marine	TBD	TBD		1	5.035
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	FFG 62	Apr 2026	40	18	Jun 2021		
FY 2021	FFG 63	Jan 2027	40	18	Mar 2022		
FY 2022	FFG 64	Jun 2028	40	18	Aug 2023		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2128 / FFG-Frigate			
Equipment Item: Situation Awareness EO/IR						PARM Code: PEO IWS 2E	
P-35 Category	FY 2020		FY 2021		FY 2022		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
Major Hardware	1	2.906	1	2.865	1	2.857	
System Engineering		0.555		0.463		0.383	
Technical Engineering Services		0.097		0.099		0.085	
Other Costs		1.012		0.269		0.428	
Total	1	4.570	1	3.696	1	3.753	
Description: Situation Awareness Electro-Optical/Infrared (EO/IR) system provides the capability to classify, identify and assist in determining intent of conventional, asymmetrical and advanced contacts/threats. EO/IR improves Situational Awareness (SA) and supports Anti-terrorism/Force Protection (AT/FP), Intelligence, Surveillance and Reconnaissance (ISR), Navigation (NAV), Intelligence (INTEL), Anti-Surface Warfare (SUW), Air Warfare (AW) and Anti-Ship Missile Defense (ASMD) missions while providing Long Range HD Visible/Infra-Red and Laser Range Finding sensors and capability for controlling, displaying and recording sensor imagery							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	FFG 62	Ball Aerospace & Technologies	C/FFP	TBD	New	1	2.906
FY 2021	FFG 63	Ball Aerospace & Technologies	C/FFP	TBD		1	2.865
FY 2022	FFG 64	Ball Aerospace & Technologies	C/FP	TBD		1	2.857
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	FFG 62	Apr 2026	40	12	Dec 2021		
FY 2021	FFG 63	Jan 2027	40	12	Sep 2022		
FY 2022	FFG 64	Jun 2028	40	12	Feb 2024		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2128 / FFG-Frigate			
Equipment Item: NIXIE AN/SLQ-25E						PARM Code: PMS 415	
P-35 Category	FY 2020		FY 2021		FY 2022		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
Major Hardware	1	1.877	1	1.828	1	2.014	
System Engineering		0.268		0.123		0.059	
Technical Engineering Services		0.442		0.323		0.239	
Other Costs		0.753		0.228		0.240	
Total	1	3.340	1	2.502	1	2.552	
Description: The Torpedo Countermeasures Transmitting Set AN/SLQ-25E (NIXIE) is a passive, electro-acoustic decoy system used to provide deceptive countermeasures against acoustic homing torpedoes.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	FFG 62	TBD	TBD	TBD		1	1.877
FY 2021	FFG 63	TBD	TBD	TBD		1	1.828
FY 2022	FFG 64	TBD	TBD	TBD		1	2.014
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	FFG 62	Apr 2026	40	18	Jun 2021		
FY 2021	FFG 63	Jan 2027	40	18	Mar 2022		
FY 2022	FFG 64	Jun 2028	40	18	Aug 2023		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																																
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2128 / FFG-Frigate																																		
Equipment Item: Aircraft Ship Integrated Securing and Traversing System (ASIST)						PARM Code: PMA 251																																
P-35 Category	FY 2020		FY 2021		FY 2022																																	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																																
Major Hardware	1	4.974	1	5.033	1	5.093																																
System Engineering		0.516		0.712		0.548																																
Technical Engineering Services		2.970		2.702		0.361																																
Other Costs		1.646		1.859		0.069																																
Total	1	10.106	1	10.306	1	6.071																																
Description: Aircraft Ship Integrated Secure and Traverse (ASIST) provides safe recovery, securing, straightening, traversing, stowing, and launching of MH-60R and MQ-8C.																																						
Contract Data: <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Prime Contractor</th> <th>Contract Method/Type</th> <th>Award Date</th> <th>New/Option</th> <th>Quantity (Each)</th> <th>Unit Cost (\$ M)</th> </tr> <tr> <td>FY 2020</td> <td>FFG 62</td> <td>INDAL</td> <td>C/FFP</td> <td>Sep 2021</td> <td>New</td> <td>1</td> <td>4.974</td> </tr> <tr> <td>FY 2021</td> <td>FFG 63</td> <td>INDAL</td> <td>C/FFP</td> <td>TBD</td> <td></td> <td>1</td> <td>5.033</td> </tr> <tr> <td>FY 2022</td> <td>FFG 64</td> <td>INDAL</td> <td>C/FFP</td> <td>TBD</td> <td></td> <td>1</td> <td>5.093</td> </tr> </table>							Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2020	FFG 62	INDAL	C/FFP	Sep 2021	New	1	4.974	FY 2021	FFG 63	INDAL	C/FFP	TBD		1	5.033	FY 2022	FFG 64	INDAL	C/FFP	TBD		1	5.093
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																															
FY 2020	FFG 62	INDAL	C/FFP	Sep 2021	New	1	4.974																															
FY 2021	FFG 63	INDAL	C/FFP	TBD		1	5.033																															
FY 2022	FFG 64	INDAL	C/FFP	TBD		1	5.093																															
Delivery Date: <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Earliest Ship Delivery Date</th> <th>Months Required Before Delivery</th> <th>Production Leadtime</th> <th>Required Award Date</th> </tr> <tr> <td>FY 2020</td> <td>FFG 62</td> <td>Apr 2026</td> <td>21</td> <td>24</td> <td>Jul 2022</td> </tr> <tr> <td>FY 2021</td> <td>FFG 63</td> <td>Jan 2027</td> <td>21</td> <td>24</td> <td>Apr 2023</td> </tr> <tr> <td>FY 2022</td> <td>FFG 64</td> <td>Jun 2028</td> <td>21</td> <td>24</td> <td>Sep 2024</td> </tr> </table>							Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date	FY 2020	FFG 62	Apr 2026	21	24	Jul 2022	FY 2021	FFG 63	Jan 2027	21	24	Apr 2023	FY 2022	FFG 64	Jun 2028	21	24	Sep 2024								
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FY 2020	FFG 62	Apr 2026	21	24	Jul 2022																																	
FY 2021	FFG 63	Jan 2027	21	24	Apr 2023																																	
FY 2022	FFG 64	Jun 2028	21	24	Sep 2024																																	
Competition/Second Source Initiatives: N/A																																						
Remarks: FY20 cost increase reflects updated cost estimate based on first of class qualifications and testing. FY21 cost increase due to continuation of non-recurring integration/test scope and subsequent update to documentation.																																						

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2128 / FFG-Frigate			
Equipment Item: Internal Communication (IC) Voice						PARM Code: PEO SHIPS AM	
P-35 Category	FY 2020		FY 2021		FY 2022		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
Major Hardware	1	1.758	1	1.811	1	1.793	
System Engineering		0.185		0.166		0.174	
Technical Engineering Services		0.853		0.818		0.622	
Other Costs		0.626		0.645		0.489	
Total	1	3.422	1	3.440	1	3.078	

Description:
IC Voice provides shipboard internal communication consisting of the announcing system, telephones, and wireless radios.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	FFG 62	TBD	TBD	TBD		1	1.758
FY 2021	FFG 63	TBD	TBD	TBD		1	1.811
FY 2022	FFG 64	TBD	TBD	TBD		1	1.793

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2020	FFG 62	Apr 2026	40	12	Dec 2021
FY 2021	FFG 63	Jan 2027	40	12	Sep 2022
FY 2022	FFG 64	Jun 2028	40	12	Feb 2024

Competition/Second Source Initiatives:
N/A

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																																
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2128 / FFG-Frigate																																		
Equipment Item: MK 41 Vertical Launch System (VLS)- 32 Cell						PARM Code: PEO IWS 3L																																
P-35 Category	FY 2020		FY 2021		FY 2022																																	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																																
Major Hardware	1	16.129	1	22.488	1	21.849																																
System Engineering		2.922		2.922		0.527																																
Other Costs		0.630		0.630		0.324																																
Total	1	19.681	1	26.040	1	22.700																																
Description: The MK 41 Vertical Launch System (VLS) is a Modular, Below Deck Missile Launcher supporting Multiple Warfighting Mission Areas to include AW/ASW/BMD/Land Attack/Strike. It will be a 32 cell launching system that will support FFG(X) warfare and AW requirement for self defense. VLS costs have increased as the system is currently being procured under a Unilateral Contract Agreement until contract negotiations are completed.																																						
Contract Data: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th style="text-align: center;">Program Year</th> <th style="text-align: center;">Hull</th> <th style="text-align: center;">Prime Contractor</th> <th style="text-align: center;">Contract Method/Type</th> <th style="text-align: center;">Award Date</th> <th style="text-align: center;">New/Option</th> <th style="text-align: center;">Quantity (Each)</th> <th style="text-align: center;">Unit Cost (\$ M)</th> </tr> <tr> <td style="text-align: center;">FY 2020</td> <td style="text-align: center;">FFG 62</td> <td style="text-align: center;">Lockheed Martin</td> <td style="text-align: center;">C/FFP</td> <td style="text-align: center;">May 2020</td> <td></td> <td style="text-align: center;">1</td> <td style="text-align: right;">16.129</td> </tr> <tr> <td style="text-align: center;">FY 2021</td> <td style="text-align: center;">FFG 63</td> <td style="text-align: center;">Lockheed Martin</td> <td style="text-align: center;">C/FFP</td> <td style="text-align: center;">May 2021</td> <td></td> <td style="text-align: center;">1</td> <td style="text-align: right;">22.488</td> </tr> <tr> <td style="text-align: center;">FY 2022</td> <td style="text-align: center;">FFG 64</td> <td style="text-align: center;">Lockheed Martin</td> <td style="text-align: center;">C/FFP</td> <td style="text-align: center;">May 2022</td> <td></td> <td style="text-align: center;">1</td> <td style="text-align: right;">21.849</td> </tr> </table>							Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2020	FFG 62	Lockheed Martin	C/FFP	May 2020		1	16.129	FY 2021	FFG 63	Lockheed Martin	C/FFP	May 2021		1	22.488	FY 2022	FFG 64	Lockheed Martin	C/FFP	May 2022		1	21.849
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																															
FY 2020	FFG 62	Lockheed Martin	C/FFP	May 2020		1	16.129																															
FY 2021	FFG 63	Lockheed Martin	C/FFP	May 2021		1	22.488																															
FY 2022	FFG 64	Lockheed Martin	C/FFP	May 2022		1	21.849																															
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FY 2021	FFG 63	Jan 2027	40	24	Sep 2021																																	
FY 2022	FFG 64	Jun 2028	40	24	Feb 2023																																	
Competition/Second Source Initiatives: N/A																																						
Remarks: Testing and Integration costs are captured under the Aegis Weapons System (AWS). Cost increases in FY20 and FY21 are due to receiving updated cost estimates based on the loss of cost sharing initiatives and higher unit costs due to the new profile, and contract alignment with the support activities and other shipbuilding programs.																																						

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2128 / FFG-Frigate			
Equipment Item: MK 48 Gun Weapon System (GWS)						PARM Code: PEO IWS 3C	
P-35 Category	FY 2020		FY 2021		FY 2022		
	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>	
Major Hardware	1	11.278	1	11.145	1	10.975	
System Engineering		1.347		1.139		0.752	
Technical Engineering Services		0.431		0.331		0.253	
Other Costs		1.782		1.007		0.805	
Total	1	14.838	1	13.622	1	12.785	
Description: The MK 48 MOD 2 Gun Weapon System (GWS) is fully integrated with MK 160 MOD 18 Gun Computer System w/ MK 20 MOD 1 Electro Optical Sight System and MK 110 MOD 0 57mm gun. The MK 160 Gun Fire Control System (GFCS) is the standard USN gun fire control system; the MK 20 Electro-Optical Sensor System (EOSS) is the standard gun optical sight used for gun engagements; and the MK 110 is an automated 57mm gun system used for surface and air engagements of hostile targets.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity <i>(Each)</i>	Unit Cost <i>(\$ M)</i>
FY 2020	FFG 62	Various	C/FFP	Various	Various	1	11.278
FY 2021	FFG 63	Various	C/FFP	Various	Various	1	11.145
FY 2022	FFG 64	Various	C/CPFF	Various	Various	1	10.975
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	FFG 62	Apr 2026	40	24	Dec 2020		
FY 2021	FFG 63	Jan 2027	40	24	Sep 2021		
FY 2022	FFG 64	Jun 2028	40	24	Feb 2023		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																																
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1				P-1 Line Item Number / Title: 2128 / FFG-Frigate																																		
Equipment Item: Rolling Airframe Missile (RAM) Guided Missile Launching System- 21 Cell						PARM Code: PEO IWS 11																																
P-35 Category	FY 2020		FY 2021		FY 2022																																	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																																
Major Hardware	1	10.310	1	10.412	1	11.108																																
System Engineering		0.775		0.785		0.816																																
Technical Engineering Services		0.372		0.372		0.307																																
Other Costs		0.797		0.884		0.974																																
Total	1	12.254	1	12.453	1	13.205																																
Description: The Rolling Airframe Missile (RAM) Guided Missile Weapon System (GMWS) is a lightweight, fast reaction, high-firepower, passive, dual-mode (IR and RF) system. The RAM GMWS is a point defense weapon system, designed to engage and destroy incoming Anti-Ship Cruise Missiles (ASCM), helicopters, and fixed wing aircraft, and to engage and intercept surface craft. FFG(X) uses the 21-cell variant.																																						
Contract Data: <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Prime Contractor</th> <th>Contract Method/Type</th> <th>Award Date</th> <th>New/Option</th> <th>Quantity (Each)</th> <th>Unit Cost (\$ M)</th> </tr> <tr> <td>FY 2020</td> <td>FFG 62</td> <td>Raytheon</td> <td>SS/FFP</td> <td>Dec 2020</td> <td>Option</td> <td>1</td> <td>10.310</td> </tr> <tr> <td>FY 2021</td> <td>FFG 63</td> <td>Raytheon</td> <td>SS/FFP</td> <td>Dec 2020</td> <td>Option</td> <td>1</td> <td>10.412</td> </tr> <tr> <td>FY 2022</td> <td>FFG 64</td> <td>Raytheon</td> <td>SS/FFP</td> <td>Dec 2021</td> <td>Option</td> <td>1</td> <td>11.108</td> </tr> </table>							Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2020	FFG 62	Raytheon	SS/FFP	Dec 2020	Option	1	10.310	FY 2021	FFG 63	Raytheon	SS/FFP	Dec 2020	Option	1	10.412	FY 2022	FFG 64	Raytheon	SS/FFP	Dec 2021	Option	1	11.108
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FY 2022	FFG 64	Jun 2028	40	24	Feb 2023																																	
Competition/Second Source Initiatives: N/A																																						
Remarks: FY20 RAM costs have increased based on timing of procurements with other programs. FY22 RAM cost increase over inflation from FY21 due to changes in ship profile and loss of quantity buy opportunity to meet ship construction schedule.																																						

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Exhibit P-10, Advance Procurement Requirements Analysis (page 1 - Budget Funding Justification): PB 2022 Navy							Date: May 2021			
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1					P-1 Line Item Number / Title: 2128 / FFG-Frigate					
First System (2022) Award Date: April 2022		First System (2022) Completion Date: October 2027			Interval Between Systems: 12 Months					
Cost Elements		Production Leadtime (Months)	When Required* (Months)	FY 2020 (\$ M)	FY 2021 (\$ M)	FY 2022 (\$ M)	FY 2023 (\$ M)	FY 2024 (\$ M)	FY 2025 (\$ M)	FY 2026 (\$ M)
BASIC CONSTRUCTION										
FFG 73 LLTM		-	-	-	-	0.000	-	-	-	-
FFG 76 LLTM		-	-	-	-	0.000	-	-	-	-
<i>Total: BASIC CONSTRUCTION</i>				-	-	-	-	-	-	-
ELECTRONICS										
FFG 65		32-44	Various	-	-	59.181	-	-	-	-
FFG 66		32-44	Various	-	-	9.919	-	-	-	-
FFG 67		-	-	-	-	0.000	-	-	-	-
FFG 68		-	-	-	-	0.000	-	-	-	-
FFG 69		-	-	-	-	0.000	-	-	-	-
FFG 70		-	-	-	-	0.000	-	-	-	-
FFG 71		-	-	-	-	0.000	-	-	-	-
FFG 72		-	-	-	-	0.000	-	-	-	-
FFG 73		-	-	-	-	0.000	-	-	-	-
FFG 74		-	-	-	-	0.000	-	-	-	-
FFG 75		-	-	-	-	0.000	-	-	-	-
FFG 76		-	-	-	-	0.000	-	-	-	-
<i>Total: ELECTRONICS</i>				-	-	69.100	-	-	-	-
Total Advance Procurement/Obligation Authority				-	-	69.100				
*Note: "When Required" is the number of months required before ship delivery.										

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Exhibit P-10, Advance Procurement Requirements Analysis (page 2 - Budget Funding Justification): PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity: 1611N / 02 / 1	P-1 Line Item Number / Title: 2128 / FFG-Frigate
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Cost Elements	FY 2022						
	Production Leadtime <i>(Months)</i>	When Required* <i>(Months)</i>	Unit Cost <i>(\$ M)</i>	Contract Forecast Date	2022 Qty <i>(Each)</i>	For FY	Total Cost Request <i>(\$ M)</i>
BASIC CONSTRUCTION							
FFG 73 LLTM	-	-	-		-		0.000
FFG 76 LLTM	-	-	-		-		0.000
Total: BASIC CONSTRUCTION							-
ELECTRONICS							
FFG 65	32-44	Various	-	Apr 2022	-	2023	59.181
FFG 66	32-44	Various	-	Apr 2022	-	2023	9.919
FFG 67	-	-	-		-		0.000
FFG 68	-	-	-		-		0.000
FFG 69	-	-	-		-		0.000
FFG 70	-	-	-		-		0.000
FFG 71	-	-	-		-		0.000
FFG 72	-	-	-		-		0.000
FFG 73	-	-	-		-		0.000
FFG 74	-	-	-		-		0.000
FFG 75	-	-	-		-		0.000
FFG 76	-	-	-		-		0.000
Total: ELECTRONICS							69.100
Total Advance Procurement/Obligation Authority							69.100

Description:
ELECTRONICS: Equipment AP is required to fund GFE material that has a long-lead procurement time that does not meet shipbuilding contract construction delivery requirements. Material includes Enterprise Air Search Radar AN/SPY-6(V)3 and Surface Electronic Weapon Improvement Program (SEWIP) BLK II SLQ-32(V)6.

*Note: "When Required" is the number of months required before ship delivery.

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy										Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 03: Amphibious Ships / BSA 1: Amphibious Ships							P-1 Line Item Number / Title: 3010 / LPD Flight II					
ID Code (A=Service Ready, B=Not Service Ready): A				Program Elements for Code B Items: N/A				Other Related Program Elements: 0604454N				
Line Item MDAP/MAIS Code: 542												
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (<i>Units in Each</i>)	1	-	1	-	-	-	-	-	-	-	-	-
Gross/Weapon System Cost (<i>\$ in Millions</i>)	1,819.600	0.000	1,957.637	0.000	0.000	0.000	-	-	-	-	-	-
Less PY Advance Procurement (<i>\$ in Millions</i>)	14.000	-	247.100	-	-	-	-	-	-	-	-	-
Less Cost To Complete (<i>\$ in Millions</i>)	5.600	-	-	-	-	-	-	-	-	-	-	-
Less Subsequent Year Full Funding (<i>\$ in Millions</i>)	-	-	60.636	-	-	-	-	-	-	-	-	-
Less Prior Year Full Funding (<i>\$ in Millions</i>)	-	-	524.100	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (<i>\$ in Millions</i>)	1,800.000	0.000	1,125.801	0.000	0.000	0.000	-	-	-	-	-	-
Plus Subsequent Year Full Funding (<i>\$ in Millions</i>)	-	-	-	60.636	-	60.636	-	-	-	-	-	-
Plus Prior Year FF (<i>\$ in Millions</i>)	-	524.100	-	-	-	-	-	-	-	-	-	-
Full Funding TOA (<i>\$ in Millions</i>)	1,800.000	524.100	1,125.801	60.636	-	60.636	-	-	-	-	-	-
Plus CY Advance Procurement (<i>\$ in Millions</i>)	261.100	-	2.000	-	-	-	-	-	-	-	-	-
Plus Cost To Complete (<i>\$ in Millions</i>)	-	0.000	-	-	-	-	-	-	-	-	-	-
Total Obligation Authority (<i>\$ in Millions</i>)	2,061.100	524.100	1,127.801	60.636	0.000	60.636	-	-	-	-	-	-
<i>(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)</i>												
Plus Outfitting and Post Delivery (<i>\$ in Millions</i>)	-	0.000	-	-	-	-	-	-	-	-	-	-
Total (<i>\$ in Millions</i>)	2,061.100	524.100	1,127.801	60.636	-	60.636	-	-	-	-	-	-
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	1,819.600	-	1,957.637	-	-	-	-	-	-	-	-	-

Description:

The LPD 17 Flight II is expected to functionally replace LSD-41 Class ships and LSD-49 Class ships for embark, transport, control, insert, sustainment, and extract of Marine Air-Ground Task Force elements and supporting forces by helicopters, landing craft, and amphibious vehicles. Program is on track to support FY27 retirement of LSDs.

Per signed Acquisition Decision Memorandum, LPD 17 Flight II will meet the Capabilities Development Document for LX(R) and shall subsume all previous LX(R) efforts. LPD 30 will be the first ship of LPD 17 Flight II.

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy				Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 03: Amphibious Ships / BSA 1: Amphibious Ships			P-1 Line Item Number / Title: 3010 / LPD Flight II		
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: N/A		Other Related Program Elements: 0604454N	
Line Item MDAP/MAIS Code: 542					
Characteristics:		Systems:			
Length Overall	-	208.5 m	684 ft	Electronics	
Beam		31.9 m	105 ft	-Mission Systems	
Displacement		25.3 lmt	24.9 klt		
Draft		7.0 m	23 ft		
Production Status:		LPD FLT II 30		LPD FLT II 31	
Contract Award Date		Mar 2019		Apr 2020	
Months to Completion					
a) Award to Delivery		71 months		82 months	
b) Construction Start to Delivery		59 months		58 months	
Delivery Date		Feb 2025		Feb 2027	
Completion Of Fitting Out		Nov 2025		Sep 2027	
Obligation Work Limit Date		Oct 2026		Aug 2028	
Design Schedule		Start / Issue		Complete / Response	
Issue Date for TLR		N/A		N/A	
Issue Date for TLS		N/A		N/A	
Preliminary Design		Mar 2015		Jun 2016	
Contract Design		Jun 2016		Jun 2017	
Detail Design		Dec 2018		Mar 2020	
Request for Proposals		Jul 2018		Aug 2018	
Design Agent		Huntington Ingalls Industries			
Classification of Cost Estimate: CLASS C					
Justification: The FY 2022 request is for the final increment of incremental funding for LPD 31.					

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1		P-1 Line Item Number / Title: 3010 / LPD Flight II		
Cost Categories (†) indicates the presence of a P-8a	FY 2018		FY 2021	
	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>
Plan Costs	1		1	
Basic Construction/Conversion		1,476.291		1,530.981
Change Orders		22.985		35.000
Electronics ^(†)		211.048		258.116
Hull, Mechanical, and Electrical (HM&E) ^(†)		16.400		21.222
Ordnance ^(†)		83.876		101.517
Other Cost		9.000		10.801
Total Ship Estimate		1,819.600		1,957.637
Less Advance Procurement FY 2016		14.000		-
Less Advance Procurement FY 2019		-		247.100
Less Subsequent Full Funding FY 2022		-		60.636
Less Cost to Complete FY 2025		5.600		-
Less Prior Year Full Funding FY 2020		-		524.100
Net P-1 Funding		1,800.000		1,125.801
Remarks: Change Orders Increase: Increase in change orders for LPD 31 is due to adding the following items to LPD 31: Advanced Training Domain, Ship Board Surveillance System, a space/weight reservation for the Commercial Broadband Satellite Program, and N-30 Fire Insulation requirements.				

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Exhibit P-27, Ship Production Schedule: PB 2022 Navy			Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1			P-1 Line Item Number / Title: 3010 / LPD Flight II		
Ship	Shipbuilder	Fiscal Year	Contract Award	Start of Construction	Delivery Date
LPD FLT II 30	HUNTINGTON INGALLS INDUSTRIES	2018	Mar 2019	Mar 2020	Feb 2025
LPD FLT II 31	HUNTINGTON INGALLS INDUSTRIES	2021	Apr 2020	Apr 2022	Feb 2027

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1		P-1 Line Item Number / Title: 3010 / LPD Flight II		
Electronics	FY 2018		FY 2021	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
P-35 Items				
Mission Systems	1	42.938	1	45.566
C4ISR	1	78.348	1	88.402
Ship Self Defense System (SSDS)	1	19.492	1	23.685
Cooperative Engagement Capability (CEC)	1	10.852	1	11.516
Interrogator System (IFF)	1	5.915	1	6.277
AN/SLQ-32(V)6 Surface Electronic Warfare Improvement Program (SEWIP)	1	12.771	1	13.553
P-35 Items Subtotal		170.316		188.999
Major Items				
Advanced Training Domain (ATD)	1	1.251	1	1.328
AN/WSN-7(RLGN)	1	3.882	1	4.120
Nulka Decoy Launching System (DLS)	1	1.419	1	1.506
Amphibious Assault Direction System (AADS)	1	1.038	1	1.102
Torpedo Countermeasures Transmitting Set (Nixie)	1	4.336	1	4.601
RADIAC	1	0.103	1	0.109
AN/UQN-10	1	0.204	1	0.216
DHYSL	1	0.373	1	0.396
Electronic Charting Display and Information Systems-Navy (ECDIS-N)	1	1.752	1	1.859
Major Items Subtotal		14.358		15.237
Other Cost Elements				
Miscellaneous Electronics		22.239		49.492
IWS CSI		4.135		4.388
Other Cost Elements Subtotal		26.374		53.880
Total Electronics		211.048		258.116
Remarks:				

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1		P-1 Line Item Number / Title: 3010 / LPD Flight II		
Hull, Mechanical, and Electrical (HM&E)	FY 2018		FY 2021	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Major Items				
Boats	1	0.616	1	0.654
Forklift Trucks	7	2.070	7	2.196
Chemical Warfare Detector	1	0.246	1	0.261
Military Payroll System	1	0.586	1	0.622
Integrated Condition Assessment System (ICAS)	1	0.216	1	0.229
Oily Water Separator	1	0.294	1	0.312
Plastic Waste Processing EQP	1	0.714	1	0.758
Unclassified Video System (UVS)	1	2.394	1	2.541
Major Items Subtotal		7.136		7.573
Other Cost Elements				
Miscellaneous HM&E		9.264		13.649
Other Cost Elements Subtotal		9.264		13.649
Total Hull, Mechanical, and Electrical (HM&E)		16.400		21.222

Remarks:

Miscellaneous HM&E:

Cost increases for LPD 31 reflect changes to HM&E network to Interior Communications Data Network "ICDN", changes to the Navy Engineering Control System "ECS" software and ECS console tech refresh.

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy				Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1			P-1 Line Item Number / Title: 3010 / LPD Flight II		
Ordnance	FY 2018		FY 2021		
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	
P-35 Items					
RAM BLOCK II	2	20.511	2	21.766	
MK 46 GUN	2	8.350	2	8.861	
AN/SPQ-9B Radar Set	1	6.571	1	6.974	
EASR	1	32.976	1	37.168	
P-35 Items Subtotal		68.408		74.769	
Major Items					
ASGSI/HOSS/MWS Fit Control & Inst Land Sys	1	3.252	1	3.451	
ORDNANCE HANDLING EQUIPMENT	1	0.423	1	0.449	
Next Generation Surface Search Radar (NGSSR)	1	2.758	1	2.927	
Major Items Subtotal		6.433		6.827	
Other Cost Elements					
MISCELLANEOUS ORDNANCE		9.035		19.921	
Other Cost Elements Subtotal		9.035		19.921	
Total Ordnance		83.876		101.517	

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1				P-1 Line Item Number / Title: 3010 / LPD Flight II			
Equipment Item: Mission Systems						PARM Code: N/A	
P-35 Category	FY 2018		FY 2021				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	40.902	1	43.406			
Other Costs		2.036		2.160			
Total	1	42.938	1	45.566			
Description: Description: Mission Systems is a microcomputer-based integration of shipboard control electronics; Engineering Control System (ECS), Ship Control System (SCS), Hull Mechanical & Electrical (HM&E) Network, Navigation Network. Mission systems and associated integration will be provided by Government supplied material and services							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2018	LPD FLT II 30	Various	SS/FFP	Jan 2020	Various	1	40.902
FY 2021	LPD FLT II 31	Various	SS/FFP	TBD	Various	1	43.406
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2018	LPD FLT II 30	Feb 2025	37	24	Jan 2020		
FY 2021	LPD FLT II 31	Feb 2027	37	24	Jan 2022		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																									
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1				P-1 Line Item Number / Title: 3010 / LPD Flight II																											
Equipment Item: C4ISR						PARM Code: N/A																									
P-35 Category	FY 2018		FY 2021																												
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																											
Major Hardware	1	41.102	1	47.618																											
Spares		0.941		1.999																											
Technical Engineering Services		6.510		7.166																											
Ancillary Equipment		0.246		0.261																											
Documentation and Systems Engineering		5.024		5.332																											
Other Appropriate Costs		9.523		10.106																											
Turnkey		15.002		15.920																											
Total	1	78.348	1	88.402																											
Description: C4ISR systems provide the link between the ship, the command hierarchy, and other units of the operating forces. The capability is provided by a turn-key Radio Communication System (line-of sight, beyond line of sight, and SATCOM), CANES, GPNTS, and associated AIT/SOVT services support provided at the shipbuilding location. The increase between LPD 30 to LPD 31 is due to inflation. Resolution of obsolescence (iRCS) and fielding of new CANES variant (hardware 2.0) proposed by PEO C4I is still pending change impact assessments and adjudication.																															
Contract Data: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Prime Contractor</th> <th>Contract Method/Type</th> <th>Award Date</th> <th>New/Option</th> <th>Quantity (Each)</th> <th>Unit Cost (\$ M)</th> </tr> <tr> <td>FY 2018</td> <td>LPD FLT II 30</td> <td>Various</td> <td>Various</td> <td>TBD</td> <td>Various</td> <td style="text-align: center;">1</td> <td style="text-align: right;">41.102</td> </tr> <tr> <td>FY 2021</td> <td>LPD FLT II 31</td> <td>Various</td> <td>Various</td> <td>TBD</td> <td>Various</td> <td style="text-align: center;">1</td> <td style="text-align: right;">47.618</td> </tr> </table>								Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2018	LPD FLT II 30	Various	Various	TBD	Various	1	41.102	FY 2021	LPD FLT II 31	Various	Various	TBD	Various	1	47.618
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																								
FY 2018	LPD FLT II 30	Various	Various	TBD	Various	1	41.102																								
FY 2021	LPD FLT II 31	Various	Various	TBD	Various	1	47.618																								
Delivery Date: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th>Program Year</th> <th>Hull</th> <th>Earliest Ship Delivery Date</th> <th>Months Required Before Delivery</th> <th>Production Leadtime</th> <th>Required Award Date</th> </tr> <tr> <td>FY 2018</td> <td>LPD FLT II 30</td> <td>Feb 2025</td> <td style="text-align: center;">16</td> <td style="text-align: center;">16</td> <td style="text-align: center;">Jun 2022</td> </tr> <tr> <td>FY 2021</td> <td>LPD FLT II 31</td> <td>Feb 2027</td> <td style="text-align: center;">16</td> <td style="text-align: center;">16</td> <td style="text-align: center;">Jun 2024</td> </tr> </table>								Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date	FY 2018	LPD FLT II 30	Feb 2025	16	16	Jun 2022	FY 2021	LPD FLT II 31	Feb 2027	16	16	Jun 2024						
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date																										
FY 2018	LPD FLT II 30	Feb 2025	16	16	Jun 2022																										
FY 2021	LPD FLT II 31	Feb 2027	16	16	Jun 2024																										
Competition/Second Source Initiatives: N/A																															
Remarks: FY21 estimate for LPD 30 is based on LPD30 estimate. FY22 estimate has been updated to incorporate cost increases due to CANES (HW 1.2) hardware used on LPD 28-30 being obsolete and contract options expired in CY19. The additional hardware cost is associated with the incorporation of defense-in-depth cybersecurity improvements and the resolution of COTS obsolescence incorporated in the next hardware baseline (HW 2.0) being fielded on all CANES ships for the US Navy in the timeframe that LPD 31 equipment will be procured. The radio communication hardware costs increases are associated with Crypto modernization mandates and new crypto equipment from NSA. There is no alternative to use of newer, more costly crypto for secure communications. There are additional hardware costs for in radio communication systems associated with new contracts with the uncertainty in contract pricing that is included in PARM estimates.																															

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1				P-1 Line Item Number / Title: 3010 / LPD Flight II			
Equipment Item: Ship Self Defense System (SSDS)						PARM Code: N/A	
P-35 Category	FY 2018		FY 2021				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	13.104	1	15.905			
Spares		0.909		1.465			
Technical Engineering Services		0.564		0.599			
Other Costs		2.291		2.931			
Documentation and Systems Engineering		2.624		2.785			
Total	1	19.492	1	23.685			
Description: Ship Self Defense System Mark 2 is microcomputer-based, self-defense coordination system that integrates and automates multiple sensors, self-defense weapons, and softkill systems to provide quick reaction combat capability against anti-ship cruise missile threats.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2018	LPD FLT II 30	TBD	TBD	TBD	New	1	13.104
FY 2021	LPD FLT II 31	TBD	TBD	TBD	New	1	15.905
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2018	LPD FLT II 30	Feb 2025	17	13	Aug 2022		
FY 2021	LPD FLT II 31	Feb 2027	17	13	Aug 2024		
Competition/Second Source Initiatives: N/A							
Remarks: FY21 LPD 31 cost estimate is based on LPD30. FY22 estimate has been updated to incorporate cost increases due to Command Table (TI-12H) hardware used on LPD 28-30 being obsolete and contract options expired in FY2020. The additional hardware cost is associated with new contracts with associated uncertainty in contract pricing, new spares, and integration necessary to resolve obsolescence for LPD 31 and deliver TI-16 TR (tech refresh) or the next variant MOD (x).							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1				P-1 Line Item Number / Title: 3010 / LPD Flight II			
Equipment Item: Cooperative Engagement Capability (CEC)						PARM Code: N/A	
P-35 Category	FY 2018		FY 2021				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	8.177	1	8.677			
Technical Engineering Services		0.435		0.462			
Documentation and Systems Engineering		0.498		0.528			
Other Costs		1.742		1.849			
Total	1	10.852	1	11.516			
Description: Cooperative Engagement Capability (CEC) coordinates all anti-warfare sensors into single, real time, fire control quality composite track which improves battle force air defense. Production of Planar Array Antenna Assembly (PAAA) production has restarted.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2018	LPD FLT II 30	Raytheon	SS/FFP	Various	Various	1	8.177
FY 2021	LPD FLT II 31	Raytheon	SS/FFP	Various	Various	1	8.677
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2018	LPD FLT II 30	Feb 2025	24	18	Aug 2021		
FY 2021	LPD FLT II 31	Feb 2027	24	18	Aug 2023		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1				P-1 Line Item Number / Title: 3010 / LPD Flight II			
Equipment Item: Interrogator System (IFF)						PARM Code: N/A	
P-35 Category	FY 2018		FY 2021				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	5.232	1	5.552			
Spares		0.146		0.155			
Technical Engineering Services		0.117		0.124			
Other Costs		0.271		0.288			
Documentation and Systems Engineering		0.149		0.158			
Total	1	5.915	1	6.277			
Description: The Transponder Set is an Automatic Identification and Monitoring System (AIMS) Identification Friend or Foe (IFF) system that receives interrogation signals from air, surface, and land IFF - equipped units and automatically replies with a coded response signal that provides ownship position and identification.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2018	LPD FLT II 30	TBD	TBD	Various	New	1	5.232
FY 2021	LPD FLT II 31	TBD	TBD	Various	New	1	5.552
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2018	LPD FLT II 30	Feb 2025	6	30	Feb 2022		
FY 2021	LPD FLT II 31	Feb 2027	6	30	Feb 2024		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021																									
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1				P-1 Line Item Number / Title: 3010 / LPD Flight II																											
Equipment Item: AN/SLQ-32(V)6 Surface Electronic Warfare Improvement Program (SEWIP)						PARM Code: N/A																									
P-35 Category	FY 2018		FY 2021																												
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)																											
Major Hardware	1	10.767	1	11.426																											
Spares		0.382		0.405																											
Technical Engineering Services		0.142		0.151																											
Other Costs		0.641		0.680																											
Documentation and Systems Engineering		0.839		0.891																											
Total	1	12.771	1	13.553																											
Description: The AN/SLQ-32(V)6 Surface Electronic Warfare Improvement Program (SEWIP) is a shipboard system that provides a full suite of Electronic Warfare capabilities designed to protect against anti-cruise ship missile threats.																															
Contract Data: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th style="text-align: center;">Program Year</th> <th style="text-align: center;">Hull</th> <th style="text-align: center;">Prime Contractor</th> <th style="text-align: center;">Contract Method/Type</th> <th style="text-align: center;">Award Date</th> <th style="text-align: center;">New/Option</th> <th style="text-align: center;">Quantity (Each)</th> <th style="text-align: center;">Unit Cost (\$ M)</th> </tr> <tr> <td style="text-align: center;">FY 2018</td> <td style="text-align: center;">LPD FLT II 30</td> <td style="text-align: center;">VARIOUS</td> <td style="text-align: center;">SS/FFP</td> <td style="text-align: center;">Aug 2020</td> <td style="text-align: center;">Various</td> <td style="text-align: center;">1</td> <td style="text-align: right;">10.767</td> </tr> <tr> <td style="text-align: center;">FY 2021</td> <td style="text-align: center;">LPD FLT II 31</td> <td style="text-align: center;">TBD</td> <td style="text-align: center;">TBD</td> <td style="text-align: center;">TBD</td> <td></td> <td style="text-align: center;">1</td> <td style="text-align: right;">11.426</td> </tr> </table>								Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)	FY 2018	LPD FLT II 30	VARIOUS	SS/FFP	Aug 2020	Various	1	10.767	FY 2021	LPD FLT II 31	TBD	TBD	TBD		1	11.426
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)																								
FY 2018	LPD FLT II 30	VARIOUS	SS/FFP	Aug 2020	Various	1	10.767																								
FY 2021	LPD FLT II 31	TBD	TBD	TBD		1	11.426																								
Delivery Date: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <tr> <th style="text-align: center;">Program Year</th> <th style="text-align: center;">Hull</th> <th style="text-align: center;">Earliest Ship Delivery Date</th> <th style="text-align: center;">Months Required Before Delivery</th> <th style="text-align: center;">Production Leadtime</th> <th style="text-align: center;">Required Award Date</th> </tr> <tr> <td style="text-align: center;">FY 2018</td> <td style="text-align: center;">LPD FLT II 30</td> <td style="text-align: center;">Feb 2025</td> <td style="text-align: center;">24</td> <td style="text-align: center;">24</td> <td style="text-align: center;">Feb 2021</td> </tr> <tr> <td style="text-align: center;">FY 2021</td> <td style="text-align: center;">LPD FLT II 31</td> <td style="text-align: center;">Feb 2027</td> <td style="text-align: center;">24</td> <td style="text-align: center;">24</td> <td style="text-align: center;">Feb 2023</td> </tr> </table>								Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date	FY 2018	LPD FLT II 30	Feb 2025	24	24	Feb 2021	FY 2021	LPD FLT II 31	Feb 2027	24	24	Feb 2023						
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date																										
FY 2018	LPD FLT II 30	Feb 2025	24	24	Feb 2021																										
FY 2021	LPD FLT II 31	Feb 2027	24	24	Feb 2023																										
Competition/Second Source Initiatives: N/A																															

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1				P-1 Line Item Number / Title: 3010 / LPD Flight II			
Equipment Item: RAM BLOCK II						PARM Code: N/A	
P-35 Category	FY 2018		FY 2021				
	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>			
Major Hardware	2	17.935	2	19.033			
Spares		0.129		0.137			
Technical Engineering Services		0.302		0.320			
Other Costs		1.020		1.082			
Documentation and Systems Engineering		1.125		1.194			
Total	2	20.511	2	21.766			
Description: The Rolling Airframe Missile (RAM) Block 2 system is a short-range, fast-reaction, high-firepower, lightweight weapon designed to destroy incoming anti-ship cruise missiles.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity <i>(Each)</i>	Unit Cost <i>(\$ M)</i>
FY 2018	LPD FLT II 30	Raytheon	C/FFP	Oct 2016	Option	2	8.968
FY 2021	LPD FLT II 31	TBD	TBD	TBD		2	9.517
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2018	LPD FLT II 30	Feb 2025	22	24	Apr 2021		
FY 2021	LPD FLT II 31	Feb 2027	22	24	Apr 2023		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1				P-1 Line Item Number / Title: 3010 / LPD Flight II			
Equipment Item: MK 46 GUN						PARM Code: N/A	
P-35 Category	FY 2018		FY 2021				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	2	8.130	2	8.628			
Technical Engineering Services		0.220		0.233			
Total	2	8.350	2	8.861			
Description: The MK 46 Gun is a remotely operated naval gun system using a high velocity cannon and second-generation thermal day-night sight for close-in ship's protection.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2018	LPD FLT II 30	General Dynamics	C/FFP	TBD	New	2	4.065
FY 2021	LPD FLT II 31	TBD	TBD	TBD		2	4.314
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2018	LPD FLT II 30	Feb 2025	24	18	Aug 2021		
FY 2021	LPD FLT II 31	Feb 2027	24	18	Aug 2023		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1				P-1 Line Item Number / Title: 3010 / LPD Flight II			
Equipment Item: AN/SPQ-9B Radar Set						PARM Code: N/A	
P-35 Category	FY 2018		FY 2021				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	4.200	1	4.457			
Spares		0.388		0.412			
Technical Engineering Services		0.707		0.751			
Other Costs		0.776		0.823			
Documentation and Systems Engineering		0.500		0.531			
Total	1	6.571	1	6.974			
Description: The AN/SPQ-9B is a high resolution, X-band, narrow beam radar that provides both air and surface tracking information.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2018	LPD FLT II 30	DRS LAUREL TECHNOLOGIES	C/FFP	Jul 2018	Option	1	4.200
FY 2021	LPD FLT II 31	TBD	TBD	TBD		1	4.457
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2018	LPD FLT II 30	Feb 2025	24	24	Feb 2021		
FY 2021	LPD FLT II 31	Feb 2027	24	24	Feb 2023		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1				P-1 Line Item Number / Title: 3010 / LPD Flight II			
Equipment Item: EASR						PARM Code: N/A	

P-35 Category	FY 2018		FY 2021	
	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>
Major Hardware	1	20.907	1	24.361
Spares		1.093		1.160
Technical Engineering Services		3.259		3.458
Other Costs		6.437		6.830
Documentation and Systems Engineering		1.280		1.359
Total	1	32.976	1	37.168

Description:
The AN/SPY-6(V)2 Enterprise Air Surveillance Radar (EASR) is the Air Search Radar.

Contract Data:

Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity <i>(Each)</i>	Unit Cost <i>(\$ M)</i>
FY 2018	LPD FLT II 30	Raytheon	C/FFP	Nov 2020	Option	1	20.907
FY 2021	LPD FLT II 31	TBD	TBD	TBD	New	1	24.361

Delivery Date:

Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date
FY 2018	LPD FLT II 30	Feb 2025	18	27	May 2021
FY 2021	LPD FLT II 31	Feb 2027	18	27	May 2023

Competition/Second Source Initiatives:
N/A

Remarks:
FY21 estimate was based on escalation of early program estimates. The updated cost reflects contract pricing and additional tech refresh kits for obsolescence resolution that is anticipated at the time that the LPD 31 system will be procured.

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Exhibit P-10, Advance Procurement Requirements Analysis (page 1 - Budget Funding Justification): PB 2022 Navy						Date: May 2021				
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1				P-1 Line Item Number / Title: 3010 / LPD Flight II						
First System (2022) Award Date: June 2023		First System (2022) Completion Date: August 2028			Interval Between Systems: 0 Months					
Cost Elements		Production Leadtime <i>(Months)</i>	When Required* <i>(Months)</i>	FY 2020 (\$ M)	FY 2021 (\$ M)	FY 2022 (\$ M)	FY 2023 (\$ M)	FY 2024 (\$ M)	FY 2025 (\$ M)	FY 2026 (\$ M)
LPD 32 Advance Procurement										
Basic Construction		-	-	-	1.000	0.000	-	-	-	-
Total: LPD 32 Advance Procurement				-	1.000	-	-	-	-	-
LPD 33 Advance Procurement										
Basic Construction		-	-	-	1.000	0.000	-	-	-	-
Total: LPD 33 Advance Procurement				-	1.000	-	-	-	-	-
Total Advance Procurement/Obligation Authority				-	2.000	-	-	-	-	-

*Note: "When Required" is the number of months required before ship delivery.

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Exhibit P-10, Advance Procurement Requirements Analysis (page 2 - Budget Funding Justification): PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1	P-1 Line Item Number / Title: 3010 / LPD Flight II
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Cost Elements	FY 2022						
	Production Leadtime <i>(Months)</i>	When Required* <i>(Months)</i>	Unit Cost <i>(\$ M)</i>	Contract Forecast Date	2022 Qty <i>(Each)</i>	For FY	Total Cost Request <i>(\$ M)</i>
LPD 32 Advance Procurement							
Basic Construction	-	-	-		-		0.000
Total: LPD 32 Advance Procurement							-
LPD 33 Advance Procurement							
Basic Construction	-	-	-		-		0.000
Total: LPD 33 Advance Procurement							-
Total Advance Procurement/Obligation Authority							-

Description:
Reflects \$2.000 million transfer to Advance Procurement (AP) for LPD 32 and LPD 33 from the Department of Defense Appropriations Act, 2021. FY 2021 AP funds support procurement of Long Lead Time Materials for LPD 32 and LPD 33.

*Note: "When Required" is the number of months required before ship delivery.

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy										Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 03: Amphibious Ships / BSA 1: Amphibious Ships							P-1 Line Item Number / Title: 3039 / Expeditionary Sea Base (ESB)					
ID Code (A=Service Ready, B=Not Service Ready): A			Program Elements for Code B Items: N/A					Other Related Program Elements: N/A				
Line Item MDAP/MAIS Code: N/A												
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (<i>Units in Each</i>)	7	-	-	-	-	-	-	-	-	-	-	-
Gross/Weapon System Cost (<i>\$ in Millions</i>)	4,118.068	0.000	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Less PY Advance Procurement (<i>\$ in Millions</i>)	179.700	-	-	-	-	-	-	-	-	-	-	-
Less Cost To Complete (<i>\$ in Millions</i>)	38.000	-	-	-	-	-	-	-	-	-	-	-
Less Subsequent Year Full Funding (<i>\$ in Millions</i>)	162.500	-	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (<i>\$ in Millions</i>)	3,737.868	0.000	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Plus Subsequent Year Full Funding (<i>\$ in Millions</i>)	162.500	-	-	-	-	-	-	-	-	-	-	-
Full Funding TOA (<i>\$ in Millions</i>)	3,900.368	-	-	-	-	-	-	-	-	-	-	-
Plus CY Advance Procurement (<i>\$ in Millions</i>)	179.700	-	73.000	-	-	-	-	-	-	-	-	-
Plus Cost To Complete (<i>\$ in Millions</i>)	-	38.000	-	-	-	-	-	-	-	-	-	-
Total Obligation Authority (<i>\$ in Millions</i>)	4,080.068	38.000	73.000	0.000	0.000	0.000	-	-	-	-	-	-
<i>(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)</i>												
Plus Outfitting and Post Delivery (<i>\$ in Millions</i>)	114.932	11.557	15.167	15.449	-	15.449	-	-	-	-	-	-
Total (<i>\$ in Millions</i>)	4,195.000	49.557	88.167	15.449	-	15.449	-	-	-	-	-	-
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	588.295	-	-	-	-	-	-	-	-	-	-	-

Description:

The Expeditionary Mobile Base (ESB) (formerly MLP Afloat Forward Staging Base (AFSB)) will serve as a dedicated Naval Afloat Forward Staging Base, optimized to support naval assets in a variety of missions rather than independently modifying ships-of-opportunity as required to meet these roles. The ESB retains sealift capabilities inherent to the Class through cargo transportation and distribution, but provides enhanced aviation, berthing, small boat handling, and command and control capabilities to meet a broader mission set. The ESB provides the Combatant Commanders flexibility to respond to immediate threats and host task organized forces, including Airborne Mine Countermeasures and Special Forces to confront irregular challenges and counter-terrorism. This includes enhanced logistics and UNREP capability (receive only) and C4I capability to support future missions.

Note:

1) The amounts in the Prior Years column includes the NDSF MPF,F MLP BLI 00401 Procurement Costs for Expeditionary Transport Dock (ESD) 1, ESD 2, and ESB 3 as well as SCN BLI 3039 for the ESB 4, 5, 6, and 7.

2) The Outfitting and Post Delivery amounts in the Prior Years column represent NDSF BLI 5000 for ESD 1, ESD 2, and ESB 3 as well as SCN BLI 5110 for the ESB 4 and 5.

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy		Date: May 2021																																																																																																										
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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy		Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1		P-1 Line Item Number / Title: 3039 / Expeditionary Sea Base (ESB)		
Cost Categories	FY 2018		FY 2019	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Plan Costs	1		1	
Basic Construction/Conversion		569.929		578.639
Change Orders		6.000		6.000
Electronics		41.415		42.409
Hull, Mechanical, and Electrical (HM&E)		13.416		14.712
Other Cost		4.240		5.240
Total Ship Estimate		635.000		647.000
Net P-1 Funding		635.000		647.000

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LI 3039 - Expeditionary Sea Base (ESB)
Navy

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy								Date: May 2021				
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 03: Amphibious Ships / BSA 1: Amphibious Ships						P-1 Line Item Number / Title: 3041 / LHA Replacement						
ID Code (A=Service Ready, B=Not Service Ready): A			Program Elements for Code B Items: N/A					Other Related Program Elements: 0604567N				
Line Item MDAP/MAIS Code: 333												
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (<i>Units in Each</i>)	3	-	-	-	-	-	-	-	-	-	-	-
Gross/Weapon System Cost (<i>\$ in Millions</i>)	10,296.259	0.000	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Less PY Advance Procurement (<i>\$ in Millions</i>)	1,148.123	-	-	-	-	-	-	-	-	-	-	-
Less Cost To Complete (<i>\$ in Millions</i>)	258.488	-	-	-	-	-	-	-	-	-	-	-
Less Subsequent Year Full Funding (<i>\$ in Millions</i>)	5,001.088	-	-	-	-	-	-	-	-	-	-	-
Less Prior Year Full Funding (<i>\$ in Millions</i>)	-	0.000	-	-	-	-	-	-	-	-	-	-
Less Hurricane (<i>\$ in Millions</i>)	202.000	-	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (<i>\$ in Millions</i>)	3,686.560	0.000	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Plus Subsequent Year Full Funding (<i>\$ in Millions</i>)	5,001.088	-	-	-	-	-	-	-	-	-	-	-
Plus Prior Year FF (<i>\$ in Millions</i>)	-	-	500.000	68.637	-	68.637	-	-	-	-	-	-
Full Funding TOA (<i>\$ in Millions</i>)	8,687.648	-	500.000	68.637	-	68.637	-	-	-	-	-	-
Plus CY Advance Procurement (<i>\$ in Millions</i>)	1,498.123	-	-	-	-	-	-	-	-	-	-	-
Plus Cost To Complete (<i>\$ in Millions</i>)	247.788	-	-	-	-	-	-	-	-	-	-	-
Plus Hurricane (<i>\$ in Millions</i>)	202.000	-	-	-	-	-	-	-	-	-	-	-
Total Obligation Authority (<i>\$ in Millions</i>)	10,635.559	0.000	500.000	68.637	0.000	68.637	-	-	-	-	-	-
(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)												
Plus Outfitting and Post Delivery (<i>\$ in Millions</i>)	167.901	12.961	17.810	16.024	-	16.024	-	-	-	-	-	-
Total (<i>\$ in Millions</i>)	10,803.460	12.961	517.810	84.661	-	84.661	-	-	-	-	-	-
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	3,432.086	-	-	-	-	-	-	-	-	-	-	-
Description: The LHA(R) Program replaces the Tarawa Class (LHA 1) Amphibious Assault Ships and the retiring Wasp Class (LHD 1) Amphibious Assault Class Ships. The LHA(R) class program ensures that the Amphibious Fleet remains capable of Expeditionary Warfare well into the 21st Century and provide for an affordable and sustainable amphibious ship development program. Provides forward presence and power projection as an integral part of joint, interagency, and multinational maritime expeditionary forces. Operates for sustained periods in transit to and operations in an Amphibious Objective Area to include the embarkation, deployment, and landing of a Marine Landing Force and supporting forces by helicopters and tilt rotors supported by Joint Strike Fighters F-35B. LHA(R) Flight 0 is considered a transitional increment intended to increase the aviation capabilities of amphibious assault ships. The LHA(R) Flight 1 design continues the incremental development of amphibious assault ships by adding a well deck, and increasing flight deck capacity by reducing the footprint of the island and adding a sponson. LHA(R) Flight 0 consisted of two ships, LHA 6 and LHA 7. LHA(R) Flight 1 is the second increment in the LHA 6 Class with LHA 8 being the first ship of Flight 1. The Flight 1 ships maintain an aviation centric capability with the addition of a well deck that will accommodate two Landing Craft, Air Cushions. LHA 9 provided Advance Procurement funds in FY19. FY20 NDAA and Appropriations Bill provided incremental funding authority. FY21, FY22, and FY24 provide the remaining LHA 9 Full Funding to complete construction of this FY23 ship.												

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Justification: FY 2022 requests \$68.6 million of incremental full funding for LHA 9 (FY 2023 ship)																																																																																																																																																		

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1	P-1 Line Item Number / Title: 3041 / LHA Replacement	
Cost Categories	FY 2017	
	Qty (Each)	Total Cost (\$ M)
Plan Costs	1	329.093
Basic Construction/Conversion		2,770.836
Change Orders		108.490
Electronics		314.754
Hull, Mechanical, and Electrical (HM&E)		63.184
Ordnance		158.708
Other Cost		95.186
Total Ship Estimate		3,840.251
Less Advance Procurement FY 2015		29.093
Less Advance Procurement FY 2016		476.038
Less Subsequent Full Funding FY 2018		1,706.701
Less Cost to Complete FY 2023		10.700
Net P-1 Funding		1,617.719

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LI 3041 - LHA Replacement
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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy										Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 03: Amphibious Ships / BSA 1: Amphibious Ships							P-1 Line Item Number / Title: 3043 / Expeditionary Fast Transport (EPF)					
ID Code (A=Service Ready, B=Not Service Ready): A				Program Elements for Code B Items: N/A				Other Related Program Elements: N/A				
Line Item MDAP/MAIS Code: N/A												
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity <i>(Units in Each)</i>	10	-	1	-	-	-	-	-	-	-	-	-
Gross/Weapon System Cost <i>(\$ in Millions)</i>	2,116.725	0.000	260.000	0.000	0.000	0.000	-	-	-	-	-	-
Less PY Advance Procurement <i>(\$ in Millions)</i>	-	-	-	-	-	-	-	-	-	-	-	-
Less Cost To Complete <i>(\$ in Millions)</i>	160.090	-	-	-	-	-	-	-	-	-	-	-
Less Program Support <i>(\$ in Millions)</i>	2.732	-	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) <i>(\$ in Millions)</i>	1,953.903	0.000	260.000	0.000	0.000	0.000	-	-	-	-	-	-
Plus CY Advance Procurement <i>(\$ in Millions)</i>	-	-	-	-	-	-	-	-	-	-	-	-
Plus Cost To Complete <i>(\$ in Millions)</i>	61.090	49.000	50.000	-	-	-	-	-	-	-	-	-
Plus Program Support <i>(\$ in Millions)</i>	2.732	-	-	-	-	-	-	-	-	-	-	-
Total Obligation Authority <i>(\$ in Millions)</i>	2,017.725	49.000	310.000	0.000	0.000	0.000	-	-	-	-	-	-
<i>(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)</i>												
Plus Outfitting and Post Delivery <i>(\$ in Millions)</i>	86.598	2.403	5.432	11.203	-	11.203	-	-	-	-	-	-
Total <i>(\$ in Millions)</i>	2,104.323	51.403	315.432	11.203	-	11.203	-	-	-	-	-	-
Gross/Weapon System Unit Cost <i>(\$ in Millions)</i>	211.673	-	260.000	-	-	-	-	-	-	-	-	-
Description: Future joint forces will be responsive, deployable, agile, versatile, lethal, survivable, and sustainable. The nation will need lift assets that can provide for assured access, decrease predictability and dwell time, and have the capacity to quickly deliver troops and equipment together in a manner that provides for unit integrity. Expeditionary Fast Transport (EPF) (formerly Joint High Speed Vessel) will provide combatant commanders high-speed intra-theater sealift with inherent cargo handling capability and the agility to achieve positional advantage over operational distances. Not limited to major ports, the EPF will be able to operate in austere port environments. EPFs 14 and 15 will have modifications to conduct a Role 2 Enhanced (R2E) Medical Transport mission which will include enhanced medical capabilities to support embarked Medical Military Detachment (MILDET) teams while retaining the ability to perform high-speed intra-theater sealift.												

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy				Date: May 2021																																																	
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 03: Amphibious Ships / BSA 1: Amphibious Ships			P-1 Line Item Number / Title: 3043 / Expeditionary Fast Transport (EPF)																																																		
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Design Agent																																																					
<u>Classification of Cost Estimate:</u> CLASS C																																																					

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy **Date:** May 2021

Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1 **P-1 Line Item Number / Title:** 3043 / Expeditionary Fast Transport (EPF)

Cost Categories (†) indicates the presence of a P-8a	FY 2018		FY 2019		FY 2021	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Plan Costs	1		1		1	
Basic Construction/Conversion		237.513		238.943		235.000
Change Orders		3.200		3.257		2.000
Electronics ^(†)		16.800		16.800		17.500
Hull, Mechanical, and Electrical (HM&E) ^(†)		11.487		8.000		3.000
Other Cost		6.000		7.000		2.500
Total Ship Estimate		275.000		274.000		260.000
Less Cost to Complete FY 2020		-		49.000		-
Less Cost to Complete FY 2021		50.000		-		-
Net P-1 Funding		225.000		225.000		260.000

Remarks:

- 1) Basic Construction increases from the FY 2021 President's Budget request on EPF 13 reflect definitization of a modification in support of the unmanned logistics prototype conversion.
- 2) Basic Construction increases from the FY 2021 President's Budget request on EPF 14 reflect definitization of a significant modification for enhanced medical capability.

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LI 3043 - Expeditionary Fast Transport (EPF)
Navy

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity:		P-1 Line Item Number / Title:		
1611N / 03 / 1		3043 / Expeditionary Fast Transport (EPF)		
Electronics	FY 2019		FY 2021	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
P-35 Items				
C4ISR	1	12.990	1	13.577
P-35 Items Subtotal		12.990		13.577
Major Items				
VISUAL LANDING AIDE SUITE	1	3.062	1	3.153
MISC ELECTRONICS		0.748		0.770
Major Items Subtotal		3.810		3.923
Total Electronics		16.800		17.500

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity:		P-1 Line Item Number / Title:		
1611N / 03 / 1		3043 / Expeditionary Fast Transport (EPF)		
Hull, Mechanical, and Electrical (HM&E)	FY 2019		FY 2021	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Major Items				
ENGINEERING SERVICES		4.130		0.862
SUPSHIP MATERIAL SERVICES		1.217		0.400
LOGISTICS SUPPORT SERVICES		1.126		0.500
TEST AND INSTRUMENTATION		0.589		0.300
EQUIPMENT		0.938		0.938
Major Items Subtotal		8.000		3.000
Total Hull, Mechanical, and Electrical (HM&E)		8.000		3.000

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1				P-1 Line Item Number / Title: 3043 / Expeditionary Fast Transport (EPF)			
Equipment Item: C4ISR						PARM Code: 3Z (NAVWAR)	
P-35 Category	FY 2019		FY 2021				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	1	7.429	1	7.851			
Spares		0.683		0.703			
System Engineering		2.622		2.700			
Technical Engineering Services		1.121		1.154			
Other Costs		1.135		1.169			
Total	1	12.990	1	13.577			
Description: The Command, Control, Communications, Computers, Intelligence, Surveillance and Reconnaissance (C4ISR) system provides the line between the ship, the command hierarchy and other units of the operation force. The C4ISR Suite consists of a Network Suite (MSC-CANES, ADNS and CENTRIXS-M), CBSP, UHF SATCOM Antenna, UHF/VHF LOS Suite and UHF SATCOM Radios, TVS-TVT, IA and RCS.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2019	EPF 14	Various	Various	Jan 2019	Various	1	7.429
FY 2021	EPF 15	Various	Various	Jun 2021	Various	1	7.851
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2019	EPF 14	Jun 2023	0	0	Jun 2023		
FY 2021	EPF 15	Feb 2024	0	0	Feb 2024		
Competition/Second Source Initiatives: N/A							
Remarks: Multiple systems comprise the C4ISR with varying delivery dates and lead times. NAVWAR PMW760 is the coordinating activity for the C4ISR System.							

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy										Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 03: Amphibious Ships / BSA 1: Amphibious Ships							P-1 Line Item Number / Title: 3036 / LPD-17					
ID Code (A=Service Ready, B=Not Service Ready): A			Program Elements for Code B Items: N/A					Other Related Program Elements: N/A				
Line Item MDAP/MAIS Code: N/A												
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (<i>Units in Each</i>)	13	-	-	-	-	-	-	-	-	-	-	-
Gross/Weapon System Cost (<i>\$ in Millions</i>)	21,383.591	0.000	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Less PY Advance Procurement (<i>\$ in Millions</i>)	1,635.736	-	-	-	-	-	-	-	-	-	-	-
Less Cost To Complete (<i>\$ in Millions</i>)	2,139.707	-	-	-	-	-	-	-	-	-	-	-
Less Subsequent Year Full Funding (<i>\$ in Millions</i>)	869.394	-	-	-	-	-	-	-	-	-	-	-
Less Prior Year Full Funding (<i>\$ in Millions</i>)	1,000.000	-	-	-	-	-	-	-	-	-	-	-
Less Hurricane (<i>\$ in Millions</i>)	1,623.280	-	-	-	-	-	-	-	-	-	-	-
Less Transfer (<i>\$ in Millions</i>)	279.031	-	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (<i>\$ in Millions</i>)	13,836.443	0.000	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Plus Subsequent Year Full Funding (<i>\$ in Millions</i>)	869.394	-	-	-	-	-	-	-	-	-	-	-
Plus Prior Year FF (<i>\$ in Millions</i>)	1,000.000	-	-	-	-	-	-	-	-	-	-	-
Full Funding TOA (<i>\$ in Millions</i>)	15,705.837	-	-	-	-	-	-	-	-	-	-	-
Plus CY Advance Procurement (<i>\$ in Millions</i>)	1,635.736	-	-	-	-	-	-	-	-	-	-	-
Plus Cost To Complete (<i>\$ in Millions</i>)	2,050.608	-	30.578	53.682	-	53.682	-	-	-	-	-	-
Plus Transfer (<i>\$ in Millions</i>)	279.031	-	-	-	-	-	-	-	-	-	-	-
Plus Hurricane (<i>\$ in Millions</i>)	1,623.280	-	-	-	-	-	-	-	-	-	-	-
Plus Hurricane Supplemental (OF & PD) (<i>\$ in Millions</i>)	25.970	-	-	-	-	-	-	-	-	-	-	-
Total Obligation Authority (<i>\$ in Millions</i>)	21,294.492	0.000	30.578	53.682	0.000	53.682	-	-	-	-	-	-
<i>(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)</i>												
Plus Outfitting and Post Delivery (<i>\$ in Millions</i>)	992.561	18.665	21.424	58.251	-	58.251	-	-	-	-	-	-
Total (<i>\$ in Millions</i>)	22,313.023	18.665	52.002	111.933	-	111.933	-	-	-	-	-	-
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	1,644.892	-	-	-	-	-	-	-	-	-	-	-
Description: Functional replacement for LKA 113, LPD 4, LSD 36, and LST 1179 classes of Amphibious Ships in embarking, transporting, and landing elements of a Marine landing force in an assault by helicopters, landing craft, amphibious vehicles, and by a combination of these methods to conduct primary amphibious warfare missions.												

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy			Date: May 2021																																																																													
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 03: Amphibious Ships / BSA 1: Amphibious Ships			P-1 Line Item Number / Title: 3036 / LPD-17																																																																													
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: N/A		Other Related Program Elements: N/A																																																																												
Line Item MDAP/MAIS Code: N/A																																																																																
Characteristics: - <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 30%;">Length Overall</td> <td style="width: 35%;">208.5 m</td> <td style="width: 35%;">684 ft</td> </tr> <tr> <td>Beam</td> <td>31.9 m</td> <td>105 ft</td> </tr> <tr> <td>Displacement</td> <td>25.3 lmt</td> <td>24.9 klt</td> </tr> <tr> <td>Draft</td> <td>7.0 m</td> <td>23 ft</td> </tr> </table> <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 30%;">Production Status:</td> <td style="width: 35%;">LPD 28</td> <td style="width: 35%;">LPD 29</td> </tr> <tr> <td>Contract Award Date</td> <td>Dec 2016</td> <td>Feb 2018</td> </tr> <tr> <td>Months to Completion</td> <td></td> <td></td> </tr> <tr> <td> a) Award to Delivery</td> <td>61 months</td> <td>65 months</td> </tr> <tr> <td> b) Construction Start to Delivery</td> <td>61 months</td> <td>60 months</td> </tr> <tr> <td>Delivery Date</td> <td>Jan 2022</td> <td>Jul 2023</td> </tr> <tr> <td>Completion Of Fitting Out</td> <td>Oct 2022</td> <td>Mar 2024</td> </tr> <tr> <td>Obligation Work Limit Date</td> <td>Sep 2023</td> <td>Feb 2025</td> </tr> </table> <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 30%;"><u>Design Schedule</u></td> <td style="width: 20%;"><u>Start / Issue</u></td> <td style="width: 20%;"><u>Complete / Response</u></td> <td style="width: 20%;"><u>Reissue</u></td> <td style="width: 30%;"><u>Reissue Complete / Response</u></td> </tr> <tr> <td>Issue Date for TLR</td> <td>N/A</td> <td>Sep 1988</td> <td></td> <td></td> </tr> <tr> <td>Issue Date for TLS</td> <td>N/A</td> <td>N/A</td> <td></td> <td></td> </tr> <tr> <td>Preliminary Design</td> <td>Jan 1993</td> <td>Nov 1993</td> <td></td> <td></td> </tr> <tr> <td>Contract Design</td> <td>Dec 1993</td> <td>Mar 1996</td> <td></td> <td></td> </tr> <tr> <td>Detail Design</td> <td>Dec 1996</td> <td>Jul 2002</td> <td></td> <td></td> </tr> <tr> <td>Request for Proposals</td> <td>N/A</td> <td>N/A</td> <td></td> <td></td> </tr> <tr> <td>Design Agent</td> <td></td> <td></td> <td></td> <td></td> </tr> </table> <u>Classification of Cost Estimate:</u> CLASS C					Length Overall	208.5 m	684 ft	Beam	31.9 m	105 ft	Displacement	25.3 lmt	24.9 klt	Draft	7.0 m	23 ft	Production Status:	LPD 28	LPD 29	Contract Award Date	Dec 2016	Feb 2018	Months to Completion			a) Award to Delivery	61 months	65 months	b) Construction Start to Delivery	61 months	60 months	Delivery Date	Jan 2022	Jul 2023	Completion Of Fitting Out	Oct 2022	Mar 2024	Obligation Work Limit Date	Sep 2023	Feb 2025	<u>Design Schedule</u>	<u>Start / Issue</u>	<u>Complete / Response</u>	<u>Reissue</u>	<u>Reissue Complete / Response</u>	Issue Date for TLR	N/A	Sep 1988			Issue Date for TLS	N/A	N/A			Preliminary Design	Jan 1993	Nov 1993			Contract Design	Dec 1993	Mar 1996			Detail Design	Dec 1996	Jul 2002			Request for Proposals	N/A	N/A			Design Agent				
Length Overall	208.5 m	684 ft																																																																														
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Request for Proposals	N/A	N/A																																																																														
Design Agent																																																																																

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1		P-1 Line Item Number / Title: 3036 / LPD-17		
Cost Categories	FY 2016		FY 2017	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Plan Costs	1		1	
Basic Construction/Conversion		1,478.976		1,435.821
Change Orders		35.000		29.000
Electronics		208.342		243.676
Hull, Mechanical, and Electrical (HM&E)		15.826		21.494
Ordnance		64.023		107.530
Other Cost		4.891		7.000
Total Ship Estimate		1,807.058		1,844.521
Less Advance Procurement FY 2013		242.471		-
Less Cost to Complete FY 2021		30.578		-
Less Cost to Complete FY 2022		-		53.682
Less Cost to Complete FY 2023		-		4.839
Less Prior Year Full Funding FY 2015		1,000.000		-
Net P-1 Funding		534.009		1,786.000

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Exhibit P-27, Ship Production Schedule: PB 2022 Navy				Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 03 / 1			P-1 Line Item Number / Title: 3036 / LPD-17		
Ship	Shipbuilder	Fiscal Year	Contract Award	Start of Construction	Delivery Date
LPD 28	HUNTINGTON INGALLS INDUSTRIES	2016	Dec 2016	Dec 2016	Jan 2022
LPD 29	HUNTINGTON INGALLS INDUSTRIES	2017	Feb 2018	Jul 2018	Jul 2023

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy

Date: May 2021

Appropriation / Budget Activity / Budget Sub Activity:
1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost

P-1 Line Item Number / Title:
5025 / TAO Fleet Oiler

ID Code (A=Service Ready, B=Not Service Ready): A

Program Elements for Code B Items: N/A

Other Related Program Elements: N/A

Line Item MDAP/MAIS Code: P452

Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (Units in Each)	4	2	-	1	-	1	-	-	-	-	-	-
Gross/Weapon System Cost (\$ in Millions)	2,422.960	1,076.261	0.000	668.184	0.000	668.184	-	-	-	-	-	-
Less PY Advance Procurement (\$ in Millions)	148.145	75.046	-	-	-	-	-	-	-	-	-	-
Less Cost To Complete (\$ in Millions)	145.749	-	-	-	-	-	-	-	-	-	-	-
Less Affordability Initiatives (\$ in Millions)	-	20.000	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (\$ in Millions)	2,129.066	981.215	0.000	668.184	0.000	668.184	-	-	-	-	-	-
Plus CY Advance Procurement (\$ in Millions)	223.191	73.000	-	76.012	-	76.012	-	-	-	-	-	-
Plus Cost To Complete (\$ in Millions)	15.449	3.700	59.900	66.700	-	66.700	-	-	-	-	-	-
Plus Affordability Initiatives (\$ in Millions)	-	-	20.000	-	-	-	-	-	-	-	-	-
Total Obligation Authority (\$ in Millions)	2,367.706	1,057.915	79.900	810.896	0.000	810.896	-	-	-	-	-	-
(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)												
Plus Outfitting and Post Delivery (\$ in Millions)	4.573	11.949	29.814	41.113	-	41.113	-	-	-	-	-	-
Total (\$ in Millions)	2,372.279	1,069.864	109.714	852.009	-	852.009	-	-	-	-	-	-
Gross/Weapon System Unit Cost (\$ in Millions)	605.740	538.131	-	668.184	-	668.184	-	-	-	-	-	-

Description:

T-AO 205 John Lewis Fleet Oiler Class will recapitalize the existing T-AO 187 fleet oiler class. The Navy's Combat Logistics Force (CLF) oilers supply fuel and dry cargo to Navy ships at sea. The T-AO Class will operate as shuttle ships from resupply posts to customer ships. Additionally, in conjunction with a T-AKE, they will accompany and stay on-station with a Carrier Strike Group (CSG) to provide fuel as required to customer ships.

Characteristics:

T-AO

Length Overall
Beam
Displacement
Draft

746 ft
106 ft
22,515 MT
33.5 ft

(Lightship)
(Design)

Production Status:

	T-AO 205	T-AO 206	T-AO 207	T-AO 208	T-AO 209	T-AO 210	T-AO 211
Contract Award Date	Jun 2016	Mar 2018	Dec 2018	Dec 2018	Mar 2020	Mar 2020	Jun 2022
Months to Completion							
a) Award to Delivery	69 months	55 months	55 months	60 months	56 months	60 months	43 months
b) Construction Start to Delivery	42 months	34 months	31 months	31 months	30 months	29 months	26 months
Delivery Date	Mar 2022	Oct 2022	Jul 2023	Dec 2023	Nov 2024	Mar 2025	Jan 2026
Completion Of Fitting Out	Jun 2022	Jan 2023	Oct 2023	Mar 2024	Feb 2025	Jun 2025	Apr 2026
Obligation Work Limit Date	May 2023	Dec 2023	Sep 2024	Feb 2025	Jan 2026	May 2026	Mar 2027

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost			P-1 Line Item Number / Title: 5025 / TAO Fleet Oiler	
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: N/A		Other Related Program Elements: N/A
Line Item MDAP/MAIS Code: P452				
<u>Design Schedule</u>	<u>Start / Issue</u>	<u>Complete / Response</u>	<u>Reissue</u>	<u>Reissue Complete / Response</u>
Issue Date for TLR	N/A	N/A		
Issue Date for TLS	N/A	N/A		
Preliminary Design	N/A	N/A		
Contract Design	N/A	N/A		
Detail Design	Jun 2016	Sep 2018		
Request for Proposals	Jun 2015	Dec 2015		
Design Agent				
<u>Classification of Cost Estimate:</u>				
Justification: The FY 2021 enacted includes congressional rescissions of \$73.0 million to T-AO Advance Procurement. However, these rescissions are not reflected on the P-40. FY 2022 requests funding for one T-AO Fleet Oiler and advance procurement for a FY 2023 Fleet Oiler.				

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1	P-1 Line Item Number / Title: 5025 / TAO Fleet Oiler
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Cost Categories ([†]) indicates the presence of a P-8a	FY 2016		FY 2018		FY 2019		FY 2020		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Plan Costs	1	102.121	1	-	2	-	2	-	1	-
Basic Construction/Conversion		617.867		547.046		960.643		963.118		616.606
Change Orders		10.828		5.891		9.500		9.643		6.166
Electronics ([†])		30.108		28.230		57.590		58.739		30.556
Hull, Mechanical, and Electrical (HM&E) ([†])		14.573		14.287		24.276		24.761		14.856
Other Cost		-		-		-		20.000		-
Total Ship Estimate		775.497		595.454		1,052.009		1,076.261		668.184
Less Advance Procurement FY 2017		-		73.077		-		-		-
Less Advance Procurement FY 2018		-		-		75.068		-		-
Less Advance Procurement FY 2019		-		-		-		75.046		-
Less Subsequent Full Funding FY 2021		-		-		-		20.000		-
Less Cost to Complete FY 2019		15.449		-		-		-		-
Less Cost to Complete FY 2020		-		3.700		-		-		-
Less Cost to Complete FY 2021		42.500		17.400		-		-		-
Less Cost to Complete FY 2022		23.358		43.342		-		-		-
Net P-1 Funding		694.190		457.935		976.941		981.215		668.184

Remarks:
The Other Cost category for the FY 2020 T-AO program includes \$20.0 million for affordability initiatives as appropriated in the Consolidated Appropriations Act, 2021.

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Exhibit P-27, Ship Production Schedule: PB 2022 Navy				Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1			P-1 Line Item Number / Title: 5025 / TAO Fleet Oiler		
Ship	Shipbuilder	Fiscal Year	Contract Award	Start of Construction	Delivery Date
T-AO 205	GD NASSCO	2016	Jun 2016	Sep 2018	Mar 2022
T-AO 206	GD NASSCO	2018	Mar 2018	Dec 2019	Oct 2022
T-AO 207	GD NASSCO	2019	Dec 2018	Dec 2020	Jul 2023
T-AO 208	GD NASSCO	2019	Dec 2018	May 2021	Dec 2023
T-AO 209	GD NASSCO	2020	Mar 2020	May 2022	Nov 2024
T-AO 210	GD NASSCO	2020	Mar 2020	Oct 2022	Mar 2025
T-AO 211	TBD	2022	Jun 2022	Nov 2023	Jan 2026

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1		P-1 Line Item Number / Title: 5025 / TAO Fleet Oiler		
Electronics	FY 2020		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
P-35 Items				
Radio Communication System (RCS) TURNKEY	2	13.448	1	6.996
P-35 Items Subtotal		13.448		6.996
Major Items				
Consolidated Afloat Networks and Enterprise Services (CANES)	2	6.886	1	3.582
Digital Modular Radio (DMR)	2	6.340	1	3.298
Commercial Broadband Satellite Program (CBSP)	2	4.026	1	2.094
AN/SLQ-25 NIXIE	2	3.996	1	2.079
AN/USQ-155 Tactical Variant Switch (TVS)	2	2.760	1	1.436
Major Items Subtotal		24.008		12.489
Other Cost Elements				
Minor Systems		21.283		11.071
Other Cost Elements Subtotal		21.283		11.071
Total Electronics		58.739		30.556

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity:		P-1 Line Item Number / Title:		
1611N / 05 / 1		5025 / TAO Fleet Oiler		
Hull, Mechanical, and Electrical (HM&E)	FY 2020		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Major Items				
Engineering Services	2	19.776	1	11.701
Logistics Support Services		1.597		1.662
SOLAS Variant RIB (Qty 2 per ship)		1.648		0.587
Material Handling Equipment		1.294		0.674
Shipboard Automated Maintenance Module (SAMM)		0.446		0.232
Major Items Subtotal		24.761		14.856
Total Hull, Mechanical, and Electrical (HM&E)		24.761		14.856

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1				P-1 Line Item Number / Title: 5025 / TAO Fleet Oiler			
Equipment Item: Radio Communication System (RCS) TURNKEY						PARM Code: N/A	
P-35 Category	FY 2020		FY 2022				
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)			
Major Hardware	2	2.514	1	1.309			
Ancillary Equipment		0.218		0.113			
Technical Engineering Services		2.100		1.092			
Ship Installation		7.784		4.049			
Program Management		0.832		0.433			
Total	2	13.448	1	6.996			
Description: The Radio Communication System (RCS) consists of the subsystems that provide data and voice communications across the RF spectrum. The RCS will be comprised of subsystems provided from various sources, including NAVWAR Program of Record systems, commercial systems, and associated ancillary equipment that can be obtained through the stock system and bought commercially. These subsystems will be integrated into one system and will include the automated and manual patching equipment required to configure these subsystems. The subsystems included in the RCS include the High Frequency System, Digital Modular Radio (DMR) VHF/UHF Line of Sight and UHF SATCOM voice, Naval Modular Automated Communications System (NAVMACS), Battle Force Tactical Network (BFTN), Tactical Variant Switch (TVS), Tactical Voice Terminal (TVT), Advanced Digital Networks System (ADNS), Commercial Broadband Satellite Program (CBSP), Fleet Broadcast System, Navy Order wire (NOW) Terminals, OE-570D/WSC UHF SATCOM Antenna, Portable Communications Equipment and Cryptologic equipment. The subsystems are integrated by NIWC LANT at the C4I- Ship Integration Facility (C-SIF) with the proper interfaces to operate as an overall system. The RCS subsystems and interfaces will be tested prior to shipment for installation onboard the T-AO ships.							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2020	T-AO 209	General Dynamics	C/FFP	Feb 2019	Option	2	1.257
FY 2022	T-AO 211	TBD	TBD	TBD		1	1.309
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2020	T-AO 209	Nov 2024	7	14	Feb 2023		
FY 2022	T-AO 211	Jan 2026	7	14	Apr 2024		
Competition/Second Source Initiatives: N/A							
Remarks: FY 2020 includes T-AO 209 and T-AO 210							

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Exhibit P-10, Advance Procurement Requirements Analysis (page 1 - Budget Funding Justification): PB 2022 Navy						Date: May 2021				
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1				P-1 Line Item Number / Title: 5025 / TAO Fleet Oiler						
First System (2022) Award Date: January 2021		First System (2022) Completion Date: January 2022			Interval Between Systems: 12 Months					
Cost Elements		Production Leadtime <i>(Months)</i>	When Required* <i>(Months)</i>	FY 2020 <i>(\$ M)</i>	FY 2021 <i>(\$ M)</i>	FY 2022 <i>(\$ M)</i>	FY 2023 <i>(\$ M)</i>	FY 2024 <i>(\$ M)</i>	FY 2025 <i>(\$ M)</i>	FY 2026 <i>(\$ M)</i>
Basic Construction/Conversion										
Propulsion, Auxiliary, Machinery, and Components ⁽¹³⁾		12-24	Various	73.000	-	76.012	-	-	-	-
Total: Basic Construction/Conversion				73.000	-	76.012	-	-	-	-
Hull, Mechanical, and Electrical (HM&E)										
Class Engineering Efforts		-	-	-	-	0.000	-	-	-	-
Total: Hull, Mechanical, and Electrical (HM&E)				-	-	-	-	-	-	-
Electronics										
Digital Modular Radio (DMR) ⁽¹⁴⁾		-	-	-	-	0.000	-	-	-	-
AN/SLQ-25 NIXIE		-	-	-	-	0.000	-	-	-	-
Total: Electronics				-	-	-	-	-	-	-
Total Advance Procurement/Obligation Authority				73.000	-	76.012				

*Note: "When Required" is the number of months required before ship delivery.

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Exhibit P-10, Advance Procurement Requirements Analysis (page 2 - Budget Funding Justification): PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1	P-1 Line Item Number / Title: 5025 / TAO Fleet Oiler
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Cost Elements	FY 2022						
	Production Leadtime (Months)	When Required* (Months)	Unit Cost (\$ M)	Contract Forecast Date	2022 Qty (Each)	For FY	Total Cost Request (\$ M)
Basic Construction/Conversion							
Propulsion, Auxiliary, Machinery, and Components ⁽¹³⁾	12-24	Various	-	Mar 2022	-	2023	76.012
<i>Total: Basic Construction/Conversion</i>							<i>76.012</i>
Hull, Mechanical, and Electrical (HM&E)							
<i>Total: Hull, Mechanical, and Electrical (HM&E)</i>							-
Electronics							
Digital Modular Radio (DMR) ⁽¹⁴⁾	-	-	-		-		0.000
AN/SLQ-25 NIXIE	-	-	-		-		0.000
<i>Total: Electronics</i>							-
Total Advance Procurement/Obligation Authority							76.012

Description:
test words

*Note: "When Required" is the number of months required before ship delivery.

Footnotes:

⁽¹³⁾ Funding to procure Contractor furnished Long Lead Time Materials (LLTM) and engineering related activities.

⁽¹⁴⁾ Funding to procure Government furnished Long Lead Time Materials (LLTM) and engineering related activities.

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy										Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost							P-1 Line Item Number / Title: 5030 / TAGOS Surtass Ships					
ID Code (A=Service Ready, B=Not Service Ready): A				Program Elements for Code B Items: N/A				Other Related Program Elements: N/A				
Line Item MDAP/MAIS Code: 180												
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (<i>Units in Each</i>)	-	-	-	1	-	1	-	-	-	-	-	-
Gross/Weapon System Cost (<i>\$ in Millions</i>)	0.000	0.000	0.000	434.384	0.000	434.384	-	-	-	-	-	-
Less PY Advance Procurement (<i>\$ in Millions</i>)	-	-	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (<i>\$ in Millions</i>)	0.000	0.000	0.000	434.384	0.000	434.384	-	-	-	-	-	-
Plus CY Advance Procurement (<i>\$ in Millions</i>)	-	-	-	-	-	-	-	-	-	-	-	-
Total Obligation Authority (<i>\$ in Millions</i>)	0.000	0.000	0.000	434.384	0.000	434.384	-	-	-	-	-	-
<i>(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)</i>												
Plus Outfitting and Post Delivery (<i>\$ in Millions</i>)	-	0.000	-	-	-	-	-	-	-	-	-	-
Total (<i>\$ in Millions</i>)	-	-	-	434.384	-	434.384	-	-	-	-	-	-
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	-	-	-	434.384	-	434.384	-	-	-	-	-	-

Description:
This is a new start effort in FY 2022.

Auxiliary General Ocean Surveillance (T-AGOS) ships gather underwater acoustical data to support the mission of the Integrated Undersea Surveillance System (IUSS) by providing a ship platform capable of theater anti-submarine acoustic passive and active surveillance. T-AGOS ships are operated by Military Sealift Command to support the anti-submarine warfare mission of the commanders of the Atlantic and Pacific Fleets. The two current classes of surveillance ships use Surveillance Towed-Array Sensor System (SURTASS) equipment to gather undersea acoustic data. The ships also carry electronic equipment to process and transmit that data via satellite to shore stations for evaluation. Funding will support recapitalization of the four Small Waterplane Area Twin Hull (SWATH) T-19 Class and one SWATH T-23 Class ships.

Characteristics:	T-AGOS(X)
Length Overall	TBD
Beam	TBD
Displacement	TBD
Draft	TBD

Production Status:	T-AGOS 2201
Contract Award Date	Aug 2022
Months to Completion	
a) Award to Delivery	48 months
b) Construction Start to Delivery	30 months
Delivery Date	Aug 2026
Completion Of Fitting Out	Nov 2026
Obligation Work Limit Date	Oct 2027

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost		P-1 Line Item Number / Title: 5030 / TAGOS Surtass Ships		
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: N/A		Other Related Program Elements: N/A
Line Item MDAP/MAIS Code: 180				
<u>Design Schedule</u>	<u>Start / Issue</u>	<u>Complete / Response</u>	<u>Reissue</u>	<u>Reissue Complete / Response</u>
Issue Date for TLR	N/A	N/A		
Issue Date for TLS	N/A	N/A		
Preliminary Design	Mar 2019	Jun 2020		
Contract Design	Jun 2020	Jun 2021		
Detail Design	Aug 2022	Feb 2024		
Request for Proposals	Aug 2021	Aug 2022		
Design Agent	TBD			
<u>Classification of Cost Estimate:</u>				

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy		Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1		P-1 Line Item Number / Title: 5030 / TAGOS Surtass Ships	
Cost Categories ^(†) indicates the presence of a P-8a	FY 2022		
	Qty <i>(Each)</i>	Total Cost <i>(\$ M)</i>	
Plan Costs	1	30.000	
Basic Construction/Conversion		297.100	
Change Orders		8.913	
Electronics ^(†)		81.000	
Hull, Mechanical, and Electrical (HM&E)		17.371	
Total Ship Estimate		434.384	
Net P-1 Funding		434.384	

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Exhibit P-27, Ship Production Schedule: PB 2022 Navy				Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1			P-1 Line Item Number / Title: 5030 / TAGOS Surtass Ships		
Ship	Shipbuilder	Fiscal Year	Contract Award	Start of Construction	Delivery Date
T-AGOS 2201	TBD	2022	Aug 2022	Feb 2024	Aug 2026

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy		Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1		P-1 Line Item Number / Title: 5030 / TAGOS Surtass Ships	
Electronics		FY 2022	
		Qty (Each)	Total Cost (\$ M)
P-35 Items			
C4I		1	28.000
SURTASS		4	53.000
P-35 Items Subtotal			81.000
Total Electronics			81.000

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1				P-1 Line Item Number / Title: 5030 / TAGOS Surtass Ships			
Equipment Item: C4I						PARM Code: N/A	
P-35 Category				FY 2022			
				Qty (Each)	Total Cost (\$ M)		
Major Hardware				1	7.808		
Ancillary Equipment					17.811		
Other Costs					2.381		
Total				1	28.000		
Description: The C4I includes Global Command & Control system - Maritime (GCCS-M) via Consolidated Afloat Networks and Enterprise Services (CANES), Global Positioning Navigation and Timing Service (GPNTS), Automatic Identification System AN/URN-31(V)1/(V)2, Identification Friend or Foe (IFF) AN/APX-123(V)3, Commercial Broadband Satellite Program (CBSP), Digital Modular Radio (DMR) UHF Line of Site and UHF SATCOM, Command and Control Office Information Exchange (C2OIX), Tactical Variant Switch, SI and HF communications, Military Sealift Command Network Management System, and Advanced Digital Networks System (ADNS).							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2022	T-AGOS 2201	TBD	TBD	TBD		1	7.808
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2022	T-AGOS 2201	Aug 2026	12	12	Aug 2024		
Competition/Second Source Initiatives: N/A							

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Exhibit P-35, Major Ship Component Fact Sheet: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1				P-1 Line Item Number / Title: 5030 / TAGOS Surtass Ships			
Equipment Item: SURTASS						PARM Code: N/A	
P-35 Category				FY 2022			
				Qty (Each)		Total Cost (\$ M)	
Major Hardware				1		15.620	
Ancillary Equipment						3.130	
System Engineering						3.240	
Active Array				1		14.040	
Active Handling System				1		11.580	
Passive Handling System				1		5.390	
Total				4		53.000	
Description: The SURTASS mission system is a Navy Acquisition Program of Record (POR) and T-AGOS is the host platform for the Surveillance Towed Array Sensor System (SURTASS). SURTASS is comprised of active and passive acoustic arrays, a processing segment, and a command, control, communications, computer, combat systems, and intelligence (C5I) segment to deliver tactical mission data to fleet and intelligence community users. SURTASS mission system will be procured to outfit the T-AGOS(X).							
Contract Data:							
Program Year	Hull	Prime Contractor	Contract Method/Type	Award Date	New/Option	Quantity (Each)	Unit Cost (\$ M)
FY 2022	T-AGOS 2201	TBD	TBD	Various		1	15.620
Delivery Date:							
Program Year	Hull	Earliest Ship Delivery Date	Months Required Before Delivery	Production Leadtime	Required Award Date		
FY 2022	T-AGOS 2201	Aug 2026	12	12	Aug 2024		
Competition/Second Source Initiatives: N/A							

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy **Date:** May 2021

Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost	P-1 Line Item Number / Title: 5035 / Towing, Salvage, and Rescue Ship (ATS)
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ID Code (A=Service Ready, B=Not Service Ready): A	Program Elements for Code B Items: N/A	Other Related Program Elements: N/A
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Line Item MDAP/MAIS Code: N/A

Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (<i>Units in Each</i>)	3	2	2	2	-	2	-	-	-	-	-	-
Gross/Weapon System Cost (<i>\$ in Millions</i>)	231.560	150.282	157.790	183.800	0.000	183.800	-	-	-	-	-	-
Less PY Advance Procurement (<i>\$ in Millions</i>)	-	-	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (<i>\$ in Millions</i>)	231.560	150.282	157.790	183.800	0.000	183.800	-	-	-	-	-	-
Plus CY Advance Procurement (<i>\$ in Millions</i>)	-	-	-	-	-	-	-	-	-	-	-	-
Total Obligation Authority (<i>\$ in Millions</i>)	231.560	150.282	157.790	183.800	0.000	183.800	-	-	-	-	-	-
<i>(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)</i>												
Plus Outfitting and Post Delivery (<i>\$ in Millions</i>)	-	2.546	9.991	17.595	-	17.595	-	-	-	-	-	-
Total (<i>\$ in Millions</i>)	231.560	152.828	167.781	201.395	-	201.395	-	-	-	-	-	-
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	77.187	75.141	78.895	91.900	-	91.900	-	-	-	-	-	-

Description:

The Navy requires ocean-going towing, salvage, and rescue capabilities to support Fleet operations. The Navy's current capabilities are provided by four T-ATF 166 Class Fleet tugs and four T-ARS 50 Class salvage ships which will reach the end of their expected service lives beginning in 2020 and 2025, respectively. The Towing, Salvage and Rescue Ships (T-ATS 6 Navajo Class) will recapitalize the current Fleet tugs and salvage ships with a common hull that is capable of performing the missions of the retiring T-ATF and T-ARS classes.

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy						Date: May 2021																																																																	
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Line Item MDAP/MAIS Code: N/A																																																																							
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Justification: Due to yard capacity and schedule challenges, the Navy has elected to not award the final three ship options to Gulf Island Shipyards (now owned by Bollinger). The Navy intends to enter sole source negotiations with Austal for the remaining ships to assist in the transition to a steel capable shipyard.																																																																							

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost		P-1 Line Item Number / Title: 5035 / Towing, Salvage, and Rescue Ship (ATS)
ID Code (A=Service Ready, B=Not Service Ready): A	Program Elements for Code B Items: N/A	Other Related Program Elements: N/A
Line Item MDAP/MAIS Code: N/A		

Footnotes:
(1) Due to yard capacity and schedule challenges, the Navy has elected to not award the final three ship options to Gulf Island Shipyards (now owned by Bollinger). The Navy intends to enter sole source negotiations with Austal for the remaining ships to assist in the transition to a steel capable shipyard.

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity:
1611N / 05 / 1

P-1 Line Item Number / Title:
5035 / Towing, Salvage, and Rescue Ship (ATS)

Cost Categories	FY 2016		FY 2018		FY 2019		FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Plan Costs	1		1		1		2		2		2	
Basic Construction/Conversion		63.742		62.842		66.639		130.097		137.102		157.606
Change Orders		3.200		2.145		2.008		2.602		2.809		4.366
Electronics		5.033		5.779		5.862		11.300		11.525		11.756
Hull, Mechanical, and Electrical (HM&E)		3.025		5.323		5.962		6.283		6.354		10.072
Total Ship Estimate		75.000		76.089		80.471		150.282		157.790		183.800
Net P-1 Funding		75.000		76.089		80.471		150.282		157.790		183.800

Remarks:

The estimated unit cost increases in FY22 are based on analysis of production man-hours for ships under construction as well as significant cost escalations involving material/commodity quotes that shipyards are experiencing as the post-COVID economy picks up

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Exhibit P-27, Ship Production Schedule: PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1	P-1 Line Item Number / Title: 5035 / Towing, Salvage, and Rescue Ship (ATS)
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Ship	Shipbuilder	Fiscal Year	Contract Award	Start of Construction	Delivery Date
T-ATS 6	Gulf Island Shipyards	2016	Mar 2018	Jun 2019	Aug 2022
T-ATS 7	Gulf Island Shipyards	2018	Apr 2019	Oct 2019	Dec 2022
T-ATS 8	Gulf Island Shipyards	2019	Apr 2019	Feb 2020	Apr 2023
T-ATS 9	Gulf Island Shipyards	2020	Apr 2020	Jun 2021	Jun 2024
T-ATS 10	Gulf Island Shipyards	2020	Apr 2020	Oct 2021	Oct 2024
T-ATS 11 ⁽¹⁾	Austal USA	2021	Sep 2021	Sep 2022	Nov 2024
T-ATS 12	Austal USA	2021	Sep 2021	Jan 2023	Apr 2025
T-ATS 13	Austal USA	2022	Mar 2022	Jun 2023	Jul 2025
T-ATS 14	Austal USA	2022	Mar 2022	Oct 2023	Nov 2025

Footnotes:

⁽¹⁾ Due to yard capacity and schedule challenges, the Navy has elected to not award the final three ship options to Gulf Island Shipyards (now owned by Bollinger). The Navy intends to enter sole source negotiations with Austal for the remaining ships to assist in the transition to a steel capable shipyard.

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy										Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost							P-1 Line Item Number / Title: 5100 / LCU 1700					
ID Code (A=Service Ready, B=Not Service Ready): A				Program Elements for Code B Items: N/A				Other Related Program Elements: N/A				
Line Item MDAP/MAIS Code: N/A												
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (<i>Units in Each</i>)	3	4	5	4	-	4	-	-	-	-	-	-
Gross/Weapon System Cost (<i>\$ in Millions</i>)	75.468	83.670	87.395	67.928	0.000	67.928	-	-	-	-	-	-
Less PY Advance Procurement (<i>\$ in Millions</i>)	-	-	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (<i>\$ in Millions</i>)	75.468	83.670	87.395	67.928	0.000	67.928	-	-	-	-	-	-
Plus CY Advance Procurement (<i>\$ in Millions</i>)	-	-	-	-	-	-	-	-	-	-	-	-
Total Obligation Authority (<i>\$ in Millions</i>)	75.468	83.670	87.395	67.928	0.000	67.928	-	-	-	-	-	-
<i>(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)</i>												
Plus Outfitting and Post Delivery (<i>\$ in Millions</i>)	-	-	6.692	9.095	-	9.095	-	-	-	-	-	-
Total (<i>\$ in Millions</i>)	75.468	83.670	94.087	77.023	-	77.023	-	-	-	-	-	-
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	25.156	20.918	17.479	16.982	-	16.982	-	-	-	-	-	-

Description:
 The Landing Craft, Utility (LCU) 1700 program provides heavy lift capability to transport personnel, weapons, equipment, and cargo from the ship to shore and shore to shore across the range of military operations (ROMO). LCU 1700 will be able to conduct 24 hours/day operations for up to 10 days for continuous landing of troops, equipment, and supplies; provide support for missions requiring persistence such as riverine sustainment, surveillance or port clearing; and execute missions to reinforce, reposition, and resupply forces over a wide operating area.

LCU 1700 provides the functional replacement for the LCU 1610 class of landing craft, all of which have significantly exceeded their 25 year service life, the average age is approaching 50 years old.

Note:
 Notional Characteristics based on Government Preliminary Design.
 Production Status dates provided are based on awarded contract delivery dates.

Commandant's Planning Guidance (CPG) issued in July 2019 identified Force Design as USMC number one priority. Force Design 2030 is developing force structure to meet future USMC warfighting requirements, including up to 40 LCU 1700 Class craft.

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy						Date: May 2021																																									
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost				P-1 Line Item Number / Title: 5100 / LCU 1700																																											
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: N/A			Other Related Program Elements: N/A																																										
Line Item MDAP/MAIS Code: N/A																																															
<table style="width:100%; border-collapse: collapse;"> <tr> <td style="width:15%;">Characteristics:</td> <td style="width:15%;">LCU</td> <td colspan="6"></td> </tr> <tr> <td>Length Overall</td> <td>139 ft</td> <td colspan="6"></td> </tr> <tr> <td>Beam</td> <td>31 ft</td> <td colspan="6"></td> </tr> <tr> <td>Displacement</td> <td>428 Tons</td> <td colspan="6"></td> </tr> <tr> <td>Draft</td> <td>7.3 ft</td> <td colspan="6"></td> </tr> </table>								Characteristics:	LCU							Length Overall	139 ft							Beam	31 ft							Displacement	428 Tons							Draft	7.3 ft						
Characteristics:	LCU																																														
Length Overall	139 ft																																														
Beam	31 ft																																														
Displacement	428 Tons																																														
Draft	7.3 ft																																														
Production Status:		LCU 1700	LCU 1701	LCU 1702	LCU 1703	LCU 1704	LCU 1705	LCU 1706																																							
Contract Award Date		Mar 2018	Feb 2019	Feb 2019	Apr 2020	Apr 2020	Apr 2020	Apr 2020																																							
Months to Completion																																															
a) Award to Delivery		45 months	39 months	42 months	32 months	35 months	38 months	41 months																																							
b) Construction Start to Delivery		22 months	21 months	21 months	20 months	20 months	20 months	20 months																																							
Delivery Date		Dec 2021	May 2022	Aug 2022	Dec 2022	Mar 2023	Jun 2023	Sep 2023																																							
Completion Of Fitting Out		Jun 2022	Jun 2022	Jan 2023	Jan 2023	Dec 2023	Dec 2023	Dec 2023																																							
Obligation Work Limit Date		May 2023	May 2023	Dec 2023	Dec 2023	Nov 2024	Nov 2024	Nov 2024																																							
Production Status:		LCU 1707	LCU 1708	LCU 1709	LCU 1710	LCU 1711	LCU 1712	LCU 1713																																							
Contract Award Date		Aug 2021	Aug 2021	Aug 2021	Aug 2021	Aug 2021	Mar 2022	Mar 2022																																							
Months to Completion																																															
a) Award to Delivery		27 months	29 months	30 months	31 months	32 months	27 months	28 months																																							
b) Construction Start to Delivery		20 months	19 months	17 months	15 months	14 months	13 months	12 months																																							
Delivery Date		Nov 2023	Jan 2024	Feb 2024	Mar 2024	Apr 2024	Jun 2024	Jul 2024																																							
Completion Of Fitting Out		Dec 2023	May 2024	May 2024	May 2024	May 2024	Dec 2024	Dec 2024																																							
Obligation Work Limit Date		Nov 2024	Apr 2025	Apr 2025	Apr 2025	Apr 2025	Nov 2025	Nov 2025																																							
Production Status:		LCU 1714	LCU 1715																																												
Contract Award Date		Mar 2022	Mar 2022																																												
Months to Completion																																															
a) Award to Delivery		30 months	32 months																																												
b) Construction Start to Delivery		12 months	12 months																																												
Delivery Date		Sep 2024	Nov 2024																																												
Completion Of Fitting Out		Dec 2024	Dec 2024																																												
Obligation Work Limit Date		Nov 2025	Nov 2025																																												
Design Schedule		Start / Issue		Complete / Response	Reissue	Reissue Complete / Response																																									
Issue Date for TLR		N/A		N/A																																											
Issue Date for TLS		N/A		N/A																																											
Preliminary Design		Mar 2014		May 2015																																											
Contract Design		Jun 2015		Jun 2016																																											
Detail Design		Apr 2018		May 2020																																											
Request for Proposals		Feb 2017		May 2017																																											

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost		P-1 Line Item Number / Title: 5100 / LCU 1700		
ID Code (A=Service Ready, B=Not Service Ready): A	Program Elements for Code B Items: N/A		Other Related Program Elements: N/A	
Line Item MDAP/MAIS Code: N/A				
<u>Design Schedule</u> Design Agent <u>Classification of Cost Estimate:</u>	<u>Start / Issue</u> Swiftships LLC	<u>Complete / Response</u>	<u>Reissue</u>	<u>Reissue Complete / Response</u>

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1	P-1 Line Item Number / Title: 5100 / LCU 1700
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Cost Categories	FY 2016		FY 2019		FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Plan Costs	1	5.000	2	-	4	-	5	-	4	-
Basic Construction/Conversion		19.018		33.163		67.673		62.862		52.615
Change Orders		1.600		1.500		3.170		2.799		2.263
Electronics		3.890		3.768		6.761		11.607		8.307
Hull, Mechanical, and Electrical (HM&E)		2.360		1.589		3.060		3.922		2.388
Other Cost		2.080		1.500		3.006		6.205		2.355
Total Ship Estimate		33.948		41.520		83.670		87.395		67.928
Net P-1 Funding		33.948		41.520		83.670		87.395		67.928

Remarks:

Beginning in FY 2021, Basic Construction pricing is based off contractor Firm-Fix Pricing.
FY 2016 Electronics cost includes non-recurring engineering costs. FY 2021 Electronics cost is higher in FY21 due to Radio Communication Suite costs based on the number of hulls supported each year.
The FY 2021 higher Other cost is due to additional SUPSHIP resources to provide on-site Navy personnel for oversight, testing and quality assurance inspections in order to ensure the Program remains on schedule and within budget. The shipbuilder is anticipated to have five craft under construction during FY 2021.

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Exhibit P-27, Ship Production Schedule: PB 2022 Navy					Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1			P-1 Line Item Number / Title: 5100 / LCU 1700		
Ship	Shipbuilder	Fiscal Year	Contract Award	Start of Construction	Delivery Date
LCU 1700	Swiftships LLC	2016	Mar 2018	Feb 2020	Dec 2021
LCU 1701	Swiftships LLC	2019	Feb 2019	Aug 2020	May 2022
LCU 1702	Swiftships LLC	2019	Feb 2019	Nov 2020	Aug 2022
LCU 1703	Swiftships LLC	2020	Apr 2020	Apr 2021	Dec 2022
LCU 1704	Swiftships LLC	2020	Apr 2020	Jul 2021	Mar 2023
LCU 1705	Swiftships LLC	2020	Apr 2020	Oct 2021	Jun 2023
LCU 1706	Swiftships LLC	2020	Apr 2020	Jan 2022	Sep 2023
LCU 1707	Swiftships LLC	2021	Aug 2021	Mar 2022	Nov 2023
LCU 1708	Swiftships LLC	2021	Aug 2021	Jun 2022	Jan 2024
LCU 1709	Swiftships LLC	2021	Aug 2021	Sep 2022	Feb 2024
LCU 1710	Swiftships LLC	2021	Aug 2021	Dec 2022	Mar 2024
LCU 1711	Swiftships LLC	2021	Aug 2021	Feb 2023	Apr 2024
LCU 1712	Swiftships LLC	2022	Mar 2022	May 2023	Jun 2024
LCU 1713	Swiftships LLC	2022	Mar 2022	Jul 2023	Jul 2024
LCU 1714	Swiftships LLC	2022	Mar 2022	Sep 2023	Sep 2024
LCU 1715	Swiftships LLC	2022	Mar 2022	Nov 2023	Nov 2024

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy								Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost						P-1 Line Item Number / Title: 5110 / Outfitting				
ID Code (A=Service Ready, B=Not Service Ready): A			Program Elements for Code B Items: N/A			Other Related Program Elements: N/A				
Line Item MDAP/MAIS Code: N/A										
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Full Funding TOA - Outfitting (\$ in Millions)	669.953	189.325	225.861	219.732	-	-	-	-	-	-
Full Funding TOA - Post Delivery (\$ in Millions)	922.565	501.866	520.748	430.555	-	-	-	-	-	-
Full Funding TOA - First Destination (\$ in Millions)	42.022	4.801	5.396	5.420	-	-	-	-	-	-
Total Obligation Authority (\$ in Millions)	1,634.540	695.992	752.005	655.707	-	-	-	-	-	-

Description:
 Outfitting (OF) funds are used to acquire on board repair parts, other secondary items, equipage, recreation items, precommissioning crew support and general use consumables furnished to the shipbuilder or the fitting-out activity to fill the ship's initial allowances as defined by the baseline coordinated shipboard allowance list (COSAL). The program also budgets for contractor-furnished spares, a lead-time away from delivery. The program ensures operational readiness of ships undergoing new construction, conversion, service life extension program (SLEP), and nuclear refueling. It ensures these ships receive their full allowances of spare parts and equipment which are vitally required to support the shipboard maintenance process; ensures ships are equipped with operating space items (tools, test equipment, damage control), personnel safety and survivability commodities for successful completion of builder sea trials; supports shipboard maintenance and thereby achieving the OPNAV-directed supply readiness goals for material on board ship at delivery. SCN funding for the initial fill of allowance list items are limited to those items on the COSAL and authorized requirements through the Obligation Work Limiting Date (OWLD). While most outfitting funds are executed prior to ships' completion of fitting out dates, some outfitting funding may be required in the fiscal year following the scheduled Delivery Date.

Post Delivery (PD) funding covers the repair of government-responsible items which were believed to have been complete to standard and/or operable at delivery, as well as funding to conduct tests and trials after delivery. It is essential to deliver to the Fleet complete ships, free from both contractor and government responsible deficiencies, capable of supporting the Navy's mission. The Post Shakedown Availability (PSA) is a shipyard availability assigned to commence after delivery and to be completed prior to the expiration of the SCN OWLD. It is during this time that acceptance and final contract trials deficiencies will be corrected. The purpose of the PSA is to correct new construction deficiencies found during the shakedown period; to correct contractor and government responsible deficiencies previously authorized; and accomplishment of other improvements or class items as authorized. Funding is used for corrections authorized by the ship's Program Manager as a result of builders' trials (pre-delivery), acceptance or underway trials, final contract trials, trial board items, and correction of production-related defects or deficiencies which develop during the post delivery period. Although the majority of post delivery funding occurs after ships' delivery dates, some funding is required prior to the delivery date in preparation for post delivery events.

First Destination Transportation (FDT). FDT delivers material from a procurement source to the first point of use or storage which may be in the Continental United States or overseas. The procurement source may be a supplier outside of the Department of Defense (DoD) supply system or a DoD activity which fabricates new material. FDT also includes transportation from point of origin to point of use or storage when Navy-owned material or equipment is provided to a contractor incident to a Research and Development project or a system/equipment acquisition or modification.

Note:
 DDG 51 Outfitting: OF funding phased to support ship requirements through Obligation Work Limiting Date (OWLD) for Hulls 118, 120-124, and 127.

DDG 51 Post Delivery: Pearl Harbor Homeported hulls (DDG 118, DDG 120, and DDG 121) incur higher PD execution costs for PSA and event support due to location based requirements.

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Exhibit P-29, Outfitting: PB 2022 Navy

Date: May 2021

Appropriation / Budget Activity / Budget Sub Activity:
1611N / 05 / 1P-1 Line Item Number / Title:
5110 / Outfitting

Ship Class	Hull Number	Program Year	Contract Award	Start of Const.	Delivery Date	CFO	PSA Start	PSA Finish	OWL Date	Prior Years	FY 2020	FY 2021	FY 2022	To Complete	Total
CO CLASS	826	2021	Oct 2020	Oct 2020	Oct 2027	Oct 2027	Mar 2029	Sep 2029	Mar 2030	-	-	-	-	-	-
CO CLASS Total										-	-	-	-	-	-
CVN	78	2008	Sep 2008	Aug 2005	May 2017	Jul 2017	Jul 2018	Oct 2019	Oct 2021	151.998	8.814	-	-	-	-
CVN	79	2013	Jun 2015	Feb 2011	Jun 2024	Feb 2025	Mar 2025	Sep 2025	Jan 2026	2.961	1.005	38.539	41.981	-	-
CVN	80	2018	Jan 2019	Jan 2019	Mar 2028	May 2028	Oct 2028	Mar 2029	Apr 2029	-	-	-	-	-	-
CVN Total										154.959	9.819	38.539	41.981	-	-
CVN	81	2020	Jan 2019	Jan 2019	Feb 2032	Apr 2032	Sep 2032	Feb 2033	Mar 2033	-	-	-	-	-	-
CVN Total										-	-	-	-	-	-
VIRGINIA	790	2013	Dec 2008	Mar 2013	Sep 2018	Sep 2018	Mar 2019	Jun 2021	Jun 2021	20.678	-	-	-	-	-
VIRGINIA	791	2013	Dec 2008	Sep 2013	Oct 2019	Oct 2019	Jul 2020	Aug 2021	Aug 2021	17.860	-	-	-	-	-
VIRGINIA	792	2014	Apr 2014	May 2014	Apr 2020	Apr 2020	Jul 2021	Feb 2022	Feb 2022	17.983	-	-	-	-	-
VIRGINIA	793	2014	Apr 2014	Sep 2014	Oct 2021	Oct 2021	Apr 2022	Jul 2022	Sep 2022	16.299	2.160	0.316	-	-	-
VIRGINIA	794	2015	Apr 2014	Apr 2015	Nov 2021	Nov 2021	May 2022	Aug 2022	Oct 2022	17.452	4.141	0.707	-	-	-
VIRGINIA	795	2015	Apr 2014	Sep 2015	Apr 2022	Apr 2022	Jul 2022	Oct 2022	Mar 2023	14.927	3.142	0.953	-	-	-
VIRGINIA	796	2016	Apr 2014	Mar 2016	Dec 2022	Dec 2022	Feb 2023	May 2023	Nov 2023	10.613	5.464	2.407	0.455	-	-
VIRGINIA	797	2016	Apr 2014	Sep 2016	Apr 2023	Apr 2023	Aug 2023	Nov 2023	Mar 2024	10.349	3.882	3.406	0.900	-	-
VIRGINIA	798	2017	Apr 2014	Mar 2017	Nov 2023	Nov 2023	Mar 2024	Jun 2024	Oct 2024	-	10.081	5.487	0.990	-	-
VIRGINIA	799	2017	Apr 2014	Sep 2017	Mar 2024	Mar 2024	Aug 2024	Nov 2024	Feb 2025	-	7.841	6.153	3.173	-	-
VIRGINIA	800	2018	Apr 2014	Mar 2018	Oct 2024	Oct 2024	Feb 2025	May 2025	Sep 2025	-	-	10.509	6.153	-	-
VIRGINIA	801	2018	Apr 2014	Sep 2018	Feb 2025	Feb 2025	May 2025	Aug 2025	Jan 2026	-	-	-	5.121	-	-
VIRGINIA	802	2019	Dec 2019	Sep 2019	Jun 2025	Jun 2025	Oct 2025	Feb 2026	May 2026	-	-	-	-	-	-
VIRGINIA	803	2019	Dec 2019	Mar 2020	Apr 2026	Apr 2026	Aug 2026	Jan 2027	Mar 2027	-	-	-	-	-	-
VIRGINIA	804	2020	Dec 2019	Sep 2020	Oct 2026	Oct 2026	Feb 2027	Jun 2027	Sep 2027	-	-	-	-	-	-
VIRGINIA	805	2020	Dec 2019	Jul 2021	May 2027	May 2027	Sep 2027	Jan 2028	Apr 2028	-	-	-	-	-	-
VIRGINIA	806	2021	Dec 2019	Dec 2021	Nov 2027	Nov 2027	Mar 2028	Jul 2028	Oct 2028	-	-	-	-	-	-
VIRGINIA	807	2021	Dec 2019	May 2022	May 2028	May 2028	Sep 2028	Jan 2029	Apr 2029	-	-	-	-	-	-
VIRGINIA	808	2022	Dec 2019	Oct 2022	Nov 2028	Nov 2028	Mar 2029	Jul 2029	Oct 2029	-	-	-	-	-	-
VIRGINIA	809	2022	Dec 2019	Mar 2023	May 2029	May 2029	Sep 2029	Jan 2030	Apr 2030	-	-	-	-	-	-
VIRGINIA	810	2023	Dec 2019	Sep 2023	Aug 2029	Aug 2029	Dec 2029	Apr 2030	Jul 2030	-	-	-	-	-	-
VIRGINIA	811	2023	Dec 2023	Mar 2024	Feb 2030	Feb 2030	Jun 2030	Oct 2030	Jan 2031	-	-	-	-	-	-
VIRGINIA	812	2024	Dec 2023	Sep 2024	Aug 2030	Aug 2030	Dec 2030	Apr 2031	Jul 2031	-	-	-	-	-	-
VIRGINIA Total										126.161	36.711	29.938	16.792	-	-
CVN-RCOH	73	2016	Aug 2017	Aug 2017	Aug 2022	Oct 2022	Aug 2022	Aug 2023	Sep 2023	22.320	21.661	9.368	7.334	-	-
CVN-RCOH	74	2020	Jan 2021	May 2021	Aug 2025	Dec 2025	Nov 2025	Dec 2025	Nov 2026	-	-	1.342	6.776	-	-
CVN-RCOH	75	2025	May 2025	May 2025	Mar 2029	May 2029	Mar 2029	Mar 2030	Apr 2030	-	-	-	-	-	-
CVN-RCOH Total										22.320	21.661	10.710	14.110	-	-
DDG 1000	1000	2007	Feb 2008	Feb 2009	Apr 2020	Jan 2021	Jul 2021	Dec 2021	Dec 2021	66.527	1.237	-	-	-	-
DDG 1000	1001	2007	Sep 2011	Mar 2010	Mar 2022	Apr 2022	Aug 2022	Mar 2023	Mar 2023	22.035	1.732	-	-	-	-
DDG 1000	1002	2009	Sep 2011	Apr 2012	Apr 2024	May 2024	Oct 2024	Mar 2025	Apr 2025	9.524	0.986	0.717	-	-	-

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Exhibit P-29, Outfitting: PB 2022 Navy

Date: May 2021

Appropriation / Budget Activity / Budget Sub Activity:
1611N / 05 / 1P-1 Line Item Number / Title:
5110 / Outfitting

Ship Class	Hull Number	Program Year	Contract Award	Start of Const.	Delivery Date	CFO	PSA Start	PSA Finish	OWL Date	Prior Years	FY 2020	FY 2021	FY 2022	To Complete	Total
DDG 1000 Total										98.086	3.955	0.717	-	-	-
DDG	118	2013	Jun 2013	Aug 2015	Mar 2021	Sep 2021	May 2022	Aug 2022	Aug 2022	20.162	0.547	1.815	1.040	-	-
DDG	120	2013	Mar 2014	Sep 2016	Mar 2022	Jul 2022	Feb 2023	Jun 2023	Jun 2023	22.238	13.597	4.400	1.561	-	-
DDG	119	2014	Mar 2014	Jul 2015	Apr 2020	Sep 2020	Mar 2021	Jul 2021	Aug 2021	0.491	0.013	2.078	-	-	-
DDG	121	2015	Jun 2013	Apr 2016	Nov 2021	Mar 2022	Sep 2022	Jan 2023	Feb 2023	18.315	3.348	1.822	2.767	-	-
DDG	122	2015	Jun 2013	Sep 2017	Jan 2023	May 2023	Dec 2023	Mar 2024	Apr 2024	-	5.212	12.509	3.125	-	-
DDG	123	2016	Jun 2013	Jan 2017	Jul 2022	Nov 2022	Jun 2023	Oct 2023	Oct 2023	-	15.054	5.897	1.806	-	-
DDG	124	2016	Jun 2013	Jul 2018	Sep 2023	Jan 2024	Jul 2024	Oct 2024	Dec 2024	-	-	14.720	4.041	-	-
DDG	127	2016	Sep 2017	Apr 2019	Mar 2024	Jul 2024	Jan 2025	May 2025	Jun 2025	-	-	6.275	2.076	-	-
DDG	125	2017	Jun 2013	May 2018	Apr 2023	Aug 2023	May 2024	Jul 2024	Jul 2024	-	-	-	5.438	-	-
DDG	126	2017	Jun 2013	Mar 2020	Jun 2024	Oct 2024	Apr 2025	Aug 2025	Sep 2025	-	-	-	-	-	-
DDG	128	2018	Sep 2018	Apr 2020	Oct 2024	Feb 2025	Aug 2025	Dec 2025	Jan 2026	-	-	-	-	-	-
DDG	129	2018	Sep 2018	Jan 2021	Jul 2025	Nov 2025	May 2026	Sep 2026	Oct 2026	-	-	-	-	-	-
DDG	130	2019	Sep 2018	Dec 2020	Jul 2025	Nov 2025	May 2026	Sep 2026	Oct 2026	-	-	-	-	-	-
DDG	131	2019	Sep 2018	Mar 2022	Apr 2026	Aug 2026	Jan 2027	May 2027	Jul 2027	-	-	-	-	-	-
DDG	132	2019	Dec 2018	Jun 2021	May 2026	Sep 2026	Mar 2027	Jun 2027	Aug 2027	-	-	-	-	-	-
DDG	133	2020	Sep 2018	Oct 2022	Nov 2026	Mar 2027	Aug 2027	Dec 2027	Feb 2028	-	-	-	-	-	-
DDG	134	2020	Sep 2018	Apr 2022	Nov 2026	Mar 2027	Sep 2027	Jan 2028	Feb 2028	-	-	-	-	-	-
DDG	135	2020	Jun 2020	May 2023	Jun 2027	Oct 2027	Apr 2028	Aug 2028	Sep 2028	-	-	-	-	-	-
DDG	136	2021	Sep 2018	Jan 2023	Jun 2027	Oct 2027	Apr 2028	Aug 2028	Sep 2028	-	-	-	-	-	-
DDG	137	2021	Sep 2018	Oct 2023	Nov 2027	Mar 2028	Sep 2028	Dec 2028	Feb 2029	-	-	-	-	-	-
DDG	138	2022	Sep 2018	Feb 2024	Apr 2028	Aug 2028	May 2029	Sep 2029	Oct 2029	-	-	-	-	-	-
DDG	139	2023	Jun 2023	Jul 2024	Jul 2028	Nov 2028	May 2029	Sep 2029	Oct 2029	-	-	-	-	-	-
DDG	140	2023	Jun 2023	Jul 2024	Jul 2028	Nov 2028	May 2029	Sep 2029	Oct 2029	-	-	-	-	-	-
DDG	141	2024	Jun 2023	Jul 2025	Jul 2029	Nov 2029	May 2030	Sep 2030	Oct 2030	-	-	-	-	-	-
DDG	142	2024	Jun 2023	Jul 2025	Jul 2029	Nov 2029	May 2030	Sep 2030	Oct 2030	-	-	-	-	-	-
DDG	143	2025	Jun 2023	Jul 2026	Jul 2030	Nov 2030	May 2031	Sep 2031	Oct 2031	-	-	-	-	-	-
DDG	144	2025	Jun 2023	Jul 2026	Jul 2030	Nov 2030	May 2031	Sep 2031	Oct 2031	-	-	-	-	-	-
DDG	145	2026	Jun 2023	Jul 2027	Jul 2031	Nov 2031	May 2032	Sep 2032	Oct 2032	-	-	-	-	-	-
DDG	147	2026	Jun 2023	Jul 2027	Jul 2031	Nov 2031			Oct 2032	-	-	-	-	-	-
DDG	146	2027	Jun 2023	Jul 2028	Jul 2032	Nov 2032	May 2033	Sep 2033	Oct 2033	-	-	-	-	-	-
DDG Total										61.206	37.771	49.516	21.854	-	-
LCS	11	2012	Mar 2012	Aug 2013	Aug 2018	Jun 2019	Oct 2019	May 2020	May 2020	6.258	-	-	-	-	-
LCS	14	2013	Mar 2013	Feb 2014	Feb 2018	Aug 2018	Apr 2019	Nov 2019	Mar 2020	6.148	-	-	-	-	-
LCS	16	2013	Mar 2013	Sep 2014	Apr 2018	Feb 2019	Oct 2019	May 2020	Jun 2020	5.942	-	-	-	-	-
LCS	13	2013	Mar 2013	Feb 2014	Aug 2018	Apr 2019	Jul 2019	Mar 2020	Mar 2020	5.329	-	-	-	-	-
LCS	15	2013	Mar 2013	Dec 2014	Feb 2019	Feb 2020	Apr 2020	Dec 2020	Jan 2021	5.857	-	-	-	-	-
LCS	18	2014	Mar 2014	Mar 2015	Aug 2018	May 2019	Feb 2020	Aug 2020	Sep 2020	5.494	-	-	-	-	-
LCS	20	2014	Mar 2014	Feb 2016	Jun 2019	Dec 2019	Jun 2020	Sep 2020	Nov 2020	6.882	0.500	-	-	-	-

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Exhibit P-29, Outfitting: PB 2022 Navy											Date: May 2021				
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1								P-1 Line Item Number / Title: 5110 / Outfitting							
Ship Class	Hull Number	Program Year	Contract Award	Start of Const.	Delivery Date	CFO	PSA Start	PSA Finish	OWL Date	Prior Years	FY 2020	FY 2021	FY 2022	To Complete	Total
LCS	17	2014	Mar 2014	Aug 2015	Jul 2019	May 2020	Aug 2020	Mar 2021	Sep 2021	5.984	-	-	-	-	-
LCS	19	2014	Mar 2014	Aug 2016	Feb 2020	Dec 2020	Mar 2021	Sep 2021	Sep 2021	4.792	0.286	0.800	-	-	-
LCS	22	2015	Mar 2015	Dec 2016	Feb 2020	Jun 2020	Mar 2021	May 2021	May 2021	6.028	0.231	0.400	-	-	-
LCS	24	2015	Mar 2015	Jul 2017	Jun 2020	Feb 2021	Jul 2021	Sep 2021	Jan 2022	4.145	2.119	0.800	-	-	-
LCS	21	2015	Mar 2015	Feb 2017	Oct 2021	Jul 2022	Dec 2022	Feb 2023	Jun 2023	3.861	2.696	-	1.006	-	-
LCS	26	2016	Mar 2016	Jan 2018	Dec 2020	Sep 2021	Jan 2022	Apr 2022	Aug 2022	1.277	6.824	1.007	-	-	-
LCS	23	2016	Nov 2015	Sep 2017	Dec 2021	Jul 2022	Dec 2022	Feb 2023	Jun 2023	5.501	1.117	-	1.007	-	-
LCS	25	2016	Mar 2016	Feb 2018	May 2022	Dec 2022	May 2023	Jul 2023	Nov 2023	2.653	2.656	0.072	1.220	-	-
LCS	28	2017	Jun 2017	Oct 2018	Jun 2021	Mar 2022	Oct 2022	Dec 2022	Feb 2023	1.328	3.850	1.658	0.986	-	-
LCS	30	2017	Oct 2017	May 2019	Dec 2021	Sep 2022	Feb 2023	Apr 2023	Aug 2023	0.400	-	5.079	0.995	-	-
LCS	27	2017	Oct 2017	Nov 2018	Oct 2022	Jun 2023	Jan 2024	Mar 2024	May 2024	-	-	5.235	2.196	-	-
LCS	32	2018	Sep 2018	Dec 2019	Jul 2022	Apr 2023	Sep 2023	Nov 2023	Mar 2024	-	-	5.831	1.930	-	-
LCS	34	2018	Sep 2018	Jun 2020	Feb 2023	Nov 2023	Apr 2024	Jun 2024	Oct 2024	-	-	3.476	4.389	-	-
LCS	29	2018	Sep 2018	Jun 2019	May 2023	Jan 2024	Jun 2024	Aug 2024	Dec 2024	-	-	3.300	2.082	-	-
LCS	36	2019	Dec 2018	Jan 2021	Sep 2023	Jun 2024	Nov 2024	Jan 2025	May 2025	-	-	-	2.872	-	-
LCS	31	2019	Jan 2019	Jun 2020	Apr 2024	Dec 2024	May 2025	Jul 2025	Nov 2025	-	-	-	2.646	-	-
LCS	38	2019	Dec 2018	Jul 2021	Apr 2024	Jan 2025	Jun 2025	Aug 2025	Dec 2025	-	-	-	-	-	-
LCS Total										77.879	20.279	27.658	21.329	-	-
LPD Flight II	30	2018	Mar 2019	Mar 2020	Feb 2025	Nov 2025	Apr 2026	Sep 2026	Oct 2026	-	-	-	-	-	-
LPD Flight II	31	2021	Apr 2020	Apr 2022	Feb 2027	Sep 2027	Feb 2028	Aug 2028	Aug 2028	-	-	-	-	-	-
LPD Flight II	32	2023	Jun 2023	Oct 2023	Aug 2028	Apr 2029	Sep 2029	Mar 2030	Mar 2030	-	-	-	-	-	-
LPD Flight II	33	2025	Apr 2025	Apr 2025	Feb 2030	Sep 2030	Feb 2031	Aug 2031	Aug 2031	-	-	-	-	-	-
LPD Flight II Total										-	-	-	-	-	-
LPD	28	2016	Dec 2016	Dec 2016	Jan 2022	Oct 2022	Mar 2023	Aug 2023	Sep 2023	-	18.458	8.124	1.557	-	-
LPD	29	2017	Feb 2018	Jul 2018	Jul 2023	Mar 2024	Sep 2024	Jan 2025	Feb 2025	-	-	-	16.231	-	-
LPD Total										-	18.458	8.124	17.788	-	-
ESB	4	2014	Dec 2014	Oct 2015	Feb 2018	Jun 2018	Mar 2019	Jul 2019	Mar 2020	15.381	-	-	-	-	-
ESB	5	2016	Dec 2016	Jan 2017	Nov 2019	Feb 2020	Aug 2020	Dec 2020	Aug 2021	8.927	5.400	-	-	-	-
ESB	6	2018	Aug 2019	Jun 2020	Sep 2022	Dec 2022	May 2023	Aug 2023	Nov 2023	-	-	13.167	4.601	-	-
ESB	7	2019	Aug 2019	Jan 2022	Jan 2024	Apr 2024	Aug 2024	Nov 2024	Mar 2025	-	-	-	4.486	-	-
ESB Total										24.308	5.400	13.167	9.087	-	-
LHA	7	2011	May 2012	Jul 2013	Feb 2020	Nov 2020	Jan 2021	Mar 2021	Oct 2021	47.262	1.600	-	-	-	-
LHA	8	2017	Jun 2017	Oct 2018	Feb 2025	Oct 2025	Dec 2025	Feb 2026	Sep 2026	-	-	1.000	16.024	-	-
LHA Total										47.262	1.600	1.000	16.024	-	-
EPF	11	2015	Sep 2016	Jan 2017	Dec 2019	Mar 2020	Jan 2021	May 2021	Jul 2021	3.928	-	-	-	-	-
EPF	12	2016	Sep 2016	Sep 2017	Sep 2020	Dec 2020	Feb 2021	Apr 2021	Nov 2021	3.084	-	-	-	-	-
EPF	13	2018	Mar 2019	Aug 2019	Jul 2022	Oct 2022	Apr 2023	Jun 2023	Sep 2023	-	-	3.000	2.166	-	-
EPF	14	2019	Mar 2019	Oct 2020	Jun 2023	Sep 2023	Mar 2024	May 2024	Aug 2024	-	-	0.400	5.113	-	-
EPF Total										7.012	-	3.400	7.279	-	-

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Exhibit P-29, Outfitting: PB 2022 Navy

Date: May 2021

Appropriation / Budget Activity / Budget Sub Activity:
1611N / 05 / 1P-1 Line Item Number / Title:
5110 / Outfitting

Ship Class	Hull Number	Program Year	Contract Award	Start of Const.	Delivery Date	CFO	PSA Start	PSA Finish	OWL Date	Prior Years	FY 2020	FY 2021	FY 2022	To Complete	Total
LAW	1	2022	Sep 2022	Jun 2023	Mar 2026	Oct 2026	Jun 2027	Sep 2027	Sep 2027	-	-	-	-	-	-
LAW	2	2023	Jun 2023	Dec 2023	Jun 2026	Oct 2026	Feb 2027	Mar 2027	Sep 2027	-	-	-	-	-	-
LAW	3	2024	Jul 2024	Jan 2025	Jan 2027	Apr 2027	Aug 2027	Sep 2027	Mar 2028	-	-	-	-	-	-
LAW	4	2024	Jul 2024	Apr 2025	Apr 2027	Jul 2027	Nov 2027	Dec 2027	Jun 2028	-	-	-	-	-	-
LAW	5	2025	Aug 2025	Feb 2026	Feb 2028	May 2028	Sep 2028	Oct 2028	Apr 2029	-	-	-	-	-	-
LAW	6	2025	Aug 2025	May 2026	May 2028	Aug 2028	Dec 2028	Jan 2029	Jul 2029	-	-	-	-	-	-
LAW	7	2025	Aug 2025	Aug 2026	Aug 2028	Nov 2028	Mar 2029	Apr 2029	Oct 2029	-	-	-	-	-	-
LAW	8	2026	Aug 2026	Feb 2027	Feb 2029	May 2029	Sep 2029	Oct 2029	Apr 2030	-	-	-	-	-	-
LAW	9	2026	Aug 2026	May 2027	May 2029	Aug 2029	Dec 2029	Jan 2030	Jul 2030	-	-	-	-	-	-
LAW	10	2026	Aug 2026	Aug 2027	Aug 2029	Nov 2029	Mar 2030	Apr 2030	Oct 2030	-	-	-	-	-	-
LAW Total										-	-	-	-	-	-
T-AO	205	2016	Jun 2016	Sep 2018	Mar 2022	Jun 2022	Oct 2022	Jan 2023	May 2023	4.573	11.355	2.946	-	-	-
T-AO	206	2018	Mar 2018	Dec 2019	Oct 2022	Jan 2023	May 2023	Aug 2023	Dec 2023	-	0.594	14.839	2.907	-	-
T-AO	207	2019	Dec 2018	Dec 2020	Jul 2023	Oct 2023	Feb 2024	May 2024	Sep 2024	-	-	1.208	12.169	-	-
T-AO	208	2019	Dec 2018	May 2021	Dec 2023	Mar 2024	Jul 2024	Oct 2024	Feb 2025	-	-	-	8.052	-	-
T-AO	209	2020	Mar 2020	May 2022	Nov 2024	Feb 2025	Jun 2025	Sep 2025	Jan 2026	-	-	-	-	-	-
T-AO	210	2020	Mar 2020	Oct 2022	Mar 2025	Jun 2025	Oct 2025	Jan 2026	May 2026	-	-	-	-	-	-
T-AO	211	2022	Jun 2022	Nov 2023	Jan 2026	Apr 2026	Aug 2026	Nov 2026	Mar 2027	-	-	-	-	-	-
T-AO	212	2023	Jan 2023	Mar 2024	Jun 2026	Sep 2026	Jan 2027	Apr 2027	Aug 2027	-	-	-	-	-	-
T-AO	213	2024	Jan 2024	Mar 2025	Jun 2027	Sep 2027	Jan 2028	Apr 2028	Aug 2028	-	-	-	-	-	-
T-AO Total										4.573	11.949	18.993	23.128	-	-
T-AGOS	2201	2022	Aug 2022	Feb 2024	Aug 2026	Nov 2026	May 2027	Jul 2027	Oct 2027	-	-	-	-	-	-
T-AGOS	2401	2024	Feb 2024	Oct 2024	Jan 2027	Apr 2027	Oct 2027	Dec 2027	Mar 2028	-	-	-	-	-	-
T-AGOS	2501	2025	Feb 2025	Oct 2025	Jan 2028	Apr 2028	Oct 2028	Dec 2028	Mar 2029	-	-	-	-	-	-
T-AGOS	2601	2026	Feb 2026	Oct 2026	Jan 2029	Apr 2029	Oct 2029	Dec 2029	Mar 2030	-	-	-	-	-	-
T-AGOS Total										-	-	-	-	-	-
T-ATS	6	2016	Mar 2018	Jun 2019	Aug 2022	Sep 2022	Sep 2022	Sep 2022	Aug 2023	-	2.116	2.022	-	-	-
T-ATS	7	2018	Apr 2019	Oct 2019	Dec 2022	Jan 2023	Jan 2023	Jan 2023	Dec 2023	-	0.430	3.484	0.168	-	-
T-ATS	8	2019	Apr 2019	Feb 2020	Apr 2023	May 2023	May 2023	May 2023	Apr 2024	-	-	0.542	3.476	-	-
T-ATS	9	2020	Apr 2020	Jun 2021	Jun 2024	Jul 2024	Jul 2024	Jul 2024	Jun 2025	-	-	0.676	2.951	-	-
T-ATS	10	2020	Apr 2020	Oct 2021	Oct 2024	Nov 2024	Nov 2024	Nov 2024	Oct 2025	-	-	-	3.644	-	-
T-ATS	11	2021	Sep 2021	Sep 2022	Nov 2024	Dec 2024	Dec 2024	Dec 2024	Nov 2025	-	-	-	0.269	-	-
T-ATS	12	2021	Sep 2021	Jan 2023	Apr 2025	May 2025	May 2025	May 2025	Apr 2026	-	-	-	-	-	-
T-ATS	13	2022	Mar 2022	Jun 2023	Jul 2025	Aug 2025	Aug 2025	Aug 2025	Jul 2026	-	-	-	-	-	-
T-ATS	14	2022	Mar 2022	Oct 2023	Nov 2025	Dec 2026	Jan 2027	Jan 2027	Nov 2027	-	-	-	-	-	-
T-ATS Total										-	2.546	6.724	10.508	-	-
T-AGS/AGOR	67	2018	Oct 2020	Mar 2019	May 2025	Aug 2025	Jun 2026	Jul 2026	Jul 2026	-	-	-	-	-	-
T-AGS/AGOR Total										-	-	-	-	-	-
MTS	711	2017	May 2017	May 2017	Apr 2022	Apr 2022			Mar 2023	12.110	4.800	-	-	-	-

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Date: May 2021

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1611N / 05 / 1P-1 Line Item Number / Title:
5110 / Outfitting

Ship Class	Hull Number	Program Year	Contract Award	Start of Const.	Delivery Date	CFO	PSA Start	PSA Finish	OWL Date	Prior Years	FY 2020	FY 2021	FY 2022	To Complete	Total
MTS Total										12.110	4.800	-	-	-	-
LCU	1700	2016	Mar 2018	Feb 2020	Dec 2021	Jun 2022	Jul 2022	Jul 2022	May 2023	-	-	1.110	-	-	-
LCU	1701	2019	Feb 2019	Aug 2020	May 2022	Jun 2022	Jul 2022	Jul 2022	May 2023	-	-	1.068	0.042	-	-
LCU	1702	2019	Feb 2019	Nov 2020	Aug 2022	Jan 2023	Feb 2023	Feb 2023	Dec 2023	-	-	-	1.115	-	-
LCU	1703	2020	Apr 2020	Apr 2021	Dec 2022	Jan 2023	Feb 2023	Feb 2023	Dec 2023	-	-	-	1.115	-	-
LCU	1704	2020	Apr 2020	Jul 2021	Mar 2023	Dec 2023	Jan 2024	Jan 2024	Nov 2024	-	-	-	1.115	-	-
LCU	1705	2020	Apr 2020	Oct 2021	Jun 2023	Dec 2023	Jan 2024	Jan 2024	Nov 2024	-	-	-	1.115	-	-
LCU	1706	2020	Apr 2020	Jan 2022	Sep 2023	Dec 2023	Jan 2024	Jan 2024	Nov 2024	-	-	-	0.857	-	-
LCU	1707	2021	Aug 2021	Mar 2022	Nov 2023	Dec 2023	Jan 2024	Jan 2024	Nov 2024	-	-	-	-	-	-
LCU	1708	2021	Aug 2021	Jun 2022	Jan 2024	May 2024	Jun 2024	Jun 2024	Apr 2025	-	-	-	-	-	-
LCU	1709	2021	Aug 2021	Sep 2022	Feb 2024	May 2024	Jun 2024	Jun 2024	Apr 2025	-	-	-	-	-	-
LCU	1710	2021	Aug 2021	Dec 2022	Mar 2024	May 2024	Jun 2024	Jun 2024	Apr 2025	-	-	-	-	-	-
LCU	1711	2021	Aug 2021	Feb 2023	Apr 2024	May 2024	Jun 2024	Jun 2021	Apr 2025	-	-	-	-	-	-
LCU	1712	2022	Mar 2022	May 2023	Jun 2024	Dec 2024	Jan 2025	Jan 2025	Nov 2025	-	-	-	-	-	-
LCU	1713	2022	Mar 2022	Jul 2023	Jul 2024	Dec 2024	Jan 2025	Jan 2025	Nov 2025	-	-	-	-	-	-
LCU	1714	2022	Mar 2022	Sep 2023	Sep 2024	Dec 2024	Jan 2025	Jan 2025	Nov 2025	-	-	-	-	-	-
LCU	1715	2022	Mar 2022	Nov 2023	Nov 2024	Dec 2024	Jan 2025	Jan 2025	Nov 2025	-	-	-	-	-	-
LCU	1716	2023	Mar 2023	Jan 2024	Jan 2025	Aug 2025	Sep 2025	Sep 2025	Jul 2026	-	-	-	-	-	-
LCU	1717	2023	Mar 2023	Mar 2024	Mar 2025	Aug 2025	Sep 2025	Sep 2025	Jul 2026	-	-	-	-	-	-
LCU	1718	2023	Mar 2023	May 2024	May 2025	Aug 2025	Sep 2025	Sep 2025	Jul 2026	-	-	-	-	-	-
LCU	1719	2023	Mar 2023	Jul 2024	Jul 2025	Aug 2025	Sep 2025	Sep 2025	Jul 2026	-	-	-	-	-	-
LCU	1720	2023	Mar 2023	Aug 2024	Aug 2025	Jun 2026	Jul 2026	Jul 2026	May 2027	-	-	-	-	-	-
LCU	1721	2024	Mar 2024	Jan 2025	Jan 2026	Jun 2026	Jul 2026	Jul 2026	May 2027	-	-	-	-	-	-
LCU	1722	2024	Mar 2024	Mar 2025	Mar 2026	Jun 2026	Jul 2026	Jul 2026	May 2027	-	-	-	-	-	-
LCU	1723	2024	Mar 2024	May 2025	May 2026	Jun 2026	Jul 2026	Jul 2026	May 2027	-	-	-	-	-	-
LCU	1724	2024	Mar 2024	Jul 2025	Jul 2026	Apr 2027	May 2027	May 2027	Mar 2028	-	-	-	-	-	-
LCU	1725	2024	Mar 2024	Aug 2025	Aug 2026	Apr 2027	May 2027	May 2027	Mar 2028	-	-	-	-	-	-
LCU	1726	2025	Mar 2025	Jan 2026	Jan 2027	Apr 2027	May 2027	May 2027	Mar 2028	-	-	-	-	-	-
LCU	1727	2025	Mar 2025	Mar 2026	Mar 2027	Apr 2027	May 2027	May 2027	Mar 2028	-	-	-	-	-	-
LCU	1728	2025	Mar 2025	May 2026	May 2027	Feb 2028	Mar 2028	Mar 2028	Jan 2029	-	-	-	-	-	-
LCU	1729	2025	Mar 2025	Jul 2026	Jul 2027	Feb 2028	Mar 2028	Mar 2028	Jan 2029	-	-	-	-	-	-
LCU	1730	2025	Mar 2025	Sep 2026	Sep 2027	Feb 2028	Mar 2028	Mar 2028	Jan 2029	-	-	-	-	-	-
LCU	1731	2026	Mar 2026	Jan 2027	Jan 2028	Feb 2028	Mar 2028	Mar 2028	Jan 2029	-	-	-	-	-	-
LCU	1732	2026	Mar 2026	Mar 2027	Mar 2028	Oct 2028	Nov 2028	Nov 2028	Sep 2029	-	-	-	-	-	-
LCU Total										-	-	2.178	5.359	-	-
LCAC	101	2015	Dec 2012	Mar 2015	Aug 2020	Apr 2022	Dec 2022	Mar 2023	Mar 2023	2.025	-	-	-	-	-
LCAC	102	2015	Mar 2015	Sep 2016	May 2021	Nov 2021	Feb 2022	May 2022	Oct 2022	1.037	-	-	-	-	-
LCAC	103	2015	Mar 2015	Nov 2016	Sep 2021	Nov 2021	Apr 2022	Jun 2022	Oct 2022	-	1.037	-	-	-	-
LCAC	104	2016	Mar 2016	Feb 2017	Nov 2021	Jan 2022	May 2022	Aug 2022	Dec 2022	-	1.037	-	-	-	-

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Date: May 2021

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Ship Class	Hull Number	Program Year	Contract Award	Start of Const.	Delivery Date	CFO	PSA Start	PSA Finish	OWL Date	Prior Years	FY 2020	FY 2021	FY 2022	To Complete	Total
LCAC	106	2016	Mar 2016	Oct 2017	Dec 2021	Jan 2022	Jul 2022	Sep 2022	Dec 2022	-	0.718	0.319	-	-	-
LCAC	105	2016	Mar 2016	May 2017	Apr 2022	Apr 2022	Sep 2022	Dec 2022	Mar 2023	-	1.037	-	-	-	-
LCAC	107	2016	Mar 2016	Apr 2018	Jul 2022	Oct 2022	Sep 2022	Dec 2022	Sep 2023	-	-	1.037	-	-	-
LCAC	108	2016	Mar 2016	Jul 2018	Oct 2022	Oct 2022	Feb 2023	May 2023	Sep 2023	-	-	1.057	-	-	-
LCAC	109	2017	Apr 2020	Dec 2018	Dec 2022	Feb 2023	Aug 2023	Nov 2023	Jan 2024	-	-	1.057	-	-	-
LCAC	110	2018	Apr 2020	Mar 2019	Feb 2023	Feb 2023	Sep 2023	Dec 2023	Jan 2024	-	-	1.057	-	-	-
LCAC	111	2018	Apr 2020	Aug 2019	Apr 2023	Jul 2023	Jan 2024	Apr 2024	Jun 2024	-	-	0.430	0.627	-	-
LCAC	112	2018	Apr 2020	Feb 2020	Jul 2023	Jul 2023	Feb 2024	May 2024	Jun 2024	-	-	-	1.057	-	-
LCAC	113	2018	Apr 2020	Aug 2020	Sep 2023	Nov 2023	May 2024	Aug 2024	Oct 2024	-	-	-	1.079	-	-
LCAC	114	2018	Apr 2020	Dec 2020	Nov 2023	Nov 2023	Jun 2024	Sep 2024	Oct 2024	-	-	-	1.079	-	-
LCAC	115	2018	Apr 2020	Jan 2021	Jan 2024	Apr 2024	Oct 2024	Jan 2025	Mar 2025	-	-	-	1.079	-	-
LCAC	116	2018	Apr 2020	Sep 2021	Apr 2024	Apr 2024	Nov 2024	Mar 2025	Mar 2025	-	-	-	0.390	-	-
LCAC	117	2019	Apr 2020	Dec 2021	Jun 2024	Aug 2024	Feb 2025	Jun 2025	Jul 2025	-	-	-	-	-	-
LCAC	118	2019	Apr 2020	Apr 2022	Aug 2024	Aug 2024	Mar 2025	Jul 2025	Jul 2025	-	-	-	-	-	-
LCAC	119	2019	Apr 2020	Jul 2022	Nov 2024	Jan 2025	Jul 2025	Nov 2025	Dec 2025	-	-	-	-	-	-
LCAC	120	2019	Apr 2020	Sep 2022	Jan 2025	Jan 2025	Aug 2025	Dec 2025	Dec 2025	-	-	-	-	-	-
LCAC	121	2019	Apr 2020	Dec 2022	Mar 2025	Jun 2025	Dec 2025	Mar 2026	May 2026	-	-	-	-	-	-
LCAC	122	2019	Apr 2020	Feb 2023	May 2025	Jun 2025	Jan 2026	Apr 2026	May 2026	-	-	-	-	-	-
LCAC	123	2020	Apr 2020	May 2023	Aug 2025	Oct 2025	Apr 2026	Aug 2026	Sep 2026	-	-	-	-	-	-
LCAC	124	2022	Mar 2022	Jul 2023	Oct 2025	Oct 2025	May 2026	Aug 2026	Sep 2026	-	-	-	-	-	-
LCAC	125	2022	Mar 2022	Sep 2023	Dec 2025	Mar 2026	Sep 2026	Dec 2026	Feb 2027	-	-	-	-	-	-
LCAC	126	2023	Mar 2023	Dec 2023	Mar 2026	Mar 2026	Oct 2026	Jan 2027	Feb 2027	-	-	-	-	-	-
LCAC	127	2023	Mar 2023	Feb 2024	May 2026	May 2026	Jan 2027	Apr 2027	Apr 2027	-	-	-	-	-	-
LCAC	128	2024	Mar 2024	Dec 2024	Mar 2027	Jun 2027	Dec 2027	Mar 2028	May 2028	-	-	-	-	-	-
LCAC	129	2024	Mar 2024	Mar 2025	Jun 2027	Jun 2027	Jan 2028	Apr 2028	May 2028	-	-	-	-	-	-
LCAC	130	2025	Mar 2025	Dec 2025	Mar 2028	Jun 2028	Dec 2028	Mar 2029	May 2029	-	-	-	-	-	-
LCAC	131	2025	Mar 2025	Feb 2026	Jun 2028	Jun 2028	Jan 2029	Apr 2029	May 2029	-	-	-	-	-	-
LCAC	132	2026	Mar 2026	Dec 2026	Mar 2029	Jun 2029	Dec 2028	Mar 2030	May 2030	-	-	-	-	-	-
LCAC	133	2026	Mar 2026	Feb 2027	Jun 2029	Jun 2029	Jan 2030	Apr 2030	May 2030	-	-	-	-	-	-
LCAC Total										3.062	3.829	4.957	5.311	-	-
LCAC SLEP	86	2017	Sep 2018	Jan 2019	Dec 2020	Jan 2021	Feb 2021	Feb 2021	Dec 2021	0.238	-	-	-	-	-
LCAC SLEP	87	2017	Sep 2018	Jun 2019	May 2021	Jun 2021	Sep 2021	Oct 2021	May 2022	-	0.242	-	-	-	-
LCAC SLEP	77	2017	Sep 2018	Nov 2019	Sep 2021	Oct 2021	Nov 2021	Dec 2021	Sep 2022	-	0.242	-	-	-	-
LCAC SLEP Total										0.238	0.484	-	-	-	-
YP SLEP	683	2017	May 2020	Jan 2021	Aug 2021	Nov 2021			Oct 2022	0.047	-	-	-	-	-
YP SLEP	690	2017	Jan 2021	Mar 2021	Aug 2021	Nov 2021			Oct 2022	0.047	-	-	-	-	-
YP SLEP	684	2017	May 2020	May 2021	Nov 2021	Feb 2022			Jan 2023	-	0.048	-	-	-	-
YP SLEP	698	2017	Jan 2021	Mar 2021	Nov 2021	Feb 2022			Jan 2023	0.047	-	-	-	-	-
YP SLEP	691	2017	Jun 2021	Aug 2021	Jan 2022	Apr 2022			Mar 2023	-	0.049	-	-	-	-

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Exhibit P-29, Outfitting: PB 2022 Navy

Date: May 2021

Appropriation / Budget Activity / Budget Sub Activity:
1611N / 05 / 1

P-1 Line Item Number / Title:
5110 / Outfitting

Ship Class	Hull Number	Program Year	Contract Award	Start of Const.	Delivery Date	CFO	PSA Start	PSA Finish	OWL Date	Prior Years	FY 2020	FY 2021	FY 2022	To Complete	Total
YP SLEP	700	2017	Jun 2021	Sep 2021	Mar 2022	Jun 2022			May 2023	-	0.048	-	-	-	-
YP SLEP Total										0.141	0.145	-	-	-	-
PUBS	0	2010								30.636	9.918	10.240	9.182	-	-
PUBS Total										30.636	9.918	10.240	9.182	-	-
Full Funding TOA - Outfitting Total										669.953	189.325	225.861	219.732	-	-

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Exhibit P-30, Delivery: PB 2022 Navy

Date: May 2021

Appropriation / Budget Activity / Budget Sub Activity:
1611N / 05 / 1P-1 Line Item Number / Title:
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Ship Class	Hull Number	Program Year	Contract Award	Start of Const.	Delivery Date	CFO	PSA Start	PSA Finish	OWL Date	Prior Years	FY 2020	FY 2021	FY 2022	To Complete	Total
CVN	78	2008	Sep 2008	Aug 2005	May 2017	Jul 2017	Jul 2018	Oct 2019	Oct 2021	305.407	137.294	78.800	-	-	-
CVN	79	2013	Jun 2015	Feb 2011	Jun 2024	Feb 2025	Mar 2025	Sep 2025	Jan 2026	-	-	-	-	-	-
CVN	80	2018	Jan 2019	Jan 2019	Mar 2028	May 2028	Oct 2028	Mar 2029	Apr 2029	-	-	-	-	-	-
CVN Total										305.407	137.294	78.800	-	-	-
CVN	81	2020	Jan 2019	Jan 2019	Feb 2032	Apr 2032	Sep 2032	Feb 2033	Mar 2033	-	-	-	-	-	-
CVN Total										-	-	-	-	-	-
VIRGINIA	790	2013	Dec 2008	Mar 2013	Sep 2018	Sep 2018	Mar 2019	Jun 2021	Jun 2021	80.073	49.452	-	-	-	-
VIRGINIA	791	2013	Dec 2008	Sep 2013	Oct 2019	Oct 2019	Jul 2020	Aug 2021	Aug 2021	10.576	36.822	-	-	-	-
VIRGINIA	792	2014	Apr 2014	May 2014	Apr 2020	Apr 2020	Jul 2021	Feb 2022	Feb 2022	3.442	10.397	-	-	-	-
VIRGINIA	793	2014	Apr 2014	Sep 2014	Oct 2021	Oct 2021	Apr 2022	Jul 2022	Sep 2022	-	8.054	39.787	-	-	-
VIRGINIA	794	2015	Apr 2014	Apr 2015	Nov 2021	Nov 2021	May 2022	Aug 2022	Oct 2022	-	-	37.236	-	-	-
VIRGINIA	795	2015	Apr 2014	Sep 2015	Apr 2022	Apr 2022	Jul 2022	Oct 2022	Mar 2023	-	-	35.668	4.607	-	-
VIRGINIA	796	2016	Apr 2014	Mar 2016	Dec 2022	Dec 2022	Feb 2023	May 2023	Nov 2023	-	-	-	31.290	-	-
VIRGINIA	797	2016	Apr 2014	Sep 2016	Apr 2023	Apr 2023	Aug 2023	Nov 2023	Mar 2024	-	-	-	14.595	-	-
VIRGINIA	798	2017	Apr 2014	Mar 2017	Nov 2023	Nov 2023	Mar 2024	Jun 2024	Oct 2024	-	-	-	4.170	-	-
VIRGINIA	799	2017	Apr 2014	Sep 2017	Mar 2024	Mar 2024	Aug 2024	Nov 2024	Feb 2025	-	-	-	-	-	-
VIRGINIA	800	2018	Apr 2014	Mar 2018	Oct 2024	Oct 2024	Feb 2025	May 2025	Sep 2025	-	-	-	-	-	-
VIRGINIA	801	2018	Apr 2014	Sep 2018	Feb 2025	Feb 2025	May 2025	Aug 2025	Jan 2026	-	-	-	-	-	-
VIRGINIA	802	2019	Dec 2019	Sep 2019	Jun 2025	Jun 2025	Oct 2025	Feb 2026	May 2026	-	-	-	-	-	-
VIRGINIA	803	2019	Dec 2019	Mar 2020	Apr 2026	Apr 2026	Aug 2026	Jan 2027	Mar 2027	-	-	-	-	-	-
VIRGINIA	804	2020	Dec 2019	Sep 2020	Oct 2026	Oct 2026	Feb 2027	Jun 2027	Sep 2027	-	-	-	-	-	-
VIRGINIA	805	2020	Dec 2019	Jul 2021	May 2027	May 2027	Sep 2027	Jan 2028	Apr 2028	-	-	-	-	-	-
VIRGINIA	806	2021	Dec 2019	Dec 2021	Nov 2027	Nov 2027	Mar 2028	Jul 2028	Oct 2028	-	-	-	-	-	-
VIRGINIA	807	2021	Dec 2019	May 2022	May 2028	May 2028	Sep 2028	Jan 2029	Apr 2029	-	-	-	-	-	-
VIRGINIA	808	2022	Dec 2019	Oct 2022	Nov 2028	Nov 2028	Mar 2029	Jul 2029	Oct 2029	-	-	-	-	-	-
VIRGINIA	809	2022	Dec 2019	Mar 2023	May 2029	May 2029	Sep 2029	Jan 2030	Apr 2030	-	-	-	-	-	-
VIRGINIA Total										94.091	104.725	112.691	54.662	-	-
CVN-RCOH	73	2016	Aug 2017	Aug 2017	Aug 2022	Oct 2022	Aug 2022	Aug 2023	Sep 2023	-	-	6.187	25.012	-	-
CVN-RCOH	74	2020	Jan 2021	May 2021	Aug 2025	Dec 2025	Nov 2025	Dec 2025	Nov 2026	-	-	-	-	-	-
CVN-RCOH Total										-	-	6.187	25.012	-	-
DDG 1000	1000	2007	Feb 2008	Feb 2009	Apr 2020	Jan 2021	Jul 2021	Dec 2021	Dec 2021	172.515	51.587	30.747	-	-	-
DDG 1000	1001	2007	Sep 2011	Mar 2010	Mar 2022	Apr 2022	Aug 2022	Mar 2023	Mar 2023	25.498	4.243	4.019	36.150	-	-
DDG 1000	1002	2009	Sep 2011	Apr 2012	Apr 2024	May 2024	Oct 2024	Mar 2025	Apr 2025	0.500	2.408	-	4.270	-	-
DDG 1000 Total										198.513	58.238	34.766	40.420	-	-
DDG	118	2013	Jun 2013	Aug 2015	Mar 2021	Sep 2021	May 2022	Aug 2022	Aug 2022	-	6.653	44.958	1.004	-	-
DDG	120	2013	Mar 2014	Sep 2016	Mar 2022	Jul 2022	Feb 2023	Jun 2023	Jun 2023	-	-	-	47.343	-	-
DDG	119	2014	Mar 2014	Jul 2015	Apr 2020	Sep 2020	Mar 2021	Jul 2021	Aug 2021	-	33.629	2.335	-	-	-
DDG	121	2015	Jun 2013	Apr 2016	Nov 2021	Mar 2022	Sep 2022	Jan 2023	Feb 2023	-	-	35.502	13.181	-	-
DDG	122	2015	Jun 2013	Sep 2017	Jan 2023	May 2023	Dec 2023	Mar 2024	Apr 2024	-	-	-	9.506	-	-

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DDG	123	2016	Jun 2013	Jan 2017	Jul 2022	Nov 2022	Jun 2023	Oct 2023	Oct 2023	-	-	-	36.692	-	-
DDG	124	2016	Jun 2013	Jul 2018	Sep 2023	Jan 2024	Jul 2024	Oct 2024	Dec 2024	-	-	-	-	-	-
DDG	127	2016	Sep 2017	Apr 2019	Mar 2024	Jul 2024	Jan 2025	May 2025	Jun 2025	-	-	-	-	-	-
DDG	125	2017	Jun 2013	May 2018	Apr 2023	Aug 2023	May 2024	Jul 2024	Jul 2024	-	-	-	-	-	-
DDG	126	2017	Jun 2013	Mar 2020	Jun 2024	Oct 2024	Apr 2025	Aug 2025	Sep 2025	-	-	-	-	-	-
DDG	128	2018	Sep 2018	Apr 2020	Oct 2024	Feb 2025	Aug 2025	Dec 2025	Jan 2026	-	-	-	-	-	-
DDG	129	2018	Sep 2018	Jan 2021	Jul 2025	Nov 2025	May 2026	Sep 2026	Oct 2026	-	-	-	-	-	-
DDG	130	2019	Sep 2018	Dec 2020	Jul 2025	Nov 2025	May 2026	Sep 2026	Oct 2026	-	-	-	-	-	-
DDG	131	2019	Sep 2018	Mar 2022	Apr 2026	Aug 2026	Jan 2027	May 2027	Jul 2027	-	-	-	-	-	-
DDG	132	2019	Dec 2018	Jun 2021	May 2026	Sep 2026	Mar 2027	Jun 2027	Aug 2027	-	-	-	-	-	-
DDG	133	2020	Sep 2018	Oct 2022	Nov 2026	Mar 2027	Aug 2027	Dec 2027	Feb 2028	-	-	-	-	-	-
DDG	134	2020	Sep 2018	Apr 2022	Nov 2026	Mar 2027	Sep 2027	Jan 2028	Feb 2028	-	-	-	-	-	-
DDG	135	2020	Jun 2020	May 2023	Jun 2027	Oct 2027	Apr 2028	Aug 2028	Sep 2028	-	-	-	-	-	-
DDG	136	2021	Sep 2018	Jan 2023	Jun 2027	Oct 2027	Apr 2028	Aug 2028	Sep 2028	-	-	-	-	-	-
DDG	137	2021	Sep 2018	Oct 2023	Nov 2027	Mar 2028	Sep 2028	Dec 2028	Feb 2029	-	-	-	-	-	-
DDG	138	2022	Sep 2018	Feb 2024	Apr 2028	Aug 2028	May 2029	Sep 2029	Oct 2029	-	-	-	-	-	-
DDG	139	2023	Jun 2023	Jul 2024	Jul 2028	Nov 2028	May 2029	Sep 2029	Oct 2029	-	-	-	-	-	-
DDG	140	2023	Jun 2023	Jul 2024	Jul 2028	Nov 2028	May 2029	Sep 2029	Oct 2029	-	-	-	-	-	-
DDG	141	2024	Jun 2023	Jul 2025	Jul 2029	Nov 2029	May 2030	Sep 2030	Oct 2030	-	-	-	-	-	-
DDG	142	2024	Jun 2023	Jul 2025	Jul 2029	Nov 2029	May 2030	Sep 2030	Oct 2030	-	-	-	-	-	-
DDG	143	2025	Jun 2023	Jul 2026	Jul 2030	Nov 2030	May 2031	Sep 2031	Oct 2031	-	-	-	-	-	-
DDG	144	2025	Jun 2023	Jul 2026	Jul 2030	Nov 2030	May 2031	Sep 2031	Oct 2031	-	-	-	-	-	-
DDG	145	2026	Jun 2023	Jul 2027	Jul 2031	Nov 2031	May 2032	Sep 2032	Oct 2032	-	-	-	-	-	-
DDG	146	2027	Jun 2023	Jul 2028	Jul 2032	Nov 2032	May 2033	Sep 2033	Oct 2033	-	-	-	-	-	-
DDG Total										-	40.282	82.795	107.726	-	-
LCS	11	2012	Mar 2012	Aug 2013	Aug 2018	Jun 2019	Oct 2019	May 2020	May 2020	42.996	0.172	-	-	-	-
LCS	14	2013	Mar 2013	Feb 2014	Feb 2018	Aug 2018	Apr 2019	Nov 2019	Mar 2020	38.686	2.580	-	-	-	-
LCS	16	2013	Mar 2013	Sep 2014	Apr 2018	Feb 2019	Oct 2019	May 2020	Jun 2020	35.356	4.458	-	-	-	-
LCS	13	2013	Mar 2013	Feb 2014	Aug 2018	Apr 2019	Jul 2019	Mar 2020	Mar 2020	37.253	0.609	-	-	-	-
LCS	15	2013	Mar 2013	Dec 2014	Feb 2019	Feb 2020	Apr 2020	Dec 2020	Jan 2021	16.781	15.676	-	-	-	-
LCS	18	2014	Mar 2014	Mar 2015	Aug 2018	May 2019	Feb 2020	Aug 2020	Sep 2020	23.470	12.467	-	-	-	-
LCS	20	2014	Mar 2014	Feb 2016	Jun 2019	Dec 2019	Jun 2020	Sep 2020	Nov 2020	11.909	7.512	-	-	-	-
LCS	17	2014	Mar 2014	Aug 2015	Jul 2019	May 2020	Aug 2020	Mar 2021	Sep 2021	12.116	33.532	-	-	-	-
LCS	19	2014	Mar 2014	Aug 2016	Feb 2020	Dec 2020	Mar 2021	Sep 2021	Sep 2021	0.799	23.463	41.173	-	-	-
LCS	22	2015	Mar 2015	Dec 2016	Feb 2020	Jun 2020	Mar 2021	May 2021	May 2021	3.701	16.245	14.675	-	-	-
LCS	24	2015	Mar 2015	Jul 2017	Jun 2020	Feb 2021	Jul 2021	Sep 2021	Jan 2022	0.032	12.615	22.245	-	-	-
LCS	21	2015	Mar 2015	Feb 2017	Oct 2021	Jul 2022	Dec 2022	Feb 2023	Jun 2023	-	1.228	28.562	8.445	-	-
LCS	26	2016	Mar 2016	Jan 2018	Dec 2020	Sep 2021	Jan 2022	Apr 2022	Aug 2022	-	0.901	30.906	0.821	-	-
LCS	23	2016	Nov 2015	Sep 2017	Dec 2021	Jul 2022	Dec 2022	Feb 2023	Jun 2023	-	2.402	-	28.581	-	-

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LCS	25	2016	Mar 2016	Feb 2018	May 2022	Dec 2022	May 2023	Jul 2023	Nov 2023	-	-	-	7.242	-	-
LCS	28	2017	Jun 2017	Oct 2018	Jun 2021	Mar 2022	Oct 2022	Dec 2022	Feb 2023	-	-	11.279	19.852	-	-
LCS	30	2017	Oct 2017	May 2019	Dec 2021	Sep 2022	Feb 2023	Apr 2023	Aug 2023	-	-	-	26.146	-	-
LCS	27	2017	Oct 2017	Nov 2018	Oct 2022	Jun 2023	Jan 2024	Mar 2024	May 2024	-	-	-	1.098	-	-
LCS	32	2018	Sep 2018	Dec 2019	Jul 2022	Apr 2023	Sep 2023	Nov 2023	Mar 2024	-	-	-	12.828	-	-
LCS	34	2018	Sep 2018	Jun 2020	Feb 2023	Nov 2023	Apr 2024	Jun 2024	Oct 2024	-	-	-	-	-	-
LCS	29	2018	Sep 2018	Jun 2019	May 2023	Jan 2024	Jun 2024	Aug 2024	Dec 2024	-	-	-	-	-	-
LCS	36	2019	Dec 2018	Jan 2021	Sep 2023	Jun 2024	Nov 2024	Jan 2025	May 2025	-	-	-	-	-	-
LCS	31	2019	Jan 2019	Jun 2020	Apr 2024	Dec 2024	May 2025	Jul 2025	Nov 2025	-	-	-	-	-	-
LCS	38	2019	Dec 2018	Jul 2021	Apr 2024	Jan 2025	Jun 2025	Aug 2025	Dec 2025	-	-	-	-	-	-
LCS Total										223.099	133.860	148.840	105.013	-	-
LPD Flight II	30	2018	Mar 2019	Mar 2020	Feb 2025	Nov 2025	Apr 2026	Sep 2026	Oct 2026	-	-	-	-	-	-
LPD Flight II	31	2021	Apr 2020	Apr 2022	Feb 2027	Sep 2027	Feb 2028	Aug 2028	Aug 2028	-	-	-	-	-	-
LPD Flight II Total										-	-	-	-	-	-
LPD	27	2012	Jul 2012	Aug 2012	Sep 2017	Apr 2018	Nov 2018	Aug 2019	Aug 2019	51.636	-	-	-	-	-
LPD	28	2016	Dec 2016	Dec 2016	Jan 2022	Oct 2022	Mar 2023	Aug 2023	Sep 2023	-	0.207	13.300	39.700	-	-
LPD	29	2017	Feb 2018	Jul 2018	Jul 2023	Mar 2024	Sep 2024	Jan 2025	Feb 2025	-	-	-	0.763	-	-
LPD Total										51.636	0.207	13.300	40.463	-	-
ESB	4	2014	Dec 2014	Oct 2015	Feb 2018	Jun 2018	Mar 2019	Jul 2019	Mar 2020	10.623	1.011	-	-	-	-
ESB	5	2016	Dec 2016	Jan 2017	Nov 2019	Feb 2020	Aug 2020	Dec 2020	Aug 2021	4.730	5.146	2.000	-	-	-
ESB	6	2018	Aug 2019	Jun 2020	Sep 2022	Dec 2022	May 2023	Aug 2023	Nov 2023	-	-	-	6.362	-	-
ESB	7	2019	Aug 2019	Jan 2022	Jan 2024	Apr 2024	Aug 2024	Nov 2024	Mar 2025	-	-	-	-	-	-
ESB Total										15.353	6.157	2.000	6.362	-	-
LHA	7	2011	May 2012	Jul 2013	Feb 2020	Nov 2020	Jan 2021	Mar 2021	Oct 2021	28.409	11.361	16.810	-	-	-
LHA	8	2017	Jun 2017	Oct 2018	Feb 2025	Oct 2025	Dec 2025	Feb 2026	Sep 2026	-	-	-	-	-	-
LHA Total										28.409	11.361	16.810	-	-	-
EPF	11	2015	Sep 2016	Jan 2017	Dec 2019	Mar 2020	Jan 2021	May 2021	Jul 2021	1.429	0.845	-	-	-	-
EPF	12	2016	Sep 2016	Sep 2017	Sep 2020	Dec 2020	Feb 2021	Apr 2021	Nov 2021	-	1.558	1.199	-	-	-
EPF	13	2018	Mar 2019	Aug 2019	Jul 2022	Oct 2022	Apr 2023	Jun 2023	Sep 2023	-	-	0.833	2.967	-	-
EPF	14	2019	Mar 2019	Oct 2020	Jun 2023	Sep 2023	Mar 2024	May 2024	Aug 2024	-	-	-	0.957	-	-
EPF Total										1.429	2.403	2.032	3.924	-	-
LAW	1	2022	Sep 2022	Jun 2023	Mar 2026	Oct 2026	Jun 2027	Sep 2027	Sep 2027	-	-	-	-	-	-
LAW	2	2023	Jun 2023	Dec 2023	Jun 2026	Oct 2026	Feb 2027	Mar 2027	Sep 2027	-	-	-	-	-	-
LAW	3	2024	Jul 2024	Jan 2025	Jan 2027	Apr 2027	Aug 2027	Sep 2027	Mar 2028	-	-	-	-	-	-
LAW	4	2024	Jul 2024	Apr 2025	Apr 2027	Jul 2027	Nov 2027	Dec 2027	Jun 2028	-	-	-	-	-	-
LAW	5	2025	Aug 2025	Feb 2026	Feb 2028	May 2028	Sep 2028	Oct 2028	Apr 2029	-	-	-	-	-	-
LAW	6	2025	Aug 2025	May 2026	May 2028	Aug 2028	Dec 2028	Jan 2029	Jul 2029	-	-	-	-	-	-
LAW	7	2025	Aug 2025	Aug 2026	Aug 2028	Nov 2028	Mar 2029	Apr 2029	Oct 2029	-	-	-	-	-	-
LAW	8	2026	Aug 2026	Feb 2027	Feb 2029	May 2029	Sep 2029	Oct 2029	Apr 2030	-	-	-	-	-	-

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LAW	9	2026	Aug 2026	May 2027	May 2029	Aug 2029	Dec 2029	Jan 2030	Jul 2030	-	-	-	-	-	-
LAW	10	2026	Aug 2026	Aug 2027	Aug 2029	Nov 2029	Mar 2030	Apr 2030	Oct 2030	-	-	-	-	-	-
LAW Total										-	-	-	-	-	-
T-AO	205	2016	Jun 2016	Sep 2018	Mar 2022	Jun 2022	Oct 2022	Jan 2023	May 2023	-	-	10.821	14.623	-	-
T-AO	206	2018	Mar 2018	Dec 2019	Oct 2022	Jan 2023	May 2023	Aug 2023	Dec 2023	-	-	-	3.362	-	-
T-AO	207	2019	Dec 2018	Dec 2020	Jul 2023	Oct 2023	Feb 2024	May 2024	Sep 2024	-	-	-	-	-	-
T-AO	208	2019	Dec 2018	May 2021	Dec 2023	Mar 2024	Jul 2024	Oct 2024	Feb 2025	-	-	-	-	-	-
T-AO	209	2020	Mar 2020	May 2022	Nov 2024	Feb 2025	Jun 2025	Sep 2025	Jan 2026	-	-	-	-	-	-
T-AO	210	2020	Mar 2020	Oct 2022	Mar 2025	Jun 2025	Oct 2025	Jan 2026	May 2026	-	-	-	-	-	-
T-AO	211	2022	Jun 2022	Nov 2023	Jan 2026	Apr 2026	Aug 2026	Nov 2026	Mar 2027	-	-	-	-	-	-
T-AO Total										-	-	10.821	17.985	-	-
T-ATS	6	2016	Mar 2018	Jun 2019	Aug 2022	Sep 2022	Sep 2022	Sep 2022	Aug 2023	-	-	2.833	1.530	-	-
T-ATS	7	2018	Apr 2019	Oct 2019	Dec 2022	Jan 2023	Jan 2023	Jan 2023	Dec 2023	-	-	0.434	3.211	-	-
T-ATS	8	2019	Apr 2019	Feb 2020	Apr 2023	May 2023	May 2023	May 2023	Apr 2024	-	-	-	2.346	-	-
T-ATS	9	2020	Apr 2020	Jun 2021	Jun 2024	Jul 2024	Jul 2024	Jul 2024	Jun 2025	-	-	-	-	-	-
T-ATS	10	2020	Apr 2020	Oct 2021	Oct 2024	Nov 2024	Nov 2024	Nov 2024	Oct 2025	-	-	-	-	-	-
T-ATS	11	2021	Sep 2021	Sep 2022	Nov 2024	Dec 2024	Dec 2024	Dec 2024	Nov 2025	-	-	-	-	-	-
T-ATS	12	2021	Sep 2021	Jan 2023	Apr 2025	May 2025	May 2025	May 2025	Apr 2026	-	-	-	-	-	-
T-ATS	13	2022	Mar 2022	Jun 2023	Jul 2025	Aug 2025	Aug 2025	Aug 2025	Jul 2026	-	-	-	-	-	-
T-ATS	14	2022	Mar 2022	Oct 2023	Nov 2025	Dec 2026	Jan 2027	Jan 2027	Nov 2027	-	-	-	-	-	-
T-ATS Total										-	-	3.267	7.087	-	-
T-AGS/AGOR	67	2018	Oct 2020	Mar 2019	May 2025	Aug 2025	Jun 2026	Jul 2026	Jul 2026	-	-	-	-	-	-
T-AGS/AGOR Total										-	-	-	-	-	-
LCU	1700	2016	Mar 2018	Feb 2020	Dec 2021	Jun 2022	Jul 2022	Jul 2022	May 2023	-	-	1.531	-	-	-
LCU	1701	2019	Feb 2019	Aug 2020	May 2022	Jun 2022	Jul 2022	Jul 2022	May 2023	-	-	1.231	-	-	-
LCU	1702	2019	Feb 2019	Nov 2020	Aug 2022	Jan 2023	Feb 2023	Feb 2023	Dec 2023	-	-	1.231	-	-	-
LCU	1703	2020	Apr 2020	Apr 2021	Dec 2022	Jan 2023	Feb 2023	Feb 2023	Dec 2023	-	-	0.501	0.730	-	-
LCU	1704	2020	Apr 2020	Jul 2021	Mar 2023	Dec 2023	Jan 2024	Jan 2024	Nov 2024	-	-	0.010	1.221	-	-
LCU	1705	2020	Apr 2020	Oct 2021	Jun 2023	Dec 2023	Jan 2024	Jan 2024	Nov 2024	-	-	0.010	1.221	-	-
LCU	1706	2020	Apr 2020	Jan 2022	Sep 2023	Dec 2023	Jan 2024	Jan 2024	Nov 2024	-	-	-	0.534	-	-
LCU	1707	2021	Aug 2021	Mar 2022	Nov 2023	Dec 2023	Jan 2024	Jan 2024	Nov 2024	-	-	-	0.010	-	-
LCU	1708	2021	Aug 2021	Jun 2022	Jan 2024	May 2024	Jun 2024	Jun 2024	Apr 2025	-	-	-	0.010	-	-
LCU	1709	2021	Aug 2021	Sep 2022	Feb 2024	May 2024	Jun 2024	Jun 2024	Apr 2025	-	-	-	0.010	-	-
LCU	1710	2021	Aug 2021	Dec 2022	Mar 2024	May 2024	Jun 2024	Jun 2024	Apr 2025	-	-	-	-	-	-
LCU	1711	2021	Aug 2021	Feb 2023	Apr 2024	May 2024	Jun 2024	Jun 2021	Apr 2025	-	-	-	-	-	-
LCU	1712	2022	Mar 2022	May 2023	Jun 2024	Dec 2024	Jan 2025	Jan 2025	Nov 2025	-	-	-	-	-	-
LCU	1713	2022	Mar 2022	Jul 2023	Jul 2024	Dec 2024	Jan 2025	Jan 2025	Nov 2025	-	-	-	-	-	-
LCU	1714	2022	Mar 2022	Sep 2023	Sep 2024	Dec 2024	Jan 2025	Jan 2025	Nov 2025	-	-	-	-	-	-
LCU	1715	2022	Mar 2022	Nov 2023	Nov 2024	Dec 2024	Jan 2025	Jan 2025	Nov 2025	-	-	-	-	-	-

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LCU	1716	2023	Mar 2023	Jan 2024	Jan 2025	Aug 2025	Sep 2025	Sep 2025	Jul 2026	-	-	-	-	-	-
LCU	1717	2023	Mar 2023	Mar 2024	Mar 2025	Aug 2025	Sep 2025	Sep 2025	Jul 2026	-	-	-	-	-	-
LCU	1718	2023	Mar 2023	May 2024	May 2025	Aug 2025	Sep 2025	Sep 2025	Jul 2026	-	-	-	-	-	-
LCU	1719	2023	Mar 2023	Jul 2024	Jul 2025	Aug 2025	Sep 2025	Sep 2025	Jul 2026	-	-	-	-	-	-
LCU	1720	2023	Mar 2023	Aug 2024	Aug 2025	Jun 2026	Jul 2026	Jul 2026	May 2027	-	-	-	-	-	-
LCU	1721	2024	Mar 2024	Jan 2025	Jan 2026	Jun 2026	Jul 2026	Jul 2026	May 2027	-	-	-	-	-	-
LCU	1722	2024	Mar 2024	Mar 2025	Mar 2026	Jun 2026	Jul 2026	Jul 2026	May 2027	-	-	-	-	-	-
LCU	1723	2024	Mar 2024	May 2025	May 2026	Jun 2026	Jul 2026	Jul 2026	May 2027	-	-	-	-	-	-
LCU	1724	2024	Mar 2024	Jul 2025	Jul 2026	Apr 2027	May 2027	May 2027	Mar 2028	-	-	-	-	-	-
LCU	1725	2024	Mar 2024	Aug 2025	Aug 2026	Apr 2027	May 2027	May 2027	Mar 2028	-	-	-	-	-	-
LCU	1726	2025	Mar 2025	Jan 2026	Jan 2027	Apr 2027	May 2027	May 2027	Mar 2028	-	-	-	-	-	-
LCU	1727	2025	Mar 2025	Mar 2026	Mar 2027	Apr 2027	May 2027	May 2027	Mar 2028	-	-	-	-	-	-
LCU	1728	2025	Mar 2025	May 2026	May 2027	Feb 2028	Mar 2028	Mar 2028	Jan 2029	-	-	-	-	-	-
LCU	1729	2025	Mar 2025	Jul 2026	Jul 2027	Feb 2028	Mar 2028	Mar 2028	Jan 2029	-	-	-	-	-	-
LCU	1730	2025	Mar 2025	Sep 2026	Sep 2027	Feb 2028	Mar 2028	Mar 2028	Jan 2029	-	-	-	-	-	-
LCU	1731	2026	Mar 2026	Jan 2027	Jan 2028	Feb 2028	Mar 2028	Mar 2028	Jan 2029	-	-	-	-	-	-
LCU	1732	2026	Mar 2026	Mar 2027	Mar 2028	Oct 2028	Nov 2028	Nov 2028	Sep 2029	-	-	-	-	-	-
LCU	1733	2026	Mar 2026	May 2027	May 2028	Oct 2028	Nov 2028	Nov 2028	Sep 2029	-	-	-	-	-	-
LCU	1734	2026	Mar 2026	Jul 2027	Jul 2028	Oct 2028	Nov 2028	Nov 2028	Sep 2029	-	-	-	-	-	-
LCU	1735	2026	Mar 2026	Sep 2027	Sep 2028	Oct 2028	Nov 2028	Nov 2028	Sep 2029	-	-	-	-	-	-
LCU	1736	2026	Mar 2026	Nov 2027	Nov 2028	Apr 2029	May 2029	May 2029	Mar 2030	-	-	-	-	-	-
LCU	1737	2026	Mar 2026	Jan 2028	Jan 2029	Apr 2029	May 2029	May 2029	Mar 2030	-	-	-	-	-	-
LCU	1738	2026	Mar 2026	Mar 2028	Mar 2029	Apr 2029	May 2029	May 2029	Mar 2030	-	-	-	-	-	-
LCU Total										-	-	4.514	3.736	-	-
LCAC	101	2015	Dec 2012	Mar 2015	Aug 2020	Apr 2022	Dec 2022	Mar 2023	Mar 2023	4.608	0.485	0.054	1.562	-	-
LCAC	102	2015	Mar 2015	Sep 2016	May 2021	Nov 2021	Feb 2022	May 2022	Oct 2022	0.010	0.485	0.781	2.703	-	-
LCAC	103	2015	Mar 2015	Nov 2016	Sep 2021	Nov 2021	Apr 2022	Jun 2022	Oct 2022	0.010	0.485	0.581	2.899	-	-
LCAC	104	2016	Mar 2016	Feb 2017	Nov 2021	Jan 2022	May 2022	Aug 2022	Dec 2022	-	1.264	0.442	1.831	-	-
LCAC	106	2016	Mar 2016	Oct 2017	Dec 2021	Jan 2022	Jul 2022	Sep 2022	Dec 2022	-	1.475	0.442	1.832	-	-
LCAC	105	2016	Mar 2016	May 2017	Apr 2022	Apr 2022	Sep 2022	Dec 2022	Mar 2023	-	1.480	0.442	1.831	-	-
LCAC	107	2016	Mar 2016	Apr 2018	Jul 2022	Oct 2022	Sep 2022	Dec 2022	Sep 2023	-	1.105	0.442	1.349	-	-
LCAC	108	2016	Mar 2016	Jul 2018	Oct 2022	Oct 2022	Feb 2023	May 2023	Sep 2023	-	0.485	0.441	1.564	-	-
LCAC	109	2017	Apr 2020	Dec 2018	Dec 2022	Feb 2023	Aug 2023	Nov 2023	Jan 2024	-	-	-	1.053	-	-
LCAC	110	2018	Apr 2020	Mar 2019	Feb 2023	Feb 2023	Sep 2023	Dec 2023	Jan 2024	-	-	-	0.643	-	-
LCAC	111	2018	Apr 2020	Aug 2019	Apr 2023	Jul 2023	Jan 2024	Apr 2024	Jun 2024	-	-	-	0.345	-	-
LCAC	112	2018	Apr 2020	Feb 2020	Jul 2023	Jul 2023	Feb 2024	May 2024	Jun 2024	-	-	-	0.401	-	-
LCAC	113	2018	Apr 2020	Aug 2020	Sep 2023	Nov 2023	May 2024	Aug 2024	Oct 2024	-	-	-	0.152	-	-
LCAC	114	2018	Apr 2020	Dec 2020	Nov 2023	Nov 2023	Jun 2024	Sep 2024	Oct 2024	-	-	-	-	-	-
LCAC	115	2018	Apr 2020	Jun 2021	Jan 2024	Apr 2024	Oct 2024	Jan 2025	Mar 2025	-	-	-	-	-	-

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Exhibit P-30, Delivery: PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1	P-1 Line Item Number / Title: 5110 / Outfitting
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Ship Class	Hull Number	Program Year	Contract Award	Start of Const.	Delivery Date	CFO	PSA Start	PSA Finish	OWL Date	Prior Years	FY 2020	FY 2021	FY 2022	To Complete	Total
LCAC	116	2018	Apr 2020	Sep 2021	Apr 2024	Apr 2024	Nov 2024	Mar 2025	Mar 2025	-	-	-	-	-	-
LCAC	117	2019	Apr 2020	Dec 2021	Jun 2024	Aug 2024	Feb 2025	Jun 2025	Jul 2025	-	-	-	-	-	-
LCAC	118	2019	Apr 2020	Apr 2022	Aug 2024	Aug 2024	Mar 2025	Jul 2025	Jul 2025	-	-	-	-	-	-
LCAC	119	2019	Apr 2020	Jul 2022	Nov 2024	Jan 2025	Jul 2025	Nov 2025	Dec 2025	-	-	-	-	-	-
LCAC	120	2019	Apr 2020	Sep 2022	Jan 2025	Jan 2025	Aug 2025	Dec 2025	Dec 2025	-	-	-	-	-	-
LCAC	121	2019	Apr 2020	Dec 2022	Mar 2025	Jun 2025	Dec 2025	Mar 2026	May 2026	-	-	-	-	-	-
LCAC	122	2019	Apr 2020	Feb 2023	May 2025	Jun 2025	Jan 2026	Apr 2026	May 2026	-	-	-	-	-	-
LCAC	123	2020	Apr 2020	May 2023	Aug 2025	Oct 2025	Apr 2026	Aug 2026	Sep 2026	-	-	-	-	-	-
LCAC	124	2022	Mar 2022	Jul 2023	Oct 2025	Oct 2025	May 2026	Aug 2026	Sep 2026	-	-	-	-	-	-
LCAC	125	2022	Mar 2022	Sep 2023	Dec 2025	Mar 2026	Sep 2026	Dec 2026	Feb 2027	-	-	-	-	-	-
LCAC	126	2023	Mar 2023	Dec 2023	Mar 2026	Mar 2026	Oct 2026	Jan 2027	Feb 2027	-	-	-	-	-	-
LCAC	127	2023	Mar 2023	Feb 2024	May 2026	May 2026	Jan 2027	Apr 2027	Apr 2027	-	-	-	-	-	-
LCAC	128	2024	Mar 2024	Dec 2024	Mar 2027	Jun 2027	Dec 2027	Mar 2028	May 2028	-	-	-	-	-	-
LCAC	129	2024	Mar 2024	Mar 2025	Jun 2027	Jun 2027	Jan 2028	Apr 2028	May 2028	-	-	-	-	-	-
LCAC	130	2025	Mar 2025	Dec 2025	Mar 2028	Jun 2028	Dec 2028	Mar 2029	May 2029	-	-	-	-	-	-
LCAC	131	2025	Mar 2025	Feb 2026	Jun 2028	Jun 2028	Jan 2029	Apr 2029	May 2029	-	-	-	-	-	-
LCAC	132	2026	Mar 2026	Dec 2026	Mar 2029	Jun 2029	Dec 2028	Mar 2030	May 2030	-	-	-	-	-	-
LCAC	133	2026	Mar 2026	Feb 2027	Jun 2029	Jun 2029	Jan 2030	Apr 2030	May 2030	-	-	-	-	-	-
LCAC Total										4.628	7.264	3.625	18.165	-	-
LCAC SLEP	86	2017	Sep 2018	Jan 2019	Dec 2020	Jan 2021	Feb 2021	Feb 2021	Dec 2021	-	0.075	-	-	-	-
LCAC SLEP	87	2017	Sep 2018	Jun 2019	May 2021	Jun 2021	Sep 2021	Oct 2021	May 2022	-	-	0.150	-	-	-
LCAC SLEP	77	2017	Sep 2018	Nov 2019	Sep 2021	Oct 2021	Nov 2021	Dec 2021	Sep 2022	-	-	0.150	-	-	-
LCAC SLEP Total										-	0.075	0.300	-	-	-
Full Funding TOA - Post Delivery Total										922.565	501.866	520.748	430.555	-	-

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy										Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost							P-1 Line Item Number / Title: 5112 / Ship to Shore Connector					
ID Code (A=Service Ready, B=Not Service Ready): A				Program Elements for Code B Items: N/A				Other Related Program Elements: N/A				
Line Item MDAP/MAIS Code: N/A												
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (<i>Units in Each</i>)	22	1	-	2	-	2	-	-	-	-	-	-
Gross/Weapon System Cost (<i>\$ in Millions</i>)	1,567.104	65.000	0.000	156.738	0.000	156.738	-	-	-	-	-	-
Less PY Advance Procurement (<i>\$ in Millions</i>)	-	-	-	-	-	-	-	-	-	-	-	-
Less Cost To Complete (<i>\$ in Millions</i>)	14.500	-	-	-	-	-	-	-	-	-	-	-
Less Previously Appropriated RDT&E,N (<i>\$ in Millions</i>)	23.700	-	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (<i>\$ in Millions</i>)	1,528.904	65.000	0.000	156.738	0.000	156.738	-	-	-	-	-	-
Plus CY Advance Procurement (<i>\$ in Millions</i>)	-	-	-	-	-	-	-	-	-	-	-	-
Plus Cost To Complete (<i>\$ in Millions</i>)	14.500	-	-	-	-	-	-	-	-	-	-	-
Plus Previously Appropriated RDT&E,N (<i>\$ in Millions</i>)	23.700	-	-	-	-	-	-	-	-	-	-	-
Total Obligation Authority (<i>\$ in Millions</i>)	1,567.104	65.000	0.000	156.738	0.000	156.738	-	-	-	-	-	-
<i>(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)</i>												
Plus Outfitting and Post Delivery (<i>\$ in Millions</i>)	9.566	11.093	8.582	23.476	-	23.476	-	-	-	-	-	-
Total (<i>\$ in Millions</i>)	1,576.670	76.093	8.582	180.214	-	180.214	-	-	-	-	-	-
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	71.232	65.000	-	78.369	-	78.369	-	-	-	-	-	-

Description:

The Ship to Shore Connector (SSC) program provides the capability to rapidly move assault forces within the littoral operational environment to accomplish Unified Command Plan (UCP) missions and ensures the Joint Force Commander's (JFCDR's) ability to conduct amphibious operations and operate over the high water mark, including movement over ice, mud, rivers, swamps and marshes. SSC provides the functional replacement for the Landing Craft, Air Cushion (LCAC) Class of ships, which began reaching extended service life in 2015.

The Department of Defense Appropriations Act, 2015 directed that the Department complete LCAC 101 in the Shipbuilding and Conversion, Navy Appropriation. LCAC 101 is partially financed with \$23.7M of FY 13/FY 14 R&D funding.

Note:
The procurement quantity in the Prior Years column changed from the FY 2021 budget submission to reflect actual contract awards.

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy					Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost				P-1 Line Item Number / Title: 5112 / Ship to Shore Connector			
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: N/A			Other Related Program Elements: N/A		
Line Item MDAP/MAIS Code: N/A							
Characteristics:	Aluminum	Systems:					
Length Overall	91.8 ft	Hull, Mechanical, and Electrical					
Beam	48.3 ft	(HM&E)					
Displacement	180.57 metric tons	-MT7 Engines, Skirt & Composite Components					
Draft	N/A						
Production Status:	LCAC 102	LCAC 103	LCAC 104	LCAC 105	LCAC 106	LCAC 107	LCAC 108
Contract Award Date	Mar 2015	Mar 2015	Mar 2016	Mar 2016	Mar 2016	Mar 2016	Mar 2016
Months to Completion							
a) Award to Delivery	74 months	78 months	68 months	73 months	69 months	76 months	79 months
b) Construction Start to Delivery	56 months	58 months	57 months	59 months	50 months	51 months	51 months
Delivery Date	May 2021	Sep 2021	Nov 2021	Apr 2022	Dec 2021	Jul 2022	Oct 2022
Completion Of Fitting Out	Nov 2021	Nov 2021	Jan 2022	Apr 2022	Jan 2022	Oct 2022	Oct 2022
Obligation Work Limit Date	Oct 2022	Oct 2022	Dec 2022	Mar 2023	Dec 2022	Sep 2023	Sep 2023
Production Status:	LCAC 109	LCAC 110	LCAC 111	LCAC 112	LCAC 113	LCAC 114	LCAC 115
Contract Award Date	Apr 2020	Apr 2020	Apr 2020	Apr 2020	Apr 2020	Apr 2020	Apr 2020
Months to Completion							
a) Award to Delivery	32 months	34 months	36 months	39 months	41 months	43 months	45 months
b) Construction Start to Delivery	48 months	47 months	44 months	41 months	37 months	35 months	31 months
Delivery Date	Dec 2022	Feb 2023	Apr 2023	Jul 2023	Sep 2023	Nov 2023	Jan 2024
Completion Of Fitting Out	Feb 2023	Feb 2023	Jul 2023	Jul 2023	Nov 2023	Nov 2023	Apr 2024
Obligation Work Limit Date	Jan 2024	Jan 2024	Jun 2024	Jun 2024	Oct 2024	Oct 2024	Mar 2025
Production Status:	LCAC 116	LCAC 117	LCAC 118	LCAC 119	LCAC 120	LCAC 121	LCAC 122
Contract Award Date	Apr 2020	Apr 2020	Apr 2020	Apr 2020	Apr 2020	Apr 2020	Apr 2020
Months to Completion							
a) Award to Delivery	48 months	50 months	52 months	55 months	57 months	59 months	61 months
b) Construction Start to Delivery	31 months	30 months	28 months	28 months	28 months	27 months	27 months
Delivery Date	Apr 2024	Jun 2024	Aug 2024	Nov 2024	Jan 2025	Mar 2025	May 2025
Completion Of Fitting Out	Apr 2024	Aug 2024	Aug 2024	Jan 2025	Jan 2025	Jun 2025	Jun 2025
Obligation Work Limit Date	Mar 2025	Jul 2025	Jul 2025	Dec 2025	Dec 2025	May 2026	May 2026
Production Status:	LCAC 123	LCAC 124	LCAC 125				
Contract Award Date	Apr 2020	Mar 2022	Mar 2022				
Months to Completion							
a) Award to Delivery	64 months	43 months	45 months				
b) Construction Start to Delivery	27 months	27 months	27 months				
Delivery Date	Aug 2025	Oct 2025	Dec 2025				
Completion Of Fitting Out	Oct 2025	Oct 2025	Mar 2026				
Obligation Work Limit Date	Sep 2026	Sep 2026	Feb 2027				

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost		P-1 Line Item Number / Title: 5112 / Ship to Shore Connector		
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: N/A		Other Related Program Elements: N/A
Line Item MDAP/MAIS Code: N/A				
<u>Design Schedule</u>	<u>Start / Issue</u>	<u>Complete / Response</u>	<u>Reissue</u>	<u>Reissue Complete / Response</u>
Issue Date for TLR	N/A	N/A		
Issue Date for TLS	N/A	N/A		
Preliminary Design	Apr 2008	May 2009		
Contract Design	May 2009	Jul 2010		
Detail Design	Jul 2012	Sep 2014		
Request for Proposals	May 2011	Jul 2012		
Design Agent	NAVSEA/TEXTRON,INC			
<u>Classification of Cost Estimate:</u>				

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1	P-1 Line Item Number / Title: 5112 / Ship to Shore Connector
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Cost Categories (†) indicates the presence of a P-8a	FY 2015		FY 2016		FY 2017		FY 2018		FY 2019		FY 2020		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Plan Costs	3		5		1		7		6		1		2	
Basic Construction/Conversion		152.103		191.718		93.152		407.115		393.811		50.800		112.213
Change Orders		6.425		5.109		1.564		9.989		7.671		0.898		2.728
Electronics		4.057		8.500		20.200		38.800		8.500		-		4.811
Hull, Mechanical, and Electrical (HM&E) (†)		12.547		4.000		7.550		58.291		77.315		13.302		32.103
Ordnance		0.010		0.015		-		-		-		-		-
Other Cost		22.658		1.288		5.601		8.537		20.578		-		4.883
Total Ship Estimate		197.800		210.630		128.067		522.732		507.875		65.000		156.738
Less Cost to Complete FY 2018		5.100		-		-		-		-		-		-
Less Cost to Complete FY 2019		9.400		-		-		-		-		-		-
Less RD TEN FY 2013		21.486		-		-		-		-		-		-
Less RD TEN FY 2014		2.214		-		-		-		-		-		-
Net P-1 Funding		159.600		210.630		128.067		522.732		507.875		65.000		156.738

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Exhibit P-27, Ship Production Schedule: PB 2022 Navy					Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1			P-1 Line Item Number / Title: 5112 / Ship to Shore Connector		
Ship	Shipbuilder	Fiscal Year	Contract Award	Start of Construction	Delivery Date
LCAC 102	TEXTRON, INC	2015	Mar 2015	Sep 2016	May 2021
LCAC 103	TEXTRON, INC	2015	Mar 2015	Nov 2016	Sep 2021
LCAC 104	TEXTRON, INC	2016	Mar 2016	Feb 2017	Nov 2021
LCAC 105	TEXTRON, INC	2016	Mar 2016	May 2017	Apr 2022
LCAC 106	TEXTRON, INC	2016	Mar 2016	Oct 2017	Dec 2021
LCAC 107	TEXTRON, INC	2016	Mar 2016	Apr 2018	Jul 2022
LCAC 108	TEXTRON, INC	2016	Mar 2016	Jul 2018	Oct 2022
LCAC 109	TEXTRON, INC	2017	Apr 2020	Dec 2018	Dec 2022
LCAC 110	TEXTRON, INC	2018	Apr 2020	Mar 2019	Feb 2023
LCAC 111	TEXTRON, INC	2018	Apr 2020	Aug 2019	Apr 2023
LCAC 112	TEXTRON, INC	2018	Apr 2020	Feb 2020	Jul 2023
LCAC 113	TEXTRON, INC	2018	Apr 2020	Aug 2020	Sep 2023
LCAC 114	TEXTRON, INC	2018	Apr 2020	Dec 2020	Nov 2023
LCAC 115	TEXTRON, INC	2018	Apr 2020	Jun 2021	Jan 2024
LCAC 116	TEXTRON, INC	2018	Apr 2020	Sep 2021	Apr 2024
LCAC 117	TEXTRON, INC	2019	Apr 2020	Dec 2021	Jun 2024
LCAC 118	TEXTRON, INC	2019	Apr 2020	Apr 2022	Aug 2024
LCAC 119	TEXTRON, INC	2019	Apr 2020	Jul 2022	Nov 2024
LCAC 120	TEXTRON, INC	2019	Apr 2020	Sep 2022	Jan 2025
LCAC 121	TEXTRON, INC	2019	Apr 2020	Dec 2022	Mar 2025
LCAC 122	TEXTRON, INC	2019	Apr 2020	Feb 2023	May 2025
LCAC 123	TEXTRON, INC	2020	Apr 2020	May 2023	Aug 2025
LCAC 124	TEXTRON, INC	2022	Mar 2022	Jul 2023	Oct 2025
LCAC 125	TEXTRON, INC	2022	Mar 2022	Sep 2023	Dec 2025

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Exhibit P-8a, Analysis of Ship Cost Estimates: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1		P-1 Line Item Number / Title: 5112 / Ship to Shore Connector		
Hull, Mechanical, and Electrical (HM&E)	FY 2020		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Major Items				
MT7 Engines, Skirt & Composite Components	4	13.302	8	32.103
Major Items Subtotal		13.302		32.103
Total Hull, Mechanical, and Electrical (HM&E)		13.302		32.103
Remarks: Total quantity refers to procurement of MT7 engines.				

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy										Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost							P-1 Line Item Number / Title: 5113 / Service Craft					
ID Code (A=Service Ready, B=Not Service Ready): A				Program Elements for Code B Items: N/A				Other Related Program Elements: N/A				
Line Item MDAP/MAIS Code: N/A												
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (<i>Units in Each</i>)	53	6	7	4	-	4	-	-	-	-	-	-
Gross/Weapon System Cost (<i>\$ in Millions</i>)	335.386	56.289	244.147	67.866	0.000	67.866	-	-	-	-	-	-
Less PY Advance Procurement (<i>\$ in Millions</i>)	-	-	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (<i>\$ in Millions</i>)	335.386	56.289	244.147	67.866	0.000	67.866	-	-	-	-	-	-
Plus CY Advance Procurement (<i>\$ in Millions</i>)	-	-	-	-	-	-	-	-	-	-	-	-
Total Obligation Authority (<i>\$ in Millions</i>)	335.386	56.289	244.147	67.866	0.000	67.866	-	-	-	-	-	-
<i>(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)</i>												
Plus Outfitting and Post Delivery (<i>\$ in Millions</i>)	3.000	-	-	-	-	-	-	-	-	-	-	-
Total (<i>\$ in Millions</i>)	338.386	56.289	244.147	67.866	-	67.866	-	-	-	-	-	-
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	6.328	9.382	34.878	16.967	-	16.967	-	-	-	-	-	-

Description:
 The US Navy owns and operates 354 Service Craft consisting of 33 different classes of craft at 56 different commands and activities throughout the world. Service Craft provide critical support to carriers, submarines, and other Navy vessels through port operations, ship maintenance, testing, and training missions. Nearly half of the Service Craft inventory is over 40 years of age. The Service Craft budget supports the acquisition of replacement craft as follows:

Auxiliary Floating Dry Dock Medium (AFDM) is utilized to dry dock surface ships and submarines (including CG, DDG, LCS, LSD, and SSN) in order to perform maintenance availabilities.

Auxiliary Personnel Lighter - Small (APL(S)) provides crew messing, duty crew berthing, and administrative spaces to ships during CNO maintenance availabilities and improves the quality-of-life for sailors over legacy barges.

Harbor Tug (YT) provides critical vessel towing, escort, personnel transfer, and emergency services to carriers, ships, and submarines. The YT program replaces aging YTB tugboats in the Northwest Region, Yokosuka, and Portsmouth Naval Shipyard and is required to meet port operations mission requirements.

Small Harbor Tug (YTL) is utilized by port operations for ship assist, towing, and escort of smaller Navy ships and craft.

Fuel Oil Barge (YON) carries liquid petroleum products for refueling ships. YON barges greatly reduce the risk of a major fuel oil spill. Many existing YONs are 50 to 60 years old and of single hull construction. The new YONs will be double-hulled and will meet the requirements of the Oil Pollution Act of 1990 (OPA-90).

Waste Oil Barge (YWO) offloads oily waste water from ships for transport and processing. The YWO will be double-hulled and have piping and other systems specifically designed for transferring oily waste. The YWO program will replace 66 to 78 year old barges that were not specifically designed to transport oily waste and are in extremely poor condition.

Covered Lighter (YFN) transports ordnance and sensitive equipment, and cargo that requires protection from the weather.

Open Lighter (YC) transports cargo/equipment and serves as a work platform for ship maintenance.

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost		P-1 Line Item Number / Title: 5113 / Service Craft
ID Code (A=Service Ready, B=Not Service Ready): A	Program Elements for Code B Items: N/A	Other Related Program Elements: N/A
Line Item MDAP/MAIS Code: N/A		
Repair, Berthing and Messing Barge (YRBM) provides crew messing, duty crew berthing, and administrative spaces for small to mid-size ships and submarines during CNO maintenance availabilities.		

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy					Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost				P-1 Line Item Number / Title: 5113 / Service Craft			
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: N/A			Other Related Program Elements: N/A		
Line Item MDAP/MAIS Code: N/A							
Characteristics:	Hull Various	Multiple Craft					
Length Overall	Various	Various					
Beam	Various	Various					
Displacement	Various	Various					
Draft	Various	Various					
Production Status:	YT 809	YT 810	YT 811	APL 67	APL 68	YT 812	YON 1801
Contract Award Date	Jul 2018	Jul 2018	Jul 2018	Sep 2018	Sep 2018	Dec 2018	Sep 2021
Months to Completion							
a) Award to Delivery	34 months	34 months	38 months	33 months	34 months	36 months	15 months
b) Construction Start to Delivery	26 months	19 months	19 months	26 months	26 months	20 months	12 months
Delivery Date	May 2021	May 2021	Sep 2021	Jun 2021	Jul 2021	Dec 2021	Dec 2022
Completion Of Fitting Out	Aug 2021	Aug 2021	Dec 2021	Sep 2021	Oct 2021	Mar 2022	Mar 2023
Obligation Work Limit Date	Jul 2022	Jul 2022	Nov 2022	Aug 2022	Sep 2022	Feb 2023	Feb 2024
Production Status:	YWO 05	YWO 04	APL 69	YT 813	YTL 815	YTL 816	YON 1901
Contract Award Date	Oct 2021	Oct 2021	Feb 2019	Mar 2019	Sep 2019	Feb 2020	Sep 2021
Months to Completion							
a) Award to Delivery	14 months	17 months	31 months	36 months	27 months	24 months	18 months
b) Construction Start to Delivery	11 months	11 months	28 months	19 months	15 months	12 months	12 months
Delivery Date	Dec 2022	Mar 2023	Sep 2021	Mar 2022	Dec 2021	Feb 2022	Mar 2023
Completion Of Fitting Out	Mar 2023	Jun 2023	Dec 2021	Jun 2022	Mar 2022	May 2022	Jun 2023
Obligation Work Limit Date	Feb 2024	May 2024	Nov 2022	May 2023	Feb 2023	Apr 2023	May 2024
Production Status:	APL 70	YON 2001	YON 2002	YC 2002	YC 2001	YWO 2001	APL 71
Contract Award Date	Mar 2020	Sep 2021	Sep 2021	Oct 2021	Oct 2021	Oct 2021	Mar 2021
Months to Completion							
a) Award to Delivery	21 months	23 months	26 months	10 months	12 months	16 months	16 months
b) Construction Start to Delivery	19 months	13 months	13 months	7 months	8 months	7 months	13 months
Delivery Date	Dec 2021	Aug 2023	Nov 2023	Aug 2022	Oct 2022	Feb 2023	Jul 2022
Completion Of Fitting Out	Mar 2022	Nov 2023	Feb 2024	Nov 2022	Jan 2023	May 2023	Oct 2022
Obligation Work Limit Date	Feb 2023	Oct 2024	Jan 2025	Oct 2023	Dec 2023	Apr 2024	Sep 2023
Production Status:	YRBM 57	YFN 2101	YFN 2102	YC 2101	YT 814	AFDM 15	YRBM 2201
Contract Award Date	Sep 2021	Oct 2021	Oct 2021	Oct 2021	Dec 2021	Dec 2021	Feb 2022
Months to Completion							
a) Award to Delivery	20 months	8 months	12 months	12 months	14 months	28 months	16 months
b) Construction Start to Delivery	14 months	7 months	7 months	7 months	12 months	21 months	13 months
Delivery Date	May 2023	Jun 2022	Oct 2022	Oct 2022	Feb 2023	Apr 2024	Jun 2023
Completion Of Fitting Out	Aug 2023	Sep 2022	Jan 2023	Jan 2023	May 2023	Jul 2024	Sep 2023
Obligation Work Limit Date	Jul 2024	Aug 2023	Dec 2023	Dec 2023	Apr 2024	Jun 2025	Aug 2024
Production Status:	APL 72	YC 2201	YC 2202				
Contract Award Date	Feb 2022	Mar 2022	Mar 2022				
Months to Completion							
a) Award to Delivery	17 months	7 months	9 months				
b) Construction Start to Delivery	13 months	6 months	6 months				
Delivery Date	Jul 2023	Oct 2022	Dec 2022				
Completion Of Fitting Out	Oct 2023	Jan 2023	Mar 2023				
Obligation Work Limit Date	Sep 2024	Dec 2023	Feb 2024				

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U-5113 - Service Craft
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P-1 Line #28

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy			Date: May 2021																																														
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost			P-1 Line Item Number / Title: 5113 / Service Craft																																														
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: N/A		Other Related Program Elements: N/A																																													
Line Item MDAP/MAIS Code: N/A																																																	
<table border="0" style="width: 100%;"> <tr> <td style="width: 35%;"><u>Design Schedule</u></td> <td style="width: 15%;"><u>Start / Issue</u></td> <td style="width: 15%;"><u>Complete / Response</u></td> <td style="width: 15%;"><u>Reissue</u></td> <td style="width: 20%;"><u>Reissue Complete / Response</u></td> </tr> <tr> <td>Issue Date for TLR</td> <td>N/A</td> <td>N/A</td> <td></td> <td></td> </tr> <tr> <td>Issue Date for TLS</td> <td>N/A</td> <td>N/A</td> <td></td> <td></td> </tr> <tr> <td>Preliminary Design</td> <td>N/A</td> <td>N/A</td> <td></td> <td></td> </tr> <tr> <td>Contract Design</td> <td>N/A</td> <td>N/A</td> <td></td> <td></td> </tr> <tr> <td>Detail Design</td> <td>N/A</td> <td>N/A</td> <td></td> <td></td> </tr> <tr> <td>Request for Proposals</td> <td>N/A</td> <td>N/A</td> <td></td> <td></td> </tr> <tr> <td>Design Agent</td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td colspan="5"><u>Classification of Cost Estimate:</u></td> </tr> </table>					<u>Design Schedule</u>	<u>Start / Issue</u>	<u>Complete / Response</u>	<u>Reissue</u>	<u>Reissue Complete / Response</u>	Issue Date for TLR	N/A	N/A			Issue Date for TLS	N/A	N/A			Preliminary Design	N/A	N/A			Contract Design	N/A	N/A			Detail Design	N/A	N/A			Request for Proposals	N/A	N/A			Design Agent					<u>Classification of Cost Estimate:</u>				
<u>Design Schedule</u>	<u>Start / Issue</u>	<u>Complete / Response</u>	<u>Reissue</u>	<u>Reissue Complete / Response</u>																																													
Issue Date for TLR	N/A	N/A																																															
Issue Date for TLS	N/A	N/A																																															
Preliminary Design	N/A	N/A																																															
Contract Design	N/A	N/A																																															
Detail Design	N/A	N/A																																															
Request for Proposals	N/A	N/A																																															
Design Agent																																																	
<u>Classification of Cost Estimate:</u>																																																	
Justification: Auxiliary Personnel Lighter - Small (APL(S)) provides critical berthing and messing facilities for sailors while their ships are in port for availabilities and Inter-Deployment Training Cycles (IDTC). Thirteen (13) of the seventeen (17) APLs in service were built from 1944-1946; they do not meet current safety standards, cannot be easily configured for dual-gender crews, and cannot support modern computer equipment. The APL program replaces outdated and dilapidated transport ships and will greatly improve sailors' quality of life, improve safety, and reduce crew berthing and messing costs during ship maintenance availabilities. Open Lighter (YC) and Covered Lighter (YFN) barges are required to replace the oldest YC and YFN in the Fleet, which are over 50 years old and have become unaffordable to overhaul due to degradation of hull structure. Yard Repair, Berthing and Messing (YRBM) barges will replace legacy barges that are 50-80 years old, do not meet current safety standards, are not dual gender-compatible, and cannot support modern computer systems.																																																	

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1	P-1 Line Item Number / Title: 5113 / Service Craft
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Cost Categories	FY 2016		FY 2017		FY 2018		FY 2019		FY 2020		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Plan Costs	2		3		5		5		6		7		4	
Basic Construction/Conversion		28.180		62.132		61.074		69.721		55.161		236.175		66.747
Change Orders		1.400		2.260		1.220		1.391		0.528		5.727		0.699
Hull, Mechanical, and Electrical (HM&E)		0.397		0.800		0.700		0.950		0.600		2.245		0.420
Total Ship Estimate		29.977		65.192		62.994		72.062		56.289		244.147		67.866
Net P-1 Funding		29.977		65.192		62.994		72.062		56.289		244.147		67.866

Remarks:

FY 16 Craft:
2 YT: \$29.977
TOTAL: \$29.977

FY 17 Craft:
1 APL: \$39.000
2 YT: \$26.192
TOTAL: \$65.192

FY 18 Craft:
1 YT: \$13.660
2 YWO: \$ 6.000
1 YON: \$ 4.334
1 APL: \$39.000
TOTAL: \$62.994

FY 19 Craft:
1 APL: \$39.808
1 YON \$4.490
1 YT: \$13.660
2 YTL: \$14.104
TOTAL: \$72.062

FY 20 Craft:
2 YC: \$ 3.264
1 YWO: \$ 3.338
2 YON: \$ 9.780
1 APL: \$39.907
TOTAL: \$56.289

FY 21 Craft:
1 APL: \$41.104

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1		P-1 Line Item Number / Title: 5113 / Service Craft
2 YFN \$5.816 1 AFDM: \$155.000 1 YC: \$1.731 1 YRBM: \$19.300 YRBM: \$7.000 (YRBM-56 Conversion) 1 YT: \$14.196 TOTAL: \$244.147 FY22 Craft: 1 APL: \$44.143 2 YC: \$3.837 1 YRBM: \$19.886 TOTAL: \$67.866		

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Exhibit P-27, Ship Production Schedule: PB 2022 Navy					Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1			P-1 Line Item Number / Title: 5113 / Service Craft		
Ship	Shipbuilder	Fiscal Year	Contract Award	Start of Construction	Delivery Date
YT 809	Dakota Creek Industries, Inc.	2016	Jul 2018	Mar 2019	May 2021
YT 810	Dakota Creek Industries, Inc.	2017	Jul 2018	Oct 2019	May 2021
YT 811	Dakota Creek Industries, Inc.	2017	Jul 2018	Feb 2020	Sep 2021
APL 67	VT Halter Marine Inc.	2017	Sep 2018	Apr 2019	Jun 2021
APL 68	VT Halter Marine Inc.	2018	Sep 2018	May 2019	Jul 2021
YT 812	Dakota Creek Industries, Inc.	2018	Dec 2018	Apr 2020	Dec 2021
YON 1801	TBD	2018	Sep 2021	Dec 2021	Dec 2022
YWO 05	TBD	2018	Oct 2021	Jan 2022	Dec 2022
YWO 04	TBD	2018	Oct 2021	Apr 2022	Mar 2023
APL 69	VT Halter Marine Inc.	2019	Feb 2019	May 2019	Sep 2021
YT 813	Dakota Creek Industries, Inc.	2019	Mar 2019	Aug 2020	Mar 2022
YTL 815	Modutech Marine, Inc	2019	Sep 2019	Sep 2020	Dec 2021
YTL 816	Modutech Marine, Inc	2019	Feb 2020	Feb 2021	Feb 2022
YON 1901	TBD	2019	Sep 2021	Mar 2022	Mar 2023
APL 70	VT Halter Marine Inc.	2020	Mar 2020	May 2020	Dec 2021
YON 2001	TBD	2020	Sep 2021	Jul 2022	Aug 2023
YON 2002	TBD	2020	Sep 2021	Oct 2022	Nov 2023
YC 2002	TBD	2020	Oct 2021	Jan 2022	Aug 2022
YC 2001	TBD	2020	Oct 2021	Feb 2022	Oct 2022
YWO 2001	TBD	2020	Oct 2021	Jul 2022	Feb 2023
APL 71	VT Halter Marine Inc.	2021	Mar 2021	Jun 2021	Jul 2022
YRBM 57	TBD	2021	Sep 2021	Mar 2022	May 2023
YFN 2101	TBD	2021	Oct 2021	Nov 2021	Jun 2022
YFN 2102	TBD	2021	Oct 2021	Mar 2022	Oct 2022
YC 2101	TBD	2021	Oct 2021	Mar 2022	Oct 2022
YT 814	TBD	2021	Dec 2021	Feb 2022	Feb 2023
AFDM 15	TBD	2021	Dec 2021	Jul 2022	Apr 2024
YRBM 2201	TBD	2022	Feb 2022	May 2022	Jun 2023
APL 72	VT Halter Marine Inc.	2022	Feb 2022	Jun 2022	Jul 2023
YC 2201	TBD	2022	Mar 2022	Apr 2022	Oct 2022
YC 2202	TBD	2022	Mar 2022	Jun 2022	Dec 2022

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy										Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost							P-1 Line Item Number / Title: 5139 / LCAC SLEP					
ID Code (A=Service Ready, B=Not Service Ready): A				Program Elements for Code B Items: N/A				Other Related Program Elements: N/A				
Line Item MDAP/MAIS Code: N/A												
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (<i>Units in Each</i>)	67	-	3	2	-	2	-	-	-	-	-	-
Gross/Weapon System Cost (<i>\$ in Millions</i>)	1,418.405	0.000	56.461	32.712	0.000	32.712	-	-	-	-	-	-
Less PY Advance Procurement (<i>\$ in Millions</i>)	27.900	-	-	-	-	-	-	-	-	-	-	-
Less Cost To Complete (<i>\$ in Millions</i>)	14.000	-	-	-	-	-	-	-	-	-	-	-
Less Subsequent Year Full Funding (<i>\$ in Millions</i>)	1.774	-	-	-	-	-	-	-	-	-	-	-
Less Hurricane (<i>\$ in Millions</i>)	19.800	-	-	-	-	-	-	-	-	-	-	-
Less Transfer (<i>\$ in Millions</i>)	1.500	-	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (<i>\$ in Millions</i>)	1,353.431	0.000	56.461	32.712	0.000	32.712	-	-	-	-	-	-
Plus Subsequent Year Full Funding (<i>\$ in Millions</i>)	1.774	-	-	-	-	-	-	-	-	-	-	-
Full Funding TOA (<i>\$ in Millions</i>)	1,355.205	-	56.461	32.712	-	32.712	-	-	-	-	-	-
Plus CY Advance Procurement (<i>\$ in Millions</i>)	27.900	-	-	-	-	-	-	-	-	-	-	-
Plus Cost To Complete (<i>\$ in Millions</i>)	14.000	-	-	-	-	-	-	-	-	-	-	-
Plus Transfer (<i>\$ in Millions</i>)	1.500	-	-	-	-	-	-	-	-	-	-	-
Plus Hurricane (<i>\$ in Millions</i>)	19.800	-	-	-	-	-	-	-	-	-	-	-
Total Obligation Authority (<i>\$ in Millions</i>)	1,418.405	0.000	56.461	32.712	0.000	32.712	-	-	-	-	-	-
<i>(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)</i>												
Plus Outfitting and Post Delivery (<i>\$ in Millions</i>)	13.283	0.559	0.300	-	-	-	-	-	-	-	-	-
Total (<i>\$ in Millions</i>)	1,431.688	0.559	56.761	32.712	-	32.712	-	-	-	-	-	-
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	21.170	-	18.820	16.356	-	16.356	-	-	-	-	-	-

Description:

Landing Craft, Air Cushion (LCAC) transports weapon systems, equipment, cargo and personnel of the assault elements of the Marine Air/Ground Task Force from ship to shore and across the beach. The LCAC Service Life Extension Program (SLEP) and LCAC Extended Service Life Extension Program (ESLEP) extends the craft service life past the original twenty years.

The LCAC SLEP program incorporates the following modifications and enhancements: upgrade from the TF40B engines to the ETF40B engines; repair corrosion damage; replace obsolete electronics; upgrade C4N suite; and replace deep skirt.

The LCAC ESLEP program incorporates the following modifications and enhancements: repairs corrosion damage; replaces obsolete electronics, upgrades C4N including cyber hardening; and replaces the deep skirt.

Note:
ESLEP contract award for FY 2021 delayed from Nov 2020 to Feb 2021 due to program being a new start in FY 2021. Program could not award a contract during the FY 2021 continuing resolution.

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy						Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost				P-1 Line Item Number / Title: 5139 / LCAC SLEP			
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: N/A			Other Related Program Elements: N/A		
Line Item MDAP/MAIS Code: N/A							
Hull number changes in the FY 2021 craft executing the E-SLEP from FY 2021 to the FY 2022 request were a result of changing operational commitments in the Fleet as to which craft could be made available to go through E-SLEP first. For the FY 2022 E-SLEP, the program will identify the craft planned for E-SLEP induction closer to planned contract award.							
Characteristics:		Air Cushion					
Length Overall		91.8 ft (on cushion)					
Beam		49.2 ft (on cushion)					
Displacement		106 tons					
Draft		None (air cushion)					
Production Status:		LCAC SLEP 86	LCAC SLEP 87	LCAC SLEP 77	LCAC ESLEP 45	LCAC ESLEP 31	LCAC ESLEP 48
Contract Award Date		Sep 2018	Sep 2018	Sep 2018	Feb 2021	Feb 2021	Feb 2021
Months to Completion							
a) Award to Delivery		27 months	32 months	36 months	25 months	29 months	33 months
b) Construction Start to Delivery		23 months	23 months	22 months	20 months	20 months	20 months
Delivery Date		Dec 2020	May 2021	Sep 2021	Mar 2023	Jul 2023	Nov 2023
Completion Of Fitting Out		Jan 2021	Jun 2021	Oct 2021	Apr 2023	Aug 2023	Dec 2023
Obligation Work Limit Date		Dec 2021	May 2022	Sep 2022	Mar 2024	Jul 2024	Nov 2024
Production Status:		LCAC ESLEP TBD22B					
Contract Award Date		Jan 2022					
Months to Completion							
a) Award to Delivery		27 months					
b) Construction Start to Delivery		18 months					
Delivery Date		Apr 2024					
Completion Of Fitting Out		May 2024					
Obligation Work Limit Date		Apr 2025					
Design Schedule		Start / Issue	Complete / Response	Reissue	Reissue Complete / Response		
Issue Date for TLR		N/A	N/A				
Issue Date for TLS		N/A	N/A				
Preliminary Design		N/A	N/A				
Contract Design		N/A	N/A				
Detail Design		N/A	N/A				
Request for Proposals		May 2020	Jul 2020				
Design Agent		Landing Craft Planning Yard					
Classification of Cost Estimate: N/A							

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy	Date: May 2021
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Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1	P-1 Line Item Number / Title: 5139 / LCAC SLEP
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Cost Categories	FY 2017		FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Plan Costs	3		3		2	
Basic Construction/Conversion		35.925		30.216		18.971
Electronics		6.542		2.225		1.378
Hull, Mechanical, and Electrical (HM&E)		33.850		17.515		11.347
Other Cost		3.983		6.505		1.016
Total Ship Estimate		80.300		56.461		32.712
Net P-1 Funding		80.300		56.461		32.712

Remarks:

FY17 is the last year of LCAC SLEP and FY21 is the first year of LCAC ESLEP. LCAC ESLEPs are structured to cost effectively gain more service life out of existing LCAC SLEP craft.

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Exhibit P-27, Ship Production Schedule: PB 2022 Navy				Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1			P-1 Line Item Number / Title: 5139 / LCAC SLEP		
Ship	Shipbuilder	Fiscal Year	Contract Award	Start of Construction	Delivery Date
LCAC SLEP 86	Tecnico	2017	Sep 2018	Jan 2019	Dec 2020
LCAC SLEP 87	Tecnico	2017	Sep 2018	Jun 2019	May 2021
LCAC SLEP 77	Tecnico	2017	Sep 2018	Nov 2019	Sep 2021
LCAC ESLEP 45	Walashek	2021	Feb 2021	Jul 2021	Mar 2023
LCAC ESLEP 31	Walashek	2021	Feb 2021	Nov 2021	Jul 2023
LCAC ESLEP 48	Walashek	2021	Feb 2021	Mar 2022	Nov 2023
LCAC ESLEP TBD22A	TBD	2022	Jan 2022	Jun 2022	Dec 2023
LCAC ESLEP TBD22B	TBD	2022	Jan 2022	Oct 2022	Apr 2024

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy										Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost						P-1 Line Item Number / Title: 5201 / Auxiliary Vessels (Used Sealift)						
ID Code (A=Service Ready, B=Not Service Ready): A				Program Elements for Code B Items: 0208036N				Other Related Program Elements: N/A				
Line Item MDAP/MAIS Code: N/A												
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (<i>Units in Each</i>)	-	-	2	5	-	5	-	-	-	-	-	-
Gross/Weapon System Cost (<i>\$ in Millions</i>)	0.000	0.000	60.000	299.900	0.000	299.900	-	-	-	-	-	-
Less PY Advance Procurement (<i>\$ in Millions</i>)	-	-	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (<i>\$ in Millions</i>)	0.000	0.000	60.000	299.900	0.000	299.900	-	-	-	-	-	-
Plus CY Advance Procurement (<i>\$ in Millions</i>)	-	-	-	-	-	-	-	-	-	-	-	-
Total Obligation Authority (<i>\$ in Millions</i>)	0.000	0.000	60.000	299.900	0.000	299.900	-	-	-	-	-	-
<i>(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)</i>												
Total (<i>\$ in Millions</i>)	-	-	60.000	299.900	-	299.900	-	-	-	-	-	-
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	-	-	30.000	59.980	-	59.980	-	-	-	-	-	-
Description:												
This funding is required to recapitalize the U.S. surge sealift fleet. The principal enabler for recapitalizing the U.S. surge sealift fleet is through ship procurements and upgrades of used commercial vessels via the RRF Recapitalization program.												
Strategic sealift is a key enabler of U.S. power projection and supports the afloat movement of equipment and supplies for major ground combat operations. During surge deployments, approximately 90 percent of Marine Corps and Army combat equipment is transported by sea, under the operation of U.S. Transportation Command (USTRANSCOM). The Ready Reserve Force (RRF), managed by the U.S. Department of Transportation's (DOT) Maritime Administration (MARAD), is the primary force of the surge sealift fleet.												
As a key element of strategic sealift, the RRF is specifically structured to transport military unit equipment and supplies during the initial surge of U.S. combat forces deploying anywhere in the world. This fleet consists of 41 government-owned ships with an average age of 46 years; a majority of these ships have already exceeded their expected service life, limiting their supportability and adding significant risk to their operational readiness. As a result, the Joint Force requires 18 ships over the next 5 years to replace anticipated sealift capacity losses.												
Characteristics:												
Length Overall	-	850 ft										
Beam	-											
Displacement	-											
Draft	-											
Production Status:												
Contract Award Date	AUX 2101	AUX 2102	AUX 2201	AUX 2202	AUX 2203	AUX 2204	AUX 2205					
Dec 2021	Dec 2021	Dec 2021	Mar 2022	May 2022	May 2022	Jul 2022	Jul 2022					
Months to Completion												
a) Award to Delivery	0 months	0 months	0 months	0 months	0 months	0 months	0 months					
b) Construction Start to Delivery	0 months	0 months	0 months	0 months	0 months	0 months	0 months					
Delivery Date	Dec 2021	Dec 2021	Mar 2022	May 2022	May 2022	Jul 2022	Jul 2022					
Completion Of Fitting Out												
Obligation Work Limit Date												

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy			Date: May 2021																																														
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost		P-1 Line Item Number / Title: 5201 / Auxiliary Vessels (Used Sealift)																																															
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: 0208036N		Other Related Program Elements: N/A																																													
Line Item MDAP/MAIS Code: N/A																																																	
<table border="0" style="width: 100%;"> <thead> <tr> <th style="text-align: left;"><u>Design Schedule</u></th> <th style="text-align: left;"><u>Start / Issue</u></th> <th style="text-align: left;"><u>Complete / Response</u></th> <th style="text-align: left;"><u>Reissue</u></th> <th style="text-align: left;"><u>Reissue Complete / Response</u></th> </tr> </thead> <tbody> <tr> <td>Issue Date for TLR</td> <td>N/A</td> <td>N/A</td> <td></td> <td></td> </tr> <tr> <td>Issue Date for TLS</td> <td>N/A</td> <td>N/A</td> <td></td> <td></td> </tr> <tr> <td>Preliminary Design</td> <td>N/A</td> <td>N/A</td> <td></td> <td></td> </tr> <tr> <td>Contract Design</td> <td>N/A</td> <td>N/A</td> <td></td> <td></td> </tr> <tr> <td>Detail Design</td> <td>N/A</td> <td>N/A</td> <td></td> <td></td> </tr> <tr> <td>Request for Proposals</td> <td>N/A</td> <td>N/A</td> <td></td> <td></td> </tr> <tr> <td>Design Agent</td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td colspan="5"><u>Classification of Cost Estimate:</u></td> </tr> </tbody> </table>					<u>Design Schedule</u>	<u>Start / Issue</u>	<u>Complete / Response</u>	<u>Reissue</u>	<u>Reissue Complete / Response</u>	Issue Date for TLR	N/A	N/A			Issue Date for TLS	N/A	N/A			Preliminary Design	N/A	N/A			Contract Design	N/A	N/A			Detail Design	N/A	N/A			Request for Proposals	N/A	N/A			Design Agent					<u>Classification of Cost Estimate:</u>				
<u>Design Schedule</u>	<u>Start / Issue</u>	<u>Complete / Response</u>	<u>Reissue</u>	<u>Reissue Complete / Response</u>																																													
Issue Date for TLR	N/A	N/A																																															
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Request for Proposals	N/A	N/A																																															
Design Agent																																																	
<u>Classification of Cost Estimate:</u>																																																	
Justification: FY 2022 requests funding for 5 used sealift vessel in the RRF recapitalization program and program management support services.																																																	

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1		P-1 Line Item Number / Title: 5201 / Auxiliary Vessels (Used Sealift)		
Cost Categories	FY 2021		FY 2022	
	Qty (Each)	Total Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Basic Construction/Conversion	2	60.000	5	299.900
Total Ship Estimate		60.000		299.900
Net P-1 Funding		60.000		299.900

Remarks:
FY 2022 requests funding for 5 used sealift vessel in the RRF recapitalization program and program management support services.

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Exhibit P-27, Ship Production Schedule: PB 2022 Navy					Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1			P-1 Line Item Number / Title: 5201 / Auxiliary Vessels (Used Sealift)		
Ship	Shipbuilder	Fiscal Year	Contract Award	Start of Construction	Delivery Date
AUX 2101	TBD	2021	Dec 2021	Dec 2021	Dec 2021
AUX 2102	TBD	2021	Dec 2021	Dec 2021	Dec 2021
AUX 2201	TBD	2022	Mar 2022	Mar 2022	Mar 2022
AUX 2202	TBD	2022	May 2022	May 2022	May 2022
AUX 2203	TBD	2022	May 2022	May 2022	May 2022
AUX 2204	TBD	2022	Jul 2022	Jul 2022	Jul 2022
AUX 2205	TBD	2022	Jul 2022	Jul 2022	Jul 2022

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy								Date: May 2021				
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost						P-1 Line Item Number / Title: 5300 / Completion of PY Shpbldg Progr						
ID Code (A=Service Ready, B=Not Service Ready): A			Program Elements for Code B Items: N/A					Other Related Program Elements: N/A				
Line Item MDAP/MAIS Code: N/A												
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity <i>(Units in Each)</i>	-	-	-	-	-	-	-	-	-	-	-	-
Gross/Weapon System Cost <i>(\$ in Millions)</i>	0.000	0.000	407.312	660.795	0.000	660.795	-	-	-	-	-	-
Less PY Advance Procurement <i>(\$ in Millions)</i>	-	-	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) <i>(\$ in Millions)</i>	0.000	0.000	407.312	660.795	0.000	660.795	-	-	-	-	-	-
Plus CY Advance Procurement <i>(\$ in Millions)</i>	-	-	-	-	-	-	-	-	-	-	-	-
LPD 17 Class <i>(\$ in Millions)</i>	-	-	30.578	53.682	-	53.682	-	-	-	-	-	-
LPD 17 FLT II <i>(\$ in Millions)</i>	-	0.000	-	-	-	-	-	-	-	-	-	-
ESB <i>(\$ in Millions)</i>	-	0.000	-	-	-	-	-	-	-	-	-	-
SSN <i>(\$ in Millions)</i>	-	0.000	-	-	-	-	-	-	-	-	-	-
LCS <i>(\$ in Millions)</i>	-	-	-	44.860	-	44.860	-	-	-	-	-	-
CVN <i>(\$ in Millions)</i>	-	-	71.000	291.000	-	291.000	-	-	-	-	-	-
CVN RCOH <i>(\$ in Millions)</i>	-	-	186.200	158.800	-	158.800	-	-	-	-	-	-
EPF <i>(\$ in Millions)</i>	-	-	50.000	-	-	-	-	-	-	-	-	-
DDG-51 <i>(\$ in Millions)</i>	-	-	9.634	45.753	-	45.753	-	-	-	-	-	-
LHA <i>(\$ in Millions)</i>	-	0.000	-	-	-	-	-	-	-	-	-	-
TAO Fleet Oiler <i>(\$ in Millions)</i>	-	-	59.900	66.700	-	66.700	-	-	-	-	-	-
Total Obligation Authority <i>(\$ in Millions)</i>	0.000	0.000	407.312	660.795	0.000	660.795	-	-	-	-	-	-
(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)												
Initial Spares <i>(\$ in Millions)</i>	-	-	-	-	-	-	-	-	-	-	-	-
Flyaway Unit Cost <i>(\$ in Millions)</i>	-	-	-	-	-	-	-	-	-	-	-	-
Total <i>(\$ in Millions)</i>	-	-	407.312	660.795	-	660.795	-	-	-	-	-	-
Gross/Weapon System Unit Cost <i>(\$ in Millions)</i>	-	-	-	-	-	-	-	-	-	-	-	-
Description: Note: General Provision Section 8073 of the Consolidated Appropriations Act, 2021 directs that funds appropriated for the Completion of Prior Year Shipbuilding Programs be merged with and available for the same purposes as the appropriation to which transferred. While FY 2021 funding is shown in this line item, funding was transferred and merged per the cited General Provision 8073. [P5 / [2127] Littoral Combat Ship (LCS)]: Funds in FY 2022 are for the Government responsible portion of the shipbuilding construction contract overrun for LCS 23, LCS 25, and LCS 26 (\$24.9M), and for the proper pricing of LCS 27, LCS 28, and LCS 30 (\$20.0M). [P5 / [2001] CVN - Carrier Replacement]: Funds in FY 2022 are for the Government responsible portion of the CVN 79 shipbuilding construction contract overrun (\$119.7M), proper pricing of CVN 79 Government Furnished Equipment (GFE) systems and Government test team (\$63.1M), to incorporate Joint Strike Fighter (JSF) into CVN 79 as mandated by the FY 2020 National Defense Authorization Act Section 124 (\$46.8M). Also includes CVN 79 single phase delivery, which includes integration and installation of GFE systems previously identified as Phase II systems and time related services (\$61.4M).												

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost		P-1 Line Item Number / Title: 5300 / Completion of PY Shpbldg Progr
ID Code (A=Service Ready, B=Not Service Ready): A	Program Elements for Code B Items: N/A	Other Related Program Elements: N/A
Line Item MDAP/MAIS Code: N/A		
[P5 / [2086] CVN RCOH]: Funds in FY 2022 are to finance the Government responsible portion of the shipbuilding construction contract overrun for CVN 73 RCOH (\$158.8M). The FY 2022 requirement has increased from FY 2021 previously programmed amount to account for the re-delivery date change from December 2021 to August 2022. [P5 / [2122] DDG-51]: Funds in FY 2022 are for the Government responsible portion of shipbuilding construction contract overrun for DDG 121/122 (\$42.2M), bridge system modifications for DDG 121/122 (\$2.4M) and bridge system modifications for DDG 123 (\$1.2M). [P5 / [3036] LPD]: Funds in FY 2022 are for the addition of the Enterprise Air Surveillance Radar (EASR) to the LPD 29 (\$43.3M), and Contract Value Change increases for the LPD 29 Ship Self Defense System (\$10.4M). [P5 / [5025] TAO Fleet Oiler]: Funds in FY 2022 are for the Government responsible portion of the shipbuilding construction contract overrun going to ceiling cost and the Economic Price Adjustment (EPA) bills for T-AO 205 (\$16.9M), increases for Change Orders for cyber security updates not provided for in the original shipbuilding construction contract for T-AO 205 (\$2.7M), and increases to both Electronics (\$2.6M) and HM&E (\$1.2M) Government Furnished Equipment for T-AO 205. Funds in FY 2022 are also for the Government responsible portion of the T-AO 206 shipbuilding construction contract overrun going to ceiling cost (\$16.7M), and the Economic Price Adjustment (EPA) bills for T-AO 206 (\$22.6M), increases for Change Orders for cyber security updates not provided in the original shipbuilding construction contract award for T-AO 206 (\$1.0M), and increases to both Electronics (\$2.0M) and H,M&E (\$1.0M) Government Furnished Equipment for T-AO 206.		

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy								Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost						P-1 Line Item Number / Title: 5300 / Completion of PY Shpbldg Progr				
ID Code (A=Service Ready, B=Not Service Ready): A				Program Elements for Code B Items: N/A			Other Related Program Elements: N/A			
Line Item MDAP/MAIS Code: N/A										
Exhibits Schedule					Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total
Exhibit Type	Title*	Subexhibits	ID CD	MDAP/MAIS Code	Quantity / Total Cost (Each) / (\$ M)	Quantity / Total Cost (Each) / (\$ M)	Quantity / Total Cost (Each) / (\$ M)	Quantity / Total Cost (Each) / (\$ M)	Quantity / Total Cost (Each) / (\$ M)	Quantity / Total Cost (Each) / (\$ M)
P-5	Ship Estimate				- / 0.000	- / 0.000	- / 407.312	- / 660.795	- / 0.000	- / 660.795
P-40	Total Gross/Weapon System Cost				- / 0.000	- / 0.000	- / 407.312	- / 660.795	- / 0.000	- / 660.795
*Title represents 1) the Number / Title for Items; 2) the Number / Title [DODIC] for Ammunition; and/or 3) the Number / Title (Modification Type) for Modifications.										
Note: Totals in this Exhibit P-40 set may not be exact or sum exactly due to rounding.										

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Exhibit P-5, Cost Analysis: PB 2022 Navy												Date: May 2021						
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1						P-1 Line Item Number / Title: 5300 / Completion of PY Shpbldg Progr						Item Number / Title [DODIC]: Ship Estimate						
ID Code (A=Service Ready, B=Not Service Ready) :										MDAP/MAIS Code:								
Resource Summary				Prior Years		FY 2020		FY 2021		FY 2022 Base		FY 2022 OCO		FY 2022 Total				
Procurement Quantity <i>(Units in Each)</i>				-		-		-		-		-		-		-		
Gross/Weapon System Cost <i>(\$ in Millions)</i>				0.000		0.000		407.312		660.795		0.000		660.795				
Less PY Advance Procurement <i>(\$ in Millions)</i>				-		-		-		-		-		-				
Net Procurement (P-1) <i>(\$ in Millions)</i>				0.000		0.000		407.312		660.795		0.000		660.795				
Plus CY Advance Procurement <i>(\$ in Millions)</i>				-		-		-		-		-		-				
Total Obligation Authority <i>(\$ in Millions)</i>				0.000		0.000		407.312		660.795		0.000		660.795				
(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)																		
Initial Spares <i>(\$ in Millions)</i>				-		-		-		-		-		-		-		
Gross/Weapon System Unit Cost <i>(\$ in Millions)</i>				-		-		-		-		-		-		-		
Note: Subtotals or Totals in this Exhibit P-5 may not be exact or sum exactly due to rounding.																		
Cost Elements	Prior Years			FY 2020			FY 2021			FY 2022 Base			FY 2022 OCO			FY 2022 Total		
	Unit Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Unit Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Unit Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Unit Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Unit Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Unit Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
Completion of PY Shipbuilding Programs - [3043] Expeditionary Fast Transport (EPF) Cost																		
1.1) Unmanned logistics prototype conversion of EPF 13	-	-	-	-	-	-	-	-	50.000	-	-	-	-	-	-	-	-	-
Subtotal: Completion of PY Shipbuilding Programs - [3043] Expeditionary Fast Transport (EPF) Cost	-	-	-	-	-	-	-	-	50.000	-	-	-	-	-	-	-	-	-
Completion of PY Shipbuilding Programs - [2127] Littoral Combat Ship (LCS) Cost																		
3.1) Government responsible portion of shipbuilding contract overrun for LCS 23, 25, 26	-	-	-	-	-	-	-	-	-	-	-	24.860	-	-	-	-	-	24.860
3.2) Proper pricing of LCS 27, LCS 28, and LCS 30	-	-	-	-	-	-	-	-	-	-	-	20.000	-	-	-	-	-	20.000
Subtotal: Completion of PY Shipbuilding Programs - [2127] Littoral Combat Ship (LCS) Cost	-	-	-	-	-	-	-	-	-	-	-	44.860	-	-	-	-	-	44.860
Completion of PY Shipbuilding Programs - [2001] CVN - Carrier Replacement Cost																		
4.1) CVN 78 Advanced Weapons Elevator efforts, Degaussing corrections, trial card resolution	-	-	-	-	-	-	-	-	71.000	-	-	-	-	-	-	-	-	-
4.2) Govt responsible portion of shpbldg	-	-	-	-	-	-	-	-	-	-	-	119.746	-	-	-	-	-	119.746

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Exhibit P-5, Cost Analysis: PB 2022 Navy												Date: May 2021						
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1						P-1 Line Item Number / Title: 5300 / Completion of PY Shpbldg Progr						Item Number / Title [DODIC]: Ship Estimate						
ID Code (A=Service Ready, B=Not Service Ready) :										MDAP/MAIS Code:								
Note: Subtotals or Totals in this Exhibit P-5 may not be exact or sum exactly due to rounding.																		
Cost Elements	Prior Years			FY 2020			FY 2021			FY 2022 Base			FY 2022 OCO			FY 2022 Total		
	Unit Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Unit Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Unit Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Unit Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Unit Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Unit Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
contract overrun for CVN 79																		
4.3) CVN 79 properly priced GFE systems and Government Test Team	-	-	-	-	-	-	-	-	-	-	-	63.060	-	-	-	-	-	63.060
4.4) Incorporate Joint Strike Fighter into CVN 79 as mandated by the FY 2020 NDAA	-	-	-	-	-	-	-	-	-	-	-	46.800	-	-	-	-	-	46.800
4.5) CVN79 Single Phase DLV integ & install of previously ID'd Phase II GFE, time rlt'd svcs	-	-	-	-	-	-	-	-	-	-	-	61.394	-	-	-	-	-	61.394
Subtotal: Completion of PY Shipbuilding Programs - [2001] CVN - Carrier Replacement Cost	-	-	-	-	-	-	-	-	71.000	-	-	291.000	-	-	-	-	-	291.000
Completion of PY Shipbuilding Programs - [2086] CVN RCOH Cost																		
5.1) Fund GFE cost increases	-	-	-	-	-	-	-	-	28.000	-	-	-	-	-	-	-	-	-
5.2) Government responsible portion of the shipbuilding contract overrun for CVN 73 RCOH	-	-	-	-	-	-	-	-	158.200	-	-	158.800	-	-	-	-	-	158.800
Subtotal: Completion of PY Shipbuilding Programs - [2086] CVN RCOH Cost	-	-	-	-	-	-	-	-	186.200	-	-	158.800	-	-	-	-	-	158.800
Completion of PY Shipbuilding Programs - [2122] DDG-51 Cost																		
6.1) Govt portion of shpbldg contract overrun and Bridge Sys Mods for DDG 121/122	-	-	-	-	-	-	-	-	9.634	-	-	44.577	-	-	-	-	-	44.577
6.2) Bridge Sys Mods for DDG 123	-	-	-	-	-	-	-	-	-	-	-	1.176	-	-	-	-	-	1.176
Subtotal: Completion of PY Shipbuilding Programs - [2122] DDG-51 Cost	-	-	-	-	-	-	-	-	9.634	-	-	45.753	-	-	-	-	-	45.753
Completion of PY Shipbuilding Programs - [3036] LPD Cost																		
7.1) Avondale Deferred Restructuring Cost contract clause for LPD 28	-	-	-	-	-	-	-	-	5.700	-	-	-	-	-	-	-	-	-

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Exhibit P-5, Cost Analysis: PB 2022 Navy													Date: May 2021					
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1							P-1 Line Item Number / Title: 5300 / Completion of PY Shpbldg Progr						Item Number / Title [DODIC]: Ship Estimate					
ID Code (A=Service Ready, B=Not Service Ready) :										MDAP/MAIS Code:								
Note: Subtotals or Totals in this Exhibit P-5 may not be exact or sum exactly due to rounding.																		
Cost Elements	Prior Years			FY 2020			FY 2021			FY 2022 Base			FY 2022 OCO			FY 2022 Total		
	Unit Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Unit Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Unit Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Unit Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Unit Cost (\$ M)	Qty (Each)	Total Cost (\$ M)	Unit Cost (\$ M)	Qty (Each)	Total Cost (\$ M)
7.2) Rolling Airframe Missile and SEWIP Block 2 for LPD 28	-	-	-	-	-	-	-	-	14.578	-	-	-	-	-	-	-	-	-
7.3) Cost Increases to Ship Self Defense System for LPD 28	-	-	-	-	-	-	-	-	10.300	-	-	-	-	-	-	-	-	-
7.4) Enterprise Air Surveillance Radar (EASR) for LPD 29	-	-	-	-	-	-	-	-	-	-	-	43.300	-	-	-	-	-	43.300
7.5) Contract Value Change Increases for LPD 29 Ship Self Defense System	-	-	-	-	-	-	-	-	-	-	-	10.382	-	-	-	-	-	10.382
Subtotal: Completion of PY Shipbuilding Programs - [3036] LPD Cost	-	-	-	-	-	-	-	-	30.578	-	-	53.682	-	-	-	-	-	53.682
Completion of PY Shipbuilding Programs - [5025] TAO Fleet Oiler Cost																		
10.1) Gov. resp. portion of shipbuilding contract overrun and EPA bills for T-AO 205	-	-	-	-	-	-	-	-	41.000	-	-	16.858	-	-	-	-	-	16.858
10.2) Gov. resp. portion of shipbuilding contract overrun and EPA bills for T-AO 206	-	-	-	-	-	-	-	-	17.400	-	-	39.342	-	-	-	-	-	39.342
10.3) Cyber Security Change Orders for T-AO 205	-	-	-	-	-	-	-	-	-	-	-	2.700	-	-	-	-	-	2.700
10.4) Cyber Security Change Orders for T-AO 206	-	-	-	-	-	-	-	-	-	-	-	1.000	-	-	-	-	-	1.000
10.5) Electronics and H,M&E GFE increases for T-AO 205	-	-	-	-	-	-	-	-	1.500	-	-	3.800	-	-	-	-	-	3.800
10.6) Electronics and H,M&E GFE increases for T-AO 206	-	-	-	-	-	-	-	-	-	-	-	3.000	-	-	-	-	-	3.000
Subtotal: Completion of PY Shipbuilding Programs - [5025] TAO Fleet Oiler Cost	-	-	-	-	-	-	-	-	59.900	-	-	66.700	-	-	-	-	-	66.700
Gross/Weapon System Cost	-	-	0.000	-	-	0.000	-	-	407.312	-	-	660.795	-	-	0.000	-	-	660.795

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy										Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost							P-1 Line Item Number / Title: 5087 / Oceanographic Ships					
ID Code (A=Service Ready, B=Not Service Ready): A				Program Elements for Code B Items: N/A				Other Related Program Elements: N/A				
Line Item MDAP/MAIS Code: N/A												
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (<i>Units in Each</i>)	4	-	-	-	-	-	-	-	-	-	-	-
Gross/Weapon System Cost (<i>\$ in Millions</i>)	470.828	0.000	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Less PY Advance Procurement (<i>\$ in Millions</i>)	-	-	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (<i>\$ in Millions</i>)	470.828	0.000	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Plus CY Advance Procurement (<i>\$ in Millions</i>)	-	-	-	-	-	-	-	-	-	-	-	-
Total Obligation Authority (<i>\$ in Millions</i>)	470.828	0.000	0.000	0.000	0.000	0.000	-	-	-	-	-	-
<i>(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)</i>												
Plus Outfitting and Post Delivery (<i>\$ in Millions</i>)	14.862	-	-	-	-	-	-	-	-	-	-	-
Total (<i>\$ in Millions</i>)	485.690	-	-	-	-	-	-	-	-	-	-	-
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	117.707	-	-	-	-	-	-	-	-	-	-	-
Description: The FY 2018 Omnibus Appropriations Act included a Congressional add for Detail Design and Construction (DD&C) efforts for one T-AGS 60 vessel. T-AGS 67 is a repeat platform of T-AGS 66 and includes the Moon Pool Launch and Recovery System (MLARS). The TAGS 67 will be capable of deep ocean and coastal surveys, oceanographic sampling and data collections of surface, midwater and ocean floor parameters, shipboard oceanographic data processing and sample analysis, and operation of remotely operated vehicles (AUVs) and hydrographic survey launches (HSLs).												
Characteristics:		T-AGS										
Length Overall		353 ft										
Beam		58 ft										
Displacement		4,888 Long Tons										
Draft		19 ft										
Production Status:		T-AGS 67										
Contract Award Date		Oct 2020										
Months to Completion												
a) Award to Delivery		55 months										
b) Construction Start to Delivery		74 months										
Delivery Date		May 2025										
Completion Of Fitting Out		Aug 2025										
Obligation Work Limit Date		Jul 2026										
Design Schedule			Start / Issue		Complete / Response		Reissue		Reissue Complete / Response			
Issue Date for TLR			Aug 1993		N/A							
Issue Date for TLS			N/A		N/A							
Preliminary Design			N/A		N/A							

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost			P-1 Line Item Number / Title: 5087 / Oceanographic Ships	
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: N/A		Other Related Program Elements: N/A
Line Item MDAP/MAIS Code: N/A				
<u>Design Schedule</u>	<u>Start / Issue</u>	<u>Complete / Response</u>	<u>Reissue</u>	<u>Reissue Complete / Response</u>
Contract Design	Sep 2018	N/A		
Detail Design	Mar 2019	N/A		
Request for Proposals	Aug 2018	N/A		
Design Agent	N/A			
Classification of Cost Estimate: N/A				

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1		P-1 Line Item Number / Title: 5087 / Oceanographic Ships
Cost Categories	FY 2018	
	Qty (Each)	Total Cost (\$ M)
Plan Costs	1	
Basic Construction/Conversion		146.756
Change Orders		3.994
Electronics		19.500
Hull, Mechanical, and Electrical (HM&E)		9.382
Other Cost		
Total Ship Estimate		179.632
Net P-1 Funding		179.632

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Exhibit P-27, Ship Production Schedule: PB 2022 Navy				Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1			P-1 Line Item Number / Title: 5087 / Oceanographic Ships		
Ship	Shipbuilder	Fiscal Year	Contract Award	Start of Construction	Delivery Date
T-AGS 67	VT HALTER MARINE, INC	2018	Oct 2020	Mar 2019	May 2025

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy										Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost							P-1 Line Item Number / Title: 5092 / Moored Training Ship					
ID Code (A=Service Ready, B=Not Service Ready): A				Program Elements for Code B Items: N/A				Other Related Program Elements: N/A				
Line Item MDAP/MAIS Code: N/A												
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (<i>Units in Each</i>)	2	-	-	-	-	-	-	-	-	-	-	-
Gross/Weapon System Cost (<i>\$ in Millions</i>)	2,186.077	0.000	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Less PY Advance Procurement (<i>\$ in Millions</i>)	824.345	-	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (<i>\$ in Millions</i>)	1,361.732	0.000	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Plus CY Advance Procurement (<i>\$ in Millions</i>)	824.345	-	-	-	-	-	-	-	-	-	-	-
Total Obligation Authority (<i>\$ in Millions</i>)	2,186.077	0.000	0.000	0.000	0.000	0.000	-	-	-	-	-	-
<i>(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)</i>												
Plus Outfitting and Post Delivery (<i>\$ in Millions</i>)	30.423	4.800	-	-	-	-	-	-	-	-	-	-
Total (<i>\$ in Millions</i>)	2,216.500	4.800	-	-	-	-	-	-	-	-	-	-
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	1,093.039	-	-	-	-	-	-	-	-	-	-	-
Description: The details of this program are CONFIDENTIAL. Characteristics: MTS-711 Length Overall 433 ft Beam 33 ft Displacement 7,500 LT Draft 27 ft Production Status: MTS- 711 ⁽¹⁾ Contract Award Date May 2017 Months to Completion a) Award to Delivery 59 months b) Construction Start to Delivery 59 months Delivery Date Apr 2022 Completion Of Fitting Out Apr 2022 Obligation Work Limit Date Mar 2023												
<u>Design Schedule</u>				<u>Start / Issue</u>		<u>Complete / Response</u>		<u>Reissue</u>		<u>Reissue Complete / Response</u>		
Issue Date for TLR				N/A		N/A						
Issue Date for TLS				Apr 2008		Jan 2015						
Preliminary Design				Jan 2012		N/A						
Contract Design				Feb 2012		N/A						

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy				Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost			P-1 Line Item Number / Title: 5092 / Moored Training Ship		
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: N/A		Other Related Program Elements: N/A	
Line Item MDAP/MAIS Code: N/A					
<u>Design Schedule</u>		<u>Start / Issue</u>		<u>Complete / Response</u>	
Detail Design		Feb 2012		N/A	
Request for Proposals		N/A		N/A	
Design Agent		ELECTRIC BOAT			
<u>Classification of Cost Estimate:</u>					
Justification: The details of this program are CONFIDENTIAL.					
Footnotes: ⁽¹⁾ The details of this program are CONFIDENTIAL.					

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy		Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1		P-1 Line Item Number / Title: 5092 / Moored Training Ship	
Cost Categories	FY 2017		
	Qty (Each)	Total Cost (\$ M)	
Design	1		46.449
Plans/Conversion			382.214
GFE			31.100
Basic Construction			404.530
Total Ship Estimate			864.293
Less Advance Procurement FY 2014			37.200
Less Advance Procurement FY 2015			64.388
Less Advance Procurement FY 2016			138.200
Net P-1 Funding			624.505

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Exhibit P-27, Ship Production Schedule: PB 2022 Navy				Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1			P-1 Line Item Number / Title: 5092 / Moored Training Ship		
Ship	Shipbuilder	Fiscal Year	Contract Award	Start of Construction	Delivery Date
MTS- 711 ⁽¹⁾	EB/NNSY	2017	May 2017	May 2017	Apr 2022
Footnotes:					
⁽¹⁾ The details of this program are CONFIDENTIAL.					

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy										Date: May 2021		
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost							P-1 Line Item Number / Title: 5212 / YP Craft Maintenance/ROH/SLEP					
ID Code (A=Service Ready, B=Not Service Ready): A				Program Elements for Code B Items: N/A				Other Related Program Elements: N/A				
Line Item MDAP/MAIS Code: N/A												
Resource Summary	Prior Years	FY 2020	FY 2021	FY 2022 Base	FY 2022 OCO	FY 2022 Total	FY 2023	FY 2024	FY 2025	FY 2026	To Complete	Total
Procurement Quantity (<i>Units in Each</i>)	12	-	-	-	-	-	-	-	-	-	-	-
Gross/Weapon System Cost (<i>\$ in Millions</i>)	43.126	0.000	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Less PY Advance Procurement (<i>\$ in Millions</i>)	-	-	-	-	-	-	-	-	-	-	-	-
Net Procurement (P-1) (<i>\$ in Millions</i>)	43.126	0.000	0.000	0.000	0.000	0.000	-	-	-	-	-	-
Plus CY Advance Procurement (<i>\$ in Millions</i>)	-	-	-	-	-	-	-	-	-	-	-	-
Total Obligation Authority (<i>\$ in Millions</i>)	43.126	0.000	0.000	0.000	0.000	0.000	-	-	-	-	-	-
<i>(The following Resource Summary rows are for informational purposes only. The corresponding budget requests are documented elsewhere.)</i>												
Plus Outfitting and Post Delivery (<i>\$ in Millions</i>)	0.426	0.145	-	-	-	-	-	-	-	-	-	-
Total (<i>\$ in Millions</i>)	43.552	0.145	-	-	-	-	-	-	-	-	-	-
Gross/Weapon System Unit Cost (<i>\$ in Millions</i>)	3.594	-	-	-	-	-	-	-	-	-	-	-
Description: Naval Academy YP (Yard Patrol) craft are utilized to train midshipmen on piloting, seamanship, navigation, and engineering. The YP Service Life Extension Program (SLEP) extends the YP 676 Class service life approximately 10 years beyond the current average vessel age of 27 years. YP SLEP work items include but are not limited to the following: hull fendering, electronic navigation system components, paint and non-skid, damaged hull sections, hatches and deck planking, various pumps (bilge, seawater cooling, fire), and galley appliances. The SLEP will also include the overhaul of the engines and transformers, and propeller repair. The required repairs will vary by craft and will be conducted at both the U.S. Coast Guard Yard in Baltimore and private small business shipyards.												
Characteristics:	YP 676 Class											
Length Overall	108 ft											
Beam	24 ft											
Displacement	173 tons											
Draft	6 ft											
Production Status:	YP SLEP 683	YP SLEP 684	YP SLEP 684	YP SLEP 698	YP SLEP 698	YP SLEP 698	YP SLEP 698	YP SLEP 691	YP SLEP 691	YP SLEP 691	YP SLEP 700	YP SLEP 700
Contract Award Date	May 2020	May 2020	May 2020	Jan 2021	Jan 2021	Jan 2021	Jan 2021	Jun 2021	Jun 2021	Jun 2021	Jun 2021	Jun 2021
Months to Completion												
a) Award to Delivery	15 months	18 months	18 months	10 months	7 months	7 months	7 months	7 months	7 months	7 months	9 months	9 months
b) Construction Start to Delivery	7 months	6 months	6 months	8 months	5 months	5 months	5 months	5 months	5 months	5 months	6 months	6 months
Delivery Date	Aug 2021	Nov 2021	Nov 2021	Nov 2021	Aug 2021	Aug 2021	Aug 2021	Jan 2022	Jan 2022	Jan 2022	Mar 2022	Mar 2022
Completion Of Fitting Out	Nov 2021	Feb 2022	Feb 2022	Feb 2022	Nov 2021	Nov 2021	Nov 2021	Apr 2022	Apr 2022	Apr 2022	Jun 2022	Jun 2022
Obligation Work Limit Date	Oct 2022	Jan 2023	Jan 2023	Jan 2023	Oct 2022	Oct 2022	Oct 2022	Mar 2023	Mar 2023	Mar 2023	May 2023	May 2023
<u>Design Schedule</u>	<u>Start / Issue</u>			<u>Complete / Response</u>		<u>Reissue</u>		<u>Reissue Complete / Response</u>				
Issue Date for TLR	N/A			N/A								
Issue Date for TLS	N/A			N/A								

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Exhibit P-40, Budget Line Item Justification: PB 2022 Navy			Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N: Shipbuilding and Conversion, Navy / BA 05: Auxiliaries, Craft, and Prior-Year Program Costs / BSA 1: Auxiliaries, Craft and Prior Yr Program Cost		P-1 Line Item Number / Title: 5212 / YP Craft Maintenance/ROH/SLEP		
ID Code (A=Service Ready, B=Not Service Ready): A		Program Elements for Code B Items: N/A		Other Related Program Elements: N/A
Line Item MDAP/MAIS Code: N/A				
<u>Design Schedule</u>	<u>Start / Issue</u>	<u>Complete / Response</u>	<u>Reissue</u>	<u>Reissue Complete / Response</u>
Preliminary Design	N/A	N/A		
Contract Design	N/A	N/A		
Detail Design	N/A	N/A		
Request for Proposals	Jun 2016	Sep 2017		
Design Agent	PEO SHIPS, PMS 325 Detachment Boston			
<u>Classification of Cost Estimate:</u> N/A				

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Exhibit P-5c, Ship Cost Analysis: PB 2022 Navy		Date: May 2021
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1		P-1 Line Item Number / Title: 5212 / YP Craft Maintenance/ROH/SLEP
Cost Categories	FY 2017	
	Qty (Each)	Total Cost (\$ M)
Basic Construction/Conversion	6	17.936
Change Orders		0.469
Electronics		1.422
Hull, Mechanical, and Electrical (HM&E)		1.461
Total Ship Estimate		21.288
Net P-1 Funding		21.288

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Exhibit P-27, Ship Production Schedule: PB 2022 Navy				Date: May 2021	
Appropriation / Budget Activity / Budget Sub Activity: 1611N / 05 / 1			P-1 Line Item Number / Title: 5212 / YP Craft Maintenance/ROH/SLEP		
Ship	Shipbuilder	Fiscal Year	Contract Award	Start of Construction	Delivery Date
YP SLEP 683	USCG YARD	2017	May 2020	Jan 2021	Aug 2021
YP SLEP 684	USCG YARD	2017	May 2020	May 2021	Nov 2021
YP SLEP 698	LYON SHIPYARD/TECNICO CORP.	2017	Jan 2021	Mar 2021	Nov 2021
YP SLEP 690	LYON SHIPYARD/TECNICO CORP.	2017	Jan 2021	Mar 2021	Aug 2021
YP SLEP 691	USCG YARD	2017	Jun 2021	Aug 2021	Jan 2022
YP SLEP 700	USCG YARD	2017	Jun 2021	Sep 2021	Mar 2022